Heavy Quarkonium in Potential Models and the Gaussian Expansion Method

Lecture notes: Hamiltonians, smearing, running coupling, the variational solution as a generalized eigenvalue problem, observables, and statistical analysis

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Abstract

These notes collect, with derivations, everything needed to set up and solve the heavy quark–antiquark bound-state problem in four families of potential models: (A) non-relativistic kinetic energy with a Cornell potential and Breit–Fermi spin-dependent corrections; (B) the same with a colour-screened confining potential; (C) the relativized model of Godfrey and Isgur; (D) the screened ("modified") Godfrey–Isgur model. The logic is: two-body kinematics and angular momentum (§2) → the interquark interaction from one-gluon exchange plus confinement and its $1/m^2$ reduction, derived from the exchange amplitude (§3) → the four Hamiltonians and where each of their factors comes from (§4) → the running coupling (§5) → smearing (§6) → the Gaussian expansion method: the generalized eigenvalue problem, the Hylleraas–Undheim–MacDonald theorem, complex-range Gaussians, parameter derivatives and basis hyperparameters (§7) → observables: annihilation widths, decay constants, E1 and M1 transitions (§8) → fitting, statistics, error analysis and the optimization of the GI running coupling, with a complete worked example (§10) → implementation and tests with reference numbers (§12). Every non-trivial formula has either a derivation in the text or an explicit "derive this" pointer, and all closed-form results quoted have been checked numerically. Three practical parts follow the physics: an intuition-first statistics primer with hand-worked toy problems (§9); an implementation blueprint for the C++ program, with libraries, layout, data structures, function signatures, algorithms in pseudocode, tests and milestones (§11); and a guide to deriving and checking every result yourself in Mathematica, including an engine with which Mathematica derives every GEM matrix element from the basis definition alone (§13).

Contents

1 Road map, conventions and notation **2**

1.1 What problem are we solving? . . . 2

1.2 Units and conventions . . . 2

1.3 How to use these notes . . . 2

2 The two-body problem **3**

2.1 Separation of the centre-of-mass motion . . . 3

2.2 Angular momentum decomposition and the radial equation . . . 3

2.3 Spin, total angular momentum and the J^{PC} classification . . . 4

3 The interquark interaction **4**

3.1 Colour factor of one-gluon exchange . . . 4
3.2 Static potential from the exchange amplitude . . . 5
3.3 Confinement . . . 5
3.4 The Breit–Fermi reduction . . . 6
3.4.1 Idea . . . 6
3.4.2 The full Breit–Fermi Hamiltonian for a vector Coulomb interaction . . . 6
3.4.3 Deriving the spin-dependent terms from the exchange amplitude . . . 6
3.5 Spin–angle algebra . . . 9
3.5.1 Constructing the replacement tensor (the method of operator equivalents) . . . 9
3.5.2 Diagonal tensor elements . . . 11
3.5.3 Off-diagonal tensor element . . . 12
3.6 Worked examples: what the spin terms do . . . 13

4 The four Hamiltonians **14**
4.1 Model A: non-relativistic Cornell + Breit–Fermi . . . 14
4.2 Model B: non-relativistic + screened confinement + Breit–Fermi . . . 15
4.3 Model C: the Godfrey–Isgur relativized model . . . 15
4.3.1 Philosophy: four relativization steps . . . 15
4.3.2 What is derived and what is chosen . . . 16
4.3.3 The Hamiltonian . . . 16
4.3.4 Where the momentum-dependent factors come from . . . 17
4.3.5 Hermiticity and boundedness . . . 18
4.3.6 Parameters . . . 18
4.4 Model D: modified (screened) Godfrey–Isgur model . . . 19
4.5 Side-by-side comparison . . . 19
4.6 Further directions (not needed for the basic project) . . . 19

5 The strong coupling α_s **19**
5.1 Renormalization-group running . . . 19
5.2 Infrared-regular forms used in practice . . . 20
5.3 From $\alpha_s(Q^2)$ to $V(r)$ . . . 21
5.4 Summary . . . 21

6 Smearing **21**
6.1 What this section is for . . . 21
6.2 Definition and basic properties . . . 22
6.3 The radial convolution formula . . . 23
6.4 Toolbox: half-line Gaussian moments . . . 24
6.5 The smeared functions, derived . . . 25
6.5.1 Coulomb: $1/r$ . . . 25
6.5.2 Contact term: $\delta^3(\boldsymbol{r})$ and $\nabla^2\tilde{G}$ . . . 25
6.5.3 Derivatives for the tensor and spin–orbit terms . . . 26

6.5.4 Linear confinement: r . . . 26
6.5.5 Screened confinement: $e^{-\mu_s r}$ . . . 27
6.6 Summary of smeared building blocks . . . 29

7 The Gaussian expansion method **29**
7.1 Idea . . . 29
7.2 The basis . . . 29
7.3 Including spin: your spinless notation and the coupled basis . . . 30
7.4 From the Schrödinger equation to the generalized eigenvalue problem . . . 30
7.4.1 Galerkin projection . . . 30
7.4.2 Reduction of the three-dimensional matrix elements . . . 31
7.4.3 Equivalence with the variational principle . . . 31
7.4.4 Properties . . . 31
7.4.5 Block structure for coupled channels . . . 32
7.5 Analytic matrix elements . . . 32
7.5.1 Master integral . . . 32
7.5.2 Overlap . . . 32
7.5.3 Kinetic energy . . . 32
7.5.4 Gaussian potentials (smeared δ, GI contact term) . . . 33
7.5.5 Error-function potentials (smeared and running Coulomb) . . . 33
7.5.6 Exponential potentials (screening) . . . 33
7.5.7 Matrix elements of smeared potentials: the transfer lemma . . . 34
7.6 Momentum space . . . 35
7.6.1 Fourier transform of a Gaussian basis function . . . 35
7.6.2 Relativistic kinetic energy . . . 36
7.7 Non-local (momentum-dependent) operators of the GI model . . . 36
7.8 Solving $\mathbf{H}\boldsymbol{c} = \mathcal{E}\mathbf{S}\boldsymbol{c}$ in practice . . . 37
7.8.1 Why $\mathbf{S}$ is ill-conditioned . . . 37
7.8.2 Reduction to a standard symmetric eigenproblem . . . 37
7.8.3 Choosing $\{n_{\max}, r_1, r_{\max}\}$ . . . 37
7.9 Observables from the eigenvector . . . 37
7.10 Complex-range Gaussians . . . 38
7.10.1 Why . . . 38
7.10.2 Definition . . . 38
7.10.3 Matrix elements by analytic continuation . . . 38
7.10.4 The HUM theorem with complex-range functions . . . 39
7.10.5 A demonstration (and why hyperparameters matter) . . . 39
7.11 Parameter derivatives in a finite basis . . . 39
7.11.1 Eigenvalues (Feynman–Hellmann for Ritz values) . . . 39
7.11.2 Eigenvectors . . . 40
7.11.3 Model-specific $\partial\mathbf{H}/\partial\lambda$ . . . 40

7.12 Basis hyperparameters: properties and optimization . . . 40
7.13 Answer key in your notation $\psi_{lm} = \sum_n c_n \phi_{nlm}$ . . . 42

8 Observables **42**
8.1 Masses, multiplets and spin-averaged quantities . . . 42
8.2 Sizes and velocities . . . 43
8.3 Annihilation decays . . . 43
8.3.1 Factorization . . . 43
8.3.2 QED anchors . . . 43
8.3.3 Colour and charge factors . . . 43
8.3.4 P- and D-wave annihilation (quoted) . . . 44
8.4 Decay constants . . . 44
8.5 Electric dipole (E1) transitions . . . 45
8.5.1 The dipole operator . . . 45
8.5.2 Rate . . . 45
8.5.3 Angular reduction, step by step . . . 45
8.6 Magnetic dipole (M1) transitions . . . 46
8.7 S–D mixing . . . 46
8.8 Beyond this section . . . 46

9 A statistics primer: the ideas behind the fitting chapter **46**
9.1 The mindset . . . 47
9.2 Random variables: the five rules you need . . . 47
9.3 The Gaussian, and why it is everywhere . . . 48
9.4 Likelihood: turning the probability around . . . 49
9.5 What "$\pm 1\sigma$" means . . . 49
9.6 Toy 2: a straight line by hand . . . 50
9.7 The geometry of least squares, and why the degrees of freedom are $N - p$ . . . 51
9.8 The χ^2 distribution: which values are normal? . . . 51
9.9 Theory floor versus scale factor: two models of what went wrong . . . 52
9.10 Fitting as walking downhill . . . 53
9.11 Beyond the parabola: profiles, bootstrap, Markov chains . . . 53
9.12 Comparing models without fooling yourself . . . 54
9.13 The error budget as a story . . . 54
9.14 Ten common mistakes . . . 55

10 Fitting, statistics and error analysis **55**
10.1 What is fitted, and to what . . . 55
10.2 From the likelihood to χ^2 . . . 56
10.2.1 Independent Gaussian errors . . . 56
10.2.2 Correlated errors . . . 57
10.2.3 Why a theory floor is unavoidable . . . 57

10.3 Finding the minimum . . . 59
10.3.1 Gradient and Hessian of χ^2 . . . 59
10.3.2 Gauss–Newton step . . . 59
10.3.3 Levenberg–Marquardt . . . 59
10.3.4 The Jacobian from Feynman–Hellmann . . . 59
10.3.5 Practicalities . . . 60
10.4 Parameter uncertainties from the curvature . . . 60
10.4.1 Derivation 1: Gaussian approximation to the likelihood . . . 60
10.4.2 Derivation 2: propagation of the data noise . . . 60
10.4.3 Why $\Delta\chi^2 = 1$: the profile of one parameter . . . 61
10.4.4 Correlations and degeneracies . . . 61
10.5 Propagating to predictions . . . 62
10.6 Beyond the quadratic approximation . . . 62
10.6.1 Profile likelihood for a derived quantity . . . 62
10.6.2 Parametric bootstrap . . . 62
10.6.3 Bayesian inference and Markov chain Monte Carlo . . . 63
10.7 Model comparison and validation . . . 64
10.7.1 Goodness of fit . . . 64
10.7.2 Nested models: likelihood-ratio test . . . 64
10.7.3 Information criteria . . . 64
10.7.4 Leave-one-out cross-validation . . . 65
10.8 Error budget . . . 65
10.9 Optimizing the running coupling of the GI model . . . 67
10.9.1 What the spectrum sees: the sensitivity kernel . . . 67
10.9.2 The infrared degeneracy: α_s at small Q versus the string tension . . . 68
10.9.3 Parametrization, constraints and label switching . . . 68
10.9.4 Penalties = priors . . . 69
10.9.5 Derivatives for the Jacobian . . . 69
10.9.6 Uncertainty band for the fitted coupling . . . 70
10.9.7 Alternative: fit a few physical parameters, use Gaussians only numerically . . . 70
10.9.8 Recommended protocol . . . 70
10.10 Worked example: Model A fitted to charmonium . . . 70
10.11 The complete workflow . . . 72

11 Implementation blueprint: the C++ program **72**
11.1 What the program must do . . . 73
11.2 Architecture . . . 73
11.3 Toolchain and libraries . . . 74
11.4 Repository layout and build . . . 75
11.5 Conventions . . . 76
11.6 From your current code to the new layout . . . 77

11.7 Core data structures . . . 77
11.8 Function catalogue . . . 79
11.9 Algorithms . . . 81
11.10 Verified implementation details . . . 85
11.11 Testing strategy . . . 88
11.12 Performance and parallelism . . . 89
11.13 Milestones . . . 89

12 Tests and reference numbers **90**
12.1 Unit tests that do not depend on the physics model . . . 90
12.2 Reference numbers for testing your code . . . 90
12.3 Literature parameter sets (recalled from memory — verify before use) . . . 92

13 Deriving and checking everything yourself in Mathematica **93**
13.1 Organizing the notebooks . . . 93
13.2 The verification toolkit . . . 94
13.3 Functions by task . . . 95
13.4 Recipes, chapter by chapter . . . 95
13.4.1 Units and the two-body problem (§1, §2) . . . 95
13.4.2 Spin and angular momentum (§3.5) . . . 96
13.4.3 Colour (§3.1, §8) . . . 96
13.4.4 Fourier transforms (§3.2, §5) . . . 97
13.4.5 Breit–Fermi (§3.4.3) . . . 97
13.4.6 Running coupling (§5) . . . 97
13.4.7 Smearing (§6) . . . 98
13.4.8 GEM: analytic matrix elements (§7.5, §7.6, §7.10) . . . 98
13.4.9 GEM: numerics, HUM and Feynman–Hellmann (§7.4–§7.11) . . . 99
13.4.10 Observables (§8) . . . 99
13.4.11 Statistics (§9, §10) . . . 99
13.4.12 Golden numbers and generated code . . . 100
13.5 Letting Mathematica derive the matrix elements itself . . . 100
13.5.1 The engine . . . 100
13.5.2 Closed forms for every l at once . . . 101
13.5.3 Comparing with the notes . . . 101
13.5.4 Potentials whose closed form depends on l . . . 102
13.5.5 Different l in bra and ket . . . 102
13.5.6 Spin-dependent matrix elements, derived end to end . . . 102
13.5.7 Momentum space: relativistic kinetic energy and GI factors . . . 103
13.5.8 Complex-range functions . . . 103
13.5.9 Closing the loop: numbers, C++, and back . . . 104
13.5.10 The generator: all kernels of Models A–D, derived, checked and written as C++ . . . 105

13.6 Pitfalls (each one met while preparing this section) 108

A Special functions and conventions **109**

B Table of integrals used **110**

C Addendum (v1.0.0): grading, decays and Bayesian inference **110**

References **111**

1 Road map, conventions and notation

1.1 What problem are we solving?

A heavy quark Q of mass m_1 and a heavy antiquark $\bar{Q}'$ of mass m_2 (c or b) bind into a colour-singlet meson. Because $m_Q \gg \Lambda_{\rm QCD}$, the relative velocity is small, $\langle v^2 \rangle \sim 0.2$–$0.3$ for $c\bar{c}$ and ~ 0.1 for $b\bar{b}$, so a description by a *Hamiltonian with an instantaneous potential* is a reasonable starting point. The meson mass is obtained as an eigenvalue,

$$H\,\Psi = M\,\Psi, \tag{1.1}$$

and all the physics is in the choice of H. The four families of H differ in (i) the kinetic energy (non-relativistic vs. $\sqrt{p^2+m^2}$), (ii) the long-distance potential (linear vs. screened), (iii) how relativistic and spin effects are included (Breit–Fermi $1/m^2$ terms vs. smeared, momentum-dependent "relativized" terms), and (iv) the treatment of the strong coupling. The Gaussian expansion method (GEM) turns any of these eigenvalue problems into a finite matrix problem $\mathbf{H}\boldsymbol{c} = M\mathbf{S}\boldsymbol{c}$.

1.2 Units and conventions

- Natural units $\hbar = c = 1$. Energies and masses in GeV, lengths in GeV^{-1}.
- Conversion: $\hbar c = 0.1973269804\,\mathrm{GeV\,fm}$, so $1\,\mathrm{GeV}^{-1} = 0.19733\,\mathrm{fm}$ and $1\,\mathrm{fm} = 5.0677\,\mathrm{GeV}^{-1}$.
- Quark masses m_1 (quark) and m_2 (antiquark); reduced mass $m_r = \frac{m_1 m_2}{m_1+m_2}$. For quarkonium $m_1 = m_2 = m$ and $m_r = m/2$.
- Relative coordinate $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$, $r = |\boldsymbol{r}|$, $\hat{r} = \boldsymbol{r}/r$; relative momentum $\boldsymbol{p}$ (defined in §2.1); $E_i(p) = \sqrt{p^2 + m_i^2}$.
- Spins $\boldsymbol{S}_i = \boldsymbol{\sigma}_i/2$, $\boldsymbol{S} = \boldsymbol{S}_1 + \boldsymbol{S}_2$; orbital $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$; total $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$.
- Spectroscopic notation $n\,^{2S+1}L_J$ with $n = 1, 2, \ldots$ (the radial wave function of level n has $n-1$ nodes), $L = S, P, D, F, \ldots$
- Primes on radial functions denote $\mathrm{d}/\mathrm{d}r$. For a radial function, $\nabla^2 f = f'' + \frac{2}{r}f' = \frac{1}{r^2}(r^2 f')'$.
- μ_s always denotes the *screening mass*; it is never the reduced mass.
- Colour generators $\boldsymbol{F}_i$ ($F^a = \lambda^a/2$ for a quark, $-\lambda^{a*}/2$ for an antiquark).

1.3 How to use these notes

Sections 2–6 are physics; §7 is the numerical method; §8 turns eigenvectors into measurable quantities. §9 explains the statistical ideas on toy problems, and §10 is how parameters are determined from data and how every error is estimated. §11 is the design of the program you will write, §12 the tests and reference numbers it must reproduce, and §13 how to check every derivation in Mathematica. Boxed "Key result" panels contain the formulas you will actually implement. "Exercise" items are derivations deliberately left as small steps; each has its answer stated in the text so you can compare.

2 The two-body problem

2.1 Separation of the centre-of-mass motion

Non-relativistic case. Start from

$$H = \frac{\boldsymbol{p}_1^2}{2m_1} + \frac{\boldsymbol{p}_2^2}{2m_2} + V(\boldsymbol{r}_1 - \boldsymbol{r}_2). \tag{2.1}$$

Define centre-of-mass and relative coordinates and their conjugate momenta,

$$\boldsymbol{R} = \frac{m_1\boldsymbol{r}_1 + m_2\boldsymbol{r}_2}{M_0}, \quad \boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2, \qquad \boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2, \quad \boldsymbol{p} = \frac{m_2\boldsymbol{p}_1 - m_1\boldsymbol{p}_2}{M_0}, \qquad M_0 = m_1 + m_2. \tag{2.2}$$

One checks $[R_i, P_j] = [r_i, p_j] = \mathrm{i}\delta_{ij}$ and $[R_i, p_j] = [r_i, P_j] = 0$, so the transformation is canonical. Inverting, $\boldsymbol{p}_1 = \frac{m_1}{M_0}\boldsymbol{P} + \boldsymbol{p}$, $\boldsymbol{p}_2 = \frac{m_2}{M_0}\boldsymbol{P} - \boldsymbol{p}$, hence

$$\frac{\boldsymbol{p}_1^2}{2m_1} + \frac{\boldsymbol{p}_2^2}{2m_2} = \Big[\frac{m_1\boldsymbol{P}^2}{2M_0^2} + \frac{\boldsymbol{P}\cdot\boldsymbol{p}}{M_0} + \frac{\boldsymbol{p}^2}{2m_1}\Big] + \Big[\frac{m_2\boldsymbol{P}^2}{2M_0^2} - \frac{\boldsymbol{P}\cdot\boldsymbol{p}}{M_0} + \frac{\boldsymbol{p}^2}{2m_2}\Big] = \frac{\boldsymbol{P}^2}{2M_0} + \frac{\boldsymbol{p}^2}{2m_r}. \tag{2.3}$$

The cross terms cancel; the Hamiltonian splits into free motion of the centre of mass and the relative problem

$$H_{\mathrm{rel}} = \frac{\boldsymbol{p}^2}{2m_r} + V(\boldsymbol{r}), \qquad M = m_1 + m_2 + E, \qquad H_{\mathrm{rel}}\psi = E\psi. \tag{2.4}$$

Relativistic kinetic energy. With $E_1 + E_2 = \sqrt{\boldsymbol{p}_1^2 + m_1^2} + \sqrt{\boldsymbol{p}_2^2 + m_2^2}$ the separation is no longer exact. We always work in the meson rest frame, $\boldsymbol{P} = 0$, so $\boldsymbol{p}_1 = -\boldsymbol{p}_2 = \boldsymbol{p}$ and

$$H = E_1(p) + E_2(p) + V, \qquad H\Psi = M\Psi. \tag{2.5}$$

This is the *spinless Salpeter* form; the eigenvalue is the meson mass itself. The non-relativistic limit follows from $E_i = m_i + p^2/2m_i - p^4/8m_i^3 + \dots$.

2.2 Angular momentum decomposition and the radial equation

In spherical coordinates

$$\boldsymbol{p}^2 = -\nabla^2 = -\frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2\frac{\partial}{\partial r}\Big) + \frac{\boldsymbol{L}^2}{r^2}. \tag{2.6}$$

For a central $V(r)$, $[H, \boldsymbol{L}^2] = [H, L_z] = 0$, so $\psi(\boldsymbol{r}) = R_{nl}(r)Y_{lm}(\hat{r})$ and $\boldsymbol{L}^2 Y_{lm} = l(l+1)Y_{lm}$. Then

$$-\frac{1}{2m_r}\Big(R'' + \frac{2}{r}R' - \frac{l(l+1)}{r^2}R\Big) + V(r)R = E\,R. \tag{2.7}$$

With $u(r) = rR(r)$ this becomes the one-dimensional problem

$$-\frac{u''}{2m_r} + \Big[V(r) + \frac{l(l+1)}{2m_r r^2}\Big]u = E\,u, \qquad u(0) = 0, \qquad \int_0^\infty u^2\,\mathrm{d}r = \int_0^\infty r^2R^2\,\mathrm{d}r = 1. \tag{2.8}$$

Exercise 2.1 (Behaviour at the origin)**.** Assume $r^2V(r) \to 0$ as $r \to 0$ (true for $1/r$, r, smeared potentials). Insert $u \sim r^s$ into (2.8); the leading power gives $s(s-1) = l(l+1)$, so $s = l+1$ (regular) or $s = -l$ (irregular, excluded). Hence $R_{nl}(r) \propto r^l$ near the origin. *Consequence for GEM:* the basis functions $r^l\mathrm{e}^{-\nu r^2}$ have the correct small-r behaviour built in.

Remark 2.2. Only $l = 0$ states have $R(0) \neq 0$. Quantities like leptonic widths ($\propto |R_{nS}(0)|^2$), the contact hyperfine splitting ($\propto |R(0)|^2$) and $E1/M1$ transitions are all moments of R; the expansion in Gaussians makes all of them analytic (§7.9, §8).

2.3 Spin, total angular momentum and the J^{PC} classification

Coupled basis. The spin states of two spin-$\frac{1}{2}$ particles are χ_{SM_S} with $S=0$ (singlet) or $S=1$ (triplet). The eigenstates of $\boldsymbol{J}^2, J_z, \boldsymbol{L}^2, \boldsymbol{S}^2$ are

$$\mathcal{Y}^{JM}_{LS}(\hat{r}) \equiv \big[Y_L(\hat{r}) \otimes \chi_S\big]_{JM} = \sum_{M_L,M_S} \langle LM_L; SM_S|JM\rangle\, Y_{LM_L}(\hat{r})\, \chi_{SM_S}, \tag{2.9}$$

with Clebsch–Gordan coefficients in the Condon–Shortley convention and coupling order $L \otimes S$. These are orthonormal on the sphere $\times$ spin space.

Discrete symmetries. For a fermion–antifermion pair,

$$P = (-1)^{L+1}, \qquad C = (-1)^{L+S} \quad \text{(only for } m_1 = m_2, \text{ i.e. a true } Q\bar{Q}\text{)}. \tag{2.10}$$

The intrinsic parity of a fermion–antifermion pair is -1; C follows from exchanging the particles (orbital $(-1)^L$, spin $(-1)^{S+1}$, fermion exchange -1).

Result 2.3 (Which channels can mix)**.** *The Hamiltonian commutes with $\boldsymbol{J}^2$, J_z, P and (for $m_1 = m_2$) C. Fix J:*

- *$S=0$, $L=J$: $P=(-1)^{J+1}$, $C=(-1)^J$.*
- *$S=1$, $L=J$: $P=(-1)^{J+1}$, $C=(-1)^{J+1}$.*
- *$S=1$, $L=J\pm1$: $P=(-1)^J$, $C=(-1)^J$ (both values of L have the same P and C).*

Therefore in quarkonium the only possible mixing is ${}^3(J-1)_J \leftrightarrow {}^3(J+1)_J$ (e.g. ${}^3S_1$–${}^3D_1$, ${}^3P_2$–${}^3F_2$), induced by the tensor force. For $m_1 \neq m_2$ (B_c, D, B) C is not a symmetry and ${}^1L_L \leftrightarrow {}^3L_L$ also mix (through the antisymmetric spin–orbit term, §3.5).

J^{PC}	channels (${}^{2S+1}L_J$)	charmonium example	bottomonium example
0^{-+}	${}^1S_0$	$\eta_c(1S), \eta_c(2S)$	$\eta_b(1S)$
1^{--}	${}^3S_1$, ${}^3D_1$	J/ψ, $\psi(2S)$, $\psi(3770)$	$\Upsilon(nS)$
1^{+-}	${}^1P_1$	$h_c(1P)$	$h_b(1P)$
$0^{++}, 1^{++}, 2^{++}$	${}^3P_0$, ${}^3P_1$, ${}^3P_2$ $(+{}^3F_2)$	$\chi_{c0,1,2}(1P)$	$\chi_{b0,1,2}(1P)$
2^{-+}	${}^1D_2$		
2^{--}, 3^{--}	${}^3D_2$; ${}^3D_3$ $(+{}^3G_3)$	$\psi_2(3823)$	$\Upsilon_2(1D)$

Quantum numbers $0^{--}, 0^{+-}, 1^{-+}, 2^{+-}, \ldots$ cannot be made from $Q\bar{Q}$ ("exotic").

Spin-independent vs. spin-dependent problem. If H contains no spin operators, $\Psi = R_{nL}Y_{LM}\chi_{SM_S}$ and all states of a given (n,L) are degenerate. The spin-dependent terms (§3.4) lift this degeneracy: fine structure (J splitting within ${}^3L_J$) and hyperfine structure (${}^1L$ vs. ${}^3L$).

3 The interquark interaction

3.1 Colour factor of one-gluon exchange

The quark–gluon vertex carries a colour matrix F^a; one-gluon exchange between particles 1 and 2 is therefore proportional to $\boldsymbol{F}_1\cdot\boldsymbol{F}_2 = \sum_a F_1^a F_2^a$. Use

$$\boldsymbol{F}_1\cdot\boldsymbol{F}_2 = \tfrac{1}{2}\Big[(\boldsymbol{F}_1+\boldsymbol{F}_2)^2 - \boldsymbol{F}_1^2 - \boldsymbol{F}_2^2\Big], \qquad \boldsymbol{F}_1^2 = \boldsymbol{F}_2^2 = C_F = \frac{N_c^2-1}{2N_c} = \frac{4}{3}. \tag{3.1}$$

In the colour singlet, $(\boldsymbol{F}_1+\boldsymbol{F}_2)^2=0$, so

$$\boxed{\langle \boldsymbol{F}_1\cdot\boldsymbol{F}_2\rangle_{\mathbf{1}} = -\frac{4}{3}} \qquad \text{(octet: } \tfrac{1}{2}[3-\tfrac{8}{3}] = +\tfrac{1}{6},\ \text{repulsive)}. \tag{3.2}$$

So in the singlet the interaction looks like QED with $e_1e_2 \to -\frac{4}{3}\cdot 4\pi\alpha_s$, i.e. attractive with strength $\kappa \equiv \frac{4}{3}\alpha_s$.

3.2 Static potential from the exchange amplitude

In the static limit the t-channel gluon propagator gives the momentum-space potential (momentum transfer $\boldsymbol{q}$)

$$V(\boldsymbol{q}) = \langle \boldsymbol{F}_1\cdot\boldsymbol{F}_2\rangle \frac{4\pi\alpha_s}{\boldsymbol{q}^2} = -\frac{16\pi}{3}\frac{\alpha_s}{\boldsymbol{q}^2}, \qquad V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}V(\boldsymbol{q}). \tag{3.3}$$

Exercise 3.1 (Fourier transform of $4\pi/q^2$)**.** Regularize with a photon/gluon mass: $\int \frac{\mathrm{d}^3q}{(2\pi)^3}\frac{4\pi\,\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}}{q^2+\lambda^2}$. Do the angular integral ($\int \mathrm{d}\Omega\,\mathrm{e}^{\mathrm{i}qr\cos\theta} = 4\pi\sin(qr)/(qr)$), then the q integral by residues; obtain $\mathrm{e}^{-\lambda r}/r$ and let $\lambda\to 0$: $\int \frac{\mathrm{d}^3q}{(2\pi)^3}\frac{4\pi}{q^2}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} = \frac{1}{r}$.

Hence the short-distance (colour-Coulomb) potential is

$$V_{\mathrm{OGE}}(r) = -\frac{4\alpha_s}{3r} = -\frac{\kappa}{r}. \tag{3.4}$$

If α_s depends on q^2 (running coupling, §5), the angular integral still applies and one obtains the general formula

$$V(r) = -\frac{4}{3}\cdot\frac{2}{\pi}\int_0^\infty \mathrm{d}q\,\frac{\sin qr}{qr}\,\alpha_s(q^2), \tag{3.5}$$

which reduces to $-4\alpha_s/3r$ for constant α_s because $\int_0^\infty \sin(qr)/q\,\mathrm{d}q = \pi/2$.

Remark 3.2 (Beyond tree level). At one loop, $V(q) = -\frac{16\pi}{3}\frac{\alpha_s(q^2)}{q^2}\left[1+\frac{a_1\alpha_s}{4\pi}+\dots\right]$ with $a_1 = \frac{31}{3}-\frac{10}{9}n_f$ in the $\overline{\mathrm{MS}}$ scheme. Potential models absorb such effects into an effective α_s; precision (pNRQCD) work keeps them.

3.3 Confinement

Lattice QCD without light quarks shows that the static energy rises linearly at large distance (area law of Wilson loops):

$$V_{\mathrm{conf}}(r) = b\,r + c, \qquad b \approx 0.18\text{–}0.20\,\mathrm{GeV}^2 \ (\text{“string tension”}), \tag{3.6}$$

and c is a constant (it absorbs self-energies and the zero of energy). The full *Cornell potential* is

$$V_0(r) = -\frac{4\alpha_s}{3r} + br + c. \tag{3.7}$$

Lorentz structure. For spin-dependent corrections it matters whether a potential is the time component of a Lorentz *vector* (like the gluon) or a Lorentz *scalar*. The standard phenomenological choice, supported by the observed P-wave fine structure and by lattice determinations of the spin-dependent potentials, is

$$V_V(r) = -\frac{4\alpha_s}{3r} \quad \text{(vector)}, \qquad V_S(r) = br + c \quad \text{(scalar)}. \tag{3.8}$$

A vector linear potential would produce a large positive long-range spin–orbit force $\propto 3b/(2m^2r)$ and spoil the χ_{cJ} ordering; a scalar one gives the Thomas-precession term $-b/(2m^2r)$, which partially cancels the short-range spin–orbit force.

String breaking and screening. With light quarks, at $r \gtrsim 1.2\,\mathrm{fm}$ the string can break: it becomes energetically favourable to create a light $q\bar{q}$ pair and form two heavy–light mesons. The static energy then saturates near twice the static–light meson energy. This is the physical motivation for the screened potentials of Models B and D (§4.2).

3.4 The Breit–Fermi reduction

3.4.1 Idea

Write the one-gluon-exchange (or scalar-exchange) scattering amplitude between Dirac spinors and expand each spinor to $O(p/m)$:

$$u(\boldsymbol{p},s)=\begin{pmatrix}\sqrt{E+m}\,\chi_s\\ \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{\sqrt{E+m}}\chi_s\end{pmatrix}\simeq\sqrt{2m}\begin{pmatrix}(1+\frac{p^2}{8m^2})\chi_s\\ \dfrac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{2m}\chi_s\end{pmatrix}\qquad(u^\dagger u=2E). \tag{3.9}$$

Collect all terms up to $O(1/m^2)$; Fourier transform to coordinate space. For QED this produces the *Breit Hamiltonian*; for QCD the only change at this order is the colour factor $\alpha_s \to \frac{4}{3}\alpha_s$ (for the vector part).

3.4.2 The full Breit–Fermi Hamiltonian for a vector Coulomb interaction

With $\kappa=\frac{4}{3}\alpha_s$ and the relative coordinates of §2.1:

$$\begin{aligned}
H_{\mathrm{BF}} &= m_1+m_2+\frac{p^2}{2m_r}-\frac{p^4}{8}\Big(\frac{1}{m_1^3}+\frac{1}{m_2^3}\Big)-\frac{\kappa}{r} && \text{kinetic + mass–velocity + Coulomb}\\
&+\frac{\pi\kappa}{2}\Big(\frac{1}{m_1^2}+\frac{1}{m_2^2}\Big)\delta^3(\boldsymbol{r}) && \text{Darwin}\\
&-\frac{\kappa}{2m_1m_2}\,p_i\Big(\frac{\delta_{ij}}{r}+\frac{r_ir_j}{r^3}\Big)p_j && \text{orbit–orbit (retardation)}\\
&+\frac{\kappa}{r^3}\Big[\Big(\frac{1}{2m_1^2}+\frac{1}{m_1m_2}\Big)\boldsymbol{L}\cdot\boldsymbol{S}_1+\Big(\frac{1}{2m_2^2}+\frac{1}{m_1m_2}\Big)\boldsymbol{L}\cdot\boldsymbol{S}_2\Big] && \text{spin–orbit}\\
&+\frac{8\pi\kappa}{3m_1m_2}\,\boldsymbol{S}_1\cdot\boldsymbol{S}_2\,\delta^3(\boldsymbol{r}) && \text{contact (hyperfine)}\\
&+\frac{\kappa}{m_1m_2r^3}\Big[3(\boldsymbol{S}_1\cdot\hat{r})(\boldsymbol{S}_2\cdot\hat{r})-\boldsymbol{S}_1\cdot\boldsymbol{S}_2\Big] && \text{tensor.}
\end{aligned} \tag{3.10}$$

Check against the limits you know: for $m_2\to\infty$ (hydrogen-like) only the $1/(2m_1^2)$ spin–orbit, Darwin and mass–velocity terms survive—the Dirac fine structure; for $m_1=m_2$ with $\kappa\to\alpha$ it is the positronium Breit Hamiltonian (without the annihilation term, which is absent for a colour singlet $Q\bar{Q}$ at this order since a single gluon cannot annihilate a singlet).

In potential models one usually keeps only the *spin-dependent* lines (spin–orbit, contact, tensor) and absorbs the spin-independent $O(v^2)$ terms (mass–velocity, Darwin, orbit–orbit) into the fitted parameters m_Q,α_s,b,c. The GI model (§4.3) instead mimics them by relativistic kinematics and momentum-dependent factors.

3.4.3 Deriving the spin-dependent terms from the exchange amplitude

This subsection derives (3.10) and its generalization below, step by step, from the one-boson-exchange amplitude. Everything follows from the expansion of the Dirac current to $O(1/m^2)$ and three Fourier transforms.

Step 1: three Fourier transforms. Start from

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\frac{4\pi}{q^2}=\frac{1}{r}\quad(\S3.2),\qquad \int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}q_jF(\boldsymbol{q})=-\mathrm{i}\partial_j\int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}F(\boldsymbol{q}).$$

Then:

$$\begin{aligned}&\text{(F1)}\quad \mathcal{F}\Big[\frac{4\pi}{q^2}\Big]=\frac{1}{r},\qquad \text{(F2)}\quad \mathcal{F}\Big[\frac{4\pi\,\mathrm{i}q_j}{q^2}\Big]=\partial_j\frac{1}{r}=-\frac{r_j}{r^3},\\ &\text{(F3)}\quad \mathcal{F}\Big[\frac{4\pi\,q_iq_j}{q^2}\Big]=-\partial_i\partial_j\frac{1}{r}=-\frac{3\hat r_i\hat r_j-\delta_{ij}}{r^3}+\frac{4\pi}{3}\delta_{ij}\delta^3(\boldsymbol{r}).\end{aligned}\tag{3.11}$$

Derivation of the last identity. For $r>0$ differentiate directly: $\partial_j\frac{1}{r}=-\frac{r_j}{r^3}$, $\partial_i\partial_j\frac{1}{r}=\frac{3r_ir_j-r^2\delta_{ij}}{r^5}$. This is traceless, but $\nabla^2\frac{1}{r}=-4\pi\delta^3(\boldsymbol{r})$; the missing trace sits at the origin and, by isotropy, is shared equally among the three diagonal components: $\partial_i\partial_j\frac{1}{r}=\frac{3\hat r_i\hat r_j-\delta_{ij}}{r^3}-\frac{4\pi}{3}\delta_{ij}\delta^3(\boldsymbol{r})$. For a general potential $V(r)$ (not only $1/r$) the same reasoning gives

$$\partial_i\partial_jV=\Big(\delta_{ij}-\hat r_i\hat r_j\Big)\frac{V'}{r}+\hat r_i\hat r_jV''=\frac{\delta_{ij}}{3}\nabla^2V+\Big(\hat r_i\hat r_j-\tfrac{1}{3}\delta_{ij}\Big)\Big(V''-\frac{V'}{r}\Big),\tag{3.12}$$

where any δ-function at the origin is contained in ∇^2V.

Step 2: normalized Dirac currents. For a quark scattered from $\boldsymbol{p}$ to $\boldsymbol{p}'=\boldsymbol{p}+\boldsymbol{q}$, with $u=\big(\sqrt{E+m}\,\chi,\ \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{\sqrt{E+m}}\chi\big)$ and unit-normalized states (divide by $\sqrt{2E}\sqrt{2E'}$):

$$\hat j^{\,0}\equiv\frac{\bar u(p')\gamma^0u(p)}{2\sqrt{EE'}}\simeq 1-\frac{q^2}{8m^2}+\frac{\mathrm{i}\,\boldsymbol{\sigma}\cdot(\boldsymbol{q}\times\boldsymbol{p})}{4m^2},\tag{3.13}$$

$$\hat{\boldsymbol{j}}\equiv\frac{\bar u(p')\boldsymbol{\gamma}u(p)}{2\sqrt{EE'}}\simeq\frac{(\boldsymbol{p}+\boldsymbol{p}')+\mathrm{i}\,\boldsymbol{\sigma}\times\boldsymbol{q}}{2m}.\tag{3.14}$$

Derivation. $\bar u'\gamma^0u=u'^{\dagger}u=N'N+\frac{(\boldsymbol{\sigma}\cdot\boldsymbol{p}')(\boldsymbol{\sigma}\cdot\boldsymbol{p})}{N'N}$ with $N=\sqrt{E+m}$; use $(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b})=\boldsymbol{a}\cdot\boldsymbol{b}+\mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b})$ and $\boldsymbol{p}'\times\boldsymbol{p}=\boldsymbol{q}\times\boldsymbol{p}$; expand $N'N\simeq 2m+\frac{p^2+p'^2}{4m}$ and $2\sqrt{EE'}\simeq 2m(1+\frac{p^2+p'^2}{4m^2})$; the spin-independent terms combine to $-\frac{p^2+p'^2-2\boldsymbol{p}\cdot\boldsymbol{p}'}{8m^2}=-\frac{q^2}{8m^2}$. For the spatial part, $\bar u'\gamma^iu=u'^{\dagger}\alpha^iu\simeq\chi'^{\dagger}\big[\sigma^i(\boldsymbol{\sigma}\cdot\boldsymbol{p})+(\boldsymbol{\sigma}\cdot\boldsymbol{p}')\sigma^i\big]\chi=(p+p')^i+\mathrm{i}(\boldsymbol{\sigma}\times\boldsymbol{q})^i$. The terms of (3.13) have names: 1 (charge), $-q^2/8m^2$ (Darwin), $\mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{q}\times\boldsymbol{p})/4m^2$ (spin–orbit charge density, which contains the Thomas precession); in (3.14), $(\boldsymbol{p}+\boldsymbol{p}')/2m$ (convection current) and $\mathrm{i}\boldsymbol{\sigma}\times\boldsymbol{q}/2m$ (spin-magnetization current). Particle 2 carries $-\boldsymbol{p}\to-\boldsymbol{p}'$, i.e. it absorbs $-\boldsymbol{q}$: replace $\boldsymbol{p}\to-\boldsymbol{p}$, $\boldsymbol{q}\to-\boldsymbol{q}$, $m\to m_2$, $\boldsymbol{\sigma}\to\boldsymbol{\sigma}_2$.

Step 3: the vector-exchange amplitude. In the colour singlet the exchange of a vector boson gives

$$V(\boldsymbol{q})=-\frac{4\pi\kappa}{q^2}\Big[\hat j_1^{\,0}\hat j_2^{\,0}-\hat{\boldsymbol{j}}_1\cdot\hat{\boldsymbol{j}}_2\Big]\equiv V_V(q)\Big[\hat j_1^{\,0}\hat j_2^{\,0}-\hat{\boldsymbol{j}}_1\cdot\hat{\boldsymbol{j}}_2\Big],\qquad V_V(q)=-\frac{4\pi\kappa}{q^2}.\tag{3.15}$$

(Feynman-gauge propagator. In Coulomb gauge the spatial part carries $\delta_{ij}-q_iq_j/q^2$. Because $\boldsymbol{q}\cdot(\boldsymbol{\sigma}\times\boldsymbol{q})=0$, the extra q_iq_j/q^2 piece never acts on a spin-magnetization current, so *all spin-dependent terms are the same in both gauges*; only the spin-independent orbit–orbit term differs, and the Coulomb-gauge form is the one in (3.10).)

Step 4: expand and Fourier transform, term by term. Write $\boldsymbol{P} = \boldsymbol{p} + \boldsymbol{p}' = 2\boldsymbol{p} + \boldsymbol{q}$ (so $\boldsymbol{q} \times \boldsymbol{P} = 2\boldsymbol{q} \times \boldsymbol{p}$). Then

$$\hat{j}_1^{\,0} \hat{j}_2^{\,0} \simeq 1 - \frac{q^2}{8}\Big(\frac{1}{m_1^2} + \frac{1}{m_2^2}\Big) + \mathrm{i}(\boldsymbol{q} \times \boldsymbol{p}) \cdot \Big(\frac{\boldsymbol{\sigma}_1}{4m_1^2} + \frac{\boldsymbol{\sigma}_2}{4m_2^2}\Big),$$

$$-\hat{\boldsymbol{j}}_1 \cdot \hat{\boldsymbol{j}}_2 = \frac{(\boldsymbol{P} + \mathrm{i}\boldsymbol{\sigma}_1 \times \boldsymbol{q}) \cdot (\boldsymbol{P} + \mathrm{i}\boldsymbol{\sigma}_2 \times \boldsymbol{q})}{4m_1 m_2} = \frac{P^2 + 2\mathrm{i}(\boldsymbol{q} \times \boldsymbol{p}) \cdot (\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_2) - (\boldsymbol{\sigma}_1 \times \boldsymbol{q}) \cdot (\boldsymbol{\sigma}_2 \times \boldsymbol{q})}{4m_1 m_2},$$

using $\boldsymbol{P} \cdot (\boldsymbol{\sigma} \times \boldsymbol{q}) = \boldsymbol{\sigma} \cdot (\boldsymbol{q} \times \boldsymbol{P}) = 2\boldsymbol{\sigma} \cdot (\boldsymbol{q} \times \boldsymbol{p})$. Multiply by $V_V(q)$ and transform:

- *Charge*: $V_V(q) \to V_V(r) = -\kappa/r$.
- *Darwin*: $-\frac{q^2}{8}(\frac{1}{m_1^2} + \frac{1}{m_2^2})V_V(q)$; since $\mathcal{F}[q^2 V(q)] = -\nabla^2 V(r)$, this gives $\frac{1}{8}(\frac{1}{m_1^2} + \frac{1}{m_2^2})\nabla^2 V_V = \frac{\pi\kappa}{2}(\frac{1}{m_1^2} + \frac{1}{m_2^2})\delta^3(\boldsymbol{r})$. ✓
- *Spin–orbit*: $\mathcal{F}[\mathrm{i}q_j V(q)] = \partial_j V(r) = \hat{r}_j V'(r)$, so $\mathrm{i}(\boldsymbol{q} \times \boldsymbol{p}) \cdot \boldsymbol{a}\, V(q) \to (\nabla V \times \boldsymbol{p}) \cdot \boldsymbol{a} = \frac{V'}{r}(\boldsymbol{r} \times \boldsymbol{p}) \cdot \boldsymbol{a} = \frac{V'}{r}\boldsymbol{L} \cdot \boldsymbol{a}$. With $\boldsymbol{\sigma} = 2\boldsymbol{S}$:

  $$\frac{V_V'}{r}\Big[\frac{\boldsymbol{L} \cdot \boldsymbol{S}_1}{2m_1^2} + \frac{\boldsymbol{L} \cdot \boldsymbol{S}_2}{2m_2^2} + \frac{\boldsymbol{L} \cdot (\boldsymbol{S}_1 + \boldsymbol{S}_2)}{m_1 m_2}\Big].$$

  The first two come from $\hat{j}^{\,0}$ (Thomas precession plus the spin–orbit charge density), the last from the spatial currents (colour-magnetic). For $V_V = -\kappa/r$ this is the spin–orbit line of (3.10). ✓
- *Spin–spin*: $-\frac{(\boldsymbol{\sigma}_1 \times \boldsymbol{q}) \cdot (\boldsymbol{\sigma}_2 \times \boldsymbol{q})}{4m_1 m_2} V_V(q)$ with $(\boldsymbol{\sigma}_1 \times \boldsymbol{q}) \cdot (\boldsymbol{\sigma}_2 \times \boldsymbol{q}) = q^2\, \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 - (\boldsymbol{\sigma}_1 \cdot \boldsymbol{q})(\boldsymbol{\sigma}_2 \cdot \boldsymbol{q})$. Using $\mathcal{F}[q^2 V] = -\nabla^2 V$, $\mathcal{F}[q_i q_j V] = -\partial_i \partial_j V$ and (3.12):

  $$\frac{1}{4m_1 m_2}\Big[\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2\, \nabla^2 V_V - \sigma_{1i}\sigma_{2j}\partial_i \partial_j V_V\Big] = \frac{1}{4m_1 m_2}\Big[\tfrac{2}{3}\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2\, \nabla^2 V_V + \tfrac{1}{3} S_{12}\Big(\frac{V_V'}{r} - V_V''\Big)\Big],$$

  where $(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r}) = \frac{1}{3}(S_{12} + \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2)$ was used. With $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 = 4\boldsymbol{S}_1 \cdot \boldsymbol{S}_2$ these are exactly (3.17) and (3.18). ✓
- *Orbit–orbit*: the $P^2/(4m_1 m_2)$ term (Coulomb-gauge version: $P^2 - (\boldsymbol{P} \cdot \hat{q})^2$) gives the retardation line of (3.10).

Step 5: scalar exchange. A scalar couples to $\bar{u}' u$ instead of $\bar{u}' \gamma^\mu u$. Repeating Step 2, $\frac{\bar{u}' u}{2\sqrt{EE'}} \simeq 1 - \frac{(\boldsymbol{p}+\boldsymbol{p}')^2}{8m^2} - \frac{\mathrm{i}\boldsymbol{\sigma} \cdot (\boldsymbol{q} \times \boldsymbol{p})}{4m^2}$: the spin–orbit term has the *opposite sign* and there is no spatial current. A scalar potential V_S therefore produces only $-\frac{V_S'}{2m_i^2 r}\boldsymbol{L} \cdot \boldsymbol{S}_i$ (pure Thomas precession) and no spin–spin or tensor force.

Result. Replacing $V_V(q)$ by the transform of any vector potential $V_V(r)$ and adding a scalar $V_S(r)$ gives the general Breit–Fermi spin-dependent Hamiltonian:

Breit–Fermi spin-dependent Hamiltonian

$$H_{SD} = H_{SS} + H_T + H_{LS}, \tag{3.16}$$

$$H_{SS} = \frac{2}{3m_1m_2}\,\nabla^2 V_V(r)\;\boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2, \tag{3.17}$$

$$H_T = \frac{1}{12\,m_1m_2}\Big(\frac{V_V'}{r} - V_V''\Big)S_{12}, \qquad S_{12} = 3(\boldsymbol{\sigma}_1\!\cdot\!\hat r)(\boldsymbol{\sigma}_2\!\cdot\!\hat r) - \boldsymbol{\sigma}_1\!\cdot\!\boldsymbol{\sigma}_2, \tag{3.18}$$

$$H_{LS} = \frac{1}{r}\Big[\Big(\frac{1}{2m_1^2} + \frac{1}{m_1m_2}\Big)V_V' - \frac{1}{2m_1^2}V_S'\Big]\boldsymbol{L}\!\cdot\!\boldsymbol{S}_1 + (1\leftrightarrow 2). \tag{3.19}$$

Equal masses: $H_{LS} = \dfrac{1}{2m^2 r}(3V_V' - V_S')\,\boldsymbol{L}\!\cdot\!\boldsymbol{S}$.

Exercise 3.3 (Reduce to one-gluon exchange)**.** Put $V_V = -\kappa/r$: $V_V' = \kappa/r^2$, $V_V'' = -2\kappa/r^3$, $\nabla^2(1/r) = -4\pi\delta^3(\boldsymbol{r})$. Show

$$H_{SS} = \frac{8\pi\kappa}{3m_1m_2}\delta^3(\boldsymbol{r})\,\boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2 = \frac{32\pi\alpha_s}{9m_1m_2}\delta^3(\boldsymbol{r})\,\boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2, \qquad H_T = \frac{\kappa}{4m_1m_2r^3}S_{12} = \frac{\alpha_s}{3m_1m_2r^3}S_{12}, \tag{3.20}$$

and, using $S_{12} = 4\big[3(\boldsymbol{S}_1\!\cdot\!\hat r)(\boldsymbol{S}_2\!\cdot\!\hat r) - \boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2\big]$, that this reproduces the last three lines of (3.10). With $V_S = br$: $V_S' = b$, giving the Thomas term $-\frac{b}{2m_i^2 r}\boldsymbol{L}\!\cdot\!\boldsymbol{S}_i$.

Remark 3.4 (Vector admixture in confinement). If the confinement has a vector part, $V_V = -\kappa/r + \xi br$, $V_S = (1-\xi)br + c$, then $\nabla^2 V_V \supset 2\xi b/r$ (long-range spin–spin) and $V_V' \supset \xi b$ (long-range colour-magnetic spin–orbit). Fits typically prefer $\xi \approx 0$ or slightly negative.

Remark 3.5 (Eichten–Feinberg form and the Gromes relation). The most general $O(1/m^2)$ spin-dependent potential from QCD can be written with four functions $V_1,\dots,V_4$ (Wilson-loop correlators). The spin–orbit part reads $\big(\frac{\boldsymbol{L}\cdot\boldsymbol{S}_1}{2m_1^2} + \frac{\boldsymbol{L}\cdot\boldsymbol{S}_2}{2m_2^2}\big)\frac{V_0'+2V_1'}{r} + \frac{\boldsymbol{L}\cdot(\boldsymbol{S}_1+\boldsymbol{S}_2)}{m_1m_2}\frac{V_2'}{r}$, and Lorentz invariance imposes the Gromes relation $V_0' + V_1' - V_2' = 0$. Our vector/scalar model corresponds to $V_2' = V_V'$, $V_1' = -V_S'$, and it satisfies the Gromes relation automatically.

3.5 Spin–angle algebra

All spin-dependent operators are rotational scalars. Within a fixed (L,S,J) they act as numbers (for S_{12}, possibly with $L \neq L'$ off-diagonal elements). We derive them.

Spin–spin. $\boldsymbol{S}^2 = \boldsymbol{S}_1^2 + \boldsymbol{S}_2^2 + 2\boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2$ with $\boldsymbol{S}_i^2 = \frac{3}{4}$:

$$\langle \boldsymbol{S}_1\!\cdot\!\boldsymbol{S}_2\rangle = \tfrac{1}{2}S(S+1) - \tfrac{3}{4} = \begin{cases} -\frac{3}{4}, & S=0 \\ +\frac{1}{4}, & S=1. \end{cases} \tag{3.21}$$

Spin–orbit. $\boldsymbol{J}^2 = \boldsymbol{L}^2 + \boldsymbol{S}^2 + 2\boldsymbol{L}\!\cdot\!\boldsymbol{S}$:

$$\langle \boldsymbol{L}\!\cdot\!\boldsymbol{S}\rangle = \tfrac{1}{2}\big[J(J+1) - L(L+1) - S(S+1)\big], \tag{3.22}$$

so for $S=1$: $\langle\boldsymbol{L}\!\cdot\!\boldsymbol{S}\rangle = L$ $(J=L+1)$, -1 $(J=L)$, $-(L+1)$ $(J=L-1)$. For unequal masses write $\boldsymbol{L}\!\cdot\!\boldsymbol{S}_1 = \frac{1}{2}\boldsymbol{L}\!\cdot\!(\boldsymbol{S}+\boldsymbol{S}_-)$, $\boldsymbol{L}\!\cdot\!\boldsymbol{S}_2 = \frac{1}{2}\boldsymbol{L}\!\cdot\!(\boldsymbol{S}-\boldsymbol{S}_-)$ with $\boldsymbol{S}_- = \boldsymbol{S}_1 - \boldsymbol{S}_2$. The operator $\boldsymbol{L}\!\cdot\!\boldsymbol{S}_-$ is antisymmetric in the spins and therefore connects $S=0$ and $S=1$ at $L=J$:

$$\big|\langle {}^1L_L|\,\boldsymbol{L}\!\cdot\!\boldsymbol{S}_-\,|{}^3L_L\rangle\big| = \sqrt{L(L+1)}. \tag{3.23}$$

So for $m_1 \neq m_2$, $H_{LS} = A(r)\,\boldsymbol{L}\!\cdot\!\boldsymbol{S} + B(r)\,\boldsymbol{L}\!\cdot\!\boldsymbol{S}_-$ with $A = \frac{1}{2}(a_1+a_2)$, $B = \frac{1}{2}(a_1-a_2)$, where $a_i(r)$ is the coefficient of $\boldsymbol{L}\!\cdot\!\boldsymbol{S}_i$ in (3.19); $B=0$ for quarkonium.

Tensor operator. Using $(\boldsymbol{\sigma}_a\cdot\hat{r})^2 = 1$, $(\boldsymbol{S}\cdot\hat{r})^2 = \frac{1}{2} + \frac{1}{2}(\boldsymbol{\sigma}_1\cdot\hat{r})(\boldsymbol{\sigma}_2\cdot\hat{r})$ and $\boldsymbol{S}^2 = \frac{3}{2} + \frac{1}{2}\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2$, one finds

$$S_{12} = 2\big[3(\boldsymbol{S}\cdot\hat{r})^2 - \boldsymbol{S}^2\big] = 6\,S_iS_j\Big(\hat{r}_i\hat{r}_j - \tfrac{1}{3}\delta_{ij}\Big). \tag{3.24}$$

Consequences: (i) S_{12} annihilates $S = 0$ states (both $(\boldsymbol{S}\cdot\hat{r})^2$ and $\boldsymbol{S}^2$ vanish on them); (ii) its diagonal $L = 0$ element vanishes (the angular average of $\hat{r}_i\hat{r}_j - \frac{1}{3}\delta_{ij}$ is zero); (iii) it is a rank-2 tensor in orbital space, so it connects L only to L and $L \pm 2$; (iv) it commutes with $\boldsymbol{S}^2$ and $\boldsymbol{J}$, and it is parity-even.

3.5.1 Constructing the replacement tensor (the method of operator equivalents)

Goal. The states $|(LS)JM\rangle$ are eigenstates of $\boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2$, but S_{12} contains $\hat{r}$, which is none of these. The plan is the same as for $\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}(\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2)$: rewrite the operator, *inside the fixed-L subspace*, in terms of operators whose eigenvalues we know. Let P_L project on fixed L (all $2L+1$ values of m). Because S_iS_j acts only on spin, $P_L\,S_iS_jT_{ij}\,P_L = S_iS_j\,P_LT_{ij}P_L$, so only the orbital factor $P_L(\hat{r}_i\hat{r}_j - \frac{1}{3}\delta_{ij})P_L$ must be replaced.

Step 1: the answer must be a polynomial in $\boldsymbol{L}$. On the $(2L+1)$-dimensional fixed-L space, L_z is diagonal and $L_\pm$ connect all m: the representation is irreducible. *Burnside's theorem* (a corollary of Schur's lemma): the polynomials in operators acting irreducibly on a finite-dimensional complex space span *all* operators on it. Hence $P_L\hat{r}_i\hat{r}_jP_L$ is some polynomial in L_x, L_y, L_z; the only question is which.

Step 2: classify by rotation properties. A vector operator obeys $[L_k, V_i] = \mathrm{i}\epsilon_{kil}V_l$; a two-index tensor obeys $[L_k, T_{ij}] = \mathrm{i}\epsilon_{kil}T_{lj} + \mathrm{i}\epsilon_{kjl}T_{il}$. Since $\boldsymbol{L}$ generates rotations of $\boldsymbol{r}$, $\hat{r}$ is a vector operator and $\hat{r}_i\hat{r}_j$ a two-index tensor. Every such tensor splits into irreducible parts ($9 = 1{+}3{+}5$):

$$T_{ij} = \underbrace{\tfrac{1}{3}\delta_{ij}T_{kk}}_{\text{rank 0}} + \underbrace{\tfrac{1}{2}(T_{ij} - T_{ji})}_{\text{rank 1}} + \underbrace{\tfrac{1}{2}(T_{ij} + T_{ji}) - \tfrac{1}{3}\delta_{ij}T_{kk}}_{\text{rank 2}}. \tag{3.25}$$

$\hat{r}_i\hat{r}_j - \frac{1}{3}\delta_{ij}$ is pure rank 2 (the components of $\hat{r}$ commute, so no antisymmetric part; $\hat{r}_i\hat{r}_i = 1$, so no trace). The replacement must also be rank 2.

Step 3: search by degree. Degree 0: only δ_{ij} (rank 0). Degree 1: L_i (one index) or $\epsilon_{ijk}L_k$ (antisymmetric, rank 1). Degree 2: L_iL_j is a two-index tensor ($\boldsymbol{L}$ is itself a vector operator). Decompose it:

- antisymmetric part $\frac{1}{2}[L_i, L_j] = \frac{\mathrm{i}}{2}\epsilon_{ijk}L_k$: this is $\boldsymbol{L}$ again, rank 1, discard. (In a commuting world L_iL_j would already be symmetric; quantum mechanically one must symmetrize.)
- trace $L_iL_i = \boldsymbol{L}^2 = L(L+1)$: rank 0, discard.
- what remains is the rank-2 candidate

$$Q_{ij} \equiv \tfrac{1}{2}(L_iL_j + L_jL_i) - \tfrac{1}{3}\delta_{ij}\boldsymbol{L}^2. \tag{3.26}$$

Step 4: uniqueness up to one constant (Wigner–Eckart). For any rank-2 operator with spherical components $T^{(2)}_q$,

$$\langle L\,m'|T^{(2)}_q|L\,m\rangle = \langle L\,m;\,2\,q\,|\,L\,m'\rangle\,\frac{\langle L\|T^{(2)}\|L\rangle}{\sqrt{2L+1}} \tag{3.27}$$

(Sakurai's convention). The double bars denote the *reduced matrix element*: one number, independent of m, m', q, which carries all the operator-specific information; the Clebsch–Gordan

coefficient carries all the orientation dependence and is the same for every rank-2 operator. (It is trace-like: $\sum_{m,m',q}|\langle Lm'|T_q|Lm\rangle|^2 = |\langle L\|T\|L\rangle|^2$ by orthonormality of the Clebsch–Gordan coefficients. Conventions for the $\sqrt{2L+1}$ differ between books; ratios do not.) Applying this to $[\hat{r}\otimes\hat{r}]^{(2)}$ and to $[L\otimes L]^{(2)}$, the Clebsch–Gordan coefficients cancel in the ratio of any pair of corresponding matrix elements, hence

$$P_L\Big(\hat{r}_i\hat{r}_j - \tfrac{1}{3}\delta_{ij}\Big)P_L = c_L\,Q_{ij}, \qquad c_L = \frac{\langle L\|\hat{r}\hat{r}\|L\rangle}{\langle L\|Q\|L\rangle}. \tag{3.28}$$

The Cartesian components follow from the spherical ones by the same linear combinations for both operators, e.g. $[\boldsymbol{A}\otimes\boldsymbol{B}]^{(2)}_0 = \frac{1}{\sqrt{6}}(3A_zB_z - \boldsymbol{A}\cdot\boldsymbol{B})$ gives $\sqrt{\frac{3}{2}}(\hat{r}_z^2 - \frac{1}{3})$ and $\sqrt{\frac{3}{2}}Q_{zz}$ (for $\boldsymbol{L}$ after symmetrization). Higher-degree polynomials in $\boldsymbol{L}$ give nothing new: their rank-2 parts are multiples of Q_{ij} inside the multiplet.

Step 5: fix the constant. Any single nonzero matrix element suffices; take $q=0$, $m=m'=L$ (the Clebsch–Gordan coefficient $\langle LL;20|LL\rangle = \sqrt{\frac{L(2L-1)}{(L+1)(2L+3)}}$ is nonzero for $L\geq 1$). *Left side:* $Y_{LL}\propto \sin^L\theta\,\mathrm{e}^{\mathrm{i}L\phi}$, so with $x=\cos\theta$ and Beta functions

$$\langle\cos^2\theta\rangle_{LL} = \frac{\int_{-1}^{1}x^2(1-x^2)^L\mathrm{d}x}{\int_{-1}^{1}(1-x^2)^L\mathrm{d}x} = \frac{B(\frac{3}{2},L+1)}{B(\frac{1}{2},L+1)} = \frac{\Gamma(\frac{3}{2})\Gamma(L+\frac{3}{2})}{\Gamma(\frac{1}{2})\Gamma(L+\frac{5}{2})} = \frac{1}{2L+3},$$

hence

$$\Big\langle \hat{r}_z^2 - \tfrac{1}{3}\Big\rangle_{LL} = -\frac{2L}{3(2L+3)}.$$

Right side: $\langle Q_{zz}\rangle_{LL} = L^2 - \frac{1}{3}L(L+1) = \frac{L(2L-1)}{3}$. Therefore

$$c_L = -\frac{2}{(2L-1)(2L+3)}. \tag{3.29}$$

Independent check with $L=1$, $|1,0\rangle$ ($Y_{10}\propto\cos\theta$): $\langle\cos^2\theta\rangle = \frac{2/5}{2/3} = \frac{3}{5}$, left side $\frac{3}{5}-\frac{1}{3} = \frac{4}{15}$; right side $c_1\langle Q_{zz}\rangle = (-\frac{2}{5})(0-\frac{2}{3}) = \frac{4}{15}$. ✓ (Explicit reduced matrix elements, Sakurai convention: $\langle L\|\hat{r}\hat{r}\|L\rangle = -\sqrt{2L+1}\sqrt{\frac{2}{3}}\sqrt{\frac{L(L+1)}{(2L-1)(2L+3)}}$, $\langle L\|Q\|L\rangle = \frac{1}{\sqrt{6}}\sqrt{(2L+1)L(L+1)(2L-1)(2L+3)}$; their ratio is c_L.)

Step 6: physical meaning. Classically the relative coordinate moves in the plane perpendicular to $\boldsymbol{L}$; averaging $\hat{r}$ over a circular orbit in that plane gives $\langle\hat{r}_i\hat{r}_j\rangle = \frac{1}{2}(\delta_{ij} - \hat{L}_i\hat{L}_j)$, i.e. $\langle\hat{r}_i\hat{r}_j - \frac{1}{3}\delta_{ij}\rangle = -\frac{1}{2}(\hat{L}_i\hat{L}_j - \frac{1}{3}\delta_{ij})$. For large L, $c_LQ_{ij}\to -\frac{1}{2L^2}L^2(\hat{L}_i\hat{L}_j - \frac{1}{3}\delta_{ij})$: the lemma is the exact quantum version of "the separation vector lies in the orbital plane". (This is Stevens' method of operator equivalents, the rank-2 analogue of the projection theorem $\langle\boldsymbol{V}\rangle\propto\langle\boldsymbol{J}\cdot\boldsymbol{V}\rangle\boldsymbol{J}/J(J+1)$.)

Limitation. Polynomials in $\boldsymbol{L}$ never change L, so the method gives only $P_LS_{12}P_L$, not the $L\to L\pm 2$ elements (§3.5.3). For a rank-1 analogue the constant can vanish: $\hat{r}$ is parity-odd and $P_L\hat{r}P_L = 0$ although Wigner–Eckart "allows" $P_L\hat{r}P_L = c\boldsymbol{L}$; $\hat{r}_i\hat{r}_j$ is parity-even, so $c_L\neq 0$.

Lemma 3.6 (Projection identity)**.** *Within a subspace of fixed L,*

$$\hat{r}_i\hat{r}_j - \tfrac{1}{3}\delta_{ij} \;\longrightarrow\; -\frac{2}{(2L-1)(2L+3)}\Big[\tfrac{1}{2}(L_iL_j + L_jL_i) - \tfrac{1}{3}\delta_{ij}\boldsymbol{L}^2\Big]. \tag{3.30}$$

(Proof: Steps 1–5 above.)

3.5.2 Diagonal tensor elements

Step 1: contract with S_iS_j. $\boldsymbol{S}$ and $\boldsymbol{L}$ commute, but components of each do not commute among themselves.

- $S_iS_jL_iL_j = (S_iL_i)(S_jL_j) = (\boldsymbol{L}\cdot\boldsymbol{S})^2$.
- $S_iS_jL_jL_i = S_iS_j(L_iL_j + [L_j, L_i]) = (\boldsymbol{L}\cdot\boldsymbol{S})^2 + \mathrm{i}\epsilon_{jik}S_iS_jL_k$. Only the antisymmetric part of S_iS_j survives the ϵ: $\epsilon_{jik}S_iS_j = \frac{1}{2}\epsilon_{jik}[S_i, S_j] = \frac{\mathrm{i}}{2}\epsilon_{jik}\epsilon_{ijl}S_l = \frac{\mathrm{i}}{2}(-2\delta_{kl})S_l = -\mathrm{i}S_k$, so the extra term is $\mathrm{i}(-\mathrm{i}S_k)L_k = \boldsymbol{L}\cdot\boldsymbol{S}$.

Hence $S_iS_jQ_{ij} = (\boldsymbol{L}\cdot\boldsymbol{S})^2 + \frac{1}{2}\boldsymbol{L}\cdot\boldsymbol{S} - \frac{1}{3}\boldsymbol{L}^2\boldsymbol{S}^2$ and, with (3.24),

$$S_{12}\Big|_L = 6c_LS_iS_jQ_{ij} = \frac{2}{(2L-1)(2L+3)}\Big[2\boldsymbol{L}^2\boldsymbol{S}^2 - 6(\boldsymbol{L}\cdot\boldsymbol{S})^2 - 3\,\boldsymbol{L}\cdot\boldsymbol{S}\Big]. \tag{3.31}$$

Step 2: insert eigenvalues. For $S = 1$: $\boldsymbol{S}^2 = 2$, $\boldsymbol{L}^2 = L(L+1)$; call $x = \langle\boldsymbol{L}\cdot\boldsymbol{S}\rangle$ and $\mathcal{B}(x) = 4L(L+1) - 6x^2 - 3x$, so $\langle S_{12}\rangle = \frac{2\mathcal{B}(x)}{(2L-1)(2L+3)}$. From $x = \frac{1}{2}[J(J+1) - L(L+1) - 2]$: $x = L$ $(J = L+1)$, $x = -1$ $(J = L)$, $x = -(L+1)$ $(J = L-1)$.

Step 3: each bracket factorizes and cancels a denominator factor.

$$\begin{aligned}
J = L+1: &\quad \mathcal{B} = 4L^2 + 4L - 6L^2 - 3L = -L(2L-1) &&\Rightarrow\ \langle S_{12}\rangle = -\frac{2L}{2L+3},\\
J = L: &\quad \mathcal{B} = 4L^2 + 4L - 3 = (2L-1)(2L+3) &&\Rightarrow\ \langle S_{12}\rangle = 2,\\
J = L-1: &\quad \mathcal{B} = (L+1)\big[4L - 6(L+1) + 3\big] = -(L+1)(2L+3) &&\Rightarrow\ \langle S_{12}\rangle = -\frac{2(L+1)}{2L-1}.
\end{aligned}$$

Result 3.7 (Diagonal tensor matrix elements)**.**

$$\langle S_{12}\rangle = \begin{cases} -\dfrac{2L}{2L+3}, & J = L+1,\\ +2, & J = L,\\ -\dfrac{2(L+1)}{2L-1}, & J = L-1, \end{cases} \qquad \textit{e.g. } {}^3P_{2,1,0}:\ -\tfrac{2}{5},\ 2,\ -4. \tag{3.32}$$

Checks: $L = 0$ *or* $S = 0$ *gives* 0*;* $\sum_J(2J+1)\langle S_{12}\rangle_J = -2L + 2(2L+1) - 2(L+1) = 0$.

3.5.3 Off-diagonal tensor element

Step 1: an algebraic identity on spin triplets. With $\boldsymbol{S}^2 = 2$, (3.24) reads $S_{12} = 6A^2 - 4$ with $A = \boldsymbol{S}\cdot\hat{r}$, the spin-1 projection on $\hat{r}$. Its eigenvalues are $-1, 0, 1$, so $(A-1)A(A+1) = 0$, i.e. $A^3 = A$ and $A^4 = A^2$ (the components of $\hat{r}$ commute with each other and with $\boldsymbol{S}$). Then

$$S_{12}^2 = 36A^4 - 48A^2 + 16 = -12A^2 + 16 = 8 - 2S_{12} \qquad\Longrightarrow\qquad S_{12}^2 + 2S_{12} - 8 = 0\ \ (S = 1), \tag{3.33}$$

so the eigenvalues of S_{12} are $+2$ (spin projection ±1 on $\hat{r}$) and -4 (projection 0). The 1×1 blocks confirm this: ${}^3L_L$ gives 2, ${}^3P_0$ gives -4.

Step 2: the 2×2 block is closed. Fix $S = 1, J, M$. S_{12} acts on spin and angles only, commutes with $\boldsymbol{J}$ and $\boldsymbol{S}^2$, and is parity-even; $L = J$ has parity opposite to $L = J\pm1$. So S_{12} maps the pair $\{|(J-1)1J\rangle, |(J+1)1J\rangle\}$ into itself:

$$S_{12}\big|_{\text{block}} = \begin{pmatrix} a & b\\ b & d\end{pmatrix}, \qquad \text{eigenvalues } 2, -4\ \Rightarrow\ a + d = -2,\quad ad - b^2 = -8.$$

Step 3: use the diagonal results. $a = -\frac{2L}{2L+3}\big|_{L=J-1} = -\frac{2(J-1)}{2J+1}$, $d = -\frac{2(L+1)}{2L-1}\big|_{L=J+1} = -\frac{2(J+2)}{2J+1}$. Trace check: $a+d=-2$ ✓. Determinant:

$$b^2 = ad+8 = \frac{4(J-1)(J+2)+8(2J+1)^2}{(2J+1)^2} = \frac{36J(J+1)}{(2J+1)^2}.$$

Result 3.8 (Off-diagonal tensor element)**.**

$$\langle (J-1)1J|\, S_{12}\, |(J+1)1J\rangle = \frac{6\sqrt{J(J+1)}}{2J+1} \qquad \Big(J=1:\ \sqrt{8}\Big). \tag{3.34}$$

The algebra fixes b^2; the sign depends on the relative phase of the two basis states. With the coupling order $[Y_L \otimes \chi_S]_J$ and Condon–Shortley phases an explicit Clebsch–Gordan calculation gives $b>0$. Masses depend only on b^2; the sign of an S–D mixing angle is convention-dependent, so use one convention throughout. In the basis $\{{}^3S_1, {}^3D_1\}$: $S_{12} = \begin{pmatrix} 0 & \sqrt{8} \\ \sqrt{8} & -2 \end{pmatrix}$ (trace -2, determinant -8). All values in this subsection were checked numerically for $L \le 3$ by building the operators explicitly.

Spin–angle factors for $S=1$, ${}^3L_J$

J	$\langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle$	$\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle$	$\langle S_{12}\rangle$
$L+1$	$\frac{1}{4}$	L	$-\frac{2L}{2L+3}$
L	$\frac{1}{4}$	-1	2
$L-1$	$\frac{1}{4}$	$-(L+1)$	$-\frac{2(L+1)}{2L-1}$

${}^1L_L$	$\langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle$	$\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle$	$\langle S_{12}\rangle$
	$-\frac{3}{4}$	0	0

For $L=0$: only ${}^3S_1$ $(\frac{1}{4},0,0)$ and ${}^1S_0$ $(-\frac{3}{4},0,0)$.

3.6 Worked examples: what the spin terms do

Example 3.9 (S-wave hyperfine splitting)**.** Treat H_{SS} in first-order perturbation theory with an unsmeared δ function. Since $\langle \delta^3(\boldsymbol{r})\rangle = |\psi(0)|^2 = |R(0)|^2/4\pi$ and $\langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle_{{}^3S_1} - \langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle_{{}^1S_0} = 1$,

$$M({}^3S_1) - M({}^1S_0) = \frac{32\pi\alpha_s}{9m_1m_2}|\psi(0)|^2 = \frac{8\alpha_s}{9m_1m_2}|R_{nS}(0)|^2. \tag{3.35}$$

Example 3.10 (P-wave fine structure)**.** For quarkonium define the radial expectation values in the $1P$ state

$$a = \Big\langle \frac{1}{2m^2 r}(3V_V' - V_S')\Big\rangle, \qquad t = \Big\langle \frac{1}{12m^2}\Big(\frac{V_V'}{r} - V_V''\Big)\Big\rangle, \qquad c_{ss} = \Big\langle \frac{2}{3m^2}\nabla^2 V_V\Big\rangle. \tag{3.36}$$

Then, to first order, with $M_{\rm cog}$ the spin-independent $1P$ mass,

$$M({}^3P_2) = M_{\rm cog} + \tfrac{1}{4}c_{ss} + a - \tfrac{2}{5}t, \qquad M({}^3P_1) = M_{\rm cog} + \tfrac{1}{4}c_{ss} - a + 2t, \tag{3.37}$$

$$M({}^3P_0) = M_{\rm cog} + \tfrac{1}{4}c_{ss} - 2a - 4t, \qquad M({}^1P_1) = M_{\rm cog} - \tfrac{3}{4}c_{ss}. \tag{3.38}$$

For pure OGE $c_{ss} \propto \langle \delta^3\rangle = 0$ in P waves.

Exercise 3.11 (Centre-of-gravity theorem)**.** Show $\sum_J (2J+1)\langle \boldsymbol{LS}\rangle_J = 0$ and $\sum_J (2J+1)\langle S_{12}\rangle_J = 0$ for fixed $L \ge 1$, $S=1$ (check: $5\cdot 1 + 3\cdot(-1) + 1\cdot(-2) = 0$; $5\cdot(-\frac{2}{5}) + 3\cdot 2 + 1\cdot(-4) = 0$). Hence $\frac{1}{9}[5M({}^3P_2) + 3M({}^3P_1) + M({}^3P_0)] = M_{\rm cog} + \frac{1}{4}c_{ss}$, and $M({}^1P_1)$ equals this weighted average if the spin–spin force is short-ranged. This is a useful test of any code.

4 The four Hamiltonians

4.1 Model A: non-relativistic Cornell + Breit–Fermi

Model A

$$H_A = m_1 + m_2 + \frac{\boldsymbol{p}^2}{2m_r} + V_V(r) + V_S(r) + H_{SD}[V_V, V_S], \qquad V_V = -\frac{4\alpha_s}{3r}, \quad V_S = br + c, \tag{4.1}$$

with H_{SD} from (3.16)–(3.19) and the contact term smeared (below). Explicitly, for quarkonium ($m_1 = m_2 = m$):

$$H_{SD} = \frac{32\pi\alpha_s}{9m^2}\,\delta_\sigma(r)\,\boldsymbol{S}_1\cdot\boldsymbol{S}_2 + \frac{1}{m^2}\Big(\frac{2\alpha_s}{r^3} - \frac{b}{2r}\Big)\boldsymbol{L}\cdot\boldsymbol{S} + \frac{\alpha_s}{3m^2r^3}S_{12}, \qquad \delta_\sigma(r) = \Big(\frac{\sigma}{\sqrt{\pi}}\Big)^3 \mathrm{e}^{-\sigma^2 r^2}. \tag{4.2}$$

Parameters: m_Q, α_s, b, c, σ.

Why smear the contact term? The δ function has two problems: (i) if H_{SS} is included in the diagonalization (rather than as a first-order perturbation), an attractive δ function (in ${}^1S_0$, $\langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle = -\frac{3}{4}$) makes the Hamiltonian unbounded below—the variational energy collapses as the basis becomes able to resolve $r \to 0$; (ii) a heavy quark is not point-like over distances $\sim 1/m$. The Gaussian δ_σ with $\int \delta_\sigma\, \mathrm{d}^3r = 1$ solves both; σ becomes a fit parameter that controls the hyperfine splittings. (Smearing is analysed systematically in §6.)

Perturbative vs. non-perturbative spin terms.

1. *Perturbative*: solve the spin-independent problem $H_0 = m_1 + m_2 + p^2/2m_r + V_0$ for R_{nL}, then add $\langle H_{SD}\rangle$ with the factors of §3.5. Simple; $1/r^3$ terms are harmless for $L \geq 1$ (finite $\langle r^{-3}\rangle$); ${}^3S_1$–${}^3D_1$ mixing must be added by hand (2×2 matrix).
2. *Non-perturbative*: diagonalize $H_0 + H_{SD}$ for each (L, S, J) channel. The spin terms then modify the wave functions (important for $E1$ transitions and for ${}^3S_1$ vs. ${}^1S_0$ wave functions at the origin). Requires a smeared contact term; with the $1/r^3$ spin–orbit and tensor terms one must check that attractive combinations do not cause collapse (e.g. ${}^3P_0$ where $\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle = -2$, $\langle S_{12}\rangle = -4$): smear them as well, or keep them perturbative.

How the parameters are typically fixed. Spin-averaged masses (e.g. $\frac{1}{4}[M(\eta_c) + 3M(J/\psi)]$ and the $1P$ centre of gravity) mostly constrain m_Q, α_s, b, c; the $1S$ hyperfine splitting fixes σ (given α_s); the χ_{cJ} splittings test the Lorentz structure.

Remark 4.1 (Useful exact relations for checking a code). *Virial theorem*: for $H = p^2/2m_r + V(r)$ and any eigenstate, $2\langle T\rangle = \langle rV'(r)\rangle$; for the Cornell potential $2\langle T\rangle = \kappa\langle r^{-1}\rangle + b\langle r\rangle$. (Derive it from $\langle [H, \boldsymbol{r}\cdot\boldsymbol{p}]\rangle = 0$.) *Feynman–Hellmann*: $\partial E_n/\partial\lambda = \langle \partial H/\partial\lambda\rangle_n$ for any parameter λ in H; in a finite basis it holds exactly as $\partial E/\partial\lambda = \boldsymbol{c}^T(\partial\mathbf{H}/\partial\lambda)\boldsymbol{c}$ (with $\mathbf{S}$ independent of λ). This gives parameter derivatives for fitting for free.

Variants of Model A you will meet in the literature. Running α_s (§5); mixed vector–scalar confinement (remark in §3.4); Gaussian or Yukawa-type hyperfine potentials; pNRQCD-motivated $O(1/m)$ and $O(1/m^2)$ spin-independent corrections; a semi-relativistic kinetic term $E_1 + E_2$ (then the model sits between A and C). Non-relativistic models with a fixed α_s per system and smeared hyperfine term are the ones used e.g. by Barnes–Godfrey–Swanson.

4.2 Model B: non-relativistic + screened confinement + Breit–Fermi

The screened potential. Replace br by a function that is linear at short distance and saturates at large distance:

$$V_S(r) = \frac{b}{\mu_s}\left(1 - \mathrm{e}^{-\mu_s r}\right) + c. \tag{4.3}$$

Exercise 4.2. Expand: $V_S = c + br - \frac{1}{2}b\mu_s r^2 + \frac{1}{6}b\mu_s^2 r^3 - \dots$ for $\mu_s r \ll 1$, and $V_S \to c + b/\mu_s$ for $\mu_s r \gg 1$. The derivative is $V_S' = b\,\mathrm{e}^{-\mu_s r}$; the "effective string tension" decays over the length $1/\mu_s$. As $\mu_s \to 0$ one recovers $br + c$.

In published fits μ_s is of order 0.1 GeV or smaller, i.e. screening lengths $1/\mu_s$ of a few fm (take the actual value from the paper you follow); the saturation value $m_1 + m_2 + c + b/\mu_s$ is then well above the low-lying states.

Model B

$$H_B = m_1 + m_2 + \frac{\boldsymbol{p}^2}{2m_r} - \frac{4\alpha_s}{3r} + \frac{b}{\mu_s}\left(1 - \mathrm{e}^{-\mu_s r}\right) + c + H_{SD}\Big[V_V = -\tfrac{4\alpha_s}{3r},\ V_S = \tfrac{b}{\mu_s}\left(1 - \mathrm{e}^{-\mu_s r}\right) + c\Big]. \tag{4.4}$$

Only the scalar (Thomas) spin–orbit term changes: $-\dfrac{V_S'}{2m_i^2 r} = -\dfrac{b\,\mathrm{e}^{-\mu_s r}}{2m_i^2 r}$. Contact and tensor terms are as in Model A. Parameters: $m_Q, \alpha_s, b, c, \sigma, \mu_s$.

Physics consequences. Screening lowers highly excited levels considerably (Li–Chao find shifts of order 100 MeV for the highest Υ states) while leaving the low-lying spectrum almost unchanged after refitting; it compresses the spectrum near threshold; it changes radial wave functions at large r, hence $E1$ rates of higher states. It is an effective description of the coupling to meson–meson channels (vacuum polarization by light quarks).

Remark 4.3 (The continuum in a screened potential). Because $V_S(\infty)$ is finite, H_B has a continuous spectrum above $E_{\mathrm{th}} = m_1 + m_2 + c + b/\mu_s$. A finite Gaussian basis turns that continuum into discrete "pseudo-states" whose energies depend on $r_{\max}$. Only eigenvalues well below E_{th} with converged rms radii are physical bound states. Test: change $r_{\max}$; pseudo-states move, bound states do not.

Other screened forms. Variants include $V_S = br\,f(\mu_s r)$ with other cut-off functions, and the "generalized screened potential model" in which the screening is built from meson–meson thresholds. All are handled identically in GEM once their Gaussian matrix elements are available (§7.5).

4.3 Model C: the Godfrey–Isgur relativized model

4.3.1 Philosophy: four relativization steps

Godfrey and Isgur start from the Breit–Fermi Hamiltonian of a vector one-gluon exchange plus scalar linear confinement and "relativize" it:

1. **Relativistic kinetic energy**: $p^2/2m_r \to E_1(p) + E_2(p)$ in the rest frame.
2. **Running coupling**: $\alpha_s \to \alpha_s(Q^2)$, parameterized as a sum of Gaussians in Q that saturates at small Q (§5).

3. **Smearing**: quarks are not point-like on the scale of their Compton wavelength; every potential is convoluted with a Gaussian of flavour-dependent width (§6). This also removes all short-distance singularities ($1/r$, δ, $1/r^3$).

4. **Momentum-dependent factors**: the $1/m$ factors of the Breit–Fermi expansion are, in a relativistic theory, really $1/E$; GI replace them by operators $(m_1 m_2/E_1 E_2)^{1/2+\epsilon}$ placed symmetrically around each potential, with small fitted exponents ϵ.

4.3.2 What is derived and what is chosen

GI is *not* derived from QCD. Godfrey and Isgur wanted a single Hamiltonian for all mesons, from the π to the Υ; for light quarks $p \sim m$ and the $1/m$ expansion of §3.4 is meaningless. They kept one rule: *for $p \ll m$ every GI term must reduce to its Breit–Fermi term*, and dressed each term with relativistic, physically motivated factors containing a few fitted parameters. Reading the Hamiltonian below is easiest with this map:

Breit–Fermi term (Model A)	GI term	modification
$m_1 + m_2 + p^2/2m_r$	$E_1(p) + E_2(p)$	do not expand
$-4\alpha_s/3r$	$(1+\frac{p^2}{E_1E_2})^{1/2}\tilde{G}(1+\frac{p^2}{E_1E_2})^{1/2}$	running α_s, smearing, current factor
$br + c$	$\tilde{S}(r)$	smearing only
$\frac{2}{3m_1m_2}\nabla^2 V_V\, \boldsymbol{S}_1\cdot\boldsymbol{S}_2$	$\tilde{H}_c$	$\tilde{G}$, Φ_{12} sandwich
$\frac{1}{12m_1m_2}(\frac{V_V'}{r} - V_V'')S_{12}$	$\tilde{H}_t$	$\tilde{G}$, Φ_{12} sandwich
spin–orbit (3.19)	$\tilde{H}_{so(v)}$, $\tilde{H}_{so(s)}$	$\tilde{G}, \tilde{S}$, $\Phi_{11}, \Phi_{22}, \Phi_{12}$

Derived (given the potential)	Model choice (fitted)
Breit–Fermi spin structure (§3.4.3)	Gaussian form of $\alpha_s(Q)$
$E_1 + E_2$ as the free energy	smearing width $\sigma_{12}(m_1, m_2)$
smeared $\tilde{G}, \tilde{S}, \nabla^2\tilde{G}$ (§6)	keeping only $(1 + p^2/E_1E_2)$ from the current
Φ at $\epsilon = 0$ = vertex normalization (below)	the exponents ϵ_x

4.3.3 The Hamiltonian

In colour-operator form GI write the confining and Coulomb parts as $H^{\rm conf} = \big[-\frac{3}{4}(c+br) + \frac{\alpha_s(r)}{r}\big]\boldsymbol{F}_1\cdot\boldsymbol{F}_2$, which for a meson ($\boldsymbol{F}_1\cdot\boldsymbol{F}_2 = -\frac{4}{3}$) is $c + br - \frac{4}{3}\alpha_s(r)/r$. After smearing ($\tilde{G}$, $\tilde{S}$; §6):

Model C: Godfrey–Isgur Hamiltonian, meson rest frame

$$H_{GI} = E_1(p) + E_2(p) + \tilde{V}_{\rm eff}(\boldsymbol{p}, \boldsymbol{r}), \qquad H_{GI}\Psi = M\Psi, \tag{4.5}$$

$$\tilde{V}_{\rm eff} = \underbrace{\Big(1+\frac{p^2}{E_1E_2}\Big)^{\frac{1}{2}}\tilde{G}(r)\Big(1+\frac{p^2}{E_1E_2}\Big)^{\frac{1}{2}}}_{\text{relativized Coulomb}} + \underbrace{\tilde{S}(r)}_{\text{confinement}} + \tilde{H}_c + \tilde{H}_t + \tilde{H}_{so(v)} + \tilde{H}_{so(s)}, \tag{4.6}$$

$$\tilde{G}(r) = -\frac{4}{3}\sum_{k=1}^{3}\frac{\alpha_k}{r}\,\mathrm{erf}(\tau_k r), \qquad \tilde{S}(r) = b\Big[\frac{\mathrm{e}^{-\sigma^2r^2}}{\sqrt{\pi}\,\sigma} + \Big(r + \frac{1}{2\sigma^2 r}\Big)\mathrm{erf}(\sigma r)\Big] + c, \tag{4.7}$$

with $\sigma \equiv \sigma_{12}$ and $\tau_k = \sigma_{12}\gamma_k/\sqrt{\sigma_{12}^2 + \gamma_k^2}$ from §6.

Spin-dependent terms. Each is the Breit–Fermi term of (3.17)–(3.19) with $V_V \to \tilde{G}$, $V_S \to \tilde{S}$, sandwiched by momentum-dependent factors. Define for a pair of quark labels $a, b \in \{1, 2\}$

$$\Phi_{ab}^{(\epsilon)}(p) = \Big(\frac{m_a m_b}{E_a(p) E_b(p)}\Big)^{\frac{1}{2}+\epsilon}, \qquad \Phi_{ab}^{(\epsilon)} \to 1 \ \text{ as } p/m \to 0. \tag{4.8}$$

Then

$$\tilde{H}_c = \Phi_{12}^{(\epsilon_c)} \Big[\frac{2}{3m_1 m_2} \nabla^2 \tilde{G}\Big] \Phi_{12}^{(\epsilon_c)} \ \boldsymbol{S}_1 \cdot \boldsymbol{S}_2, \tag{4.9}$$

$$\tilde{H}_t = \Phi_{12}^{(\epsilon_t)} \Big[\frac{1}{12 m_1 m_2} \Big(\frac{\tilde{G}'}{r} - \tilde{G}''\Big)\Big] \Phi_{12}^{(\epsilon_t)} \ S_{12}, \tag{4.10}$$

$$\tilde{H}_{so(v)} = \sum_{i=1,2} \Big\{ \Phi_{ii}^{(\epsilon_{sov})} \frac{\tilde{G}'}{2m_i^2 r} \Phi_{ii}^{(\epsilon_{sov})} + \Phi_{12}^{(\epsilon_{sov})} \frac{\tilde{G}'}{m_1 m_2 r} \Phi_{12}^{(\epsilon_{sov})} \Big\} \boldsymbol{L} \cdot \boldsymbol{S}_i, \tag{4.11}$$

$$\tilde{H}_{so(s)} = -\sum_{i=1,2} \Phi_{ii}^{(\epsilon_{sos})} \frac{\tilde{S}'}{2m_i^2 r} \Phi_{ii}^{(\epsilon_{sos})} \ \boldsymbol{L} \cdot \boldsymbol{S}_i. \tag{4.12}$$

The rule: *a term that carries $1/(m_a m_b)$ is sandwiched by Φ_{ab}.* For quarkonium all Φ's coincide, $\Phi = (m^2/E^2)^{1/2+\epsilon} = (m/E)^{1+2\epsilon}$, and the distinction disappears.

Remark 4.4 (GI's own split of the spin–orbit force). GI write the spin–orbit force as a colour-magnetic piece $\frac{4\alpha_s}{3r^3}\big(\frac{1}{m_1} + \frac{1}{m_2}\big)\big(\frac{\boldsymbol{S}_1}{m_1} + \frac{\boldsymbol{S}_2}{m_2}\big) \cdot \boldsymbol{L}$ (smeared: $\frac{\tilde{G}'}{r} \to$) plus a Thomas-precession piece $-\frac{1}{2r}\frac{\partial(\tilde{G}+\tilde{S})}{\partial r}\big(\frac{\boldsymbol{S}_1}{m_1^2} + \frac{\boldsymbol{S}_2}{m_2^2}\big) \cdot \boldsymbol{L}$. Adding them, the coefficient of $\boldsymbol{L} \cdot \boldsymbol{S}_1$ is $\frac{\tilde{G}'}{r}\big(\frac{1}{m_1^2} + \frac{1}{m_1 m_2}\big) - \frac{\tilde{G}'+\tilde{S}'}{2m_1^2 r} = \frac{\tilde{G}'}{r}\big(\frac{1}{2m_1^2} + \frac{1}{m_1 m_2}\big) - \frac{\tilde{S}'}{2m_1^2 r}$, which is exactly (3.19). (Verify this line yourself.)

4.3.4 Where the momentum-dependent factors come from

Go back to the normalized currents of §3.4.3, but *do not* expand the normalization $2\sqrt{EE'}$. Approximating $E \simeq E'$ only inside the spinor ratios,

$$\hat{j}^{\,0} \simeq 1, \qquad \hat{\boldsymbol{j}} \simeq \frac{(\boldsymbol{p}+\boldsymbol{p}') + \mathrm{i}\,\boldsymbol{\sigma} \times \boldsymbol{q}}{2\sqrt{E(p)E(p')}} \xrightarrow{\ p \ll m\ } \frac{(\boldsymbol{p}+\boldsymbol{p}') + \mathrm{i}\,\boldsymbol{\sigma} \times \boldsymbol{q}}{2m}. \tag{4.13}$$

Every $1/m$ of the Breit–Fermi expansion is really $1/\sqrt{EE'}$.

(a) The Coulomb factor. In the rest frame ($\boldsymbol{p}_2 = -\boldsymbol{p}_1$), for small momentum transfer, the convection parts of $\hat{j}_1^{\,0}\hat{j}_2^{\,0} - \hat{\boldsymbol{j}}_1 \cdot \hat{\boldsymbol{j}}_2$ give

$$1 - \frac{(\boldsymbol{p}_1 + \boldsymbol{p}_1') \cdot (\boldsymbol{p}_2 + \boldsymbol{p}_2')}{4\sqrt{E_1 E_1' E_2 E_2'}} \simeq 1 + \frac{p^2}{E_1 E_2}. \tag{4.14}$$

So the vector (Coulomb) interaction is multiplied by $(1 + p^2/E_1 E_2)$; in the non-relativistic limit $1 + p^2/m_1 m_2$, which is the same order and sign as the Breit orbit–orbit term (the exact Breit form distributes it between p^2 and $(\hat{r} \cdot \boldsymbol{p})^2$; GI keep the simple form). The factor depends on the initial momentum p and the final momentum p'. In operator language "after" stands on the left and "before" on the right, so GI split it symmetrically,

$$\Big(1 + \frac{p^2}{E_1 E_2}\Big)^{1/2} \tilde{G}(r) \Big(1 + \frac{p^2}{E_1 E_2}\Big)^{1/2}, \tag{4.15}$$

which is also what makes the operator Hermitian (p does not commute with $\tilde{G}(r)$). Expanding, $(1+p^2/E_1E_2)^{1/2}\tilde{G}(1+p^2/E_1E_2)^{1/2} \simeq \tilde{G} + \frac{1}{2m_1 m_2}\{p^2, \tilde{G}\}$. A scalar exchange couples to $\bar{u}'u$, which has no current–current term: $\tilde{S}$ gets no such factor.

(b) The spin factors Φ. The contact and tensor forces come from the spin-magnetization currents, i.e. from $(\mathrm{i}\boldsymbol{\sigma}_1 \times \boldsymbol{q}) \cdot (\mathrm{i}\boldsymbol{\sigma}_2 \times \boldsymbol{q})$ times the normalizations. Breit–Fermi has $\frac{1}{4m_1m_2}$ there; unexpanded it is

$$\frac{1}{4\sqrt{E_1E_1'E_2E_2'}} = \frac{1}{4m_1m_2} \underbrace{\Big(\frac{m_1m_2}{E_1(p')E_2(p')}\Big)^{1/2}}_{\Phi_{12}(p')} \underbrace{\Big(\frac{m_1m_2}{E_1(p)E_2(p)}\Big)^{1/2}}_{\Phi_{12}(p)}. \tag{4.16}$$

So the GI sandwich Φ_{12} [Breit–Fermi term] Φ_{12} at $\epsilon = 0$ is *exactly the relativistic normalization of the two vertices*; each side carries one momentum, hence the sandwich. The approximation $E \simeq E'$ in the spinor ratios, the $(E + m)$ factors and neglected higher terms mean the true function is not exactly $(m/E)^{1/2}$; GI parameterize this with a small exponent correction, $\Phi^{(\epsilon)} = (m_am_b/E_aE_b)^{1/2+\epsilon}$, one ϵ per type of term $(\epsilon_c, \epsilon_t, \epsilon_{sov}, \epsilon_{sos})$.

(c) Why the spin–orbit terms carry different Φ's. The coefficient of $\boldsymbol{L} \cdot \boldsymbol{S}_1$ in (3.19) contains $\frac{1}{2m_1^2}V_V'$ and $-\frac{1}{2m_1^2}V_S'$ (from particle 1's own current: Thomas precession and spin–orbit charge density) and $\frac{1}{m_1m_2}V_V'$ (from the product of both currents: colour-magnetic). A $1/(m_1m_1)$ term carries particle 1's normalization twice, i.e. Φ_{11}; a $1/(m_1m_2)$ term carries one of each, Φ_{12}. This is the rule "a term carrying $1/(m_am_b)$ is sandwiched by Φ_{ab}". For quarkonium all Φ's coincide.

(d) Thomas precession and the GI split. Two physical effects make up the spin–orbit force. *Colour-magnetic*: moving colour charges produce a magnetic field that couples to the spin; only vector exchange has it; coefficient of $\boldsymbol{L} \cdot \boldsymbol{S}_1$: $\frac{\tilde{G}'}{r}\frac{1}{m_1}(\frac{1}{m_1} + \frac{1}{m_2})$. *Thomas precession*: pure kinematics of an accelerated spinning particle; every force contributes, with the universal coefficient $-\frac{1}{2m_1^2r}\partial_r(\tilde{G}+\tilde{S})$. The sum is $\frac{\tilde{G}'}{r}(\frac{1}{2m_1^2}+\frac{1}{m_1m_2}) - \frac{\tilde{S}'}{2m_1^2r}$, i.e. (3.19) with $V_V \to \tilde{G}$, $V_S \to \tilde{S}$. This is the same mechanism as the factor $\frac{1}{2}$ in the hydrogen fine structure; for scalar confinement only the (negative) Thomas term survives, which reduces the P-wave splittings.

Non-relativistic limit (a check you should do). Set $\epsilon_x = -\frac{1}{2}$ (all $\Phi \to 1$), drop $(1 + p^2/E_1E_2)^{1/2}$, expand $E_i \simeq m_i + p^2/2m_i$ and let $\sigma_{12} \to \infty$: you recover Model A with the running coupling $G(r)$.

4.3.5 Hermiticity and boundedness

Hermiticity. $E_i(p)$ are functions of the operator p^2 and commute with each other but not with functions of r. The symmetric sandwich $f(p)\,W(r)\,f(p)$ is Hermitian when W is; never write $f(p)^2W(r)$.

Boundedness. For a single relativistic particle, $\sqrt{p^2+m^2} - \kappa/r$ is bounded below only if $\kappa \le 2/\pi$ (Herbst). Smearing replaces $1/r$ by $\mathrm{erf}(\tau r)/r$, which is finite at the origin, so H_{GI} has no such problem.

4.3.6 Parameters

$m_u = m_d, m_s, m_c, m_b$; b, c; the smearing parameters σ_0, s; the running-coupling parameters $(\alpha_k, \gamma_k)_{k=1,2,3}$ (fixed by GI; freeing them is treated in §10.9); and $\epsilon_c, \epsilon_t, \epsilon_{sov}, \epsilon_{sos}$. GI's values are listed (from memory, to be verified) in §12.3.

4.4 Model D: modified (screened) Godfrey–Isgur model

Model D is Model C with the linear confinement replaced by the screened potential (4.3) *before* smearing:

Model D

$$\tilde{S}(r) \;\longrightarrow\; \tilde{S}_{\rm scr}(r) = \frac{b}{\mu_s}\Big[1 - \widetilde{\mathrm{e}^{-\mu_s r}}(r)\Big] + c, \tag{4.17}$$

where $\widetilde{\mathrm{e}^{-\mu_s r}}$ is the Gaussian-smeared exponential, given in closed form in (6.24). $\tilde{S}_{\rm scr}$ enters the potential and, through $\tilde{S}'_{\rm scr}$, the scalar spin–orbit term $\tilde{H}_{so(s)}$. Everything else (kinetic term, $\tilde{G}$, momentum factors, contact, tensor, vector spin–orbit) is as in Model C. Parameters: those of Model C plus μ_s (and in practice a refit of $b, c, \sigma_0, s, \epsilon_x$).

Checks: $\mu_s \to 0$ reproduces Model C; for $r \gg 1/\sigma$ smearing is irrelevant and $\tilde{S}_{\rm scr} \to \frac{b}{\mu_s}(1 - \mathrm{e}^{-\mu_s r}) + c$. The remark on pseudo-states (§4.2) applies here too.

4.5 Side-by-side comparison

	A: NR+BF	B: NR+scr+BF	C: GI	D: mod. GI
Kinetic	$p^2/2m_r$	$p^2/2m_r$	$E_1 + E_2$	$E_1 + E_2$
Coulomb	$-4\alpha_s/3r$	$-4\alpha_s/3r$	$\tilde{G}$, running, smeared, $\times(1 + p^2/E_1E_2)^{1/2}$	same as C
Confinement	$br + c$	$\frac{b}{\mu_s}(1 - \mathrm{e}^{-\mu_s r}) + c$	$\tilde{S}$ (smeared linear)	$\tilde{S}_{\rm scr}$
Spin terms	BF, smeared δ	BF, smeared δ	BF with $\tilde{G}, \tilde{S}$ and Φ factors	as C with $\tilde{S}'_{\rm scr}$
Local?	yes	yes	no (p-dependent)	no
Eigenvalue	$M - m_1 - m_2$	$M - m_1 - m_2$	M	M
Extra params	σ	σ, μ_s	σ_0, s, ϵ_x	$\sigma_0, s, \epsilon_x, \mu_s$

4.6 Further directions (not needed for the basic project)

- Relativistic quasipotential models (Ebert–Faustov–Galkin): v^2/c^2 corrections including retardation and one-loop terms in a quasipotential equation.
- pNRQCD-based potentials with perturbative $V(q)$ at short distance and lattice-motivated $1/m$ terms.
- Coupled-channel (unquenched) models: explicit coupling of $Q\bar{Q}$ to $D\bar{D}$, $D\bar{D}^*, \ldots$ via the $^3P_0$ mechanism; screened models B/D are an effective substitute.

5 The strong coupling α_s

5.1 Renormalization-group running

The coupling depends on the momentum scale Q through the renormalization-group equation

$$Q^2\frac{\mathrm{d}\alpha_s}{\mathrm{d}Q^2} = -\frac{\beta_0}{4\pi}\alpha_s^2 - \frac{\beta_1}{(4\pi)^2}\alpha_s^3 - \ldots, \qquad \beta_0 = 11 - \frac{2}{3}n_f, \quad \beta_1 = 102 - \frac{38}{3}n_f, \tag{5.1}$$

with n_f the number of active quark flavours ($n_f = 3$ or 4 at quarkonium scales).

Exercise 5.1 (One-loop solution)**.** Keep only β_0; separate variables, $\mathrm{d}\alpha_s/\alpha_s^2 = -\frac{\beta_0}{4\pi}\mathrm{d}\ln Q^2$, integrate from a reference point and define Λ by the position of the pole:

$$\alpha_s^{(1)}(Q^2) = \frac{4\pi}{\beta_0 \ln(Q^2/\Lambda^2)}. \tag{5.2}$$

Two loops. Solving (5.1) iteratively to the next order in $1/\ln$, with $L = \ln(Q^2/\Lambda^2)$,

$$\alpha_s^{(2)}(Q^2) = \frac{4\pi}{\beta_0 L}\Big[1 - \frac{\beta_1}{\beta_0^2}\frac{\ln L}{L}\Big]. \tag{5.3}$$

The problem for potential models. Both forms diverge at $Q = \Lambda$ (Landau pole) and are meaningless for $Q \lesssim 1\,\mathrm{GeV}$, exactly where the bound-state wave functions live. A potential model therefore needs an infrared-regular coupling.

5.2 Infrared-regular forms used in practice

1. **Constant coupling.** $\alpha_s = \text{const}$, fitted per system ($c\bar{c}$, $b\bar{b}$). Simplest; used in Models A/B. Typical fitted values are $\alpha_s \approx 0.3$–0.6, reflecting that the potential samples $Q \sim 1$–$2\,\mathrm{GeV}$.

2. **Mass-scale-dependent constant.** Evaluate a perturbative form at a scale tied to the system, e.g. $\mu_{12} = 2m_1m_2/(m_1+m_2)$ or $\mu \sim m_Q$, so that one formula serves $c\bar{c}$, $b\bar{b}$ and $b\bar{c}$.

3. **Frozen / gluon-mass regularized one-loop.**

$$\alpha_s(Q^2) = \frac{4\pi}{\beta_0 \ln\left[(Q^2 + M_g^2)/\Lambda^2\right]}, \tag{5.4}$$

   finite at $Q = 0$ if $M_g > \Lambda$; M_g plays the role of a dynamical gluon mass (variants use $4M_g^2$).

4. **Analytic perturbation theory (Shirkov–Solovtsov).** Remove the Landau pole while keeping the correct analytic structure in Q^2 (the one-loop pole term is subtracted by a dispersion relation):

$$\alpha_s^{\mathrm{APT}}(Q^2) = \frac{4\pi}{\beta_0}\Big[\frac{1}{\ln(Q^2/\Lambda^2)} + \frac{\Lambda^2}{\Lambda^2 - Q^2}\Big]. \tag{5.5}$$

   Exercise 5.2. Show the bracket is finite at $Q^2 = \Lambda^2$ (expand $\ln(Q^2/\Lambda^2) \simeq x - x^2/2$ with $x = Q^2/\Lambda^2 - 1$; the bracket $\to \frac{1}{2}$), and that $\alpha_s^{\mathrm{APT}}(0) = 4\pi/\beta_0$ (≈ 1.4 for $n_f = 3$).

5. **Richardson's potential.** Interpolates between asymptotic freedom and linear confinement with a single scale:

$$\alpha_s^{\mathrm{R}}(Q^2) = \frac{12\pi}{(33 - 2n_f)\ln(1 + Q^2/\Lambda^2)} = \frac{4\pi}{\beta_0 \ln(1 + Q^2/\Lambda^2)}. \tag{5.6}$$

   Exercise 5.3 (Where the string tension comes from)**.** For $Q \to 0$, $\ln(1 + Q^2/\Lambda^2) \to Q^2/\Lambda^2$ so $V(q) = -\frac{16\pi}{3}\frac{\alpha_s(q^2)}{q^2} \to -\frac{16\pi}{3}\frac{4\pi\Lambda^2}{\beta_0 q^4}$. Using $\int \frac{\mathrm{d}^3q}{(2\pi)^3}\frac{\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}}{q^4} = -\frac{r}{8\pi}$ (up to an infinite constant), the long-distance potential is br with $b = \frac{8\pi\Lambda^2}{33-2n_f}$.

   $V(r)$ itself must be obtained numerically from (3.5).

6. **Godfrey–Isgur Gaussian sum.**

$$\alpha_s^{\mathrm{GI}}(Q^2) = \sum_{k=1}^{3} \alpha_k\, \mathrm{e}^{-Q^2/4\gamma_k^2}, \qquad (\alpha_1,\alpha_2,\alpha_3) = (0.25, 0.15, 0.20), \quad (\gamma_1,\gamma_2,\gamma_3) = \Big(\tfrac{1}{2}, \tfrac{\sqrt{10}}{2}, \tfrac{\sqrt{1000}}{2}\Big)\,\mathrm{GeV}. \tag{5.7}$$

It saturates at $\alpha_s(0) = \sum_k \alpha_k = 0.6$ (GI's "critical" value) and falls to zero at large Q; the three γ_k were chosen so that the sum follows the perturbative running in the intermediate region. It is not meant to be accurate at large Q (it falls off too fast), but the Q range relevant for spectroscopy is covered.

5.3 From $\alpha_s(Q^2)$ to $V(r)$

Result 5.4 (Gaussian sum ⇒ error functions)**.** *For one term* $\alpha_s(q^2) = \alpha\, \mathrm{e}^{-q^2/4\gamma^2}$,

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3}\, \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\, \frac{4\pi}{q^2}\mathrm{e}^{-q^2/4\gamma^2} = \frac{\operatorname{erf}(\gamma r)}{r}, \qquad \textit{hence} \qquad G(r) = -\frac{4}{3}\sum_k \frac{\alpha_k}{r}\operatorname{erf}(\gamma_k r). \tag{5.8}$$

Derivation. Write $\frac{1}{q^2}\mathrm{e}^{-q^2/4\gamma^2} = \int_{1/4\gamma^2}^{\infty} \mathrm{d}s\, \mathrm{e}^{-sq^2}$ and use the Gaussian transform $\int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}-sq^2} = (4\pi s)^{-3/2}\mathrm{e}^{-r^2/4s}$:

$$4\pi\int_{1/4\gamma^2}^{\infty} \mathrm{d}s\, (4\pi s)^{-3/2}\mathrm{e}^{-r^2/4s} \overset{u=r/2\sqrt{s}}{=} \frac{1}{2\sqrt{\pi}}\cdot\frac{4}{r}\int_0^{\gamma r}\mathrm{e}^{-u^2}\mathrm{d}u = \frac{\operatorname{erf}(\gamma r)}{r}. \tag{5.9}$$

(Check: $s = r^2/4u^2$, $\mathrm{d}s = -\frac{r^2}{2u^3}\mathrm{d}u$, $s^{-3/2} = \frac{8u^3}{r^3}$.) □

Note $\operatorname{erf}(\gamma r)/r \to 2\gamma/\sqrt{\pi}$ as $r \to 0$: the Gaussian cut-off in Q removes the Coulomb singularity. This is why the GI coupling is "GEM-friendly": its r-space potential has analytic Gaussian matrix elements (§7.5).

General $\alpha_s(q^2)$. Use (3.5). The integrand oscillates; practical evaluation: subtract the large-q behaviour analytically, or use Filon-type quadrature, or fit $\alpha_s(q^2) \approx \sum_k \alpha_k \mathrm{e}^{-q^2/4\gamma_k^2}$ (then the result is a sum of erf's). Tabulate $V(r)$ once on a grid if needed.

Position-space running. Some models use $\alpha_s(r)$ directly, $V_V = -\frac{4}{3}\alpha_s(r)/r$. Then V_V', V_V'', $\nabla^2 V_V$ in the spin terms must include the derivatives of $\alpha_s(r)$; the hyperfine term is no longer a pure δ function.

5.4 Summary

Form	$Q \to 0$	$Q \to \infty$	$V(r)$ closed form?	GEM matrix elements
Constant	const	const	$-4\alpha_s/3r$	analytic
One/two-loop	Landau pole	$\sim 1/\ln Q^2$	no	unusable at low Q
Frozen (M_g)	finite	$\sim 1/\ln Q^2$	no	numerical $V(r)$
APT	$4\pi/\beta_0$	$\sim 1/\ln Q^2$	no	numerical $V(r)$
Richardson	$\propto \Lambda^2/Q^2$ (confining)	$\sim 1/\ln Q^2$	no	numerical $V(r)$
GI Gaussian sum	0.6	$\to 0$ (Gaussian)	$\sum \operatorname{erf}(\gamma r)/r$	analytic

6 Smearing

6.1 What this section is for

Smearing replaces a point interaction $f(r)$ by its average over a Gaussian cloud of size $\sim 1/\sigma$ (physical motivation in §4.1 and §4.3). Which smeared functions each model needs:

Model	smeared objects needed	where derived
A, B	$\delta^3(\boldsymbol{r}) \to \rho_\sigma$ only	§6.5.2
C (GI)	$\tilde{G}$, $\nabla^2\tilde{G}$, $\tilde{G}'$, $\tilde{G}''$, $\tilde{S}$, $\tilde{S}'$	§6.5.1–§6.5.4
D (mod. GI)	as C, with $\tilde{S} \to \tilde{S}_{\rm scr}$ and $\tilde{S}'_{\rm scr}$	§6.5.5

There are two routes to a smeared function. *Momentum space* (multiply by a Gaussian form factor) is the fastest route for $1/r$ and δ. *Coordinate space* (the radial convolution formula, §6.3) is the practical route for r and $\mathrm{e}^{-\mu_s r}$, whose Fourier transforms are singular or awkward. Everything below reduces to three elementary Gaussian integrals (§6.4).

6.2 Definition and basic properties

Definition 6.1 (Smeared potential)**.** With the normalized Gaussian

$$\rho_\sigma(\boldsymbol{r}) = \frac{\sigma^3}{\pi^{3/2}}\mathrm{e}^{-\sigma^2 r^2}, \tag{6.1}$$

the smeared version of a potential f is the convolution

$$\tilde{f}(\boldsymbol{r}) = (\rho_\sigma * f)(\boldsymbol{r}) = \int \mathrm{d}^3r'\, \rho_\sigma(\boldsymbol{r}-\boldsymbol{r}')\, f(\boldsymbol{r}') = \int \mathrm{d}^3x\, \rho_\sigma(\boldsymbol{x})\, f(\boldsymbol{r}-\boldsymbol{x}). \tag{6.2}$$

The second form (substitute $\boldsymbol{x} = \boldsymbol{r}-\boldsymbol{r}'$) says: $\tilde{f}(\boldsymbol{r})$ is the average of f over points displaced from $\boldsymbol{r}$ by a Gaussian-distributed random vector $\boldsymbol{x}$.

Normalization. The Gaussian factorizes, $\mathrm{e}^{-\sigma^2r^2} = \mathrm{e}^{-\sigma^2x^2}\mathrm{e}^{-\sigma^2y^2}\mathrm{e}^{-\sigma^2z^2}$, and $\int_{-\infty}^{\infty}\mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = \sqrt{\pi}/\sigma$, so

$$\int \rho_\sigma\, \mathrm{d}^3r = \frac{\sigma^3}{\pi^{3/2}}\Big(\frac{\sqrt{\pi}}{\sigma}\Big)^3 = 1. \tag{6.3}$$

Two moments used later. Per Cartesian component the variance is $\langle x^2\rangle = \frac{1}{2\sigma^2}$ (Gaussian $\mathrm{e}^{-x^2/2s^2}$ with $s^2 = 1/2\sigma^2$). The mean distance is

$$\langle|\boldsymbol{x}|\rangle = \int_0^\infty 4\pi x^2\, x\, \rho_\sigma\, \mathrm{d}x = \frac{4\sigma^3}{\sqrt{\pi}}\int_0^\infty x^3\mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = \frac{4\sigma^3}{\sqrt{\pi}}\cdot\frac{1}{2\sigma^4} = \frac{2}{\sqrt{\pi}\,\sigma}. \tag{6.4}$$

Fourier transform of the kernel. Complete the square in each component, $-\sigma^2x^2 - \mathrm{i}q_x x = -\sigma^2\big(x + \frac{\mathrm{i}q_x}{2\sigma^2}\big)^2 - \frac{q_x^2}{4\sigma^2}$; the shifted Gaussian integral is still $\sqrt{\pi}/\sigma$. Hence

$$\rho_\sigma(\boldsymbol{q}) \equiv \int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\rho_\sigma(\boldsymbol{r}) = \frac{\sigma^3}{\pi^{3/2}}\Big(\frac{\sqrt{\pi}}{\sigma}\Big)^3\mathrm{e}^{-q^2/4\sigma^2} = \mathrm{e}^{-q^2/4\sigma^2}. \tag{6.5}$$

Convolution theorem. With $f(\boldsymbol{q}) = \int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} f(\boldsymbol{r})$,

$$\tilde{f}(\boldsymbol{q}) = \int \mathrm{d}^3r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\int \mathrm{d}^3r'\rho_\sigma(\boldsymbol{r}-\boldsymbol{r}')f(\boldsymbol{r}') = \int \mathrm{d}^3r'\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}'} f(\boldsymbol{r}')\int \mathrm{d}^3y\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{y}}\rho_\sigma(\boldsymbol{y}) = \rho_\sigma(\boldsymbol{q})\, f(\boldsymbol{q}), \tag{6.6}$$

($\boldsymbol{y} = \boldsymbol{r}-\boldsymbol{r}'$). *Smearing = multiplying the momentum-space potential by the form factor* $\mathrm{e}^{-q^2/4\sigma^2}$.

Consequence 1: widths compose. Smearing with σ_a and then with σ_b multiplies by both form factors:

$$\mathrm{e}^{-q^2/4\sigma_a^2}\,\mathrm{e}^{-q^2/4\sigma_b^2} = \mathrm{e}^{-q^2/4\tau^2}, \qquad \frac{1}{\tau^2} = \frac{1}{\sigma_a^2} + \frac{1}{\sigma_b^2}. \tag{6.7}$$

(In coordinate space: variances of independent Gaussian displacements add, and the variance is $1/2\sigma^2$.)

Consequence 2: smearing commutes with ∇. From the second form of (6.2), $\nabla_r \tilde{f} = \int \mathrm{d}^3x\, \rho_\sigma(\boldsymbol{x}) \nabla f(\boldsymbol{r}-\boldsymbol{x}) = \widetilde{\nabla f}$, and likewise $\nabla^2 \tilde{f} = \widetilde{\nabla^2 f}$.

Remark 6.2 (A trap with radial derivatives). ∇ commutes with smearing, but $\mathrm{d}/\mathrm{d}r$ does not in the naive sense. For radial f, $\nabla f = \hat{r}\, f'(r)$ is a *vector* field; smearing it and taking the radial component gives $\tilde{f}'(r)$, which is *not* the same as smearing the scalar function $f'(r)$. Example: $f = r$ has $f' = 1$, whose smearing is 1, but $\tilde{r}\,' \neq 1$ (§6.5.4). Always obtain $\tilde{f}'$ by differentiating $\tilde{f}$. The Laplacian is safe: $\nabla^2 f$ is a scalar.

GI smearing width. In GI the width depends on the pair of quarks:

$$\sigma_{12}^2 = \sigma_0^2\Big[\frac{1}{2} + \frac{1}{2}\Big(\frac{4m_1m_2}{(m_1+m_2)^2}\Big)^4\Big] + s^2\Big(\frac{2m_1m_2}{m_1+m_2}\Big)^2. \tag{6.8}$$

The first term interpolates between σ_0^2 (equal masses, $4m_1m_2/(m_1+m_2)^2 = 1$) and $\sigma_0^2/2$ (very unequal masses); the second makes the smearing radius $1/\sigma_{12}$ shrink like the Compton wavelength of the pair. Equal masses: $\sigma^2 = \sigma_0^2 + s^2m^2$.

6.3 The radial convolution formula

Why we need it. Definition (6.2) is a three-dimensional integral to be done at every r. For a radial f the angular part can be done once and for all, leaving a one-dimensional integral. This is what produces the closed forms of $\tilde{r}$ and $\widetilde{\mathrm{e}^{-\mu_s r}}$ (needed in Models C and D), and it is the numerical fallback (one 1D quadrature per r) for any smeared potential without a closed form.

Result 6.3. *For a radial function f,*

$$\tilde{f}(r) = \frac{\sigma}{\sqrt{\pi}\, r}\int_0^\infty \mathrm{d}r'\, r' f(r')\Big[\mathrm{e}^{-\sigma^2(r-r')^2} - \mathrm{e}^{-\sigma^2(r+r')^2}\Big] = \frac{\sigma}{\sqrt{\pi}\, r}\int_{-\infty}^\infty \mathrm{d}u\; u\, f(|u|)\, \mathrm{e}^{-\sigma^2(r-u)^2}. \tag{6.9}$$

Derivation, step by step. Step 1 (orient the axes). $\tilde{f}$ depends only on $r = |\boldsymbol{r}|$, so choose the z' axis of the $\boldsymbol{r}'$ integration along $\boldsymbol{r}$. With $x = \cos\theta'$,

$$|\boldsymbol{r}-\boldsymbol{r}'|^2 = r^2 + r'^2 - 2rr'x, \qquad \mathrm{d}^3r' = r'^2\mathrm{d}r'\,\mathrm{d}\phi'\,\mathrm{d}x, \quad x\in[-1,1].$$

Step 2 (the ϕ' integral). Nothing depends on ϕ' (both $f(r')$ and $|\boldsymbol{r}-\boldsymbol{r}'|$ are independent of it): $\int_0^{2\pi}\mathrm{d}\phi' = 2\pi$.

Step 3 (the x integral). Pull out the x-independent part of the exponent:

$$\int_{-1}^{1}\mathrm{d}x\, \mathrm{e}^{-\sigma^2(r^2+r'^2)+2\sigma^2rr'x} = \mathrm{e}^{-\sigma^2(r^2+r'^2)}\,\frac{\mathrm{e}^{2\sigma^2rr'} - \mathrm{e}^{-2\sigma^2rr'}}{2\sigma^2rr'}.$$

Step 4 (recombine the exponents). $-\sigma^2(r^2+r'^2) \pm 2\sigma^2rr' = -\sigma^2(r\mp r')^2$, so the angular integral is

$$\int \mathrm{d}\Omega'\, \mathrm{e}^{-\sigma^2|\boldsymbol{r}-\boldsymbol{r}'|^2} = \frac{2\pi}{2\sigma^2rr'}\Big[\mathrm{e}^{-\sigma^2(r-r')^2} - \mathrm{e}^{-\sigma^2(r+r')^2}\Big] = \frac{\pi}{\sigma^2rr'}\Big[\mathrm{e}^{-\sigma^2(r-r')^2} - \mathrm{e}^{-\sigma^2(r+r')^2}\Big].$$

Step 5 (assemble the prefactors).

$$\tilde{f}(r) = \int_0^\infty r'^2\mathrm{d}r'\, f(r')\,\frac{\sigma^3}{\pi^{3/2}}\cdot\frac{\pi}{\sigma^2rr'}[\,\ldots\,] = \frac{\sigma}{\sqrt{\pi}\, r}\int_0^\infty \mathrm{d}r'\, r' f(r')\big[\mathrm{e}^{-\sigma^2(r-r')^2} - \mathrm{e}^{-\sigma^2(r+r')^2}\big].$$

This is the first form.

Step 6 (odd extension). Define $g(u) = u\,f(|u|)$ on the whole real line; it is odd, $g(-u) = -g(u)$. In the second exponential substitute $r' = -u$:

$$\int_0^\infty g(r')\mathrm{e}^{-\sigma^2(r+r')^2}\mathrm{d}r' = \int_{-\infty}^0 g(-u)\mathrm{e}^{-\sigma^2(r-u)^2}\mathrm{d}u = -\int_{-\infty}^0 g(u)\mathrm{e}^{-\sigma^2(r-u)^2}\mathrm{d}u.$$

Subtracting it therefore *adds* the negative half-line, and the two pieces combine into

$$\int_{-\infty}^\infty g(u)\,\mathrm{e}^{-\sigma^2(r-u)^2}\mathrm{d}u,$$

which is the second form. □

What the formula means. Multiply by r:

$$r\,\tilde{f}(r) = \int_{-\infty}^\infty \mathrm{d}u\; k_\sigma(r-u)\,g(u), \qquad k_\sigma(x) = \frac{\sigma}{\sqrt{\pi}}\mathrm{e}^{-\sigma^2x^2} \quad \Big(\textstyle\int k_\sigma = 1\Big). \tag{6.10}$$

For radial functions, 3D Gaussian smearing of f is 1D Gaussian smearing of the odd function rf. This is the same reduction that turns the radial Schrödinger equation into a 1D equation for $u = rR$ with $u(0) = 0$ (oddness). Sanity check: $f = 1$ gives $g(u) = u$ and $\int k_\sigma(r-u)\,u\,\mathrm{d}u = r$, so $\tilde{1} = 1$.

6.4 Toolbox: half-line Gaussian moments

Every closed form below comes from these three integrals (for any real c):

Lemma 6.4.

$$I_0(c) \equiv \int_{-c}^\infty \mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = \frac{\sqrt{\pi}}{2\sigma}\,\mathrm{erfc}(-\sigma c) = \frac{\sqrt{\pi}}{2\sigma}\big[1+\mathrm{erf}(\sigma c)\big], \tag{6.11}$$

$$I_1(c) \equiv \int_{-c}^\infty x\,\mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = \frac{\mathrm{e}^{-\sigma^2c^2}}{2\sigma^2}, \tag{6.12}$$

$$I_2(c) \equiv \int_{-c}^\infty x^2\mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = -\frac{c\,\mathrm{e}^{-\sigma^2c^2}}{2\sigma^2} + \frac{\sqrt{\pi}}{4\sigma^3}\,\mathrm{erfc}(-\sigma c). \tag{6.13}$$

Full-line versions ($c \to \infty$): $\frac{\sqrt{\pi}}{\sigma}$, 0, $\frac{\sqrt{\pi}}{2\sigma^3}$.

Proof. I_0. Substitute $t = \sigma x$ and use the definition $\mathrm{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty \mathrm{e}^{-t^2}\mathrm{d}t$:

$$I_0 = \frac{1}{\sigma}\int_{-\sigma c}^\infty \mathrm{e}^{-t^2}\mathrm{d}t = \frac{\sqrt{\pi}}{2\sigma}\,\mathrm{erfc}(-\sigma c), \qquad \mathrm{erfc}(-z) = 2-\mathrm{erfc}(z) = 1+\mathrm{erf}(z).$$

I_1. The integrand is a total derivative, $x\,\mathrm{e}^{-\sigma^2x^2} = -\frac{1}{2\sigma^2}\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-\sigma^2x^2}$:

$$I_1 = -\frac{1}{2\sigma^2}\Big[\mathrm{e}^{-\sigma^2x^2}\Big]_{-c}^\infty = \frac{\mathrm{e}^{-\sigma^2c^2}}{2\sigma^2}.$$

I_2. By parts with $u = x$, $\mathrm{d}v = x\,\mathrm{e}^{-\sigma^2x^2}\mathrm{d}x$, so $v = -\frac{\mathrm{e}^{-\sigma^2x^2}}{2\sigma^2}$:

$$I_2 = \Big[-\frac{x\,\mathrm{e}^{-\sigma^2x^2}}{2\sigma^2}\Big]_{-c}^\infty + \frac{1}{2\sigma^2}\int_{-c}^\infty \mathrm{e}^{-\sigma^2x^2}\mathrm{d}x = -\frac{c\,\mathrm{e}^{-\sigma^2c^2}}{2\sigma^2} + \frac{1}{2\sigma^2}I_0.$$

(At the lower limit $x = -c$ the bracket is $+\frac{c\,\mathrm{e}^{-\sigma^2c^2}}{2\sigma^2}$, which is subtracted.) □

6.5 The smeared functions, derived

6.5.1 Coulomb: $1/r$

Route 1: radial formula. $f = 1/r$ gives $g(u) = u/|u| = \mathrm{sgn}(u)$. Shift $x = u - r$ and write $\mathrm{sgn}(x + r) = 2\Theta(x + r) - 1$:

$$\int_{-\infty}^{\infty} \mathrm{sgn}(u) \mathrm{e}^{-\sigma^2(r-u)^2} \mathrm{d}u = 2I_0(r) - \frac{\sqrt{\pi}}{\sigma} = \frac{\sqrt{\pi}}{\sigma}[1 + \mathrm{erf}(\sigma r)] - \frac{\sqrt{\pi}}{\sigma} = \frac{\sqrt{\pi}}{\sigma}\mathrm{erf}(\sigma r).$$

Multiply by $\frac{\sigma}{\sqrt{\pi} r}$:

$$\boxed{\widetilde{\left(\frac{1}{r}\right)} = \frac{\mathrm{erf}(\sigma r)}{r}} \tag{6.14}$$

Route 2: Gauss's law. $\mathrm{erf}(\sigma r)/r$ is the electrostatic potential (in units where a point charge gives $1/r$) of the unit charge cloud ρ_σ. Charge inside radius r:

$$Q(r) = \int_0^r 4\pi s^2 \rho_\sigma(s) \mathrm{d}s = \frac{4\sigma^3}{\sqrt{\pi}} \int_0^r s^2 \mathrm{e}^{-\sigma^2 s^2} \mathrm{d}s = \mathrm{erf}(\sigma r) - \frac{2\sigma r}{\sqrt{\pi}} \mathrm{e}^{-\sigma^2 r^2}$$

(the last integral is I_2-type: by parts as above). The field is $Q(r)/r^2$ and the potential $\phi(r) = \int_r^\infty Q(s)/s^2 \, \mathrm{d}s$. Check by differentiating the claimed answer: $\frac{\mathrm{d}}{\mathrm{d}r}\frac{\mathrm{erf}(\sigma r)}{r} = \frac{2\sigma}{\sqrt{\pi} r}\mathrm{e}^{-\sigma^2 r^2} - \frac{\mathrm{erf}(\sigma r)}{r^2} = -\frac{Q(r)}{r^2}$, and both vanish at infinity.

Route 3: momentum space. $\frac{4\pi}{q^2} \to \frac{4\pi}{q^2}\mathrm{e}^{-q^2/4\sigma^2}$, whose transform is $\mathrm{erf}(\sigma r)/r$ (Result in §5).

With the GI running coupling. Each term of $G(r) = -\frac{4}{3}\sum_k \alpha_k \, \mathrm{erf}(\gamma_k r)/r$ is already a Coulomb potential smeared with width γ_k (that is what the Gaussian in $\alpha_s(Q)$ means). Smearing it again with σ_{12} composes the widths by (6.7):

Smeared Coulomb potential with running coupling

$$\tilde{G}(r) = -\frac{4}{3}\sum_{k=1}^{3} \alpha_k \frac{\mathrm{erf}(\tau_k r)}{r}, \qquad \frac{1}{\tau_k^2} = \frac{1}{\gamma_k^2} + \frac{1}{\sigma_{12}^2} \iff \tau_k = \frac{\gamma_k \sigma_{12}}{\sqrt{\gamma_k^2 + \sigma_{12}^2}}. \tag{6.15}$$

6.5.2 Contact term: $\delta^3(\boldsymbol{r})$ and $\nabla^2 \tilde{G}$

$\tilde{\delta} = \rho_\sigma * \delta = \rho_\sigma$: the smeared δ function is the Gaussian itself (this is δ_σ of Model A). For the contact term generated by the Coulomb potential, use $\nabla^2 \frac{1}{r} = -4\pi\delta^3(\boldsymbol{r})$ and Consequence 2:

$$\nabla^2 \frac{\mathrm{erf}(\tau r)}{r} = \widetilde{\nabla^2 \frac{1}{r}} = -4\pi\rho_\tau(r) = -\frac{4\tau^3}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}. \tag{6.16}$$

Direct check: with $g = \mathrm{erf}(\tau r)/r$, $r^2 g' = \frac{2\tau r}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} - \mathrm{erf}(\tau r)$, so $(r^2 g')' = \frac{2\tau}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} - \frac{4\tau^3 r^2}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} - \frac{2\tau}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} = -\frac{4\tau^3 r^2}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}$, and $\nabla^2 g = (r^2 g')'/r^2$ reproduces (6.16). Therefore

$$\nabla^2 \tilde{G} = \frac{16\pi}{3}\sum_k \alpha_k \, \rho_{\tau_k}(r), \qquad \frac{2}{3m_1 m_2}\nabla^2 \tilde{G} = \frac{32\pi}{9m_1 m_2}\sum_k \alpha_k \frac{\tau_k^3}{\pi^{3/2}}\mathrm{e}^{-\tau_k^2 r^2}, \tag{6.17}$$

the Breit–Fermi contact term with $\alpha_s \delta^3 \to \sum_k \alpha_k \rho_{\tau_k}$: a sum of normalized Gaussians.

6.5.3 Derivatives for the tensor and spin–orbit terms

For $g_\tau(r) = \operatorname{erf}(\tau r)/r$, differentiate directly (using $\frac{\mathrm{d}}{\mathrm{d}r}\operatorname{erf}(\tau r) = \frac{2\tau}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}$):

$$g'_\tau = \frac{2\tau}{\sqrt{\pi}\, r}\mathrm{e}^{-\tau^2 r^2} - \frac{\operatorname{erf}(\tau r)}{r^2}, \tag{6.18}$$

$$g''_\tau = \nabla^2 g_\tau - \frac{2}{r}g'_\tau = -\frac{4\tau^3}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} - \frac{4\tau}{\sqrt{\pi}\, r^2}\mathrm{e}^{-\tau^2 r^2} + \frac{2\operatorname{erf}(\tau r)}{r^3}, \tag{6.19}$$

$$T_\tau(r) \equiv \frac{g'_\tau}{r} - g''_\tau = \frac{6\tau}{\sqrt{\pi}\, r^2}\mathrm{e}^{-\tau^2 r^2} - \frac{3\operatorname{erf}(\tau r)}{r^3} + \frac{4\tau^3}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}. \tag{6.20}$$

T_τ enters the tensor force, g'_τ/r the vector spin–orbit force (with $\tilde{G} = -\frac{4}{3}\sum \alpha_k g_{\tau_k}$).

Limits (derive them).

- $\tau r \gg 1$: erf $\to 1$, Gaussians vanish: $g'/r \to -1/r^3$, $T_\tau \to -3/r^3$ (the point values; with $-\kappa g$ this is the $+3\kappa/r^3$ of the unsmeared tensor force).
- $\tau r \ll 1$: use $\operatorname{erf} z = \frac{2}{\sqrt{\pi}}(z - \frac{z^3}{3} + \frac{z^5}{10} - \dots)$ and $\mathrm{e}^{-z^2} = 1 - z^2 + \frac{z^4}{2} - \dots$ In T_τ the $1/r^2$ terms cancel ($\frac{6\tau}{\sqrt{\pi}} - \frac{6\tau}{\sqrt{\pi}}$), the constants cancel ($-\frac{6\tau^3}{\sqrt{\pi}} + \frac{2\tau^3}{\sqrt{\pi}} + \frac{4\tau^3}{\sqrt{\pi}}$), and the r^2 terms give $\frac{\tau^5}{\sqrt{\pi}}(3 - \frac{3}{5} - 4)r^2$:

$$T_\tau(r) = -\frac{8\tau^5}{5\sqrt{\pi}}\, r^2 + O(r^4), \qquad \frac{g'_\tau}{r} = -\frac{4\tau^3}{3\sqrt{\pi}} + O(r^2). \tag{6.21}$$

  Both are regular at the origin: smearing removes the $1/r^3$ singularities completely.

6.5.4 Linear confinement: r

Result 6.5.

$$\tilde{r} \equiv \widetilde{(r)} = \frac{\mathrm{e}^{-\sigma^2 r^2}}{\sqrt{\pi}\,\sigma} + \Big(r + \frac{1}{2\sigma^2 r}\Big)\operatorname{erf}(\sigma r), \qquad \tilde{r}\,' = \Big(1 - \frac{1}{2\sigma^2 r^2}\Big)\operatorname{erf}(\sigma r) + \frac{\mathrm{e}^{-\sigma^2 r^2}}{\sqrt{\pi}\,\sigma r}. \tag{6.22}$$

Derivation, step by step. Step 1. $f = r$ gives $g(u) = u|u| = \operatorname{sgn}(u)\, u^2$, so by (6.9) $\tilde{r} = \frac{\sigma}{\sqrt{\pi}\, r}J$ with

$$J = \int_{-\infty}^{\infty} \operatorname{sgn}(u)\, u^2\, \mathrm{e}^{-\sigma^2(u-r)^2}\mathrm{d}u.$$

Step 2 (centre the Gaussian). Put $x = u - r$ and $\operatorname{sgn}(x + r) = 2\Theta(x + r) - 1$:

$$J = 2\int_{-r}^{\infty}(x + r)^2\mathrm{e}^{-\sigma^2 x^2}\mathrm{d}x - \int_{-\infty}^{\infty}(x + r)^2\mathrm{e}^{-\sigma^2 x^2}\mathrm{d}x.$$

Step 3 (expand $(x + r)^2 = x^2 + 2rx + r^2$ and use the toolbox with $c = r$).

$$\begin{aligned}
\int_{-r}^{\infty}(x + r)^2\mathrm{e}^{-\sigma^2 x^2}\mathrm{d}x &= I_2 + 2rI_1 + r^2 I_0 \\
&= -\frac{r\mathrm{e}^{-\sigma^2 r^2}}{2\sigma^2} + \frac{\sqrt{\pi}}{4\sigma^3}\operatorname{erfc}(-\sigma r) + \frac{r\mathrm{e}^{-\sigma^2 r^2}}{\sigma^2} + \frac{\sqrt{\pi}r^2}{2\sigma}\operatorname{erfc}(-\sigma r) \\
&= \frac{r\mathrm{e}^{-\sigma^2 r^2}}{2\sigma^2} + \Big(\frac{\sqrt{\pi}}{4\sigma^3} + \frac{\sqrt{\pi}r^2}{2\sigma}\Big)[1 + \operatorname{erf}(\sigma r)], \\
\int_{-\infty}^{\infty}(x + r)^2\mathrm{e}^{-\sigma^2 x^2}\mathrm{d}x &= \frac{\sqrt{\pi}}{2\sigma^3} + 0 + \frac{\sqrt{\pi}r^2}{\sigma}.
\end{aligned}$$

Step 4 (combine). Twice the first minus the second: the "1" in $[1+\mathrm{erf}]$ cancels exactly against the full-line integral, leaving

$$J = \frac{r\,\mathrm{e}^{-\sigma^2 r^2}}{\sigma^2} + \Big(\frac{\sqrt{\pi}}{2\sigma^3} + \frac{\sqrt{\pi}\,r^2}{\sigma}\Big)\,\mathrm{erf}(\sigma r).$$

Step 5. Multiply by $\frac{\sigma}{\sqrt{\pi}r}$: $\frac{\sigma}{\sqrt{\pi}r}\cdot\frac{r\mathrm{e}^{-\sigma^2r^2}}{\sigma^2} = \frac{\mathrm{e}^{-\sigma^2r^2}}{\sqrt{\pi}\sigma}$ and $\frac{\sigma}{\sqrt{\pi}r}\big(\frac{\sqrt{\pi}}{2\sigma^3} + \frac{\sqrt{\pi}r^2}{\sigma}\big) = \frac{1}{2\sigma^2 r} + r$. This is (6.22).

Step 6 (derivative). Differentiate term by term:

$$\frac{\mathrm{d}}{\mathrm{d}r}\frac{\mathrm{e}^{-\sigma^2r^2}}{\sqrt{\pi}\sigma} = -\frac{2\sigma r}{\sqrt{\pi}}\mathrm{e}^{-\sigma^2r^2},$$
$$\frac{\mathrm{d}}{\mathrm{d}r}\Big[\Big(r+\frac{1}{2\sigma^2 r}\Big)\,\mathrm{erf}(\sigma r)\Big] = \Big(1-\frac{1}{2\sigma^2r^2}\Big)\,\mathrm{erf}(\sigma r) + \Big(r+\frac{1}{2\sigma^2 r}\Big)\frac{2\sigma}{\sqrt{\pi}}\mathrm{e}^{-\sigma^2r^2}.$$

The last term is $\frac{2\sigma r}{\sqrt{\pi}}\mathrm{e}^{-\sigma^2r^2} + \frac{1}{\sqrt{\pi}\sigma r}\mathrm{e}^{-\sigma^2r^2}$; its first piece cancels the derivative of the first term, leaving $\tilde r\,'$ in (6.22). □

Limits, with their meaning.

- $r\to\infty$: erf $\to 1$ up to $\mathrm{e}^{-\sigma^2r^2}$, so $\tilde r \to r + \frac{1}{2\sigma^2 r}$. Meaning: $\tilde r = \langle|\boldsymbol{r}+\boldsymbol{x}|\rangle$; for $r \gg |\boldsymbol{x}|$, $|\boldsymbol{r}+\boldsymbol{x}| \simeq r + x_\parallel + \frac{x_\perp^2}{2r}$, and $\langle x_\parallel\rangle = 0$, $\langle x_\perp^2\rangle = 2\cdot\frac{1}{2\sigma^2} = \frac{1}{\sigma^2}$.
- $r\to 0$: expanding as before,

$$\tilde r = \frac{2}{\sqrt{\pi}\,\sigma} + \frac{2\sigma}{3\sqrt{\pi}}r^2 + O(r^4), \qquad \frac{\tilde r\,'}{r} \to \frac{4\sigma}{3\sqrt{\pi}}. \tag{6.23}$$

  The constant is $\langle|\boldsymbol{x}|\rangle$ (at $r=0$ we average the distance itself). The finite $\tilde r\,'/r$ means the scalar Thomas term $-b\,\tilde r\,'/(2m^2 r)$ is regular at the origin.

For GI: $\tilde S(r) = b\,\tilde r + c$, $\tilde S'(r) = b\,\tilde r\,'$ with $\sigma = \sigma_{12}$.

6.5.5 Screened confinement: $\mathrm{e}^{-\mu_s r}$

Notation: $a \equiv \frac{\mu_s}{2\sigma^2}$, $\beta \equiv \frac{\mu_s}{2\sigma}$ (so $\beta^2 = \frac{\mu_s^2}{4\sigma^2}$ and $2\sigma\beta = \mu_s$).

Result 6.6.

$$\widetilde{\mathrm{e}^{-\mu_s r}}(r) = \frac{\mathrm{e}^{\beta^2}}{2r}\Big[(r-a)\,\mathrm{e}^{-\mu_s r}\,\mathrm{erfc}(\beta-\sigma r) + (r+a)\,\mathrm{e}^{+\mu_s r}\,\mathrm{erfc}(\beta+\sigma r)\Big]. \tag{6.24}$$

Derivation, step by step. Step 1. $g(u) = u\,\mathrm{e}^{-\mu_s|u|}$. Split the real line at $u=0$ and put $u\to -u$ on the negative half (oddness):

$$\int_{-\infty}^{\infty} u\,\mathrm{e}^{-\mu_s|u|}\mathrm{e}^{-\sigma^2(r-u)^2}\mathrm{d}u = K_+ - K_-, \qquad K_\pm = \int_0^\infty u\,\mathrm{e}^{-\mu_s u}\mathrm{e}^{-\sigma^2(u\mp r)^2}\mathrm{d}u.$$

Step 2 (complete the square). Expand $\sigma^2(u-c)^2$ with $c_\pm = \pm r - a$:

$$-\sigma^2(u\mp r)^2 - \mu_s u = -\sigma^2(u-c_\pm)^2 \mp \mu_s r + \beta^2.$$

(Check: $\sigma^2(u-c_\pm)^2 = \sigma^2(u\mp r)^2 + \mu_s(u\mp r) + \beta^2$, since $2\sigma^2 a = \mu_s$ and $\sigma^2 a^2 = \beta^2$.) Hence $K_\pm = \mathrm{e}^{\beta^2\mp\mu_s r}\int_0^\infty u\,\mathrm{e}^{-\sigma^2(u-c_\pm)^2}\mathrm{d}u$.

Step 3 (shifted half-line integral, toolbox with $y = u - c$).

$$\int_0^\infty u\,\mathrm{e}^{-\sigma^2(u-c)^2}\mathrm{d}u = \int_{-c}^\infty (y+c)\mathrm{e}^{-\sigma^2 y^2}\mathrm{d}y = I_1(c) + c\,I_0(c) = \frac{\mathrm{e}^{-\sigma^2 c^2}}{2\sigma^2} + \frac{\sqrt{\pi}\,c}{2\sigma}\operatorname{erfc}(-\sigma c).$$

Step 4 (the Gaussian pieces cancel). $\sigma^2 c_\pm^2 = \sigma^2 r^2 \mp \mu_s r + \beta^2$, so the Gaussian piece of $K_\pm$ is $\mathrm{e}^{\beta^2 \mp \mu_s r}\frac{\mathrm{e}^{-\sigma^2 r^2 \pm \mu_s r - \beta^2}}{2\sigma^2} = \frac{\mathrm{e}^{-\sigma^2 r^2}}{2\sigma^2}$, *the same for both signs*; it cancels in $K_+ - K_-$:

$$K_+ - K_- = \frac{\sqrt{\pi}}{2\sigma}\mathrm{e}^{\beta^2}\Big[c_+\mathrm{e}^{-\mu_s r}\operatorname{erfc}(-\sigma c_+) - c_-\mathrm{e}^{+\mu_s r}\operatorname{erfc}(-\sigma c_-)\Big].$$

Step 5. Insert $c_+ = r - a$, $c_- = -(r+a)$, $-\sigma c_+ = \beta - \sigma r$, $-\sigma c_- = \beta + \sigma r$, and multiply by $\frac{\sigma}{\sqrt{\pi} r}$: this is (6.24). □

Checks.

- $\sigma \to \infty$ $(a, \beta \to 0)$: $\operatorname{erfc}(-\infty) = 2$, $\operatorname{erfc}(+\infty) = 0 \Rightarrow$ $\frac{1}{2r}\,r\,\mathrm{e}^{-\mu_s r}\cdot 2 = \mathrm{e}^{-\mu_s r}$.
- $\mu_s \to 0$: $\frac{1}{2r}\big[r\operatorname{erfc}(-\sigma r) + r\operatorname{erfc}(\sigma r)\big] = \frac{1}{2}\big[2 - \operatorname{erfc}(\sigma r) + \operatorname{erfc}(\sigma r)\big] = 1$.
- First order in μ_s: $\widetilde{\mathrm{e}^{-\mu_s r}} = 1 - \mu_s \tilde{r} + O(\mu_s^2)$, consistent with (6.22).
- $r \to \infty$: $\widetilde{\mathrm{e}^{-\mu_s r}} \to \mathrm{e}^{\beta^2}(1 - \frac{a}{r})\mathrm{e}^{-\mu_s r}$. Smearing an exponential tail multiplies it by $\mathrm{e}^{\mu_s^2/4\sigma^2} > 1$ (averaging a convex function raises it).

Derivative (needed for the scalar spin–orbit term of Model D). Write $\widetilde{\mathrm{e}^{-\mu_s r}} = \frac{\mathrm{e}^{\beta^2}}{2r}F(r)$ with F the bracket in (6.24). Differentiating F produces two kinds of terms: products from $\frac{\mathrm{d}}{\mathrm{d}r}[(r \mp a)\mathrm{e}^{\mp\mu_s r}]$, and Gaussians from $\frac{\mathrm{d}}{\mathrm{d}r}\operatorname{erfc}(\beta \mp \sigma r) = \pm\frac{2\sigma}{\sqrt{\pi}}\mathrm{e}^{-(\beta\mp\sigma r)^2}$. The Gaussians combine because $\mathrm{e}^{\mp\mu_s r}\mathrm{e}^{-(\beta\mp\sigma r)^2} = \mathrm{e}^{-\beta^2-\sigma^2 r^2}$ for both signs (again $2\sigma\beta = \mu_s$):

$$\text{Gaussian part of } F' = \frac{2\sigma}{\sqrt{\pi}}\mathrm{e}^{-\beta^2-\sigma^2 r^2}\big[(r-a) - (r+a)\big] = -\frac{2\mu_s}{\sqrt{\pi}\,\sigma}\mathrm{e}^{-\beta^2-\sigma^2 r^2}.$$

With $\frac{\mathrm{d}}{\mathrm{d}r}\big[\frac{F}{2r}\big] = \frac{F'}{2r} - \frac{F}{2r^2}$ and $\mathrm{e}^{\beta^2}\mathrm{e}^{-\beta^2} = 1$:

Smeared exponential and its derivative

$$\frac{\mathrm{d}}{\mathrm{d}r}\widetilde{\mathrm{e}^{-\mu_s r}} = -\frac{\widetilde{\mathrm{e}^{-\mu_s r}}}{r} - \frac{\mu_s}{\sqrt{\pi}\,\sigma\,r}\mathrm{e}^{-\sigma^2 r^2} + \frac{\mathrm{e}^{\beta^2}}{2r}\Big[(1-\mu_s(r-a))\mathrm{e}^{-\mu_s r}\operatorname{erfc}(\beta - \sigma r) + (1+\mu_s(r+a))\mathrm{e}^{\mu_s r}\operatorname{erfc}(\beta+\sigma r)\Big]. \tag{6.25}$$

Check $\sigma \to \infty$: $-\frac{\mathrm{e}^{-\mu_s r}}{r} + \frac{(1-\mu_s r)\mathrm{e}^{-\mu_s r}}{r} = -\mu_s \mathrm{e}^{-\mu_s r}$. ✓

$$\tilde{S}_{\mathrm{scr}}(r) = \frac{b}{\mu_s}\Big[1 - \widetilde{\mathrm{e}^{-\mu_s r}}(r)\Big] + c, \qquad \tilde{S}'_{\mathrm{scr}}(r) = -\frac{b}{\mu_s}\frac{\mathrm{d}}{\mathrm{d}r}\widetilde{\mathrm{e}^{-\mu_s r}}(r). \tag{6.26}$$

Both formulas were checked against the radial convolution (6.9) by quadrature and against a finite-difference derivative (agreement $\sim 10^{-10}$); $\tilde{S}'_{\mathrm{scr}}/r$ is finite at the origin.

Numerically stable form. For large σr, $\mathrm{e}^{+\mu_s r}\,\mathrm{erfc}(\beta+\sigma r)$ is a huge number times a tiny one. Use $\mathrm{erfcx}(z) = \mathrm{e}^{z^2}\,\mathrm{erfc}(z)$ and the identity $\mathrm{e}^{\beta^2}\mathrm{e}^{\mu_s r}\mathrm{e}^{-(\beta+\sigma r)^2} = \mathrm{e}^{-\sigma^2 r^2}$:

$$\widetilde{\mathrm{e}^{-\mu_s r}}(r) = \frac{1}{2r}\Big[(r-a)\,\mathrm{e}^{\beta^2-\mu_s r}\,\mathrm{erfc}(\beta-\sigma r) + (r+a)\,\mathrm{e}^{-\sigma^2 r^2}\,\mathrm{erfcx}(\beta+\sigma r)\Big], \tag{6.27}$$

and the same substitution in the derivative ($\mathrm{e}^{\beta^2}\mathrm{e}^{\mu_s r}\,\mathrm{erfc}(\beta+\sigma r) \to \mathrm{e}^{-\sigma^2 r^2}\,\mathrm{erfcx}(\beta+\sigma r)$). The first term is harmless: $\mathrm{erfc}(\beta-\sigma r) \to 2$ for large r.

6.6 Summary of smeared building blocks

Point form	Smeared form	Derivatives	Used in
$\delta^3(\boldsymbol{r})$	$\rho_\sigma = \frac{\sigma^3}{\pi^{3/2}}\mathrm{e}^{-\sigma^2 r^2}$	—	A, B, C, D
$1/r$	$\mathrm{erf}(\sigma r)/r$	g', $\nabla^2 g$, T_τ (6.20)	C, D
running α_s (GI)	$\tilde{G} = -\frac{4}{3}\sum_k \alpha_k\,\mathrm{erf}(\tau_k r)/r$	termwise; τ_k from (6.7)	C, D
r	(6.22)	$\tilde{r}\,'$ in (6.22)	C
$\mathrm{e}^{-\mu_s r}$	(6.24)	key result above	D

7 The Gaussian expansion method

7.1 Idea

We want eigenvalues and eigenfunctions of H (any of Models A–D). The variational (Rayleigh–Ritz) method expands the unknown wave function in a finite set of known functions and determines the coefficients so that the energy is stationary. The GEM (Kamimura; Hiyama, Kino, Kamimura) uses Gaussians whose ranges form a geometric progression. Why Gaussians?

1. Products of Gaussians are Gaussians, so overlaps, kinetic energies and matrix elements of r^k, $\mathrm{e}^{-\beta r^2}$, $\mathrm{erf}(\tau r)/r$, $\mathrm{e}^{-\mu_s r}$, … are analytic or reduce to one-dimensional integrals.
2. Their Fourier transforms are Gaussians, so momentum-space operators ($\sqrt{p^2+m^2}$, m/E factors) are equally easy.
3. A geometric progression of ranges covers many length scales with few functions: short-range (Coulomb, $r \lesssim 0.1\,\mathrm{fm}$) and long-range (confinement, highly excited states at several fm) structure simultaneously.
4. The factor r^l gives the correct behaviour at the origin automatically.

The price is that the basis is *non-orthogonal*, which leads to a generalized eigenvalue problem with an overlap matrix that is nearly singular. Both are easy to handle (§7.4, §7.8).

7.2 The basis

Definition 7.1 (GEM radial basis)**.** For orbital angular momentum l,

$$\phi_{nlm}(\boldsymbol{r}) = R_{nl}(r)\,Y_{lm}(\hat{r}), \qquad R_{nl}(r) = N_{nl}\,r^l \mathrm{e}^{-\nu_n r^2}, \qquad n = 1,\dots,n_{\max}, \tag{7.1}$$

$$\nu_n = \frac{1}{r_n^2}, \qquad r_n = r_1\,a^{\,n-1}, \qquad a = \Big(\frac{r_{\max}}{r_1}\Big)^{\frac{1}{n_{\max}-1}}. \tag{7.2}$$

Three numbers $\{n_{\max}, r_1, r_{\max}\}$ define the whole basis. (Hiyama et al. write $\mathrm{e}^{-(r/r_n)^2}$, i.e. the same thing.)

Exercise 7.2 (Normalization)**.** With $\int |Y_{lm}|^2 \mathrm{d}\Omega = 1$ we need $\int_0^\infty r^2 R_{nl}^2 \mathrm{d}r = 1$. Substitute $t = 2\nu r^2$ in $\int_0^\infty r^{2l+2} \mathrm{e}^{-2\nu r^2} \mathrm{d}r$ to get $\frac{\Gamma(l+3/2)}{2(2\nu)^{l+3/2}}$. Hence

$$N_{nl}^2 = \frac{2(2\nu_n)^{l+3/2}}{\Gamma(l+\frac{3}{2})} = \frac{2^{l+2}(2\nu_n)^{l+3/2}}{\sqrt{\pi}\,(2l+1)!!}, \tag{7.3}$$

using $\Gamma(l+\frac{3}{2}) = \frac{(2l+1)!!}{2^{l+1}}\sqrt{\pi}$.

7.3 Including spin: your spinless notation and the coupled basis

Spinless notation. The expansion

$$\psi_{lm}(\boldsymbol{r}) = \sum_{n=1}^{n_{\max}} c_n\, \phi_{nlm}(\boldsymbol{r}) \tag{7.4}$$

is complete for any *spin-independent* central Hamiltonian; the full state is $\psi_{lm}\chi_{SM_S}$ and the problem for each l is purely radial (the angular integral gives $\delta_{ll'}\delta_{mm'}$, §7.4).

Adding spin terms without changing the basis. All spin operators are rotational scalars, so in the coupled states $\mathcal{Y}_{LS}^{JM}$ of (2.9) they act as numbers $o_k(L,S,J)$ (Key result in §3.5). Writing $H_{SD} = \sum_k W_k(r)\hat{O}_k$ with $\hat{O}_k \in \{\boldsymbol{S}_1\cdot\boldsymbol{S}_2, \boldsymbol{L}\cdot\boldsymbol{S}, S_{12}\}$, the matrix for a fixed (L,S,J) is

$$H_{n'n}^{(LSJ)} = \langle R_{n'L}|H_0|R_{nL}\rangle + \sum_k o_k(L,S,J)\,\langle R_{n'L}|W_k|R_{nL}\rangle, \tag{7.5}$$

i.e. *radial matrices times spin–angle numbers*. Your ψ_{lm} basis therefore suffices for every channel except the tensor coupling.

Coupled channels. For $S=1$, $L = J\mp 1$ (e.g. ${}^3S_1$–${}^3D_1$) the tensor force has off-diagonal elements. Use

$$\Psi_{JM}(\boldsymbol{r}) = \sum_n c_n^{(J-1)} R_{n,J-1}(r)\mathcal{Y}_{J-1,1}^{JM} + \sum_n c_n^{(J+1)} R_{n,J+1}(r)\mathcal{Y}_{J+1,1}^{JM}, \tag{7.6}$$

which gives a 2×2 block matrix of size $2n_{\max}$ (§7.4.5). Keeping a single L is equivalent to treating the S–D mixing to zeroth order.

7.4 From the Schrödinger equation to the generalized eigenvalue problem

7.4.1 Galerkin projection

Write the eigenvalue problem as $(H-\mathcal{E})\Psi = 0$ with $\mathcal{E} = M - m_1 - m_2$ (Models A, B) or $\mathcal{E} = M$ (Models C, D). Insert the finite expansion

$$\Psi^{(N)} = \sum_{n=1}^{N} c_n \Phi_n, \qquad \Phi_n \equiv \phi_{nlm}\chi_{SM_S} \text{ or } R_{nL}\mathcal{Y}_{LS}^{JM}. \tag{7.7}$$

Since the basis is finite, $(H-\mathcal{E})\Psi^{(N)}$ cannot vanish identically. Galerkin's condition demands that this residual be orthogonal to every basis function:

$$\langle\Phi_{n'}|(H-\mathcal{E})\Psi^{(N)}\rangle = 0 \quad \forall n' \quad\Longrightarrow\quad \sum_n \Big[\underbrace{\langle\Phi_{n'}|H|\Phi_n\rangle}_{H_{n'n}} - \mathcal{E}\,\underbrace{\langle\Phi_{n'}|\Phi_n\rangle}_{S_{n'n}}\Big] c_n = 0. \tag{7.8}$$

Generalized eigenvalue problem

$$\mathbf{H}\boldsymbol{c} = \mathcal{E}\,\mathbf{S}\boldsymbol{c}, \qquad H_{n'n} = \langle\Phi_{n'}|H|\Phi_n\rangle, \qquad S_{n'n} = \langle\Phi_{n'}|\Phi_n\rangle, \qquad \det(\mathbf{H}-\mathcal{E}\mathbf{S}) = 0. \tag{7.9}$$

7.4.2 Reduction of the three-dimensional matrix elements

For $\Phi_n = \phi_{nlm}\chi$ and a spin-independent central operator $O(r, p^2)$:

$$\langle \phi_{n'l'm'}|O|\phi_{nlm}\rangle = \underbrace{\int \mathrm{d}\Omega\, Y^*_{l'm'} Y_{lm}}_{\delta_{ll'}\delta_{mm'}} \int_0^\infty r^2 \mathrm{d}r\, R_{n'l}(r)\,[O_l R_{nl}](r), \tag{7.10}$$

where O_l is the radial operator in partial wave l (for p^2: $-\partial_r^2 - \frac{2}{r}\partial_r + \frac{l(l+1)}{r^2}$; for a function of p^2 the partial-wave momentum representation of §7.6). The matrices do not depend on m: all $2l+1$ states are degenerate, as they must be. With spin and coupled states, the spin–angle integral produces the factors $o_k(L, S, J)$ and, for the tensor force between $L \neq L'$, the off-diagonal factor; the remaining integral is always radial.

7.4.3 Equivalence with the variational principle

For real symmetric $\mathbf{H}, \mathbf{S}$ define the Rayleigh quotient

$$\mathcal{E}[\boldsymbol{c}] = \frac{\langle \Psi^{(N)}|H|\Psi^{(N)}\rangle}{\langle \Psi^{(N)}|\Psi^{(N)}\rangle} = \frac{\boldsymbol{c}^T\mathbf{H}\boldsymbol{c}}{\boldsymbol{c}^T\mathbf{S}\boldsymbol{c}}. \tag{7.11}$$

Exercise 7.3. Differentiate: $\frac{\partial \mathcal{E}}{\partial c_m} = \frac{2}{\boldsymbol{c}^T\mathbf{S}\boldsymbol{c}}\big[(\mathbf{H}\boldsymbol{c})_m - \mathcal{E}(\mathbf{S}\boldsymbol{c})_m\big]$. Stationarity $\Leftrightarrow$ $\mathbf{H}\boldsymbol{c} = \mathcal{E}\mathbf{S}\boldsymbol{c}$. So Galerkin and Rayleigh–Ritz give the same equations when H is Hermitian.

7.4.4 Properties

Result 7.4. *1.* $\mathbf{S}$ *is symmetric positive definite:* $\boldsymbol{c}^T\mathbf{S}\boldsymbol{c} = \int r^2 \big(\sum_n c_n R_{nl}\big)^2 \mathrm{d}r > 0$ *for* $\boldsymbol{c} \neq 0$ *because Gaussians with distinct exponents are linearly independent.*

2. $\mathbf{H}$ *is symmetric if* H *is Hermitian and the basis is real. (For non-local GI terms, enforce it: §7.7.)*

3. All eigenvalues $\mathcal{E}_k$ *are real; eigenvectors are* $\mathbf{S}$*-orthogonal: from* $\boldsymbol{c}_b^T\mathbf{H}\boldsymbol{c}_a = \mathcal{E}_a\boldsymbol{c}_b^T\mathbf{S}\boldsymbol{c}_a$ *and the same with* $a \leftrightarrow b$*, subtract:* $(\mathcal{E}_a - \mathcal{E}_b)\boldsymbol{c}_b^T\mathbf{S}\boldsymbol{c}_a = 0$*. Normalize with* $\boldsymbol{c}_a^T\mathbf{S}\boldsymbol{c}_b = \delta_{ab}$*, which is* $\langle\Psi_a|\Psi_b\rangle = \delta_{ab}$*.*

4. Upper bounds (Hylleraas–Undheim–MacDonald, HUM). *Order the Ritz values* $\mathcal{E}_1^{(N)} \leq \cdots \leq \mathcal{E}_N^{(N)}$ *from one diagonalization. Then* $\mathcal{E}_k^{(N)} \geq \mathcal{E}_k^{\text{exact}}$ *for every* $k \leq N$*, and if the basis is enlarged by* adding *functions,* $\mathcal{E}_k^{(N+1)} \leq \mathcal{E}_k^{(N)}$*.*

Proof of 4. Min–max (Courant–Fischer). For a Hermitian H bounded below,

$$\mathcal{E}_k^{\text{exact}} = \min_{V_k} \max_{\psi \in V_k} \mathcal{E}[\psi],$$

the minimum running over all k-dimensional subspaces V_k of the Hilbert space, and the Ritz value is the same expression with V_k restricted to subspaces of the span $\mathcal{V}_N$ of the basis. A minimum over a smaller family is larger: $\mathcal{E}_k^{(N)} \geq \mathcal{E}_k^{\text{exact}}$. *Interlacing.* In an orthonormalized basis ($\mathbf{X}^T\mathbf{S}\mathbf{X} = \mathbf{1}$) the Ritz values are the eigenvalues of $\mathbf{H}' = \mathbf{X}^T\mathbf{H}\mathbf{X}$. Adding one function adds one row and column; by Cauchy's interlacing theorem the eigenvalues of the $(N+1)\times(N+1)$ matrix interlace those of its $N \times N$ submatrix: $\mathcal{E}_k^{(N+1)} \leq \mathcal{E}_k^{(N)} \leq \mathcal{E}_{k+1}^{(N+1)}$. □

Remark 7.5 (When HUM does *not* apply). (i) H unbounded below (an unsmeared attractive δ treated non-perturbatively; a spinless Salpeter Coulomb term above the critical coupling): the Ritz values have no lower limit and collapse as the basis improves. (ii) The matrix is not $\langle\phi_i|H|\phi_j\rangle$

of a *fixed* operator: e.g. the closure approximation for the GI non-local terms (§7.7) builds $\mathbf{F}\mathbf{S}^{-1}\mathbf{W}\mathbf{S}^{-1}\mathbf{F}$, which changes with the basis; the Ritz values are then not guaranteed bounds and need not decrease monotonically (use the two-step method if you need strict bounds). (iii) Non-nested basis sequences: each result is a bound, but they need not decrease (§7.12). (iv) Changing the orthogonalization threshold η changes the span. (v) The bounds refer to the eigenvalues of the *model* Hamiltonian, not to the physical masses.

7.4.5 Block structure for coupled channels

Order $\boldsymbol{c} = (c_1^{\alpha_1}, \dots, c_{n_{\max}}^{\alpha_1}, c_1^{\alpha_2}, \dots)^T$ with channels $\alpha = (L, S)$. Then

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}^{\alpha_1\alpha_1} & \mathbf{H}^{\alpha_1\alpha_2} \\ \mathbf{H}^{\alpha_2\alpha_1} & \mathbf{H}^{\alpha_2\alpha_2} \end{pmatrix}, \qquad \mathbf{S} = \begin{pmatrix} \mathbf{S}^{\alpha_1} & 0 \\ 0 & \mathbf{S}^{\alpha_2} \end{pmatrix}, \tag{7.12}$$

with off-diagonal blocks only from the tensor force: $H^{\alpha_1\alpha_2}_{n'n} = \langle S_{12} \rangle_{L_1 L_2} \int r^2 R_{n'L_1} W_T(r) R_{nL_2} \mathrm{d}r$. Channel probabilities: $P_\alpha = \boldsymbol{c}_\alpha^T \mathbf{S}^\alpha \boldsymbol{c}_\alpha$, $\sum_\alpha P_\alpha = 1$.

7.5 Analytic matrix elements

Throughout, $a \equiv \nu_{n'} + \nu_n$, the bra has l', the ket l, and $\ell \equiv \frac{l+l'}{2}$ (for diagonal blocks $\ell = l$).

7.5.1 Master integral

$$\int_0^\infty r^{2\ell+2+k} \mathrm{e}^{-ar^2} \mathrm{d}r = \frac{\Gamma(\ell + \frac{k+3}{2})}{2\, a^{\ell + \frac{k+3}{2}}} \qquad (2\ell + 3 + k > 0), \tag{7.13}$$

(substitute $t = ar^2$). Every power-law matrix element follows:

$$\langle r^k \rangle_{n'n} \equiv \int_0^\infty r^2 R_{n'l'}\, r^k\, R_{nl}\, \mathrm{d}r = N_{n'l'} N_{nl} \frac{\Gamma(\ell + \frac{k+3}{2})}{2\, a^{\ell + \frac{k+3}{2}}}. \tag{7.14}$$

Special cases: $k = -1$ (Coulomb), $k = 1$ (linear), $k = 2$ (rms radius), $k = -3$ (point spin–orbit/tensor; requires $\ell \geq \frac{1}{2}$, i.e. $l + l' \geq 1$).

7.5.2 Overlap

$$S_{n'n} = \langle r^0 \rangle_{n'n}\Big|_{l=l'} = N_{n'l} N_{nl} \frac{\Gamma(l + \frac{3}{2})}{2a^{l+3/2}} = \Big(\frac{2\sqrt{\nu_{n'}\nu_n}}{\nu_{n'} + \nu_n}\Big)^{l+\frac{3}{2}}. \tag{7.15}$$

7.5.3 Kinetic energy

Lemma 7.6. $\Big(-\partial_r^2 - \frac{2}{r}\partial_r + \frac{l(l+1)}{r^2}\Big) r^l \mathrm{e}^{-\nu r^2} = \big[2\nu(2l+3) - 4\nu^2 r^2\big] r^l \mathrm{e}^{-\nu r^2}$.

Proof. With $R = r^l \mathrm{e}^{-\nu r^2}$: $R' = (\frac{l}{r} - 2\nu r)R$, $R'' = [(\frac{l}{r} - 2\nu r)^2 - \frac{l}{r^2} - 2\nu]R$. Add $\frac{2}{r}R'$ and subtract $\frac{l(l+1)}{r^2}R$: the $1/r^2$ terms cancel ($l^2 - l + 2l - l^2 - l = 0$) and $-[R'' + \frac{2}{r}R' - \frac{l(l+1)}{r^2}R] = [2\nu(2l+3) - 4\nu^2 r^2]R$. □

Hence, using $\langle r^2 \rangle_{n'n} = S_{n'n} \frac{l+3/2}{a}$,

$$\langle R_{n'l} | p^2 | R_{nl} \rangle = S_{n'n} \Big[2\nu_n (2l+3) - 4\nu_n^2 \frac{l + \frac{3}{2}}{\nu_{n'} + \nu_n} \Big] = 2(2l+3) \frac{\nu_{n'}\nu_n}{\nu_{n'} + \nu_n} S_{n'n}, \tag{7.16}$$

Non-relativistic kinetic matrix

$$T_{n'n} = \frac{1}{2m_r}\langle R_{n'l}|p^2|R_{nl}\rangle = \frac{2l+3}{m_r}\,\frac{\nu_{n'}\nu_n}{\nu_{n'}+\nu_n}\,S_{n'n}. \tag{7.17}$$

(Symmetric in $n' \leftrightarrow n$ as it must be. A second derivation in momentum space is in §7.6.)

7.5.4 Gaussian potentials (smeared δ, GI contact term)

$$\int_0^\infty r^2 R_{n'l'}\mathrm{e}^{-\beta r^2} R_{nl}\,\mathrm{d}r = N_{n'l'}N_{nl}\frac{\Gamma(\ell+\frac{3}{2})}{2(a+\beta)^{\ell+3/2}}. \tag{7.18}$$

For $\delta_\sigma = (\sigma/\sqrt{\pi})^3\mathrm{e}^{-\sigma^2 r^2}$ multiply by $(\sigma/\sqrt{\pi})^3$ with $\beta = \sigma^2$. Note: in S waves it is $\langle\delta_\sigma\rangle$ that replaces $|\psi(0)|^2 = |R(0)|^2/4\pi$; for $l \geq 1$ it is small but non-zero.

7.5.5 Error-function potentials (smeared and running Coulomb)

Use $\frac{\mathrm{erf}(\tau r)}{r} = \frac{2}{\sqrt{\pi}}\int_0^\tau \mathrm{e}^{-t^2r^2}\mathrm{d}t$ and the Gaussian formula:

$$\Big\langle\frac{\mathrm{erf}(\tau r)}{r}\Big\rangle_{n'n} = N_{n'l'}N_{nl}\frac{\Gamma(\ell+\frac{3}{2})}{\sqrt{\pi}}\int_0^\tau \frac{\mathrm{d}t}{(a+t^2)^{\ell+3/2}}. \tag{7.19}$$

Exercise 7.7 (Closed forms)**.** $\int_0^\tau \frac{\mathrm{d}t}{(a+t^2)^{3/2}} = \frac{\tau}{a\sqrt{a+\tau^2}}$ (substitute $t = \sqrt{a}\tan\theta$). Hence for $\ell = 0$: $\langle\frac{\mathrm{erf}(\tau r)}{r}\rangle = N_{n'0}N_{n0}\frac{\tau}{2a\sqrt{a+\tau^2}}$; $\tau\to\infty$ gives $\langle 1/r\rangle = N_{n'0}N_{n0}/(2a)$. For integer ℓ use the recursion $I_{j+1} = \dfrac{2j\,I_j + \tau(a+\tau^2)^{-j-1/2}}{(2j+1)\,a}$ for $I_j = \int_0^\tau (a+t^2)^{-j-1/2}\mathrm{d}t$, $I_0 = \mathrm{arsinh}(\tau/\sqrt{a})$ (derive it by integrating $\frac{\mathrm{d}}{\mathrm{d}t}[t(a+t^2)^{-j-1/2}]$); the $\ell = 0$ case is I_1, or simply Gauss–Legendre quadrature on $[0,\tau]$ (the integrand is smooth).

The tensor and spin–orbit radial functions of GI, $\tilde{G}'/r$ and $\tilde{G}''$, are combinations of $\mathrm{erf}(\tau r)/r^3$, $\mathrm{e}^{-\tau^2r^2}/r^2$ and $\mathrm{e}^{-\tau^2r^2}$; the ones with inverse powers are handled with the same t-integral trick or by quadrature.

7.5.6 Exponential potentials (screening)

Lemma 7.8. *Let* $\mathcal{Q}_k(a,\mu) = \int_0^\infty r^k\mathrm{e}^{-ar^2-\mu r}\mathrm{d}r$. *Then*

$$\mathcal{Q}_0 = \frac{1}{2}\sqrt{\frac{\pi}{a}}\,\mathrm{erfcx}\Big(\frac{\mu}{2\sqrt{a}}\Big), \qquad \mathcal{Q}_1 = \frac{1-\mu\mathcal{Q}_0}{2a}, \qquad \mathcal{Q}_{k+2} = \frac{(k+1)\mathcal{Q}_k - \mu\,\mathcal{Q}_{k+1}}{2a}\quad (k\geq 0), \tag{7.20}$$

with $\mathrm{erfcx}(x) = \mathrm{e}^{x^2}\,\mathrm{erfc}(x)$.

Proof. $\mathcal{Q}_0$: complete the square, $ar^2+\mu r = a(r+\frac{\mu}{2a})^2 - \frac{\mu^2}{4a}$. Recursion: integrate

$$\frac{\mathrm{d}}{\mathrm{d}r}\big[r^{k+1}\mathrm{e}^{-ar^2-\mu r}\big] = \big[(k+1)r^k - 2ar^{k+2} - \mu r^{k+1}\big]\mathrm{e}^{-ar^2-\mu r}$$

over $[0,\infty)$; the boundary terms vanish for $k\geq 0$. For $\mathcal{Q}_1$ integrate $\frac{\mathrm{d}}{\mathrm{d}r}\mathrm{e}^{-ar^2-\mu r}$. □

Then for Model B

$$\Big\langle\frac{b}{\mu_s}(1-\mathrm{e}^{-\mu_s r})\Big\rangle_{n'n} = \frac{b}{\mu_s}\Big[S_{n'n}\delta_{ll'} - N_{n'l'}N_{nl}\,\mathcal{Q}_{2\ell+2}(a,\mu_s)\Big], \qquad \Big\langle\frac{\mathrm{e}^{-\mu_s r}}{r}\Big\rangle_{n'n} = N_{n'l'}N_{nl}\mathcal{Q}_{2\ell+1}(a,\mu_s) \tag{7.21}$$

(the latter for the screened Thomas term $-\frac{b}{2m^2}\frac{\mathrm{e}^{-\mu_s r}}{r}$).

Remark 7.9 (Numerics of $\mathcal{Q}_k$: run the recursion the right way). Two rules. (i) Use erfcx, never e^{x^2} and $\operatorname{erfc} x$ separately: the first overflows and the second underflows long before their product is in trouble. (ii) The upward recursion is *not* safe for all parameters. Write $x = \mu/(2\sqrt{a})$ and $q_k = a^{(k+1)/2}\mathcal{Q}_k$, which depends on x only; the recursion becomes

$$q_0 = \tfrac{\sqrt{\pi}}{2}\operatorname{erfcx}(x), \qquad q_1 = \tfrac{1}{2}(1 - 2x\,q_0), \qquad q_{k+2} = \tfrac{k+1}{2}\,q_k - x\,q_{k+1}.$$

A three-term recursion has two independent solutions. For large x the one we want is the small one: the integrand $t^k\mathrm{e}^{-t^2-2xt}$ lives at $t \lesssim k/2x$, and $q_k \approx k!/(2x)^{k+1}$ *decreases* with k, while the other solution alternates in sign and grows. Every rounding error feeds the growing solution, and the subtraction $\frac{k+1}{2}q_k - xq_{k+1}$ of two nearly equal numbers exposes it. In double precision the upward recursion loses about 2 digits by $k = 8$ at $x = 1$, 8 digits by $k = 12$ at $x = 3$, and 12 digits at $x = 5$. Such x occur for diffuse basis functions ($a \sim 10^{-3}\,\mathrm{GeV}^2$) with ordinary screening masses, and even more after smearing, which replaces a by $a' < a$.

The cure is the standard one for a small ("minimal") solution: recur the *ratio* $\rho_k = q_{k+1}/q_k$ *downward*. Dividing the recursion by q_{k+1},

$$\rho_k = \frac{k+1}{2x + 2\rho_{k+1}},$$

which only adds positive numbers. Start from $\rho_K = 0$ at a depth K well above the needed k; the error of the starting value shrinks at every downward step, so the ρ_k for small k are accurate to rounding. Then $q_{k+1} = \rho_k q_k$ from q_0. In practice: upward for $x < \frac{1}{2}$ (there both solutions have the same size and nothing is lost), downward ratios with $K = \lfloor \frac{1}{2}(20/x + \sqrt{2k})^2 \rfloor + 10$ for $x \geq \frac{1}{2}$. With this, every q_k with $k \leq 31$ has a relative error below 10^{-13} for all x (checked against 40-digit quadrature from $x = 0.01$ to $x = 1000$), and below 10^{-14} for the $k \leq 10$ that $l \leq 4$ needs. This is the function `detail::screened_moments` of the generated header (§13.5.10).

7.5.7 Matrix elements of smeared potentials: the transfer lemma

Lemma 7.10 (Smearing transfer)**.** *For a smeared radial potential $\tilde{f} = \rho_\sigma * f$,*

$$\begin{aligned} \int_0^\infty r^{2l+2}\mathrm{e}^{-ar^2}\,\tilde{f}(r)\,\mathrm{d}r &= \Big(-\frac{\partial}{\partial a}\Big)^l \left[\Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{3/2} I_f(a')\right], \\ I_f(a') &= \int_0^\infty r^2 f(r)\mathrm{e}^{-a'r^2}\mathrm{d}r, \qquad a' = \frac{a\sigma^2}{a+\sigma^2}. \end{aligned} \tag{7.22}$$

Proof. Write the radial integral as $\frac{1}{4\pi}\int \mathrm{d}^3r\, r^{2l}\mathrm{e}^{-ar^2}\,\tilde{f}$ and move the convolution onto the Gaussian factor (ρ_σ is even): $\int g\,(\rho * f) = \int (\rho * g)\, f$. For $l = 0$, $\rho_\sigma * \mathrm{e}^{-ar^2} = \big(\frac{\sigma^2}{a+\sigma^2}\big)^{3/2}\mathrm{e}^{-a'r^2}$ (multiply the Fourier transforms $\mathrm{e}^{-q^2/4a}$ and $\mathrm{e}^{-q^2/4\sigma^2}$). For $l > 0$ use $r^{2l}\mathrm{e}^{-ar^2} = (-\partial_a)^l\mathrm{e}^{-ar^2}$; the convolution does not involve a. □

With $I_{1/r} = \frac{1}{2a'}$, $I_r = \frac{1}{2a'^2}$, $I_{\mathrm{e}^{-\mu_s r}} = \mathcal{Q}_2(a', \mu_s)$ this gives closed forms for *all* smeared potentials of Models C and D (one derivative per unit of l; do it by hand for $l \leq 2$, or by automatic differentiation). It also provides an independent check of quadrature results. Multiply by $N_{n'l}N_{nl}$.

The derivatives never need to be done on $\mathcal{Q}_k$ by hand: from the definition, $\partial_a\mathcal{Q}_k = -\mathcal{Q}_{k+2}$ and $\partial_\mu\mathcal{Q}_k = -\mathcal{Q}_{k+1}$. So the transfer lemma for a screened potential gives a finite combination of $\mathcal{Q}_{2j}(a', \mu_s)$, $j \leq l+1$, with rational functions of a, σ as coefficients; for $l = 1$, for example,

$$\int_0^\infty r^4\mathrm{e}^{-ar^2}\,\widetilde{\mathrm{e}^{-\mu_s r}}\,\mathrm{d}r = \Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{3/2}\Big[\frac{3}{2(a+\sigma^2)}\,\mathcal{Q}_2(a') + \Big(\frac{\sigma^2}{a+\sigma^2}\Big)^2\mathcal{Q}_4(a')\Big],$$

where $\partial a'/\partial a = \big(\frac{\sigma^2}{a+\sigma^2}\big)^2$ was used. Both terms are positive: no cancellation.

The spin–orbit radial functions. The smeared spin–orbit and Thomas terms have the radial function $\frac{1}{r}\tilde{f}'(r)$. One could integrate by parts and use the transfer lemma twice:

$$\int_0^\infty r^{2l+1}\mathrm{e}^{-ar^2}\,\tilde{f}'\,\mathrm{d}r = -\Big[(2l+1)\,T_{l-1} - 2a\,T_l\Big], \qquad T_l = \int_0^\infty r^{2l+2}\mathrm{e}^{-ar^2}\,\tilde{f}\,\mathrm{d}r \quad (l \geq 1).$$

This is correct but numerically poor for compact basis functions. When $a \gg \sigma^2$ the Gaussian factor sees only $r \lesssim a^{-1/2}$, where the smooth $\tilde{f}$ is nearly constant; the two terms are then each of order $\tilde{f}(0)$ and cancel to leave something of relative size σ^2/a. With a up to $2\times10^4\,\mathrm{GeV}^2$ and $\sigma^2 \sim 1\,\mathrm{GeV}^2$ that costs four to five digits. The following version avoids the subtraction.

Lemma 7.11 (Spin–orbit transfer)**.** *For $l \geq 1$,*

$$\begin{aligned}\int_0^\infty r^{2l+1}\mathrm{e}^{-ar^2}\,\tilde{f}'(r)\,\mathrm{d}r &= \Big(-\frac{\partial}{\partial a}\Big)^{l-1}\left[\Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{5/2} J_f(a')\right],\\ J_f(a') &= \int_0^\infty r^3 f'(r)\,\mathrm{e}^{-a'r^2}\mathrm{d}r, \qquad a' = \frac{a\sigma^2}{a+\sigma^2}.\end{aligned} \tag{7.23}$$

Proof. Convolution commutes with derivatives, so $\nabla\tilde{f} = \rho_\sigma * \nabla f$ and $r\tilde{f}' = \boldsymbol{r}\cdot\nabla\tilde{f} = \sum_j r_j\,(\rho_\sigma * \partial_j f)$. Then

$$\int_0^\infty r^{2l+1}\mathrm{e}^{-ar^2}\,\tilde{f}'\mathrm{d}r = \frac{1}{4\pi}\int \mathrm{d}^3r\; r^{2l-2}\mathrm{e}^{-ar^2}\,r\tilde{f}' = \frac{1}{4\pi}\sum_j \int \mathrm{d}^3r\,\big[\rho_\sigma * (r^{2l-2}\mathrm{e}^{-ar^2}r_j)\big]\,\partial_j f,$$

moving the (even) ρ_σ onto the other factor as in the transfer lemma. For $l=1$: $r_j\mathrm{e}^{-ar^2} = -\frac{1}{2a}\partial_j\mathrm{e}^{-ar^2}$, so

$$\rho_\sigma * (r_j\mathrm{e}^{-ar^2}) = -\frac{1}{2a}\,\partial_j\Big[\Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{3/2}\mathrm{e}^{-a'r^2}\Big] = \frac{a'}{a}\Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{3/2} r_j\mathrm{e}^{-a'r^2} = \Big(\frac{\sigma^2}{a+\sigma^2}\Big)^{5/2} r_j\mathrm{e}^{-a'r^2}.$$

Then $\sum_j r_j\partial_j f = rf'$ and $\frac{1}{4\pi}\int\mathrm{d}^3r = \int_0^\infty r^2\mathrm{d}r$ give the $l=1$ result. For $l>1$ write $r^{2l-2}\mathrm{e}^{-ar^2} = (-\partial_a)^{l-1}\mathrm{e}^{-ar^2}$ and take the derivatives outside. □

The functions needed: $J_r = \frac{1}{2a'^2}$ (linear), $J_{1/r} = -\frac{1}{2a'}$ (Coulomb), $J_{\mathrm{e}^{-\mu_s r}} = -\mu_s\,\mathcal{Q}_3(a',\mu_s)$ (screened). The result is again a combination of terms of one sign, with no cancellation for any a. Multiply by $N_{n'l}N_{nl}$ as before. For $l=0$ the spin–orbit operator vanishes and the integral is not needed.

Remark 7.12 (Quadrature is fine too). All integrands $r^{2\ell+2}\mathrm{e}^{-ar^2}f(r)$ are smooth and decay like Gaussians. Gauss–Legendre with $\sim$ 100–200 nodes on $[0, r_{\mathrm{cut}}]$, $r_{\mathrm{cut}} = \sqrt{40/a}$, gives near machine precision for every potential in these notes. Use analytic formulas where available and quadrature as a cross-check (or vice versa).

7.6 Momentum space

7.6.1 Fourier transform of a Gaussian basis function

Convention: $\tilde{\phi}(\boldsymbol{p}) = (2\pi)^{-3/2}\int\mathrm{d}^3r\,\mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}}\phi(\boldsymbol{r})$.

Lemma 7.13 (Hobson)**.** *For a harmonic homogeneous polynomial $\mathcal{Y}_l(\boldsymbol{r})$ of degree l (e.g. the solid harmonic r^lY_{lm}) and any smooth F: $\mathcal{Y}_l(\nabla)F(r^2) = 2^lF^{(l)}(r^2)\,\mathcal{Y}_l(\boldsymbol{r})$. (Check $l=1$: $\partial_zF(r^2) = 2zF'(r^2)$.)*

Since $\phi_{nlm} = N\,\mathcal{Y}_{lm}(\boldsymbol{r})\mathrm{e}^{-\nu r^2}$ with $\mathcal{Y}_{lm} = r^l Y_{lm}$, and multiplication by r_j becomes $\mathrm{i}\partial_{p_j}$,

$$\tilde{\phi}_{nlm}(\boldsymbol{p}) = N\,\mathcal{Y}_{lm}(\mathrm{i}\nabla_p)\,(2\nu)^{-3/2}\mathrm{e}^{-p^2/4\nu} = (-\mathrm{i})^l\,N_{nl}\,(2\nu_n)^{-l-\frac{3}{2}}\,p^l\mathrm{e}^{-p^2/4\nu_n}\,Y_{lm}(\hat{p}). \tag{7.24}$$

Momentum-space radial basis

$$\tilde{R}_{nl}(p) = N_{nl}\,(2\nu_n)^{-l-\frac{3}{2}}\,p^l\mathrm{e}^{-p^2/4\nu_n}, \qquad \int_0^\infty p^2\tilde{R}_{nl}^2\mathrm{d}p = 1, \tag{7.25}$$

equivalently $\tilde{R}_{nl}(p) = \sqrt{2/\pi}\int_0^\infty r^2 j_l(pr)R_{nl}(r)\mathrm{d}r$. A Gaussian of range parameter ν in r is a Gaussian of range parameter $1/4\nu$ in p.

Exercise 7.14. Recompute $T_{n'n}$ as $\int_0^\infty p^2\tilde{R}_{n'l}\frac{p^2}{2m_r}\tilde{R}_{nl}\mathrm{d}p$ with the master integral; you must recover the coordinate-space result.

7.6.2 Relativistic kinetic energy

Any function of p^2 is diagonal in momentum space:

$$\begin{aligned} K_{n'n} &= \int_0^\infty p^2\tilde{R}_{n'l}\big[E_1(p)+E_2(p)\big]\tilde{R}_{nl}\mathrm{d}p \\ &= \frac{N_{n'l}N_{nl}}{(4\nu_{n'}\nu_n)^{l+3/2}}\int_0^\infty p^{2l+2}\mathrm{e}^{-\beta p^2}\,[E_1+E_2]\mathrm{d}p, \qquad \beta = \frac{\nu_{n'}+\nu_n}{4\nu_{n'}\nu_n}. \end{aligned} \tag{7.26}$$

With $t = \beta p^2$:

$$\int_0^\infty p^{2l+2}\mathrm{e}^{-\beta p^2}\,g(p)\mathrm{d}p = \frac{1}{2\beta^{l+3/2}}\int_0^\infty t^{l+\frac{1}{2}}\mathrm{e}^{-t}\,g(\sqrt{t/\beta})\mathrm{d}t, \tag{7.27}$$

which is ideal for generalized Gauss–Laguerre quadrature with weight $t^{l+1/2}\mathrm{e}^{-t}$ (the function $\sqrt{t/\beta+m^2}$ is smooth in t). Closed forms exist in terms of confluent hypergeometric functions, but quadrature is simpler and exact to machine precision with ~ 50 nodes.

7.7 Non-local (momentum-dependent) operators of the GI model

We need $\langle R_{n'l'}|f(p)\,W(r)\,f(p)|R_{nl}\rangle$ with $f \in \{(1+p^2/E_1E_2)^{1/2},\ \Phi_{ab}^{(\epsilon)}(p)\}$ and W a smeared radial function (times a spin–angle number). Three methods, all tested in §12.2:

(i) Coordinate space, two steps ("exact"). Define the transformed basis function

$$\chi_{nl}(r) \equiv [f(p)R_{nl}](r) = \sqrt{\tfrac{2}{\pi}}\int_0^\infty p^2 j_l(pr)\,f(p)\,\tilde{R}_{nl}(p)\,\mathrm{d}p, \tag{7.28}$$

tabulate it on radial quadrature nodes, then $\langle R_{n'l'}|fWf|R_{nl}\rangle = \int_0^\infty r^2\chi_{n'l'}(r)W(r)\chi_{nl}(r)\mathrm{d}r$. Equivalent to using the momentum-space kernel $W_{l'l}(p',p) = \frac{2}{\pi}\int r^2 j_{l'}(p'r)W(r)j_l(pr)\mathrm{d}r$ in a double p integral.

(ii) Closure in the finite basis. Insert the projector onto the basis, $\mathcal{P} = \sum_{kj}|R_k\rangle(\mathbf{S}^{-1})_{kj}\langle R_j|$, between f and W:

$$\langle R_{n'}|fWf|R_n\rangle \simeq \big[\mathbf{F}\mathbf{S}^{-1}\mathbf{W}\mathbf{S}^{-1}\mathbf{F}\big]_{n'n}, \qquad F_{kj} = \int p^2\tilde{R}_k f\tilde{R}_j\mathrm{d}p, \quad W_{kj} = \int r^2 R_k W R_j\mathrm{d}r. \tag{7.29}$$

All ingredients are analytic or one-dimensional. Exact if fR_n lies in the span; since f is bounded and smooth ($\Phi \in (0,1]$, $1 \le (1+p^2/E_1E_2)^{1/2} < \sqrt{2}$), the error decreases quickly with $n_{\max}$. *Do not form* $\mathbf{S}^{-1}$ *explicitly*; in the orthonormalized basis of §7.8 ($\mathbf{X}^T\mathbf{S}\mathbf{X} = \mathbf{1}$) the closure product is simply $\mathbf{F}'\mathbf{W}'\mathbf{F}'$ with $\mathbf{A}' = \mathbf{X}^T\mathbf{A}\mathbf{X}$.

(iii) Series in p^2. Expand $f(p)$ in powers of p^2/m^2 and use $p^2 R_{nl} = [2\nu_n(2l+3) - 4\nu_n^2 r^2]R_{nl}$ repeatedly; each term is a polynomial times a Gaussian. Exact term by term but converges poorly for $p \gtrsim m$; useful only for checking the leading $1/m^2$ terms.

Hermiticity check. Both (i) and (ii) produce symmetric matrices when done consistently; numerically, symmetrize $\mathbf{H} \to \frac{1}{2}(\mathbf{H} + \mathbf{H}^T)$ and monitor $\max|\mathbf{H} - \mathbf{H}^T|$ as a diagnostic.

7.8 Solving $\mathbf{H}\boldsymbol{c} = \mathcal{E}\mathbf{S}\boldsymbol{c}$ in practice

7.8.1 Why S is ill-conditioned

For neighbouring functions with range ratio $q = r_{n+1}/r_n = a$,

$$S_{n,n+1} = \Big(\frac{2q}{1+q^2}\Big)^{l+\frac{3}{2}} \xrightarrow{q\to 1} 1. \tag{7.30}$$

For $q = 1.2$ and $l = 0$ this is 0.975: neighbouring functions are almost parallel, so the smallest eigenvalues of $\mathbf{S}$ are tiny (10^{-10}–10^{-15} for typical bases). A direct inversion of $\mathbf{S}$ amplifies rounding errors; the eigenvalue problem itself is fine if handled as below.

7.8.2 Reduction to a standard symmetric eigenproblem

1. **Cholesky.** $\mathbf{S} = \mathbf{L}\mathbf{L}^T$; $\mathbf{H}' = \mathbf{L}^{-1}\mathbf{H}\mathbf{L}^{-T}$ (two triangular solves); solve $\mathbf{H}'\boldsymbol{c}' = \mathcal{E}\boldsymbol{c}'$; back-transform $\boldsymbol{c} = \mathbf{L}^{-T}\boldsymbol{c}'$. This is what LAPACK's `dsygv` and Eigen's `GeneralizedSelfAdjointEigenSolver` do. Works if $\mathbf{S}$ is not too close to singular.
2. **Canonical orthogonalization (robust).** Diagonalize $\mathbf{S} = \mathbf{U}\,\mathrm{diag}(s_i)\,\mathbf{U}^T$; keep only $s_i > \eta\, s_{\max}$ (e.g. $\eta = 10^{-12}$); set $\mathbf{X} = \mathbf{U}_{\mathrm{kept}}\,\mathrm{diag}(s_i^{-1/2})$. Then $\mathbf{X}^T\mathbf{S}\mathbf{X} = \mathbf{1}$, solve $\mathbf{H}'\boldsymbol{c}' = \mathcal{E}\boldsymbol{c}'$ with $\mathbf{H}' = \mathbf{X}^T\mathbf{H}\mathbf{X}$, and $\boldsymbol{c} = \mathbf{X}\boldsymbol{c}'$. Discarding near-null directions removes the linear dependence and is the recommended default.

Exercise 7.15. Show that $\boldsymbol{c} = \mathbf{X}\boldsymbol{c}'$ with orthonormal $\boldsymbol{c}'$ satisfies $\boldsymbol{c}^T\mathbf{S}\boldsymbol{c} = 1$ automatically.

7.8.3 Choosing $\{n_{\max}, r_1, r_{\max}\}$

Rules of thumb: r_1 below the shortest relevant length (Bohr radius $1/m_r\kappa$, smearing radius $1/\sigma$, $1/m$); $r_{\max}$ a few times the rms radius of the highest state wanted; $n_{\max}$ such that the ratio $q = (r_{\max}/r_1)^{1/(n_{\max}-1)}$ is about 1.2–1.4. These hyperparameters, their properties and how to optimize them are treated systematically in §7.12.

7.9 Observables from the eigenvector

With $R(r) = \sum_n c_n R_{nl}(r)$ normalized ($\boldsymbol{c}^T\mathbf{S}\boldsymbol{c} = 1$):

- S waves: $R(0) = \sum_n c_n N_{n0}$.
- $l \geq 1$: $R(r) \simeq A_l r^l$ near the origin with $A_l = \sum_n c_n N_{nl}$; hence $R'(0) = A_1$ (P waves) and $R''(0) = 2A_2$ (D waves).
- Moments: $\langle r^k\rangle = \boldsymbol{c}^T\langle r^k\rangle\boldsymbol{c}$; transition overlaps $\langle f|r^k|i\rangle = \boldsymbol{c}_f^T\langle r^k\rangle\boldsymbol{c}_i$ with $\ell = \frac{l_f + l_i}{2}$ in (7.13).
- Momentum distribution $\tilde{R}(p) = \sum_n c_n \tilde{R}_{nl}(p)$; $\langle p^2\rangle = 2m_r\,\boldsymbol{c}^T\mathbf{T}\boldsymbol{c}$.
- Non-polynomial operators such as $j_0(kr/2)$: one-dimensional quadrature of the integrand $r^2 R_f R_i\, j_0(kr/2)$.

The physics of these observables (decay widths, transitions) is in §8.

7.10 Complex-range Gaussians

7.10.1 Why

The k-th radial excitation has $k-1$ nodes. A real-Gaussian basis can only build oscillations from cancellations among many functions of similar range, which requires dense range spacing (small q) and makes $\mathbf{S}$ nearly singular. A complex-range Gaussian has an oscillation built in.

7.10.2 Definition

$$u_n(r) = r^l \mathrm{e}^{-z_n r^2}, \qquad z_n = \eta_n(1+\mathrm{i}\omega), \qquad \eta_n = \frac{1}{\rho_n^2}, \quad \rho_n \text{ geometric}, \tag{7.31}$$

$$c_n(r) \equiv \operatorname{Re} u_n = r^l \mathrm{e}^{-\eta_n r^2}\cos(\omega\eta_n r^2), \qquad s_n(r) \equiv -\operatorname{Im} u_n = r^l \mathrm{e}^{-\eta_n r^2}\sin(\omega\eta_n r^2). \tag{7.32}$$

The real functions c_n, s_n span the same space as $u_n, \bar{u}_n$. The envelope has range ρ_n and the phase at $r=\rho_n$ is ω: ω sets how many oscillations fit inside the envelope. Real Gaussians are the case $\omega = 0$ (only c_n). A mixed basis (real-range set plus complex-range set) is one linear variational space.

7.10.3 Matrix elements by analytic continuation

Lemma 7.16. *For complex z with $\operatorname{Re} z > 0$ and $\operatorname{Re} s > 0$:* $\displaystyle\int_0^\infty r^{s-1}\mathrm{e}^{-zr^2}\mathrm{d}r = \frac{\Gamma(s/2)}{2z^{s/2}}$ *with the principal branch of $z^{s/2}$.*

Proof. Both sides are analytic in z on the right half-plane (the integral converges uniformly on compact subsets) and agree for real $z>0$ by the substitution $t = zr^2$; by the identity theorem they agree everywhere on the half-plane. □

Hence every closed form of §7.5 (overlap, kinetic, r^k, Gaussians, the erf integral, the transfer lemma) holds with ν_i, ν_j replaced by complex exponents, provided all exponent sums have positive real part (true here: $\operatorname{Re}(z_i + z_j) = \eta_i + \eta_j > 0$). For the screening integrals $\mathcal{Q}_k$ one needs erfcx of a complex argument, available as the Faddeeva function, $\operatorname{erfcx}(z) = w(\mathrm{i}z)$; quadrature is an alternative.

From complex to real functions. Let O be a real operator (real local potentials, p^2, real functions of p^2) and define the *bilinear* form (no complex conjugation) $\mathcal{B}[f,g] = \int_0^\infty r^2 f O g\,\mathrm{d}r$. Since O is real, $\mathcal{B}[\bar{f},\bar{g}] = \overline{\mathcal{B}[f,g]}$. With $c = \frac{1}{2}(u+\bar{u})$ and $s = \frac{\mathrm{i}}{2}(u-\bar{u})$, expanding and using this property:

$$\begin{aligned}
\langle c_i|O|c_j\rangle &= \tfrac{1}{2}\operatorname{Re}\{\mathcal{B}[u_i,u_j] + \mathcal{B}[u_i,\bar{u}_j]\},\\
\langle s_i|O|s_j\rangle &= \tfrac{1}{2}\operatorname{Re}\{\mathcal{B}[u_i,\bar{u}_j] - \mathcal{B}[u_i,u_j]\},\\
\langle c_i|O|s_j\rangle &= -\tfrac{1}{2}\operatorname{Im}\{\mathcal{B}[u_i,u_j] - \mathcal{B}[u_i,\bar{u}_j]\}.
\end{aligned} \tag{7.33}$$

$\mathcal{B}[u_i,u_j]$ is the real-Gaussian formula with exponent sum $a = z_i + z_j$; $\mathcal{B}[u_i,\bar{u}_j]$ uses $a = z_i + \bar{z}_j$. *Derivation of the first line:* $\langle c_i|O|c_j\rangle = \frac{1}{4}(\mathcal{B}[u_i,u_j] + \mathcal{B}[u_i,\bar{u}_j] + \mathcal{B}[\bar{u}_i,u_j] + \mathcal{B}[\bar{u}_i,\bar{u}_j])$; $\mathcal{B}[\bar{u}_i,u_j] = \overline{\mathcal{B}[u_i,\bar{u}_j]}$ and $\mathcal{B}[\bar{u}_i,\bar{u}_j] = \overline{\mathcal{B}[u_i,u_j]}$, so the sum is twice the real part of the first two. The other lines follow the same way. Normalize each function by $\sqrt{S_{nn}}$ computed from (7.33). Equations (7.33) were checked for S and H (kinetic, $1/r$, r) against direct quadrature (agreement 10^{-8}, limited by the finite-difference Laplacian of the test). In momentum space u_n transforms like a

real Gaussian with $\nu \to z_n$; the relativistic kinetic matrix uses $\beta = \frac{1}{4z_i} + \frac{1}{4z_j}$ (or its conjugate partner), and $\mathrm{Re}\,\beta > 0$, so the p quadrature converges.

7.10.4 The HUM theorem with complex-range functions

Rayleigh–Ritz never asks the basis functions to be real-range Gaussians: it needs a finite set of linearly independent functions in the domain of a Hermitian H bounded below. A real-range set plus $\{c_n, s_n\}$ is such a set. Therefore:

- *One* diagonalization of the combined basis gives all levels at once, and every Ritz value is an upper bound to the corresponding exact level (min–max proof above, unchanged).
- Adding complex-range functions to an existing basis is a nested enlargement: by interlacing it can only lower *every* level. So there is no reason to reserve them for excited states; include them in every calculation, with tuned hyperparameters.
- Use the real functions c_n, s_n (real symmetric $\mathbf{H}, \mathbf{S}$), or equivalently $u_n, \bar{u}_n$ with the Hermitian inner product (conjugate the bra): same span, same Ritz values. Do *not* use the bilinear form as the inner product; that is the complex-scaling method for resonances, a non-Hermitian complex-symmetric problem with complex eigenvalues and no upper-bound property.
- Cost: c_n and s_n with the same η_n, and neighbouring ranges, overlap strongly; conditioning is worse, so canonical orthogonalization (§7.8) is mandatory.

7.10.5 A demonstration (and why hyperparameters matter)

Cornell potential, $m_r = 0.75\,\mathrm{GeV}$, $\kappa = \frac{2}{3}$, $b = 0.18\,\mathrm{GeV}^2$, S waves, all bases with 40 functions. Errors $\mathcal{E}_k^{(N)} - \mathcal{E}_k^{\mathrm{ref}}$ in GeV (reference: finite differences with Richardson extrapolation, accurate to $\sim 10^{-7}$):

basis (40 functions; ranges in GeV^{-1})	1S	2S	3S	4S	5S	6S	7S	8S
real, $(r_1, r_{\max}) = (0.05, 40)$	1e-7	9e-8	7e-8	3e-7	4e-6	6e-6	1.0e-3	2.3e-3
real, $(0.02, 50)$	1e-8	7e-9	1e-7	9.5e-6	2.9e-5	9.5e-4	9.7e-3	9.0e-3
real 20 $(0.02, 30)$ + cplx 10×2 $(1, 20)$, ω=0.5	2e-8	1e-8	1e-8	8e-9	1e-8	9e-9	7e-8	7e-7
cplx only 20×2 $(0.02, 50)$, $\omega = \frac{\pi}{2}$	6.8e-3	3.7e-3	3.4e-3	2.8e-3	3.6e-3	3.2e-3	1.1e-2	4.2e-2

Lessons: (i) all errors are positive (HUM); (ii) a mixed basis, with real Gaussians covering the short-distance (Coulomb) region and complex-range Gaussians covering the oscillatory region where excited states live, gains three orders of magnitude on 7S–8S at equal size; (iii) badly chosen complex-range parameters (complex functions at very short range, large ω) are much worse than plain real Gaussians. So ω and the two range sets are genuine hyperparameters (§7.12).

7.11 Parameter derivatives in a finite basis

7.11.1 Eigenvalues (Feynman–Hellmann for Ritz values)

Let $\mathbf{H}(\lambda)$, $\mathbf{S}(\lambda)$ depend on a parameter and $\mathbf{H}\boldsymbol{c} = \mathcal{E}\mathbf{S}\boldsymbol{c}$, $\boldsymbol{c}^T\mathbf{S}\boldsymbol{c} = 1$. Differentiate: $\mathbf{H}'\boldsymbol{c} + \mathbf{H}\boldsymbol{c}' = \mathcal{E}'\mathbf{S}\boldsymbol{c} + \mathcal{E}\mathbf{S}'\boldsymbol{c} + \mathcal{E}\mathbf{S}\boldsymbol{c}'$. Multiply by $\boldsymbol{c}^T$ and use $\boldsymbol{c}^T\mathbf{H} = \mathcal{E}\boldsymbol{c}^T\mathbf{S}$ (symmetry): the $\boldsymbol{c}'$ terms cancel, leaving

$$\boxed{\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}\lambda} = \boldsymbol{c}^T\Big(\frac{\partial \mathbf{H}}{\partial \lambda} - \mathcal{E}\frac{\partial \mathbf{S}}{\partial \lambda}\Big)\boldsymbol{c}} \tag{7.34}$$

- *Physics parameters* $(m_Q, \alpha_s, b, c, \sigma, \mu_s, \alpha_k, \gamma_k, \epsilon_x, \dots)$ do not change the basis: $\mathbf{S}' = 0$ and $\mathcal{E}' = \boldsymbol{c}^T \mathbf{H}' \boldsymbol{c}$. This is exact for the Ritz value even if the basis is not converged.
- *Basis parameters* $(r_1, r_{\max}, \omega, \dots)$ change $\mathbf{S}$ and $\mathbf{H}$; (7.34) gives the gradient for variational basis optimization.

Numerical check (Cornell, $n_{\max} = 40$): $\partial \mathcal{E}_{1S}/\partial b = \boldsymbol{c}^T \langle r \rangle \boldsymbol{c} = 1.77047279$ vs. finite difference 1.77047276; basis derivative $\partial \mathcal{E}_{3S}/\partial \ln r_1$ at $n_{\max} = 12$: 0.114401002 vs. 0.114401003.

7.11.2 Eigenvectors

Expand $\boldsymbol{c}'_a = \sum_b x_b \boldsymbol{c}_b$ in the ($\mathbf{S}$-orthonormal) eigenvectors. Multiplying the differentiated equation by $\boldsymbol{c}_b^T$ $(b \neq a)$ gives $\boldsymbol{c}_b^T(\mathbf{H}' - \mathcal{E}_a \mathbf{S}')\boldsymbol{c}_a + (\mathcal{E}_b - \mathcal{E}_a)x_b = 0$; the normalization condition gives $2\boldsymbol{c}_a^T \mathbf{S} \boldsymbol{c}'_a + \boldsymbol{c}_a^T \mathbf{S}' \boldsymbol{c}_a = 0$. Hence

$$\frac{\mathrm{d}\boldsymbol{c}_a}{\mathrm{d}\lambda} = \sum_{b \neq a} \boldsymbol{c}_b \, \frac{\boldsymbol{c}_b^T(\mathbf{H}' - \mathcal{E}_a \mathbf{S}')\boldsymbol{c}_a}{\mathcal{E}_a - \mathcal{E}_b} - \frac{1}{2}\boldsymbol{c}_a (\boldsymbol{c}_a^T \mathbf{S}' \boldsymbol{c}_a). \tag{7.35}$$

(The sum runs over all eigenvectors of the finite problem; degeneracies within one channel do not occur for the radial problem.) Observable derivatives follow by the chain rule, e.g. $\frac{\mathrm{d}|R(0)|^2}{\mathrm{d}\lambda} = 2R(0)\, \boldsymbol{v}^T \frac{\mathrm{d}\boldsymbol{c}}{\mathrm{d}\lambda}$ with $v_n = N_{n0}$. Check: $\partial |R_{1S}(0)|^2/\partial b = 3.82003567$ vs. finite difference 3.82003605 GeV.

7.11.3 Model-specific $\partial \mathbf{H}/\partial \lambda$

- c: $\partial \mathbf{H}/\partial c = \mathbf{S}$, so $\partial M/\partial c = 1$ exactly for every level (a good code test).
- b (Model A): $\langle r \rangle$; Model C: $\langle \tilde{r} \rangle$; Model B/D: ∂_b of the screened potential.
- α_k (GI): every Coulomb-generated term is linear in α_k; differentiate term by term.
- γ_k, σ_{12} (GI): through τ_k; use $\partial_\tau \frac{\mathrm{erf}(\tau r)}{r} = \frac{2}{\sqrt{\pi}} \mathrm{e}^{-\tau^2 r^2}$ (a Gaussian: analytic matrix elements), $\frac{\partial \tau_k}{\partial \gamma_k} = \frac{\sigma^3}{(\gamma_k^2 + \sigma^2)^{3/2}}$, $\frac{\partial \tau_k}{\partial \sigma} = \frac{\gamma_k^3}{(\gamma_k^2 + \sigma^2)^{3/2}}$.
- ϵ_x: $\partial_\epsilon \Phi_{ab}^{(\epsilon)} = \Phi_{ab}^{(\epsilon)} \ln \frac{m_a m_b}{E_a E_b}$, diagonal in p; use the same p quadrature as for $\mathbf{F}$.
- m_Q enters E_i, Φ, σ_{12} and the $1/m$ factors: use automatic differentiation or a central finite difference of $\mathbf{H}$ (cheap compared with diagonalization).

7.12 Basis hyperparameters: properties and optimization

The hyperparameters. $n_{\max}$, r_1, $r_{\max}$ (equivalently r_1 and the ratio q) per channel; for complex-range sets $(n_c, \rho_1, \rho_{\max}, \omega)$; the orthogonalization threshold η; for GI the quadrature settings of the p integrals. They are not physics: a converged result must not depend on them.

Property 1: variational optimization is legitimate. For fixed $n_{\max}$, every Ritz value is an upper bound (HUM), so lowering it by changing $(r_1, r_{\max}, \omega, \dots)$ can only bring it closer to the exact value. For several target states use the trace (Ky Fan) principle: $\sum_{k \leq K} \mathcal{E}_k^{(N)} \geq \sum_{k \leq K} \mathcal{E}_k^{\mathrm{exact}}$, so minimize $\sum_{k \leq K} w_k \mathcal{E}_k$ with positive weights. Gradients come from (7.34) with $\mathbf{S}' \neq 0$.

Property 2: $n_{\max}$ is not a smooth hyperparameter. With $r_1, r_{\max}$ fixed, $n_{\max} \to n_{\max}+1$ changes *all* the ranges: the bases are not nested, and HUM monotonicity does not apply. They are nested only for $n_{\max} \to 2n_{\max}-1$: then $a' = a^{1/2}$ and $r'_{2k-1} = r_1 a'^{2(k-1)} = r_k$, so every old function is kept. Cornell, $r_1 = 0.02$, $r_{\max} = 50\,\mathrm{GeV}^{-1}$, 3S level:

	nested: $n \to 2n-1$				non-nested: $n \to n+1$						
$n_{\max}$	6	11	21	41	10	11	12	13	14	15	16
$\mathcal{E}_{3S}$	7.483	1.820	1.32448	1.32402	1.545	1.820	1.404	1.408	1.492	1.339	1.358

The nested sequence decreases monotonically; the non-nested one goes up and down. Consequences: use nested doubling for convergence tests and extrapolation, and do not hand $n_{\max}$ to a generic optimizer as if it were continuous.

Property 3: conditioning. Neighbouring overlaps are $S_{n,n+1} = (2q/(1+q^2))^{l+3/2}$; the smallest eigenvalue of $\mathbf{S}$ decays rapidly as $q \to 1$ ($l = 0$, $\lambda_{\min}/\lambda_{\max}$):

q	1.1	1.2	1.3	1.5	2.0
$S_{n,n+1}$	0.993	0.976	0.950	0.887	0.716
$n_{\max} = 10$	3×10^{-13}	1×10^{-8}	2×10^{-6}	2×10^{-4}	1×10^{-2}
$n_{\max} = 20$	1×10^{-17}	2×10^{-10}	3×10^{-7}	1×10^{-4}	1×10^{-2}
$n_{\max} = 40$	(singular)	6×10^{-11}	2×10^{-7}	9×10^{-5}	1×10^{-2}

Below $q \approx 1.15$ in double precision, extra functions add almost nothing new to the span and must be removed by the threshold η. The *effective* basis dimension is the number of kept eigenvalues.

Property 4: the Coulomb cusp. Near the origin with $V \simeq -\kappa/r$ and $l = 0$, insert $R = R_0(1 + \alpha r + \dots)$ into $R'' + \frac{2}{r}R' = 2m_r(V - \mathcal{E})R$: the $1/r$ terms give $2\alpha R_0 = -2m_r\kappa R_0$, i.e. the Kato cusp $R'(0) = -m_r\kappa R(0)$. Every Gaussian has $R'(0) = 0$, so the cusp must be built from very narrow functions and the S-wave energies of a point-Coulomb potential converge slowly in r_1. Smeared potentials (GI: $\mathrm{erf}(\tau r)/r$ is regular) have no cusp. 1S errors (GeV) at $n_{\max} = 20$, $r_{\max} = 30$:

r_1 (GeV^{-1})	0.5	0.2	0.1	0.05	0.02	0.01
point Coulomb	1.8×10^{-4}	1.4×10^{-5}	1.9×10^{-6}	3.4×10^{-7}	2.4×10^{-6}	8.5×10^{-6}
smeared, $\tau = 2\,\mathrm{GeV}$	2.6×10^{-9}	2.3×10^{-9}	1.8×10^{-8}	5.9×10^{-8}	2.4×10^{-6}	1.3×10^{-5}

At fixed $n_{\max}$ there is an interior optimum in r_1 (too small wastes functions where the wave function is negligible): this is exactly what a hyperparameter optimization finds.

Property 5: basis noise in physics fits. With a *fixed* basis, every matrix element is a smooth function of the physics parameters, hence so are the Ritz values and $\chi^2(\theta)$. Re-optimizing the basis at every step of a fit makes $\chi^2(\theta)$ piecewise and noisy (Property 2), which breaks gradient-based minimizers.

Recommended protocol.

1. At a nominal parameter point θ_0, optimize the basis hyperparameters per channel by minimizing $\sum_k w_k \mathcal{E}_k$ (Property 1) over continuous parameters $(r_1, r_{\max}, \rho_1, \rho_{\max}, \omega)$ at fixed $n_{\max}$; use (7.34) or a derivative-free method (Nelder–Mead) on log-scaled variables.

2. Check convergence with the nested sequence $n_{\max} \to 2n_{\max} - 1$; take the change as the numerical error σ_{num} (§10.8).
3. Freeze the basis and fit the physics parameters (§10).
4. At the best fit $\hat{\theta}$, re-optimize the basis and repeat step 2. If observables move by more than a small fraction of their errors, refit with the new basis.
5. Report σ_{num} as part of the error budget.

Cost. Building $\mathbf{H}$ costs $O(N^2)$ matrix elements (plus a 1D quadrature each for GI terms); diagonalization costs $O(N^3)$. For $N \lesssim 100$ per channel both are negligible, so generous bases are cheap; the binding constraint is conditioning, not time.

7.13 Answer key in your notation $\psi_{lm} = \sum_n c_n \phi_{nlm}$

Derive, then compare:

C1 $N_{nl}^2 = \frac{2(2\nu_n)^{l+3/2}}{\Gamma(l+3/2)}$.

C2 $\sum_n (H_{n'n} - \mathcal{E} S_{n'n}) c_n = 0$; angular integral $\Rightarrow \delta_{ll'}\delta_{mm'}$; m-independent.

C3 $S_{n'n} = \left(\frac{2\sqrt{\nu_{n'}\nu_n}}{\nu_{n'}+\nu_n}\right)^{l+3/2}$.

C4 $-\nabla^2(r^l \mathrm{e}^{-\nu r^2} Y_{lm}) = [2\nu(2l+3) - 4\nu^2 r^2] r^l \mathrm{e}^{-\nu r^2} Y_{lm}$; $T_{n'n} = \frac{2l+3}{m_r} \frac{\nu_{n'}\nu_n}{\nu_{n'}+\nu_n} S_{n'n}$.

C5 $\langle r^k \rangle$ from (7.13).

C6 $\mathrm{erf}(\tau r)/r$ via the t-integral; Gaussian; $\mathcal{Q}_k$ (downward ratios for $x \geq \frac{1}{2}$); transfer lemmas (7.22) and (7.23).

C7 $\tilde{R}_{nl}(p) = N_{nl}(2\nu_n)^{-l-3/2} p^l \mathrm{e}^{-p^2/4\nu_n}$; $K_{n'n}$ by Gauss–Laguerre.

C8 Spin factors from §3.5; tensor coupling blocks for $L = J \mp 1$.

C9 Canonical orthogonalization; $\boldsymbol{c}^T \mathbf{S} \boldsymbol{c} = 1$; upper-bound property (min–max, interlacing).

C10 Complex-range functions: analytic continuation lemma and the real-combination rules (7.33).

C11 $\mathrm{d}\mathcal{E}/\mathrm{d}\lambda = \boldsymbol{c}^T(\mathbf{H}' - \mathcal{E}\mathbf{S}')\boldsymbol{c}$ and the eigenvector derivative (7.35).

C12 Nestedness of $n \to 2n-1$; Kato cusp $R'(0) = -m_r \kappa R(0)$.

8 Observables

Everything here needs only the eigenvalues and the GEM radial wave functions, $R = \sum_n c_n R_{nl}$ (§7.9). Formulas are leading order in v^2 and α_s unless stated; QCD corrections are quoted where they are standard.

8.1 Masses, multiplets and spin-averaged quantities

- Model A/B: $M = m_1 + m_2 + \mathcal{E}$; Model C/D: $M = \mathcal{E}$.
- Spin-averaged masses remove the spin-dependent forces at first order (centre-of-gravity theorem, §3.6):

$$\bar{M}(nS) = \tfrac{1}{4}[M(n^1S_0) + 3M(n^3S_1)], \tag{8.1}$$

$$\bar{M}(nP) = \tfrac{1}{12}[3M(n^1P_1) + 5M(n^3P_2) + 3M(n^3P_1) + M(n^3P_0)]. \tag{8.2}$$

- Hyperfine splitting $\Delta_{\rm hf}(nS) = M(n^3S_1) - M(n^1S_0)$; it measures $\langle H_{SS}\rangle$, i.e. $|R(0)|^2$ (point δ) or $\langle\delta_\sigma\rangle$ (smeared).
- Fine-structure ratio. With the first-order formulas of §3.6, $M(^3P_2) - M(^3P_1) = 2a - \frac{12}{5}t$ and $M(^3P_1) - M(^3P_0) = a + 6t$, so

$$\rho \equiv \frac{M(^3P_2) - M(^3P_1)}{M(^3P_1) - M(^3P_0)} = \frac{2a - \frac{12}{5}t}{a + 6t} \tag{8.3}$$

  depends only on t/a (tensor-to-spin–orbit ratio): a direct probe of the Lorentz structure of confinement.

8.2 Sizes and velocities

$\sqrt{\langle r^2\rangle}$ from the GEM moments; $\langle p^2\rangle = 2m_r\langle T\rangle$ and $\langle v^2\rangle = \langle p^2\rangle/m_Q^2$ per quark. The latter is the internal consistency check of a non-relativistic model ($\langle v^2\rangle \approx 0.28$ for the $1S$ state of the test model of §12.2).

8.3 Annihilation decays

8.3.1 Factorization

Annihilation of $Q\bar{Q}$ happens at distances $\sim 1/m_Q$, much smaller than the bound-state size. The decay amplitude therefore factorizes into the free annihilation amplitude at threshold times the wave function at zero separation:

$$\mathcal{A}(n^{2S+1}S_J \to f) = \psi_n(0)\,\mathcal{A}_{\rm free}(Q\bar{Q} \to f)\big|_{\rm threshold}, \qquad |\psi_n(0)|^2 = \frac{|R_n(0)|^2}{4\pi}. \tag{8.4}$$

For $L \geq 1$ the wave function vanishes at the origin; the leading term involves L derivatives, so P-wave widths $\propto |R'(0)|^2$ and D-wave widths $\propto |R''(0)|^2$, each with extra powers of $1/m_Q^2$.

8.3.2 QED anchors

For a colourless fermion–antifermion bound state with unit charges (positronium, true muonium), the standard results are

$$\begin{aligned}
\Gamma(\text{para, } {}^1S_0 \to \gamma\gamma) &= \frac{16\pi\alpha^2|\psi(0)|^2}{M^2}, \\
\Gamma(\text{ortho, } {}^3S_1 \to \gamma\gamma\gamma) &= \frac{16(\pi^2-9)}{9}\,\frac{\alpha^3|\psi(0)|^2}{m^2}, \\
\Gamma({}^3S_1 \to \ell^+\ell^-) &= \frac{16\pi}{3}\,\frac{\alpha^2|\psi(0)|^2}{M^2} \qquad (M = 2m).
\end{aligned} \tag{8.5}$$

Exercise 8.1. Insert the hydrogen-like $|\psi_{1S}(0)|^2 = (m_r\alpha)^3/\pi = m^3\alpha^3/8\pi$ and recover the textbook numbers $\Gamma(p\text{-Ps}) = \frac{1}{2}\alpha^5 m$, $\Gamma(o\text{-Ps}) = \frac{2(\pi^2-9)}{9\pi}\alpha^6 m$, and $\Gamma((\mu^+\mu^-)_{1^3S_1} \to e^+e^-) = \frac{1}{6}\alpha^5 m_\mu$.

8.3.3 Colour and charge factors

The colour-singlet state is $|\mathbf{1}\rangle = \frac{1}{\sqrt{\mathbf{N_c}}}\sum_{\mathbf{i}}|\mathbf{i}\bar{\mathbf{i}}\rangle$. Sum the free amplitudes over colours with this weight and square:

- *Photons or a virtual photon* (colour-blind, each vertex e_Q): amplitude $\propto \sum_i \frac{1}{\sqrt{N_c}} e_Q^n = \sqrt{N_c}\,e_Q^n$, so the rate is multiplied by $N_c\,e_Q^{2n}$.

- *Two gluons* ($^1S_0$, $C=+$): amplitude $\propto \frac{1}{\sqrt{N_c}}\,\mathrm{Tr}(T^aT^b) = \frac{\delta^{ab}}{2\sqrt{N_c}}$; $\sum_{ab}\left|\frac{\delta^{ab}}{2\sqrt{N_c}}\right|^2 = \frac{N_c^2-1}{4N_c} = \frac{2}{3}$.
- *Three gluons* ($^3S_1$, $C=-$): only the C-odd part of $\mathrm{Tr}(T^aT^bT^c) = \frac{1}{4}(d^{abc}+\mathrm{i}f^{abc})$ survives, i.e. $\frac{1}{4}d^{abc}$; with $\sum_{abc}(d^{abc})^2 = \frac{(N_c^2-4)(N_c^2-1)}{N_c}$, the factor is $\frac{1}{16N_c}\cdot\frac{(N_c^2-4)(N_c^2-1)}{N_c} = \frac{5}{18}$.

Gluons and photons have the same Lorentz and Bose structure at this order, so the QED anchors carry over with $\alpha^n \to \alpha_s^n$ and the colour factor.

S-wave annihilation widths (leading order; $M = 2m_Q$ in the derivation)

$$\Gamma(n^3S_1 \to \ell^+\ell^-) = N_c e_Q^2 \cdot \frac{16\pi}{3}\frac{\alpha^2|\psi(0)|^2}{M^2} = \frac{4\alpha^2 e_Q^2|R(0)|^2}{M^2}\Big(1-\frac{16\alpha_s}{3\pi}\Big), \tag{8.6}$$

$$\Gamma(n^1S_0 \to \gamma\gamma) = N_c e_Q^4 \cdot \frac{16\pi\alpha^2|\psi(0)|^2}{M^2} = \frac{12\alpha^2 e_Q^4|R(0)|^2}{M^2}\Big(1-\frac{(20-\pi^2)\alpha_s}{3\pi}\Big), \tag{8.7}$$

$$\Gamma(n^1S_0 \to gg) = \frac{2}{3}\cdot\frac{16\pi\alpha_s^2|\psi(0)|^2}{M^2} = \frac{8\alpha_s^2|R(0)|^2}{3M^2}, \tag{8.8}$$

$$\Gamma(n^3S_1 \to ggg) = \frac{5}{18}\cdot\frac{16(\pi^2-9)}{9}\frac{\alpha_s^3|\psi(0)|^2}{m_Q^2} = \frac{40(\pi^2-9)\alpha_s^3|R(0)|^2}{81\pi M^2}. \tag{8.9}$$

The first-order QCD corrections shown for $\ell^+\ell^-$ and $\gamma\gamma$ are standard; those for gg and ggg are large and scale dependent (see Kwong et al.).

Exercise 8.2. Verify the last line: $\frac{5}{18}\cdot\frac{16(\pi^2-9)}{9}\alpha_s^3\frac{|R(0)|^2}{4\pi}\frac{4}{M^2} = \frac{40(\pi^2-9)}{81\pi}\frac{\alpha_s^3|R(0)|^2}{M^2}$.

Practical ambiguities (systematic errors). Whether M means $2m_Q$ or the physical mass, the scale of α_s in the corrections, and the use of $|R(0)|^2$ versus $\langle\delta_\sigma\rangle$-type smeared quantities change predictions by tens of per cent. Treat these as part of the theory error (§10.8).

8.3.4 P- and D-wave annihilation (quoted)

With m_Q the quark mass,

$$\Gamma(^3P_0 \to \gamma\gamma) = \frac{27e_Q^4\alpha^2|R'(0)|^2}{m_Q^4}, \qquad \Gamma(^3P_2 \to \gamma\gamma) = \frac{36e_Q^4\alpha^2|R'(0)|^2}{5m_Q^4}, \tag{8.10}$$

$$\Gamma(^3P_0 \to gg) = \frac{6\alpha_s^2|R'(0)|^2}{m_Q^4}, \qquad \Gamma(^3P_2 \to gg) = \frac{8\alpha_s^2|R'(0)|^2}{5m_Q^4}, \tag{8.11}$$

$$\Gamma(^3D_1 \to \ell^+\ell^-) = \frac{25e_Q^2\alpha^2|R''(0)|^2}{2m_Q^4M^2}. \tag{8.12}$$

Consistency checks: $\Gamma(^3P_0)/\Gamma(^3P_2) = \frac{15}{4}$ in both channels, and $\Gamma(\gamma\gamma)/\Gamma(gg) = \frac{9e_Q^4\alpha^2}{2\alpha_s^2}$ for $^3P_0$, $^3P_2$ and $^1S_0$ alike (the same colour-factor ratio $\frac{N_c}{2/3} = \frac{9}{2}$).

8.4 Decay constants

Define $\langle 0|\bar{Q}\gamma^\mu Q|V(\boldsymbol{P},\varepsilon)\rangle = f_V M_V \varepsilon^\mu$ (similarly f_P with $\gamma^\mu\gamma_5$). The same factorization gives the Van Royen–Weisskopf relation

$$f_V^2 = \frac{12|\psi(0)|^2}{M_V} = \frac{3|R(0)|^2}{\pi M_V}, \qquad \Gamma(V \to \ell^+\ell^-) = \frac{4\pi\alpha^2 e_Q^2}{3}\frac{f_V^2}{M_V}. \tag{8.13}$$

Check: inserting f_V^2 in the second formula gives $\frac{16\pi\alpha^2 e_Q^2 |\psi(0)|^2}{M_V^2}$, the leptonic width above. At leading order $f_P = f_V$; lattice QCD provides precise $f_{\eta_c}, f_{J/\psi}$ for comparison.

8.5 Electric dipole (E1) transitions

8.5.1 The dipole operator

For charges e_1, e_2 at $\boldsymbol{r}_1, \boldsymbol{r}_2$ the dipole moment relative to the centre of mass is $\boldsymbol{d} = e_1\boldsymbol{r}_1 + e_2\boldsymbol{r}_2$. With $\boldsymbol{r}_1 = \frac{m_2}{m_1+m_2}\boldsymbol{r}$, $\boldsymbol{r}_2 = -\frac{m_1}{m_1+m_2}\boldsymbol{r}$:

$$\boldsymbol{d} = \langle e\rangle\,\boldsymbol{r}, \qquad \langle e\rangle = \frac{m_2 e_1 - m_1 e_2}{m_1+m_2} \xrightarrow{Q\bar{Q}:\ e_1=e_Q,\ e_2=-e_Q} e_Q. \tag{8.14}$$

8.5.2 Rate

Fermi's golden rule for spontaneous emission in the long-wavelength limit (units with $e^2 = \alpha$) gives

$$\Gamma_{E1} = \frac{4}{3}\alpha\,\langle e\rangle^2 k^3\,\frac{1}{2J+1}\sum_{M,M',q}\left|\langle n'(L'S)J'M'|r_q|n(LS)JM\rangle\right|^2, \qquad k = \frac{M_i^2 - M_f^2}{2M_i}. \tag{8.15}$$

8.5.3 Angular reduction, step by step

Write $r_q = r\,C_q^{(1)}(\hat{r})$ with $C_q^{(1)} = \sqrt{4\pi/3}\,Y_{1q}$ (Edmonds convention for reduced matrix elements).

1. Wigner–Eckart: $\sum_{M',q}|\langle J'M'|T_q|JM\rangle|^2 = \frac{|\langle J'\|T\|J\rangle|^2}{2J+1}$, independent of M; so the average over M is trivial.
2. $C^{(1)}$ acts on the orbital part only:
$$\langle (L'S)J'\|C^{(1)}\|(LS)J\rangle = (-1)^{L'+S+J+1}\sqrt{(2J+1)(2J'+1)}\begin{Bmatrix} L' & J' & S \\ J & L & 1\end{Bmatrix}\langle L'\|C^{(1)}\|L\rangle.$$
3. The orbital reduced matrix element is
$$\langle L'\|C^{(1)}\|L\rangle = (-1)^{L'}\sqrt{(2L+1)(2L'+1)}\begin{pmatrix} L' & 1 & L \\ 0 & 0 & 0\end{pmatrix},$$
$$\begin{pmatrix} L' & 1 & L \\ 0 & 0 & 0\end{pmatrix}^2 = \frac{\max(L,L')}{(2L+1)(2L'+1)} \qquad (L' = L\pm 1),$$
hence $|\langle L'\|C^{(1)}\|L\rangle|^2 = \max(L,L')$.

Combining (the factors $2J+1$ cancel):

E1 width

$$\Gamma_{E1}(i\to f+\gamma) = \frac{4}{3}\,C_{fi}\,\delta_{SS'}\,\langle e\rangle^2\alpha\,k^3\,|\langle f|r|i\rangle|^2, \tag{8.16}$$
$$C_{fi} = \max(L,L')\,(2J'+1)\begin{Bmatrix} L' & J' & S \\ J & L & 1\end{Bmatrix}^2,$$

with the radial overlap $\langle f|r|i\rangle = \int_0^\infty r^3 R_f(r)R_i(r)\mathrm{d}r = \boldsymbol{c}_f^T\langle r\rangle\boldsymbol{c}_i$ ($\ell = \frac{L+L'}{2}$). Values (checked with exact $6j$ symbols): $n^3S_1 \to n'^3P_J$: $C = \frac{2J+1}{9}$, i.e. $\Gamma = \frac{4}{27}(2J+1)\alpha e_Q^2 k^3|\langle r\rangle|^2$; $n^3P_J \to n'^3S_1$: $C = \frac{1}{3}$, $\Gamma = \frac{4}{9}\alpha e_Q^2 k^3|\langle r\rangle|^2$ for all J; $n^3D_1 \to n'^3P_{0,1,2}$: $C = \frac{2}{9}, \frac{1}{6}, \frac{1}{90}$.

Common refinements. A recoil/phase-space factor E_f/M_i is often included; a finite-size form replaces r by $\frac{3}{k}\big[\frac{kr}{2}j_0(\frac{kr}{2}) - j_1(\frac{kr}{2})\big]$, which reduces to r for $kr \ll 1$. Both are model choices; state which you use.

8.6 Magnetic dipole (M1) transitions

Operator. With $g = 2$, each quark contributes $\frac{e_i}{2m_i}\boldsymbol{\sigma}_i$. The photon is emitted at $\boldsymbol{r}_{1,2} = \pm\frac{\boldsymbol{r}}{2}$ (equal masses); expanding the plane waves $\mathrm{e}^{\mp \mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}/2}$ and keeping the part that connects S waves to S waves gives $j_0(kr/2)$:

$$\boldsymbol{\mu} = \frac{e_Q}{2m_Q}(\boldsymbol{\sigma}_1 - \boldsymbol{\sigma}_2)\, j_0\Big(\frac{kr}{2}\Big) \qquad (Q\bar{Q}). \tag{8.17}$$

Spin matrix element. $(S_{1z} - S_{2z})|1,0\rangle = |0,0\rangle$ (check on $|\uparrow\downarrow\rangle \pm |\downarrow\uparrow\rangle$), and by rotational symmetry each initial M connects to $|0,0\rangle$ through one spherical component with $|\langle\cdot\rangle|^2 = 1$. With $\boldsymbol{\sigma} = 2\boldsymbol{S}$, $\sum_q |\langle 0,0|(\sigma_1 - \sigma_2)_q|1,M\rangle|^2 = 4$ for every M.

Rate. The golden rule as for E1, $\Gamma = \frac{4}{3}\alpha k^3 \overline{\sum}|\langle f|\boldsymbol{\mu}/e|i\rangle|^2$, gives

$$\Gamma(n^3S_1 \to n'^1S_0 + \gamma) = \frac{4}{3}\,\frac{e_Q^2\alpha}{m_Q^2}\,k^3\,\Big|\Big\langle n'S\Big|j_0\Big(\frac{kr}{2}\Big)\Big|nS\Big\rangle\Big|^2, \tag{8.18}$$

and three times this for ${}^1S_0 \to {}^3S_1$ (three final states). For $n = n'$ the overlap is $\approx 1 - k^2\langle r^2\rangle/24$; for $n \neq n'$ ("hindered" transitions) it vanishes at $k \to 0$ by orthogonality and is very sensitive to the wave functions, to spin-dependent wave-function distortions and to relativistic corrections. In the literature the general form $\Gamma_{M1} = \frac{4}{3}\frac{2J'+1}{2L+1}\delta_{LL'}\delta_{S,S'\pm1}\frac{e_Q^2\alpha}{m_Q^2}k^3|\langle f|j_0(kr/2)|i\rangle|^2$ (times E_f/M_i) is used.

8.7 S–D mixing

In the coupled-channel GEM calculation the eigenvector contains both channels; the D probability is $P_D = \boldsymbol{c}_D^T\mathbf{S}^D\boldsymbol{c}_D$ and a mixing angle can be defined by $\sin^2\theta = P_D$. For a state $\cos\theta|n^3S_1\rangle - \sin\theta|n'^3D_1\rangle$ the leptonic amplitude combines $R_S(0)$ and $\frac{5}{2\sqrt{2}\,m_Q^2}R''_D(0)$ (from the ${}^3D_1$ formula above), with a convention-dependent relative sign. This is how $\psi(3770)$ acquires its leptonic width.

8.8 Beyond this section

Open-flavour strong decays (${}^3P_0$ pair-creation model, which needs the GEM wave functions of initial and final mesons) and hadronic transitions ($\psi' \to J/\psi\pi\pi$, QCD multipole expansion) are standard next steps; see Barnes–Godfrey–Swanson for a complete set of formulas used with the GI and NR models.

9 A statistics primer: the ideas behind the fitting chapter

This section is written for a reader who knows the formulas of statistics but does not yet *think* statistically. It contains no new methods. It explains, with two toy problems small enough to do by hand, what the machinery of §10 is doing and why. Every number quoted here was computed (and the coverage statements simulated) so you can reproduce them. If you read only one subsection, read §9.6: the straight-line toy contains almost every idea of the fitting chapter.

The whole fitting chapter in one sentence. *We pretend that the data were produced by the model plus Gaussian noise of known size; then the best parameters, their errors, and every test follow from one question: how probable would our data be, for each value of the parameters?*

9.1 The mindset

Data are truth plus noise. A measured mass y is not the mass. It is the true mass plus an error that would come out differently if the experiment were repeated. Physics gives us a model $\mu(\boldsymbol{\theta})$ for the truth; statistics is the bookkeeping of the noise. A statistical statement is always a statement about this imagined repetition ("if I redid everything many times, then ...") or, in the Bayesian language, about our degree of belief given the data.

Three questions. Every fit answers three different questions, and it helps to keep them apart:

1. *Estimation*: which parameter values describe the data best? (The minimum of χ^2, §9.4, §9.10.)
2. *Uncertainty*: how far could the true values be from the best ones? (The curvature of χ^2, its profile, or sampling, §9.5–§9.11.)
3. *Testing*: does the model describe the data at all, and better than another model? (The size of χ^2_{min}, information criteria, cross-validation, §9.8, §9.12.)

A good fit can have large errors (question 2 fails, question 3 is fine); a fit with tiny errors can be a bad description (question 3 fails). Never let one answer stand in for another.

Two languages, one set of numbers.

concept	frequentist reading	Bayesian reading
$L(\boldsymbol{\theta}) = p(\text{data}\,\vert\,\boldsymbol{\theta})$	likelihood of $\boldsymbol{\theta}$	likelihood (times prior = posterior)
best value	maximum-likelihood estimate $\hat{\boldsymbol{\theta}}$	posterior mode or mean
$\pm\sigma$ interval	covers the truth in 68% of repeated experiments	contains the truth with 68% posterior probability
covariance matrix	spread of $\hat{\boldsymbol{\theta}}$ over repeated experiments	width of the posterior
penalty term in χ^2	"external constraint"	prior

For a model that is linear in its parameters, with Gaussian noise and flat priors, both columns give *identical* numbers. The quarkonium fits are close to that case: in the worked example (§10.10) the curvature errors and the Markov-chain posterior widths agree to within 5%. So use whichever picture you find more intuitive; switch only when the answers differ (strongly non-linear models, parameters near a boundary).

9.2 Random variables: the five rules you need

For a random variable X with density $p(x)$: mean $\mathbb{E}[X] = \int x\,p(x)\,\mathrm{d}x$, variance $\operatorname{Var} X = \mathbb{E}[(X - \mathbb{E}X)^2]$, standard deviation $\sigma_X = \sqrt{\operatorname{Var} X}$. For two variables, covariance $\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$ and correlation $\rho = \operatorname{Cov}(X,Y)/(\sigma_X\sigma_Y) \in [-1,1]$.

(R1) *Expectation is linear*: $\mathbb{E}[aX + bY + c] = a\,\mathbb{E}X + b\,\mathbb{E}Y + c$, always (no independence needed).

(R2) *Scaling*: $\operatorname{Var}(aX + c) = a^2 \operatorname{Var} X$. Shifts do not change spreads; scales enter squared.

(R3) *Sums*: $\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)$. Independent variables have Cov = 0, so their variances add (errors add *in quadrature*). The converse is false: zero covariance does not imply independence.

(R4) *Linear maps of vectors*: if $\boldsymbol{x}$ has covariance V and $\boldsymbol{u} = \mathsf{A}\boldsymbol{x} + \boldsymbol{b}$, then

$$\mathrm{Cov}(\boldsymbol{u}) = \mathsf{A}\,\mathsf{V}\mathsf{A}^T. \tag{9.1}$$

Proof: $\boldsymbol{u} - \mathbb{E}\boldsymbol{u} = \mathsf{A}(\boldsymbol{x} - \mathbb{E}\boldsymbol{x})$, so $\mathbb{E}[(\boldsymbol{u} - \mathbb{E}\boldsymbol{u})(\boldsymbol{u} - \mathbb{E}\boldsymbol{u})^T] = \mathsf{A}\,\mathbb{E}[(\boldsymbol{x} - \mathbb{E}\boldsymbol{x})(\boldsymbol{x} - \mathbb{E}\boldsymbol{x})^T]\mathsf{A}^T$. (R2) and (R3) are special cases: take $\mathsf{A} = (a)$ or $\mathsf{A} = (1\ 1)$.

(R5) *Smooth functions*: linearize, $f(\boldsymbol{x}) \simeq f(\bar{\boldsymbol{x}}) + \boldsymbol{g}^T(\boldsymbol{x} - \bar{\boldsymbol{x}})$ with $\boldsymbol{g} = \nabla f$, and apply (R4) with $\mathsf{A} = \boldsymbol{g}^T$: $\mathrm{Var}\,f \simeq \boldsymbol{g}^T\mathsf{V}\boldsymbol{g}$. This is *all* of error propagation, (10.14). It is good as long as f is close to linear over the $\pm\sigma$ region.

Example 9.1 (A correlated systematic in your own database)**.** In the PDG file `data/pdg-2026.0.sqlite`, the three $\chi_{bJ}(1P)$ masses are listed as, e.g., $9859.44 \pm 0.42 \pm 0.31$ MeV for $\chi_{b0}(1P)$, and similarly for $J = 1, 2$. The second error comes from the $\Upsilon(2S)$ mass used to convert measured photon energies into masses; it is the *same* ± 0.31 MeV for all three states. Model each mass as $M_J = \mathrm{true}_J + \varepsilon_J + \delta$, with independent ε_J (σ_J) and one common δ (0.31 MeV). By (R3), $\mathrm{Var}\,M_J = \sigma_J^2 + 0.31^2$ and $\mathrm{Cov}(M_J, M_{J'}) = 0.31^2$ for $J \neq J'$. A splitting $M_2 - M_0$ has $\mathrm{Var} = \sigma_2^2 + \sigma_0^2 + 2(0.31)^2 - 2(0.31)^2 = \sigma_2^2 + \sigma_0^2$: the common error cancels, exactly as it should, but *only* if you put the off-diagonal 0.31^2 in the covariance V of (10.4). (Practical note: in the database the error columns contain only the first error; the common one is visible only in `display_value_text`, §11.10.)

Example 9.2 (Why averaging helps)**.** N independent measurements with the same σ:

$$\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} y_i\right) = \frac{1}{N^2}\cdot N\sigma^2 = \frac{\sigma^2}{N}$$

by (R2) and (R3). The error of the mean is $\sigma/\sqrt{N}$. If the measurements share a common systematic δ, that part does *not* average down: $\mathrm{Var} = \sigma^2/N + \sigma_\delta^2$.

9.3 The Gaussian, and why it is everywhere

One variable. $p(x) = \frac{1}{\sqrt{2\pi}\sigma}\mathrm{e}^{-(x-\mu)^2/2\sigma^2}$. The probability to fall within $\pm 1, \pm 2, \pm 3\sigma$ of the mean is 68.27%, 95.45%, 99.73%. The *central limit theorem* says that a sum of many small independent contributions is approximately Gaussian, whatever the distribution of each one. Measurement errors are such sums (many small instrumental effects), which is why the Gaussian works. It fails when one large effect dominates (a mis-assigned state, an outlier), so look at your residuals.

Many variables. $p(\boldsymbol{x}) \propto \exp\left[-\tfrac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^T\mathsf{V}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\right]$. The exponent is $-\frac{1}{2}\chi^2$. The surfaces of constant density are ellipsoids, whose axes are the eigenvectors of V and whose half-lengths are $\sqrt{\text{eigenvalues}}$; a strong correlation is a thin ellipse tilted with respect to the axes.

Whitening: every correlated problem is an uncorrelated one in disguise. Factor $\mathsf{V} = \mathsf{L}\mathsf{L}^T$ (Cholesky) and define $\boldsymbol{z} = \mathsf{L}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})$. By (R4), $\mathrm{Cov}(\boldsymbol{z}) = \mathsf{L}^{-1}\mathsf{V}\mathsf{L}^{-T} = \mathsf{1}$: the components of $\boldsymbol{z}$ are independent unit Gaussians, and $\chi^2 = (\boldsymbol{x} - \boldsymbol{\mu})^T\mathsf{V}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}) = \boldsymbol{z}^T\boldsymbol{z}$. This is why a fitting code only ever needs *whitened residuals* z_i and their derivatives (Algorithm 11.9): for independent errors $z_i = (y_i - \mu_i)/s_i$.

9.4 Likelihood: turning the probability around

Definition. The probability density of the observed data, regarded as a function of the parameters, $L(\boldsymbol{\theta}) = p(\boldsymbol{y} \mid \boldsymbol{\theta})$, is the *likelihood*. It is *not* a probability density for $\boldsymbol{\theta}$ (it need not integrate to one over $\boldsymbol{\theta}$). It answers: "how probable would *these* data have been if the parameters were $\boldsymbol{\theta}$?"

Likelihood ratios are the evidence. Only ratios matter: $L(\boldsymbol{\theta}_1)/L(\boldsymbol{\theta}_2)$ tells how much better $\boldsymbol{\theta}_1$ explains the data than $\boldsymbol{\theta}_2$. With $L \propto \mathrm{e}^{-\chi^2/2}$,

$$\Delta\chi^2 \equiv -2\ln\frac{L(\boldsymbol{\theta})}{L(\hat{\boldsymbol{\theta}})}: \qquad \frac{L}{L_{\max}} = 0.61,\ 0.14,\ 0.011 \quad \text{for} \quad \Delta\chi^2 = 1,\ 4,\ 9.$$

A parameter value one unit of χ^2 above the minimum explains the data almost as well as the best value; one that is nine units above explains them a hundred times worse.

Example 9.3 (Toy 1: the weighted mean, by hand)**.** Two measurements of the same quantity: $y_1 = 10.0 \pm 1.0$, $y_2 = 12.0 \pm 2.0$. The model is a single number μ. Then

$$\chi^2(\mu) = \frac{(10-\mu)^2}{1} + \frac{(12-\mu)^2}{4}.$$

Estimation. $\mathrm{d}\chi^2/\mathrm{d}\mu = -2(10-\mu) - \frac{1}{2}(12-\mu) = 0$ gives $\hat{\mu} = \frac{10/1+12/4}{1/1+1/4} = 10.4$: a weighted mean with weights $1/\sigma^2$. The better measurement counts four times as much. *Uncertainty.* χ^2 is an exact parabola, $\chi^2 = \chi^2_{\min} + 1.25\,(\mu-\hat{\mu})^2$, with $1.25 = \sum 1/\sigma_i^2$. It rises by one unit at $\mu - \hat{\mu} = \pm 1/\sqrt{1.25} = \pm 0.894$: $\hat{\mu} = 10.40 \pm 0.89$, smaller than either input error. The error came from the *curvature*: a sharply curved minimum means a well-determined parameter. *Testing.* $\chi^2_{\min} = 0.4^2 + 1.6^2/4 = 0.80$ with $\nu = N - p = 1$; the probability of a χ^2 this large or larger is $p = 0.37$: nothing unusual.

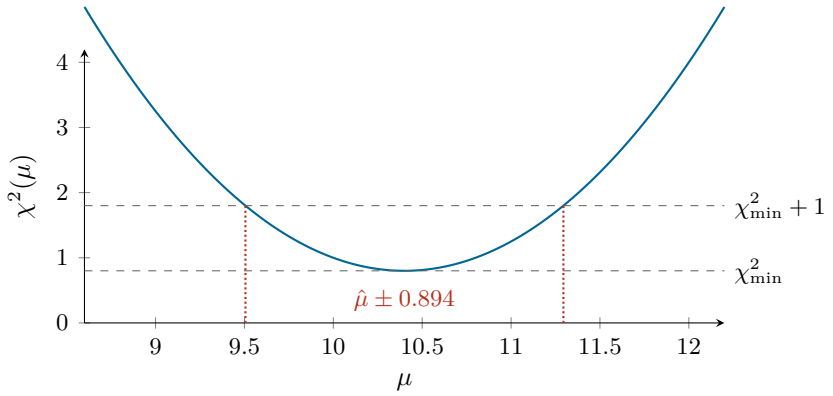


Figure 1: Toy 1. The best value is the bottom of the parabola; the 1σ interval is where χ^2 has risen by one unit.

9.5 What "$\pm 1\sigma$" means

Coverage. The frequentist meaning of "$\theta = \hat{\theta} \pm \sigma$" is a statement about the *procedure*: if the experiment were repeated many times and each time the interval $[\hat{\theta} - \sigma, \hat{\theta} + \sigma]$ were constructed in the same way, 68.3% of those intervals would contain the true value. For the straight-line toy below, a simulation with 2×10^5 repeated data sets gives 68.28% for the slope. It does *not* say that this particular interval has a 68% chance of containing the truth; that is the Bayesian statement, which here happens to give the same interval.

One parameter versus several. The rule "$\Delta\chi^2 = 1$" gives 68.3% for *one* parameter, with all others re-optimized. The region $\Delta\chi^2 \le 1$ in a plane of *two* parameters contains the true pair only 39.3% of the time (simulated: 39.5%); a 68.3% joint region needs $\Delta\chi^2 \le 2.30$ (simulated: 68.6%). The reason: the sum of squares of two unit Gaussians is less often below one than one squared Gaussian is. The table in §10.4 gives the values for up to four parameters.

Marginal versus conditional. Figure 2 shows the two ways of reading one error off a two-parameter ellipse. The *marginal* error is the half-width of the shadow of the $\Delta\chi^2 = 1$ ellipse on the axis: the other parameter is allowed to adjust. The *conditional* error is the half-width of a horizontal slice through the centre: the other parameter is frozen at its best value. Only the marginal error is an honest uncertainty; the conditional error is what you get if you vary one parameter and keep the others fixed, and it is always smaller.

9.6 Toy 2: a straight line by hand

Three points $(x, y) = (0,1), (1,2), (2,4)$, each with $\sigma = 1$, and the model $\mu(x) = a + bx$.

Estimation. $\chi^2(a,b) = \sum_i (y_i - a - bx_i)^2$. Setting both derivatives to zero gives the *normal equations* $\mathsf{F}\boldsymbol{\theta} = \mathsf{J}^T\boldsymbol{y}$ with $\mathsf{J} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\mathsf{F} = \mathsf{J}^T\mathsf{J} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \qquad \mathsf{J}^T\boldsymbol{y} = \begin{pmatrix} 7 \\ 10 \end{pmatrix},$$

$$\mathsf{C} = \mathsf{F}^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}, \qquad \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \mathsf{C}\,\mathsf{J}^T\boldsymbol{y} = \begin{pmatrix} 5/6 \\ 3/2 \end{pmatrix}.$$

Uncertainty. $\sigma_a = \sqrt{5/6} = 0.913$, $\sigma_b = \sqrt{1/2} = 0.707$, $\rho_{ab} = -3/\sqrt{15} = -0.775$. Conditional errors: $1/\sqrt{\mathsf{F}_{aa}} = 0.577$ and $1/\sqrt{\mathsf{F}_{bb}} = 0.447$.

Testing. Residuals $\boldsymbol{y} - \hat{\boldsymbol{\mu}} = (\frac{1}{6}, -\frac{1}{3}, \frac{1}{6})$, $\chi^2_{\min} = \frac{1}{6}$ with $\nu = 1$.

Why are a and b correlated? The intercept a is the value of the line at $x = 0$, at the *edge* of the data. The data pin the line down best at their centre $\bar{x} = 1$. If the slope is increased, the line pivots about $\bar{x}$ and the intercept must decrease: negative correlation. Re-parametrize around the centre, $\mu = a' + b(x - 1)$: then $\mathsf{F}' = \mathrm{diag}(3, 2)$, $a' = \frac{7}{3} \pm 0.577$, $b = \frac{3}{2} \pm 0.707$, and $\rho = 0$. *Correlations belong to the parametrization, not to the data.* In the quarkonium fits the same thing happens with m_Q and the constant c ($\rho = -0.987$, §10.10): both shift all masses, and the data only constrain a combination.

Prediction, and why correlations matter. Predict $P = \mu(3) = a + 3b = \frac{16}{3} = 5.33$. With $\boldsymbol{g} = (1, 3)$, (9.1) gives $\sigma_P^2 = \boldsymbol{g}^T\mathsf{C}\boldsymbol{g} = \frac{1}{6}(5 - 18 + 27) = \frac{7}{3}$, $\sigma_P = 1.53$. Dropping the covariance would give $\sqrt{5/6 + 9/2} = 2.31$, fifty per cent too large. In the centred parametrization, $P = a' + 2b$ and $\sigma_P^2 = \frac{1}{3} + 4 \cdot \frac{1}{2} = \frac{7}{3}$ again, now without any correlation term: *predictions do not depend on the parametrization; only the bookkeeping does.*

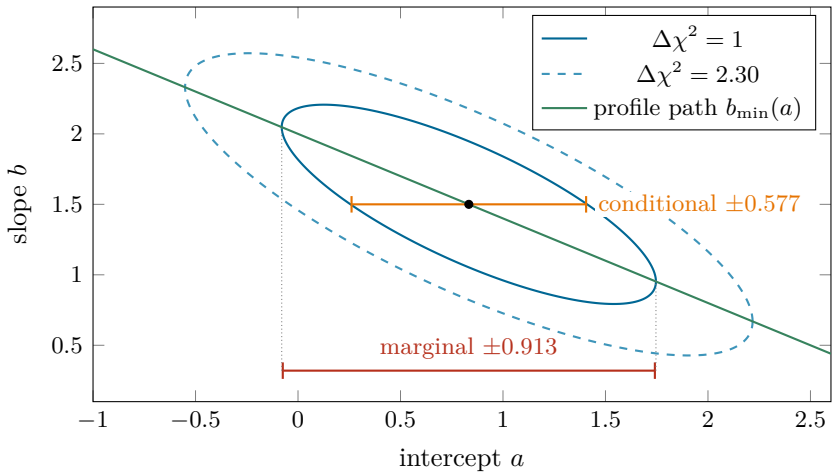


Figure 2: Toy 2 in the (a, b) plane. The tilt of the ellipses is the correlation $\rho = -0.775$. The marginal error of a is the shadow of the $\Delta\chi^2 = 1$ ellipse; the conditional error is the slice at fixed $b = \hat{b}$. The green line is the profile path: for each a, the b that minimizes χ^2; it touches the ellipse exactly at its leftmost and rightmost points, which is why the profile gives the marginal error (§10.4).

9.7 The geometry of least squares, and why the degrees of freedom are $N - p$

Work with whitened data $\boldsymbol{z} \in \mathbb{R}^N$ (one axis per datum). As $\boldsymbol{\theta}$ varies, the model predictions $\boldsymbol{\mu}(\boldsymbol{\theta})$ sweep out a p-dimensional surface; for a linear model it is a flat subspace, the span of the columns of J. Minimizing $\chi^2 = |\boldsymbol{z} - \boldsymbol{\mu}|^2$ means finding the point of this surface closest to the data: the *orthogonal projection* (Figure 3). The residual vector is perpendicular to the surface.

Now count. The noise vector has N independent unit-variance components in any orthonormal frame. Choose the frame with p axes in the model surface and $N - p$ perpendicular to it. The p components along the surface are absorbed by adjusting $\boldsymbol{\theta}$; the $N - p$ perpendicular ones remain in the residual. Hence $\mathbb{E}[\chi^2_{\min}] = N - p$: each fitted parameter "uses up" one dimension of the noise. This is the geometric content of the hat-matrix calculation in §10.2 (the hat matrix P is the projector onto the surface). In Toy 2, $N = 3$, $p = 2$: the residual lives on a line, and $\chi^2_{\min} = \frac{1}{6}$ is the square of a single Gaussian-distributed number.

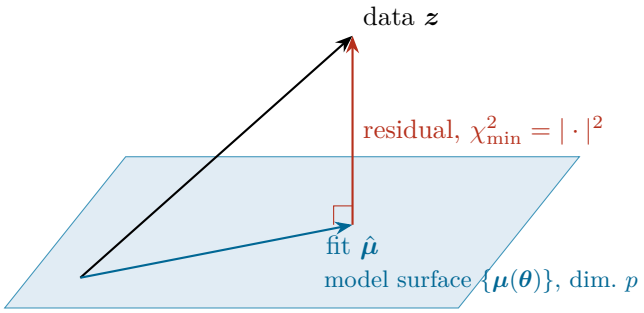


Figure 3: Least squares as orthogonal projection. Of the N noise directions, p lie in the model surface and are absorbed by the fit; $N - p$ are perpendicular and make up $\chi^2_{\min}$.

9.8 The χ^2 distribution: which values are normal?

If the model and the error model are right, $\chi^2_{\min}$ is (approximately) a sum of $\nu = N - p$ squared unit Gaussians: a χ^2_ν variable. For one unit Gaussian z, $\mathbb{E}z^2 = 1$ and $\mathbb{E}z^4 = 3$, so $\operatorname{Var} z^2 = 2$; by

(R3), χ^2_ν has mean ν and variance 2ν, i.e. χ^2/ν has mean 1 and standard deviation $\sqrt{2/\nu}$.

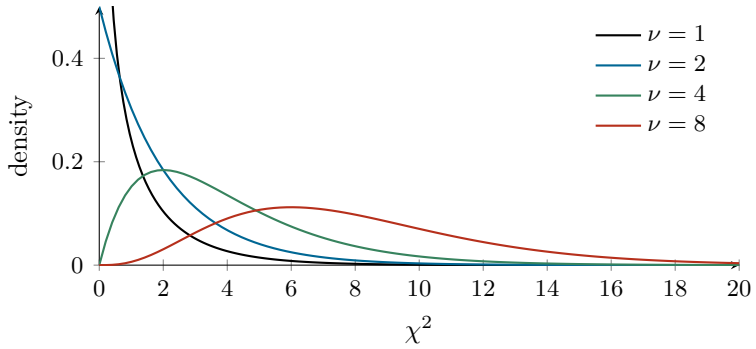


Figure 4: χ^2_ν densities. For few degrees of freedom the distribution is broad and skewed.

ν	1	2	4	8	20	100
central 95% range of χ^2/ν	0.001–5.02	0.025–3.69	0.12–2.79	0.27–2.19	0.48–1.71	0.74–1.30

Lesson for your fits. With eight charmonium states and four parameters ($\nu = 4$), any χ^2/ν between 0.12 and 2.8 is unremarkable. Do not read meaning into $\chi^2/\nu = 1.6$ versus 0.9.

p-values. $p = P(\chi^2_\nu \geq \chi^2_{\text{min}})$ is the probability of a misfit at least this bad *if* model and errors were right. A tiny p means: the model is wrong, or the errors are underestimated, or both. For quarkonium it is usually "both", and the theory floor is how we say so quantitatively.

How well is the theory floor itself known? σ_{th} is determined from ν residuals, so it is itself uncertain. If the true floor were σ, the estimate $\sqrt{\chi^2_\nu/\nu}\,\sigma$ has, for $\nu = 4$, a 16–84% range of 0.60–1.28 times σ (simulated). Turned around: the worked example's "$\sigma_{\text{th}} = 8.3\,\text{MeV}$" means "roughly 6.5–14 MeV". Two models whose fitted floors differ by 20% are not distinguishable this way.

9.9 Theory floor versus scale factor: two models of what went wrong

Take Toy 1 with much smaller, inconsistent errors: $y_1 = 10.0 \pm 0.1$, $y_2 = 12.0 \pm 0.2$. Now $\chi^2_{\text{min}} = 80$ for $\nu = 1$: the two numbers cannot both be right within their errors. The naive result $\hat{\mu} = 10.40 \pm 0.09$ is absurd. There are two standard repairs, and they encode different beliefs:

- *Scale factor* (PDG, §10.2): multiply all errors by $S = \sqrt{\chi^2_{\text{min}}/\nu} = 8.9$. Result 10.40 ± 0.80. The relative weights $(4 : 1)$ are kept. Belief: "the experimenters underestimated their errors by a common factor".
- *Additive floor*: add σ_{th} in quadrature to both errors and choose it so that $\chi^2/\nu = 1$; here $\chi^2 = 4/(0.05 + 2\sigma^2_{\text{th}}) = 1$ gives $\sigma_{\text{th}} = 1.41$. The weights become $0.504 : 0.496$ and $\hat{\mu} = 10.99 \pm 1.00$. Belief: "each point is off by a random amount of typical size σ_{th}, *independent of how precisely it was measured*".

For a potential model the scatter comes from the model, not from the measurements: a 6 keV error on the J/ψ mass does not make the model more accurate for the J/ψ. That is why §10.2 uses the additive floor. With the scale factor, the J/ψ would keep 10^4 times the weight of the $\eta_c(2S)$, and the fit would still ignore most of the spectrum.

9.10 Fitting as walking downhill

Think of $\chi^2(\boldsymbol{\theta})$ as a landscape over parameter space; the fit looks for its lowest point.

- *Gradient descent*: step in the direction of steepest descent. Safe but slow, especially in long narrow valleys, which is exactly what strongly correlated parameters produce: the gradient points across the valley, not along it, and the walker zig-zags.
- *Newton*: approximate the landscape by a paraboloid (value, gradient and curvature at the current point) and jump to the bottom of the paraboloid. Excellent near the minimum (it converges in a few steps), dangerous far from it (the paraboloid can be a poor guide and the jump can go anywhere).
- *Gauss–Newton*: Newton with the curvature estimated from first derivatives only, $\mathsf{F} = \mathsf{J}^T\mathsf{W}\mathsf{J}$ (10.7). For least squares this is almost free and almost as good.
- *Levenberg–Marquardt* (10.8): Gauss–Newton on a leash. The damping λ is the leash: with large λ the step is short and along the (rescaled) gradient; with small λ it is the full Gauss–Newton jump. After a successful step the leash is loosened ($\lambda \to \lambda/10$), after a failed one it is tightened ($\lambda \to 10\lambda$). The method therefore behaves like gradient descent far away and like Newton close by, without you having to decide when to switch.
- *Local versus global.* All of the above find *a* minimum, the one whose basin they start in. A landscape with several valleys (GI fits with ten or more parameters) needs several starting points (§10.3).

9.11 Beyond the parabola: profiles, bootstrap, Markov chains

The curvature error $\mathsf{C} = \mathsf{F}^{-1}$ assumes that χ^2 is a paraboloid. Three tools check or replace that assumption.

Profile likelihood: let the others do their best. For each fixed value of the parameter you care about, minimize χ^2 over all the others, and plot the result. In Figure 2 this is χ^2 along the green line. For a linear model the profile is an exact parabola and gives the marginal error. For a non-linear model the valley is curved; the profile follows it and gives asymmetric intervals ($\Delta\chi^2 = 1$ crossings at different distances), which the parabola cannot.

Parametric bootstrap: rerun the experiment on the computer. Take the fitted model as if it were the truth, add Gaussian noise with the data covariance, refit, and repeat a few hundred times. The spread of the refitted parameters *is* the sampling distribution of $\hat{\boldsymbol{\theta}}$, including all non-linearities, parameter boundaries and biases. It costs one fit per replica, and it answers the frequentist question directly.

Markov chain Monte Carlo: a walker that spends its time where the probability is. The Metropolis rule: propose a random step; if it goes to higher posterior probability, take it; if it goes down, take it with probability $\pi_{\text{new}}/\pi_{\text{old}}$, otherwise stay (and count the current point again). After many steps the fraction of time spent in any region equals its posterior probability.

Example 9.4 (Detailed balance with two states)**.** Let a "parameter" take two values, A and B, with target probabilities $\pi(A) = \frac{2}{3}$, $\pi(B) = \frac{1}{3}$. Always propose the other state. From A, the move to B goes downhill and is accepted with probability $\pi(B)/\pi(A) = \frac{1}{2}$; from B the move to A goes uphill and is always accepted. In equilibrium the probability flow $A \to B$ is $\pi(A) \cdot \frac{1}{2} = \frac{1}{3}$ and the flow $B \to A$ is $\pi(B) \cdot 1 = \frac{1}{3}$: they balance, so the occupation $(\frac{2}{3}, \frac{1}{3})$ is stationary. Check

by the transition matrix $T = \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix}$: $(\frac{2}{3}, \frac{1}{3})\,T = (\frac{1}{3} + \frac{1}{3},\ \frac{1}{3}) = (\frac{2}{3}, \frac{1}{3})$. The general proof in §10.6 is this argument with integrals.

Practical consequences: (i) the start is arbitrary, so discard an initial *burn-in*; (ii) successive samples are correlated (the walker moves a little each time), so n samples are worth only $n_{\text{eff}} = n/\tau_{\text{int}}$ independent ones; (iii) one walker cannot tell you that it has explored everything, so run several from different starts and check that they agree ($\hat{R}$).

method	answers	use when
curvature $\mathsf{C} = \mathsf{F}^{-1}$	local Gaussian approximation	always (cheap); sufficient if the profile is parabolic
profile likelihood	interval for one quantity, asymmetric if needed	key parameters; any doubt about non-linearity
bootstrap	frequentist sampling distribution	bias, boundaries, non-linear predictions
MCMC	full posterior, priors, marginals of anything	many parameters, priors/penalties, correlated predictions

9.12 Comparing models without fooling yourself

More parameters always lower χ^2. In Toy 2, a parabola $a + bx + cx^2$ goes exactly through all three points: $\chi^2 = 0$. Is it the better model?

- *AIC* ($\chi^2_{\min} + 2p$; the expected χ^2 on *new* data, §10.7): line $\frac{1}{6} + 4 = 4.17$, parabola $0 + 6 = 6$. The line wins.
- *BIC* ($\chi^2_{\min} + p \ln N$): line 2.36, parabola 3.30. The line wins.
- *Leave-one-out*: drop each point, refit the line through the other two, and predict the dropped point. By hand: without $(0, 1)$ the line is $y = 2x$, predicting 0 (miss $+1$); without $(1, 2)$ it is $y = 1 + 1.5x$, predicting 2.5 (miss -0.5); without $(2, 4)$ it is $y = 1 + x$, predicting 3 (miss $+1$). The one-fit formula (10.19) reproduces these exactly: the leverages are $h = (\frac{5}{6}, \frac{1}{3}, \frac{5}{6})$ (sum $= 2 = p$) and $\Delta_i/(1 - h_i) = (+1, -0.5, +1)$. The parabola cannot be tested at all: with one point removed it has three parameters for two points.

The idea behind all three: a model is good if it *predicts* data it has not seen, not if it *reproduces* data it was tuned to. AIC estimates the prediction error from the fit itself; leave-one-out measures it; BIC asks how much of the parameter space the data actually needed (an Occam factor). The leverage h_i tells you how much datum i pulled the fit towards itself: the two end points of the line ($h = \frac{5}{6}$) control the slope, the middle point ($h = \frac{1}{3}$) hardly matters.

9.13 The error budget as a story

A number without its error budget cannot be used by anybody else. Tell the story of every error source separately (§10.8):

- *Parametric (statistical)*: "given this model and this error model, the data allow the parameters to move this much". It shrinks with better data.
- *Numerical*: "our own approximations (basis size, quadrature) move the result this much". You control it; make it negligible ($\leq 1\%$ of the smallest experimental error) and say so.
- *Theory floor*: "this model misses masses by about this much". It does not shrink with better data.

- *Ambiguities*: "choices the model does not fix (which M in a width formula, the scale of α_s) move the result this much".
- *Model spread*: "other reasonable Hamiltonians give this range".

Add in quadrature only errors that are independent and roughly Gaussian (parametric and numerical, say). Keep the others separate so that a reader sees which one dominates. Example from the worked fit: for $\Gamma_{ee}(J/\psi)$ the parametric error is $\pm 1.1\,\mathrm{keV}$, the choice of M moves it by $1.3\,\mathrm{keV}$, and the QCD factor $(1 - \frac{16\alpha_s}{3\pi})$ ranges from 0.09 to about 0.4 depending on which α_s is used: the last item dominates everything, and only a separate budget shows it.

9.14 Ten common mistakes

1. Fitting with experimental errors only, when the model cannot reach that precision (§9.9).
2. Quoting conditional errors (others fixed) instead of marginal ones (Figure 2).
3. Combining errors of two predictions without their covariance, e.g. for a splitting (§9.6).
4. Over-interpreting χ^2/ν with few degrees of freedom (§9.8).
5. Reading a two-parameter $\Delta\chi^2 = 1$ contour as a 68% region (§9.5).
6. Re-optimizing the basis during the fit, which makes $\chi^2(\boldsymbol{\theta})$ noisy (§7.12).
7. Fitting states above the open-flavour threshold with the same floor as those below (§10.1).
8. Not knowing what your tool does by default: Mathematica's `NonlinearModelFit` rescales the parameter errors by $\sqrt{\chi^2/\nu}$ unless told not to (§13.4.11); Ceres reports $\frac{1}{2}\chi^2$ as its cost.
9. Running a Markov chain without convergence diagnostics, or reporting n instead of n_{eff}.
10. Adding the model spread in quadrature to the statistical error.

10 Fitting, statistics and error analysis

If the statistical way of thinking is not yet second nature, read the primer (§9) first: it explains every idea of this section on two toy problems you can do by hand. Each subsection below has an "In plain words" box that points back to it.

Up to now the parameters of each Hamiltonian were given. In the project you will *determine* them from data, together with their uncertainties, and then propagate those uncertainties to predictions. This section derives every statistical tool you need, in the order you will use them. A complete worked example (Model A fitted to the charmonium states below open-charm threshold) is in §10.10; every method below is illustrated there with numbers you can reproduce.

Notation (sans-serif = statistics matrices). Data y_i, $i = 1, \ldots, N$; model predictions $\mu_i(\boldsymbol{\theta})$ (eigenvalues, or observables computed from eigenvectors); parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)$; residuals $\Delta_i(\boldsymbol{\theta}) = y_i - \mu_i(\boldsymbol{\theta})$; data covariance V and weight matrix $\mathsf{W} = \mathsf{V}^{-1}$; Jacobian $\mathsf{J}_{ia} = \partial\mu_i/\partial\theta_a$; Fisher matrix $\mathsf{F} = \mathsf{J}^T\mathsf{W}\mathsf{J}$; parameter covariance C. Bold upright $\mathbf{H}, \mathbf{S}$ remain the GEM matrices.

10.1 What is fitted, and to what

Two kinds of parameters. They are listed in Table 1. They are treated completely differently. $\boldsymbol{\theta}$ is *inferred from data* and carries statistical uncertainty. $\boldsymbol{\xi}$ is *not physics*: it is chosen by the variational principle (§7.12), it is never fitted to data, and its only footprint on the result is a numerical error σ_{num} (§10.8).

	physics parameters $\boldsymbol{\theta}$	basis hyperparameters $\boldsymbol{\xi}$
Model A	m_Q, α_s (or κ), b, c, σ (smearing of the contact term)	per channel: $n_{\max}, r_1, r_{\max}$; complex-range set $(\rho_1, \rho_{\max}, \omega)$; threshold η
Model B	A + μ_s	same
Model C	m_q, b, c, σ_0, s, $\epsilon_c, \epsilon_{sov}, \epsilon_t, \epsilon_{sos}$; optionally $\{\alpha_k, \gamma_k\}$ (§10.9)	same + quadrature settings for the p integrals
Model D	C + μ_s	same

Table 1: Physics parameters and basis hyperparameters of the four models.

Which data. A potential model with a $Q\bar{Q}$ Fock space only knows about states that cannot decay into open flavour. Above the threshold ($D\bar{D}$ at $\approx 3.73\,\text{GeV}$, $B\bar{B}$ at $\approx 10.56\,\text{GeV}$) the coupling to meson–meson channels shifts masses by tens of MeV, which the model does not contain. Standard practice:

- fit to the well-established states *below* threshold;
- use states above threshold only as *predictions* (and expect deviations of the size of coupled-channel shifts);
- if you fit them anyway, give them a larger theory floor (below).

For charmonium this leaves the eight states of Table 2.

state	$n^{2S+1}L_J$	(L, S, J), radial index	mass (MeV)	used in fit
$\eta_c(1S)$	1^1S_0	$(0, 0, 0)$, 1st	2984.09 ± 0.30	yes
J/ψ	1^3S_1	$(0, 1, 1)$, 1st	3096.900 ± 0.006	yes
$\chi_{c0}(1P)$	1^3P_0	$(1, 1, 0)$, 1st	3415.50 ± 0.19	yes
$\chi_{c1}(1P)$	1^3P_1	$(1, 1, 1)$, 1st	3510.67 ± 0.05	yes
$h_c(1P)$	1^1P_1	$(1, 0, 1)$, 1st	3525.37 ± 0.14	yes
$\chi_{c2}(1P)$	1^3P_2	$(1, 1, 2)$, 1st	3556.17 ± 0.07	yes
$\eta_c(2S)$	2^1S_0	$(0, 0, 0)$, 2nd	3637.8 ± 0.6	yes
$\psi(2S)$	2^3S_1	$(0, 1, 1)$, 2nd	3686.097 ± 0.010	yes

Table 2: Charmonium states below the $D\bar{D}$ threshold; PDG averages as listed in pdgLive (accessed October 2026). Check the current values before your final fits.

Level identification. Each datum must be attached to a definite eigenvalue: channel (L, S, J) and radial index n. In a coupled channel ($L = J \mp 1$) the eigenvector has both components; label each eigenvalue by its dominant component, $P_D = \boldsymbol{c}_D^T \mathbf{S}^D \boldsymbol{c}_D$ (§8). During a fit the ordering can change (an S-dominated and a D-dominated level can cross as parameters move). Track levels by *maximum overlap*: at the new parameter point choose the eigenvector $\boldsymbol{c}'_b$ maximizing $|\boldsymbol{c}'^T_b \mathbf{S} \boldsymbol{c}_a|$ with the old one. With a fixed basis, $\mathbf{S}$ is the same at both points, so this is one matrix–vector product.

10.2 From the likelihood to χ^2

10.2.1 Independent Gaussian errors

Step 1 (model for the data). Write each measurement as truth plus noise, and the truth as model plus model error:

$$y_i = \mu_i(\boldsymbol{\theta}) + \varepsilon_i^{\text{exp}} + \varepsilon_i^{\text{th}}, \qquad \varepsilon_i^{\text{exp}} \sim \mathcal{N}(0, \sigma^2_{\text{exp},i}), \quad \varepsilon_i^{\text{th}} \sim \mathcal{N}(0, \sigma^2_{\text{th},i}). \tag{10.1}$$

The sum of two independent Gaussians is Gaussian with the variances added: $y_i - \mu_i \sim \mathcal{N}(0, s_i^2)$, $s_i^2 = \sigma_{\mathrm{exp},i}^2 + \sigma_{\mathrm{th},i}^2$.

Step 2 (likelihood). The probability density of the observed data as a function of $\boldsymbol{\theta}$:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi s_i^2}} \exp\Big[-\frac{(y_i - \mu_i(\boldsymbol{\theta}))^2}{2 s_i^2} \Big]. \tag{10.2}$$

Step 3 (log). Products become sums:

$$-2 \ln L(\boldsymbol{\theta}) = \underbrace{\sum_{i=1}^{N} \frac{(y_i - \mu_i(\boldsymbol{\theta}))^2}{s_i^2}}_{\chi^2(\boldsymbol{\theta})} + \sum_{i=1}^{N} \ln(2\pi s_i^2). \tag{10.3}$$

Step 4. If the s_i do not depend on $\boldsymbol{\theta}$, the second sum is a constant: *maximizing the likelihood is minimizing* χ^2. The maximum-likelihood estimate is $\hat{\boldsymbol{\theta}} = \arg\min \chi^2$.

In plain words

χ^2 measures, for each datum, by how many standard deviations the prediction misses it, squares, and adds. Minimizing it means choosing the parameters under which the observed data would have been most probable. Toy 1 (Example 9.3) does this by hand for a weighted mean.

10.2.2 Correlated errors

If the noise vector has covariance V (symmetric, positive definite), the multivariate Gaussian density is $(2\pi)^{-N/2} (\det \mathsf{V})^{-1/2} \exp[-\tfrac{1}{2} \boldsymbol{\Delta}^T \mathsf{V}^{-1} \boldsymbol{\Delta}]$, so

$$\chi^2(\boldsymbol{\theta}) = \boldsymbol{\Delta}^T \mathsf{W} \boldsymbol{\Delta}, \qquad \mathsf{W} = \mathsf{V}^{-1}, \qquad \boldsymbol{\Delta} = \boldsymbol{y} - \boldsymbol{\mu}(\boldsymbol{\theta}). \tag{10.4}$$

For diagonal $\mathsf{V} = \mathrm{diag}(s_i^2)$ this is the previous formula.

Example 10.1 (Fitting splittings instead of masses)**.** Suppose you fit differences $d_i = y_i - y_0$ to a reference state (to remove the constant c, or because the splittings are what you trust). If the y's are independent with variances s_i^2, then

$$\mathrm{Cov}(d_i, d_j) = \mathrm{Cov}(y_i - y_0,\, y_j - y_0) = \delta_{ij} s_i^2 + s_0^2,$$

because the common y_0 appears in both. So the splittings are correlated and you must use (10.4) with $\mathsf{V}_{ij} = \delta_{ij} s_i^2 + s_0^2$; ignoring the off-diagonal s_0^2 double-counts the reference state.

10.2.3 Why a theory floor is unavoidable

Experimental mass errors range from 6 keV (J/ψ) to 0.6 MeV ($\eta_c(2S)$); no potential model is accurate to 6 keV. With $s_i = \sigma_{\mathrm{exp},i}$ alone the weights $1/s_i^2$ differ by 10^4, and the fit simply reproduces the two most precise states and ignores the others. In the worked example, the fit with experimental errors only has $\chi^2 = 10\,011$ for $\nu = 4$ and residuals of 0.004 and 0.009 MeV for J/ψ and $\psi(2S)$ but -17 MeV for χ_{c0}. The theory floor σ_{th} states how well the model *can* work and restores sensible weights.

Three ways to set σ_{th}.

1. *A priori*: from the size of what the model neglects (e.g. the next order in v^2; §10.8). Usually conservative.
2. *A posteriori*: choose σ_{th} (common to all states) such that $\chi^2_{\min}/\nu = 1$, $\nu = N - p$.
3. *As a fitted nuisance parameter*: maximize the full likelihood (10.3), including the $\ln s_i^2$ term (which now depends on σ_{th}).

Options 2 and 3 are closely related, as the next derivation shows.

Result 10.2 (Maximum-likelihood theory floor)**.** *Take $s_i^2 = \sigma_{\text{exp},i}^2 + \sigma_{\text{th}}^2$ and differentiate (10.3) with respect to σ_{th}^2 at fixed $\boldsymbol{\theta}$:*

$$\frac{\partial(-2\ln L)}{\partial\sigma_{\text{th}}^2} = \sum_i \Big[-\frac{\Delta_i^2}{s_i^4} + \frac{1}{s_i^2}\Big] = 0 \quad\Longrightarrow\quad \sum_i \frac{\Delta_i^2 - s_i^2}{s_i^4} = 0.$$

When $\sigma_{\text{th}} \gg \sigma_{\text{exp}}$ all s_i are equal and this gives $\sigma_{\text{th}}^2 = \frac{1}{N}\sum_i \Delta_i^2$.

Bias of the ML estimate, and the $N - p$. For a model linear in $\boldsymbol{\theta}$ (or linearized at $\hat{\boldsymbol{\theta}}$) with equal errors σ, the fitted residuals are $\hat{\boldsymbol{\Delta}} = (1 - \mathsf{P})\boldsymbol{\varepsilon}$, where $\mathsf{P} = \mathsf{J}(\mathsf{J}^T\mathsf{J})^{-1}\mathsf{J}^T$ is the projector onto the columns of J (the "hat matrix": $\hat{\boldsymbol{\mu}} = \mathsf{P}\boldsymbol{y}$). Then

$$\mathbb{E}[\hat{\boldsymbol{\Delta}}^T\hat{\boldsymbol{\Delta}}] = \mathbb{E}[\boldsymbol{\varepsilon}^T(1-\mathsf{P})\boldsymbol{\varepsilon}] = \sigma^2\,\mathrm{tr}(1-\mathsf{P}) = \sigma^2(N-p),$$

using $(1-\mathsf{P})^T(1-\mathsf{P}) = 1 - \mathsf{P}$ (projector), $\mathbb{E}[\varepsilon_i\varepsilon_j] = \sigma^2\delta_{ij}$, and $\mathrm{tr}\,\mathsf{P} = \mathrm{tr}[(\mathsf{J}^T\mathsf{J})^{-1}\mathsf{J}^T\mathsf{J}] = \mathrm{tr}\,1_p = p$. So $\frac{1}{N}\sum\Delta_i^2$ underestimates σ^2 and the unbiased estimate is $\sigma_{\text{th}}^2 = \frac{1}{N-p}\sum_i \hat{\Delta}_i^2$, which is *exactly* the condition $\chi^2_{\min}/\nu = 1$ of option 2. Fitting the parameters "uses up" p degrees of freedom of the residuals.

Consequences.

- Once σ_{th} is tuned to give $\chi^2/\nu = 1$, the χ^2 value carries *no* goodness-of-fit information; the information is in σ_{th} itself ("this model describes the spectrum to $\pm\sigma_{\text{th}}$"). Compare models by their σ_{th} (or with a common fixed σ_{th}, §10.7).
- Determine σ_{th} iteratively: fit at a trial value, solve $\sum_i \Delta_i^2/(\sigma_{\text{exp},i}^2 + \sigma_{\text{th}}^2) = \nu$ for σ_{th} with the residuals fixed (one-dimensional root), refit; it converges in a few iterations.
- State-dependent floors are reasonable (e.g. larger for $2S$ than $1S$, or $\sigma_{\text{th},i} \propto \langle v^2\rangle_i$); but every extra choice is a modelling decision you must report.

In plain words

σ_{th} answers "how far off is this model, typically?". It is the rms residual, with $N - p$ instead of N because the fit has already bent towards the data (§9.7). It is added in quadrature, not multiplied, because the misfit comes from the model and does not care how precisely a state was measured (§9.9). And it is itself uncertain: with $\nu = 4$, by roughly $\pm 35\%$ (§9.8).

The PDG scale factor (Birge ratio). When a fit with *fixed* errors gives $\chi^2/\nu > 1$, a common practice is to multiply all parameter errors by $S = \sqrt{\chi^2_{\min}/\nu}$. This is equivalent to scaling all s_i by S, which leaves $\hat{\boldsymbol{\theta}}$ unchanged and restores $\chi^2/\nu = 1$. It is the same idea as option 2 applied multiplicatively instead of in quadrature.

10.3 Finding the minimum

10.3.1 Gradient and Hessian of χ^2

Differentiate $\chi^2 = \boldsymbol{\Delta}^T \mathsf{W} \boldsymbol{\Delta}$ with $\partial \boldsymbol{\Delta}/\partial\theta_a = -\mathsf{J}_{\cdot a}$:

$$\frac{\partial \chi^2}{\partial \theta_a} = -2\,(\mathsf{J}^T \mathsf{W} \boldsymbol{\Delta})_a, \tag{10.5}$$

$$\frac{\partial^2 \chi^2}{\partial \theta_a \partial \theta_b} = 2\,(\mathsf{J}^T \mathsf{W} \mathsf{J})_{ab} - 2 \sum_{ij} \Delta_i \mathsf{W}_{ij} \frac{\partial^2 \mu_j}{\partial \theta_a \partial \theta_b}. \tag{10.6}$$

The second term is a sum of residuals times curvatures. Near a good minimum the residuals are small and of random sign, so it is small compared with the first; dropping it gives the *Gauss–Newton Hessian* $2\mathsf{F}$, $\mathsf{F} = \mathsf{J}^T \mathsf{W} \mathsf{J}$, which is positive semi-definite by construction and needs only first derivatives.

10.3.2 Gauss–Newton step

Linearize the model around the current point, $\boldsymbol{\mu}(\boldsymbol{\theta} + \boldsymbol{\delta}) \simeq \boldsymbol{\mu}(\boldsymbol{\theta}) + \mathsf{J}\boldsymbol{\delta}$; then $\chi^2(\boldsymbol{\theta} + \boldsymbol{\delta}) \simeq (\boldsymbol{\Delta} - \mathsf{J}\boldsymbol{\delta})^T \mathsf{W} (\boldsymbol{\Delta} - \mathsf{J}\boldsymbol{\delta})$, a quadratic in $\boldsymbol{\delta}$. Setting its gradient to zero:

$$-2\mathsf{J}^T \mathsf{W}(\boldsymbol{\Delta} - \mathsf{J}\boldsymbol{\delta}) = 0 \quad \Longrightarrow \quad \mathsf{F}\,\boldsymbol{\delta} = \mathsf{J}^T \mathsf{W} \boldsymbol{\Delta}. \tag{10.7}$$

Iterate $\boldsymbol{\theta} \to \boldsymbol{\theta} + \boldsymbol{\delta}$. Gauss–Newton converges quadratically near the minimum but can overshoot far from it.

10.3.3 Levenberg–Marquardt

Add a damping term:

$$(\mathsf{F} + \lambda\,\mathsf{D})\boldsymbol{\delta} = \mathsf{J}^T \mathsf{W} \boldsymbol{\Delta}, \qquad \mathsf{D} = \mathrm{diag}(\mathsf{F}). \tag{10.8}$$

For $\lambda \to 0$ this is Gauss–Newton; for $\lambda \to \infty$, $\boldsymbol{\delta} \to \frac{1}{\lambda}\mathsf{D}^{-1}\mathsf{J}^T\mathsf{W}\boldsymbol{\Delta} = -\frac{1}{2\lambda}\mathsf{D}^{-1}\nabla\chi^2$, a short steepest-descent step (scaled per parameter by D, which makes it invariant under rescaling of individual parameters). Equation (10.8) is the exact minimizer of the linearized χ^2 under a constraint on the step length $\|\mathsf{D}^{1/2}\boldsymbol{\delta}\|$ (Lagrange multiplier λ), so it is a trust-region method. Algorithm: accept the step and set $\lambda \to \lambda/10$ if χ^2 decreased; otherwise reject and set $\lambda \to 10\lambda$. Stop when the relative change of χ^2 is below 10^{-10} or so.

In plain words

Gauss–Newton jumps to the bottom of the local paraboloid; Levenberg–Marquardt is Gauss–Newton on a leash λ that is loosened after every success and tightened after every failure (§9.10). The D scaling makes the leash the same length for $m_Q \sim 1.5$ and $\epsilon \sim 0.02$. Algorithm 11.9 in §11 is the version to implement.

10.3.4 The Jacobian from Feynman–Hellmann

For a mass, $\mu_i = \mathcal{E}_{k(i)}$ (plus $m_1 + m_2$ in Models A/B), and (7.34) with a fixed basis ($\mathbf{S}' = 0$) gives

$$\mathsf{J}_{ia} = \boldsymbol{c}_{k(i)}^T \frac{\partial \mathbf{H}}{\partial \theta_a} \boldsymbol{c}_{k(i)} \qquad (+\,\partial(m_1 + m_2)/\partial\theta_a). \tag{10.9}$$

One diagonalization per channel gives the full Jacobian (one matrix $\partial\mathbf{H}/\partial\theta_a$ per parameter, listed in §7.11), against $2p$ extra diagonalizations for central finite differences. For observables built

from eigenvectors use (7.35). For perturbative shifts $\boldsymbol{c}^T\mathbf{H}_1\boldsymbol{c}$ (as for the spin–orbit and tensor terms in Model A) the derivative is $\boldsymbol{c}^T\mathbf{H}_1'\boldsymbol{c} + 2\boldsymbol{c}^T\mathbf{H}_1\boldsymbol{c}'$ with $\boldsymbol{c}'$ from (7.35). Always check the analytic Jacobian against finite differences once (worked example: $\partial M_{J/\psi}/\partial b = \langle r \rangle = 1.88670872$ by both).

10.3.5 Practicalities

- *Fixed basis during the fit* (Property 5 of §7.12): otherwise $\chi^2(\boldsymbol{\theta})$ is not smooth.
- *Scaling and positivity.* Parameters have very different sizes ($m_Q \sim 1.5$, $b \sim 0.15$, $\epsilon_x \sim 0.02$). Fit in $\ln m_Q, \ln b, \ln \sigma, \ln \mu_s$ (positive by construction), or in units of a typical scale. Errors transform with the Jacobian of the reparametrization: if $\boldsymbol{\phi} = \boldsymbol{\phi}(\boldsymbol{\theta})$ then $\mathsf{C}_\phi = \mathsf{K}\mathsf{C}_\theta\mathsf{K}^T$, $\mathsf{K}_{ab} = \partial\phi_a/\partial\theta_b$ (derived in §10.5).
- *Global search.* For Models C/D (10+ parameters) χ^2 is multimodal. Use a global stage (multistart from a Latin hypercube, differential evolution, or CMA-ES) to find the basin, then polish with Levenberg–Marquardt. Report whether different starts converge to the same $\hat{\boldsymbol{\theta}}$.
- *Sanity bounds.* Reject parameter points where the Hamiltonian becomes unbounded or unphysical (e.g. $\mu_s < 0$, negative smearing widths); return $\chi^2 = +\infty$ there.

10.4 Parameter uncertainties from the curvature

10.4.1 Derivation 1: Gaussian approximation to the likelihood

Expand χ^2 around its minimum, where the gradient vanishes, and use the Gauss–Newton Hessian:

$$\chi^2(\boldsymbol{\theta}) \simeq \chi^2_{\min} + (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T\mathsf{F}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}). \tag{10.10}$$

Then $L \propto \mathrm{e}^{-\chi^2/2} \propto \exp\big[-\tfrac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T\mathsf{F}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})\big]$, a Gaussian in $\boldsymbol{\theta}$ whose covariance is the inverse of the matrix in the exponent:

$$\boxed{\mathsf{C} = \mathsf{F}^{-1} = (\mathsf{J}^T\mathsf{W}\mathsf{J})^{-1} = \left(\tfrac{1}{2}\nabla\nabla\chi^2\right)^{-1}} \tag{10.11}$$

(note the factor $\frac{1}{2}$ if you use the numerical Hessian of χ^2).

In plain words

A sharply curved minimum means well-determined parameters. In Toy 1 the curvature $\sum 1/\sigma_i^2 = 1.25$ gives $\sigma = 1/\sqrt{1.25} = 0.894$; in Toy 2 the inverse of F gives the whole error ellipse (§9.6). The two derivations below say the same thing in the two languages of §9.1: width of the likelihood, and spread of $\hat{\boldsymbol{\theta}}$ over repeated experiments.

10.4.2 Derivation 2: propagation of the data noise

Linearize the Gauss–Newton solution around the truth $\boldsymbol{\theta}_0$: $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 = \mathsf{A}(\boldsymbol{y} - \boldsymbol{\mu}(\boldsymbol{\theta}_0))$, $\mathsf{A} = \mathsf{F}^{-1}\mathsf{J}^T\mathsf{W}$, which is linear in the noise $\boldsymbol{\varepsilon} = \boldsymbol{y} - \boldsymbol{\mu}(\boldsymbol{\theta}_0)$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \mathsf{V} = \mathsf{W}^{-1}$. For any linear map, $\mathrm{Cov}(\mathsf{A}\boldsymbol{\varepsilon}) = \mathsf{A}\mathsf{V}\mathsf{A}^T$:

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) = \mathsf{F}^{-1}\mathsf{J}^T\mathsf{W}\,\mathsf{W}^{-1}\,\mathsf{W}\mathsf{J}\mathsf{F}^{-1} = \mathsf{F}^{-1}(\mathsf{J}^T\mathsf{W}\mathsf{J})\mathsf{F}^{-1} = \mathsf{F}^{-1}.$$

Both derivations give the same C: it is the frequentist spread of $\hat{\boldsymbol{\theta}}$ over repeated experiments *and* the width of the likelihood.

10.4.3 Why $\Delta\chi^2 = 1$: the profile of one parameter

Split $\boldsymbol{\delta} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}} = (\delta_1, \boldsymbol{\delta}_2)$, with δ_1 the parameter of interest and $\boldsymbol{\delta}_2$ the rest, and F correspondingly into blocks. Minimize the quadratic χ^2 over $\boldsymbol{\delta}_2$ at fixed δ_1:

$$\frac{\partial}{\partial \boldsymbol{\delta}_2}\left[\mathsf{F}_{11}\delta_1^2 + 2\delta_1 \mathsf{F}_{12}\boldsymbol{\delta}_2 + \boldsymbol{\delta}_2^T \mathsf{F}_{22}\boldsymbol{\delta}_2\right] = 2\mathsf{F}_{21}\delta_1 + 2\mathsf{F}_{22}\boldsymbol{\delta}_2 = 0 \quad \Rightarrow \quad \boldsymbol{\delta}_2 = -\mathsf{F}_{22}^{-1}\mathsf{F}_{21}\delta_1 .$$

Substituting back,

$$\chi^2_{\mathrm{prof}}(\theta_1) = \chi^2_{\min} + \delta_1^2(\mathsf{F}_{11} - \mathsf{F}_{12}\mathsf{F}_{22}^{-1}\mathsf{F}_{21}) = \chi^2_{\min} + \frac{\delta_1^2}{\mathsf{C}_{11}}, \tag{10.12}$$

where the last step is the block-inversion (Schur complement) formula

$$(\mathsf{F}^{-1})_{11} = (\mathsf{F}_{11} - \mathsf{F}_{12}\mathsf{F}_{22}^{-1}\mathsf{F}_{21})^{-1}.$$

Hence the profile rises by one unit exactly at $\delta_1 = \pm\sqrt{\mathsf{C}_{11}}$: *the* 68.3% *interval for one parameter, with all others re-optimized, is* $\Delta\chi^2 = 1$.

Marginal versus conditional errors. If instead you vary θ_1 with the others *fixed* at $\hat{\boldsymbol{\theta}}$, χ^2 rises as $\mathsf{F}_{11}\delta_1^2$, giving the conditional error $1/\sqrt{\mathsf{F}_{11}} \le \sqrt{\mathsf{C}_{11}}$. The two coincide only without correlations. In the worked example the conditional error on m_c is 1.7 MeV, the marginal one 21 MeV: quoting the conditional error would be a factor twelve too optimistic. Figure 2 shows the two readings on Toy 2: the marginal error is the shadow of the ellipse, the conditional error the slice through its centre.

Joint regions. For k parameters jointly, $\Delta\chi^2$ follows a χ^2_k distribution; the contour $\chi^2 = \chi^2_{\min} + \Delta$ contains the true values with probability:

k	1	2	3	4
68.27%	1.00	2.30	3.53	4.72
95.45%	4.00	6.18	8.02	9.72

10.4.4 Correlations and degeneracies

The correlation matrix $\rho_{ab} = \mathsf{C}_{ab}/\sqrt{\mathsf{C}_{aa}\mathsf{C}_{bb}}$ is the first thing to look at after a fit. $|\rho| \gtrsim 0.95$ signals a near-degeneracy. A cleaner diagnosis uses the eigen-decomposition of the (dimensionless) Fisher matrix,

$$\tilde{\mathsf{F}} = \mathsf{D}_\theta \mathsf{F} \mathsf{D}_\theta = \sum_k \lambda_k \, \boldsymbol{u}_k \boldsymbol{u}_k^T, \qquad \mathsf{D}_\theta = \mathrm{diag}(\hat{\theta}_a), \tag{10.13}$$

in relative variables $\delta\theta_a/\hat{\theta}_a$. The combination $\boldsymbol{u}_k^T(\delta\theta/\hat{\theta})$ is determined to $\pm 1/\sqrt{\lambda_k}$. Small λ_k = a direction the data barely constrain; the components of $\boldsymbol{u}_k$ tell you which parameters move together. Typical degeneracies in quarkonium fits:

- m_Q and c in Models A/B: $M = 2m_Q + c + \mathcal{E}(m_Q)$, and $\partial\mathcal{E}/\partial m_Q = -\langle T\rangle/m_Q$ (Feynman–Hellmann on $p^2/2m_r$) varies only mildly between states, so a change of m_Q is almost compensated by c. Worked example: $\rho(m_c, c) = -0.987$.
- α_s and b: both shape the potential at $r \sim 0.3$–0.6 fm, where the low states live.
- In GI: γ_k and σ_{12} enter the Coulomb-generated terms only through τ_k (§10.9); the ϵ_x with the smearing parameters.

Remedies: fix one parameter of the degenerate pair (and say so), add data that break the degeneracy (decay widths are sensitive to $|R(0)|^2$, i.e. to the short-distance potential), or add a prior (§10.6).

10.5 Propagating to predictions

Result 10.3 (Linear error propagation)**.** *Let $P(\boldsymbol{\theta})$ be a prediction (a mass of a state not in the fit, a splitting, a width) and $\boldsymbol{g} = \nabla_\theta P(\hat{\boldsymbol{\theta}})$. To first order $P(\boldsymbol{\theta}) - P(\hat{\boldsymbol{\theta}}) = \boldsymbol{g}^T(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})$, so*

$$\sigma^2(P) = \mathbb{E}[\boldsymbol{g}^T \boldsymbol{\delta}\boldsymbol{\delta}^T \boldsymbol{g}] = \boldsymbol{g}^T \mathsf{C} \boldsymbol{g}, \qquad \mathrm{Cov}(P, Q) = \boldsymbol{g}_P^T \mathsf{C} \boldsymbol{g}_Q. \tag{10.14}$$

- The gradients come from (7.34) (masses) or (7.35) (wave-function observables), or from central differences.
- Correlations matter for differences: $\sigma^2(P - Q) = \sigma^2(P) + \sigma^2(Q) - 2\,\mathrm{Cov}(P, Q)$. Always propagate the splitting as one quantity rather than combining the two individual errors.
- For a *predicted mass* to be compared with experiment, the total error is $\sqrt{\boldsymbol{g}^T \mathsf{C} \boldsymbol{g} + \sigma_{\mathrm{th}}^2}$: the theory floor applies to new states as well.
- The same formula transforms covariances under reparametrization: with $\boldsymbol{\phi}(\boldsymbol{\theta})$ and $\mathsf{K} = \partial\boldsymbol{\phi}/\partial\boldsymbol{\theta}$, $\mathsf{C}_\phi = \mathsf{K}\mathsf{C}\mathsf{K}^T$.
- If P is strongly non-linear over the 1σ region (check: compare $P(\hat{\boldsymbol{\theta}} \pm \sqrt{\mathsf{C}_{aa}}\hat{\boldsymbol{e}}_a)$ with the linear prediction), use the profile likelihood or sampling methods below.

In plain words

Linearize the prediction and apply rule (R4) of §9.2. The covariance terms are not optional: in Toy 2 the prediction at $x = 3$ has error 1.53 with them and 2.31 without (§9.6).

10.6 Beyond the quadratic approximation

10.6.1 Profile likelihood for a derived quantity

Define

$$\chi^2_{\mathrm{prof}}(\psi) = \min_{\boldsymbol{\theta}:\ P(\boldsymbol{\theta})=\psi} \chi^2(\boldsymbol{\theta}), \qquad 68\%\text{ interval: } \{\psi :\ \chi^2_{\mathrm{prof}}(\psi) \le \chi^2_{\min} + 1\}. \tag{10.15}$$

Check against (10.14). In the quadratic/linear approximation minimize $\boldsymbol{\delta}^T \mathsf{F} \boldsymbol{\delta}$ subject to $\boldsymbol{g}^T\boldsymbol{\delta} = \Delta\psi$ with a Lagrange multiplier: $\mathsf{F}\boldsymbol{\delta} = \lambda \boldsymbol{g} \Rightarrow \boldsymbol{\delta} = \lambda \mathsf{C}\boldsymbol{g}$; the constraint gives $\lambda = \Delta\psi/\boldsymbol{g}^T\mathsf{C}\boldsymbol{g}$; then $\Delta\chi^2 = \boldsymbol{\delta}^T\mathsf{F}\boldsymbol{\delta} = \lambda^2 \boldsymbol{g}^T\mathsf{C}\boldsymbol{g} = \Delta\psi^2/\boldsymbol{g}^T\mathsf{C}\boldsymbol{g}$. So $\Delta\chi^2 = 1$ at $\Delta\psi = \pm\sqrt{\boldsymbol{g}^T\mathsf{C}\boldsymbol{g}}$, as it should be. Beyond that approximation the profile gives asymmetric intervals and is reparametrization invariant. *Implementation:* for a parameter, fix it on a grid and refit the others; for a derived quantity, minimize $\chi^2 + (P(\boldsymbol{\theta}) - \psi)^2/\epsilon^2$ with small ϵ on a grid of ψ.

10.6.2 Parametric bootstrap

1. From the best fit, generate pseudo-data $\boldsymbol{y}^{*(b)} = \boldsymbol{\mu}(\hat{\boldsymbol{\theta}}) + \boldsymbol{\varepsilon}^{*(b)}$ with $\boldsymbol{\varepsilon}^{*(b)} \sim \mathcal{N}(0, \mathsf{V})$, $b = 1, \ldots, B$ ($B \sim 200$–1000).
2. Refit each replica (start at $\hat{\boldsymbol{\theta}}$: fast) $\to \hat{\boldsymbol{\theta}}^{*(b)}$; compute any prediction $P(\hat{\boldsymbol{\theta}}^{*(b)})$.
3. The spread of $\{\hat{\boldsymbol{\theta}}^{*(b)}\}$ estimates the sampling distribution of $\hat{\boldsymbol{\theta}}$: standard deviations, percentile intervals (16th and 84th percentiles for 68%), and the bias $\overline{\boldsymbol{\theta}^*} - \hat{\boldsymbol{\theta}}$.

Why it works: if the fitted model with its error model were the truth, this is literally the distribution of fit results over repeated experiments; it includes all non-linearities of the fit. The

non-parametric bootstrap (resampling the data points) is not appropriate here: the states are not exchangeable draws from one distribution.

In plain words

The bootstrap reruns the experiment on the computer, with the fitted model playing the truth; the scatter of the refits is the error, non-linearities included (§9.11).

10.6.3 Bayesian inference and Markov chain Monte Carlo

Posterior. Bayes' theorem $p(\boldsymbol{\theta}|\boldsymbol{y}) \propto L(\boldsymbol{\theta})\,\pi(\boldsymbol{\theta})$; with $L \propto \mathrm{e}^{-\chi^2/2}$,

$$-2\ln p(\boldsymbol{\theta}|\boldsymbol{y}) = \chi^2(\boldsymbol{\theta}) - 2\ln\pi(\boldsymbol{\theta}) + \text{const.} \tag{10.16}$$

Priors. Flat in $\ln\theta$ for positive scale parameters (m_Q, b, σ, μ_s); a Gaussian prior $\pi \propto \exp[-\frac{1}{2}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)^T\Lambda(\boldsymbol{\theta}-\boldsymbol{\theta}_0)]$ to encode external information, which adds $(\boldsymbol{\theta}-\boldsymbol{\theta}_0)^T\Lambda(\boldsymbol{\theta}-\boldsymbol{\theta}_0)$ to χ^2. The posterior covariance in the Gaussian approximation is then $(\mathsf{F}+\Lambda)^{-1}$ (the Hessians add): a prior is the Bayesian version of a penalty term.

In plain words

The Markov chain is a random walker that spends time in each region in proportion to its posterior probability. The two-state example of §9.11 contains the whole proof below in miniature: the probability flow from A to B equals the flow back.

Metropolis–Hastings. To draw samples from a density $\pi(\boldsymbol{\theta})$ known up to normalization:

1. at the current point $\boldsymbol{\theta}$, propose $\boldsymbol{\theta}' = \boldsymbol{\theta} + \boldsymbol{\zeta}$ with $\boldsymbol{\zeta}$ from a symmetric distribution q (e.g. Gaussian);
2. accept with probability $a(\boldsymbol{\theta},\boldsymbol{\theta}') = \min\big(1, \pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta})\big)$; otherwise stay at $\boldsymbol{\theta}$ (and record $\boldsymbol{\theta}$ again).

Proof that π is the stationary distribution. The transition density for $\boldsymbol{\theta}' \neq \boldsymbol{\theta}$ is $T(\boldsymbol{\theta}\to\boldsymbol{\theta}') = q(\boldsymbol{\theta}'-\boldsymbol{\theta})\,a(\boldsymbol{\theta},\boldsymbol{\theta}')$. Then

$$\pi(\boldsymbol{\theta})T(\boldsymbol{\theta}\to\boldsymbol{\theta}') = q(\boldsymbol{\theta}'-\boldsymbol{\theta})\min\big(\pi(\boldsymbol{\theta}),\pi(\boldsymbol{\theta}')\big),$$

which is symmetric under $\boldsymbol{\theta}\leftrightarrow\boldsymbol{\theta}'$ because q is symmetric. This is *detailed balance*, $\pi(\boldsymbol{\theta})T(\boldsymbol{\theta}\to\boldsymbol{\theta}') = \pi(\boldsymbol{\theta}')T(\boldsymbol{\theta}'\to\boldsymbol{\theta})$. Integrating over $\boldsymbol{\theta}$: $\int \pi(\boldsymbol{\theta})T(\boldsymbol{\theta}\to\boldsymbol{\theta}')\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}')\int T(\boldsymbol{\theta}'\to\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}')$ (the total probability of going somewhere from $\boldsymbol{\theta}'$, including staying, is one). So if the chain is distributed as π, it stays so; ergodicity (any region reachable) makes it converge to π from any start.

Tuning. Use a Gaussian proposal with covariance $\frac{2.38^2}{p}\mathsf{C}$, with C from the Levenberg–Marquardt fit. For Gaussian targets this scaling is known to be close to optimal; acceptance rates of 0.2–0.4 are typical (worked example: 0.30).

Convergence: Gelman–Rubin $\hat{R}$. Run $m \geq 4$ chains from over-dispersed starting points (e.g. $\hat{\boldsymbol{\theta}} \pm 3\sigma$); discard a burn-in; keep n samples per chain. For each parameter let $\bar{\theta}_j$ and s_j^2 be the mean and variance of chain j, and

$$W = \frac{1}{m}\sum_j s_j^2, \qquad B = \frac{n}{m-1}\sum_j(\bar{\theta}_j - \bar{\theta})^2, \qquad \hat{V} = \frac{n-1}{n}W + \frac{1}{n}B, \qquad \hat{R} = \sqrt{\hat{V}/W}. \tag{10.17}$$

W underestimates the posterior variance while the chains have not explored everything; $\hat{V}$ overestimates it when the starts are over-dispersed; both converge to the true variance, so $\hat{R} \to 1$. Require $\hat{R} < 1.01$.

Autocorrelation and effective sample size. Successive samples are correlated. For a stationary series with variance σ^2 and autocorrelation ρ_k,

$$\mathrm{Var}\left(\frac{1}{n}\sum_{t=1}^{n} x_t\right) = \frac{1}{n^2}\sum_{s,t}\mathrm{Cov}(x_s, x_t) = \frac{\sigma^2}{n}\Big[1 + 2\sum_{k\geq 1}\Big(1 - \frac{k}{n}\Big)\rho_k\Big] \simeq \frac{\sigma^2 \tau_{\mathrm{int}}}{n},$$

with $\tau_{\mathrm{int}} = 1 + 2\sum_k \rho_k$. The chain is worth $n_{\mathrm{eff}} = n/\tau_{\mathrm{int}}$ independent samples. Aim for $n_{\mathrm{eff}} \gtrsim 10^3$ for 68% intervals.

Posterior predictive. For every sample compute the predictions $P(\boldsymbol{\theta})$; their distribution (plus σ_{th} for masses) gives credible intervals for new states with all correlations and non-linearities included.

Cost. One posterior evaluation = one diagonalization per channel (∼ms for $N \leq 50$), so 10^5 samples take minutes for Models A/B. For GI, precompute everything that does not depend on the parameter being varied.

10.7 Model comparison and validation

10.7.1 Goodness of fit

With σ_{th} fixed independently of the fit, χ^2_{min} follows (approximately) a χ^2_ν distribution, $\nu = N - p$, and the p-value is $P(\chi^2_\nu \geq \chi^2_{\mathrm{min}})$. With σ_{th} tuned to $\chi^2/\nu = 1$ the p-value is meaningless (§10.2).

10.7.2 Nested models: likelihood-ratio test

If model B contains model A as a special case with Δp extra parameters (e.g. c free vs. $c = 0$; screened vs. linear with $\mu_s \to 0$), then under A, $\Delta\chi^2 = \chi^2_{\mathrm{min},A} - \chi^2_{\mathrm{min},B}$ follows asymptotically $\chi^2_{\Delta p}$ (Wilks' theorem). Use the *same* σ_{th} for both fits.

Boundary caveat. For $\mu_s \geq 0$ the null value $\mu_s = 0$ is on the boundary of the allowed region. Then half of the replicas would want $\mu_s < 0$ and get $\Delta\chi^2 = 0$, and the distribution is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (Chernoff): the p-value is half the naive one.

10.7.3 Information criteria

$$\mathrm{AIC} = \chi^2_{\mathrm{min}} + 2p, \qquad \mathrm{BIC} = \chi^2_{\mathrm{min}} + p \ln N. \tag{10.18}$$

Lower is better; differences below about 2 are not significant.

In plain words

More parameters always fit better; AIC, BIC and leave-one-out ask whether they also *predict* better. Toy 2 with a line and with a parabola through three points shows all three at work by hand (§9.12).

Where $2p$ comes from (AIC). Use whitened variables ($\mathsf{W}=1$) and a linear model $\boldsymbol{y}=\mathsf{J}\boldsymbol{\theta}_0+\boldsymbol{\varepsilon}$. The fit residual is $(1-\mathsf{P})\boldsymbol{\varepsilon}$, so $\mathbb{E}[\chi^2_{\rm fit}]=N-p$ (§10.2). For an *independent* new data set $\boldsymbol{y}'=\mathsf{J}\boldsymbol{\theta}_0+\boldsymbol{\varepsilon}'$, the prediction error is $\boldsymbol{y}'-\mathsf{J}\hat{\boldsymbol{\theta}}=\boldsymbol{\varepsilon}'-\mathsf{P}\boldsymbol{\varepsilon}$, so $\mathbb{E}[\chi^2_{\rm new}]=N+\operatorname{tr}\mathsf{P}=N+p$. The fit χ^2 is optimistic by $2p$ on average; AIC corrects for this and estimates the out-of-sample χ^2.

Where $p\ln N$ comes from (BIC). The Bayesian evidence is $Z=\int L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})\,\mathrm{d}^p\theta$. Laplace's method (Gaussian approximation around the peak) gives

$$Z\simeq L(\hat{\boldsymbol{\theta}})\,\pi(\hat{\boldsymbol{\theta}})\,(2\pi)^{p/2}(\det\mathsf{F})^{-1/2}.$$

Each datum adds information, so $\mathsf{F}\propto N$ and $\ln\det\mathsf{F}=p\ln N+O(1)$. Hence $-2\ln Z=\chi^2_{\rm min}+p\ln N+O(1)$: BIC approximates $-2\ln Z$, and ΔBIC approximates $-2\ln$ of the Bayes factor.

10.7.4 Leave-one-out cross-validation

Refit with state i removed and predict it; the pull $z_i=(y_i-\mu_i^{(-i)})/\sqrt{s_i^2+\sigma^2_{{\rm pred},i}}$ tests whether the model *predicts* rather than merely fits. No refits are needed to first order:

Result 10.4 (Leave-one-out from one fit)**.** *Let $h_i=w_i\,\mathsf{J}_i^T\mathsf{C}\mathsf{J}_i$ (the i-th diagonal element of the weighted hat matrix $\mathsf{W}^{1/2}\mathsf{J}\mathsf{C}\mathsf{J}^T\mathsf{W}^{1/2}$; J_i is row i of J and $w_i=1/s_i^2$). Then, to linear order,*

$$y_i-\mu_i^{(-i)}=\frac{\Delta_i}{1-h_i},\qquad z_i=\frac{\Delta_i}{s_i\sqrt{1-h_i}},\qquad \sum_i h_i=p. \tag{10.19}$$

Derivation. Removing datum i changes $\mathsf{F}\to\mathsf{F}-w_i\mathsf{J}_i\mathsf{J}_i^T$ and $\mathsf{J}^T\mathsf{W}\boldsymbol{y}\to\mathsf{J}^T\mathsf{W}\boldsymbol{y}-w_i\mathsf{J}_i y_i$. The Sherman–Morrison formula

$$\left(\mathsf{F}-w\boldsymbol{v}\boldsymbol{v}^T\right)^{-1}=\mathsf{C}+\frac{w\,\mathsf{C}\boldsymbol{v}\boldsymbol{v}^T\mathsf{C}}{1-w\boldsymbol{v}^T\mathsf{C}\boldsymbol{v}}$$

(verify by multiplying with $\mathsf{F}-w\boldsymbol{v}\boldsymbol{v}^T$) with $\boldsymbol{v}=\mathsf{J}_i$ gives $\mathsf{J}_i^T(\mathsf{F}-w_i\mathsf{J}_i\mathsf{J}_i^T)^{-1}=\mathsf{J}_i^T\mathsf{C}/(1-h_i)$. The leave-one-out prediction (linearized model) is therefore

$$\mu_i^{(-i)}=\frac{\mathsf{J}_i^T\mathsf{C}(\mathsf{J}^T\mathsf{W}\boldsymbol{y}-w_i\mathsf{J}_i y_i)}{1-h_i}=\frac{\hat\mu_i-h_i y_i}{1-h_i}\quad\Rightarrow\quad y_i-\mu_i^{(-i)}=\frac{y_i-\hat\mu_i}{1-h_i}.$$

For the variance, $\sigma^2_{{\rm pred},i}=\mathsf{J}_i^T(\mathsf{F}-w_i\mathsf{J}_i\mathsf{J}_i^T)^{-1}\mathsf{J}_i=\frac{\mathsf{J}_i^T\mathsf{C}\mathsf{J}_i}{1-h_i}=\frac{s_i^2h_i}{1-h_i}$, so $s_i^2+\sigma^2_{{\rm pred},i}=s_i^2/(1-h_i)$, giving z_i. Finally $\sum_i h_i=\operatorname{tr}(\mathsf{W}^{1/2}\mathsf{J}\mathsf{C}\mathsf{J}^T\mathsf{W}^{1/2})=\operatorname{tr}(\mathsf{C}\mathsf{F})=p$.

Interpretation. h_i is the *leverage*: how much datum i pulls the fit towards itself. $h_i\to1$ means the datum determines a parameter on its own (in the worked example $h=0.90$ for $\eta_c(1S)$, which essentially fixes the smearing σ through the hyperfine splitting). A state with $|z_i|\gtrsim2$ is one the model cannot predict from the others (χ_{c0} in the worked example). For the straight-line toy the formula is exact and can be checked by hand (§9.12): leverages $(\frac56,\frac13,\frac56)$, leave-one-out misses $(+1,-0.5,+1)$.

10.8 Error budget

Every number you quote should come with the following separate contributions.

In plain words

Each error source tells a different story (statistics of the data, your numerics, the model's limits, unfixed conventions, other models); keep them apart so a reader can see which one dominates (§9.13).

(1) Parametric (statistical) error. $\sigma_{\rm par}(P) = \sqrt{\boldsymbol{g}^T \mathsf{C} \boldsymbol{g}}$ (10.14), or from the bootstrap/posterior. It reflects how well the data constrain $\boldsymbol{\theta}$, given the error model including $\sigma_{\rm th}$.

(2) Numerical error. From nested basis doubling $n_{\max} \to 2n_{\max} - 1$ (Property 2 of §7.12): $\sigma_{\rm num} \simeq |P^{(2n-1)} - P^{(n)}|$. Repeat for the quadrature settings (GI) and the threshold η. If the sequence of nested results converges geometrically, $P_k = P_\infty + A\rho^k$, three terms determine the limit:

$$P_\infty = P_k - \frac{(P_{k+1} - P_k)^2}{P_{k+2} - 2P_{k+1} + P_k} \qquad \text{(Aitken)}. \tag{10.20}$$

Derivation: $P_{k+1} - P_k = A\rho^k(\rho - 1)$ and $P_{k+2} - 2P_{k+1} + P_k = A\rho^k(\rho - 1)^2$; their ratio squared over the second gives $A\rho^k$, which is subtracted from P_k. Requirement: $\sigma_{\rm num} \le 10^{-2} \min_i s_i$, so that numerics never move a fit. Note that wave-function observables converge more slowly than energies: the energy error is quadratic in the wave-function error (variational principle), while $|R(0)|^2$ is linear in it. In the worked example, doubling the basis moves the masses by $\le 6 \times 10^{-5}$ MeV but $|R_{1S}(0)|^2$ by 0.1%.

(3) Theory (truncation) error. What the Hamiltonian leaves out. The fitted $\sigma_{\rm th}$ is the empirical measure for masses. An a-priori estimate for an NR model is the first neglected relativistic correction to the kinetic energy, $-\sum_i p^4/8m_i^3$ ($= -p^4/4m^3$ for equal masses). Its GEM matrix elements are analytic: with $-\nabla^2\phi_n = (A_n - 4\nu_n^2 r^2)\phi_n$, $A_n = 2\nu_n(2l+3)$ (answer key C4),

$$\langle\phi_{n'}|p^4|\phi_n\rangle = \langle p^2\phi_{n'}|p^2\phi_n\rangle = N_{n'l}N_{nl}\Big[A_{n'}A_n\,\mathcal{M}_0 - 4(A_{n'}\nu_n^2 + A_n\nu_{n'}^2)\mathcal{M}_2 + 16\nu_{n'}^2\nu_n^2\,\mathcal{M}_4\Big], \tag{10.21}$$

with $\mathcal{M}_k = \int_0^\infty r^{2l+2+k} \mathrm{e}^{-(\nu_{n'}+\nu_n)r^2} \mathrm{d}r = \frac{\Gamma(l+\frac{3+k}{2})}{2(\nu_{n'}+\nu_n)^{l+(3+k)/2}}$. Only the *state-dependent* part of such a correction matters (a common shift is absorbed into m_Q or c). In the worked example the correction ranges from -36 MeV ($1P$) to -146 MeV ($\eta_c(1S)$), with an rms state-to-state spread of 39 MeV, while the fitted floor is $\sigma_{\rm th} = 8.3$ MeV. The refitted parameters absorb most of the correction, so a-priori estimates of this kind are conservative.

(4) Input and convention ambiguities. Choices that are not fixed by the model:

- M in annihilation widths: physical mass or $2m_Q$;
- the scale of α_s in QCD correction factors (vary the scale by a factor 2);
- smeared versus point contact interaction;
- the E1 recoil factor E_f/M_i and the finite-size form of the dipole operator;
- which states are in the fit.

Evaluate the prediction with each reasonable choice and quote the spread (half the envelope) as a separate error. Worked example: for $\Gamma_{ee}(J/\psi)$ at leading order, $M = M_{\rm exp}$ gives 11.6 keV and $M = 2m_c$ gives 12.9 keV. The factor $(1 - \frac{16\alpha_s}{3\pi})$ is 0.09 at the fitted potential coupling $\alpha_s = 0.53$ but 0.3–0.4 at $\alpha_s(m_c) \approx 0.35$–$0.40$: the ambiguity in this observable is far larger than the parametric error (± 1.1 keV). This is the reason ratios of widths (in which $|R(0)|^2$ and part of the corrections cancel) are more robust tests.

(5) Model dependence. The spread of a prediction across Models A–D, all fitted to the same data, is a measure of model dependence. It is not a statistical error. Report it separately; do not add it in quadrature.

Reporting. Quote $P = \bar{P} \pm \sigma_{\rm par} \pm \sigma_{\rm num} \pm \sigma_{\rm amb}$ (and the model spread separately). Add in quadrature only contributions that are independent and Gaussian-like. A useful template:

prediction	value	$\sigma_{\rm par}$	$\sigma_{\rm num}$	$\sigma_{\rm th}$ (masses)	$\sigma_{\rm amb}$	model spread
$M(1^3D_1)$	…	…	…	…	—	…
$\Gamma_{ee}(J/\psi)$	…	…	…	—	…	…

10.9 Optimizing the running coupling of the GI model

GI fixed $\alpha_s(Q^2) = \sum_{k=1}^{3} \alpha_k e^{-Q^2/4\gamma_k^2}$ with the values (5.7). Freeing it adds up to six parameters. Before doing that, find out *which part of the function $\alpha_s(Q)$ the spectrum can determine at all.* That question has a clean answer.

10.9.1 What the spectrum sees: the sensitivity kernel

Step 1 (potential from the coupling). The vector (one-gluon) potential is (§5)

$$V_V(r) = -\frac{4}{3}\int \frac{\mathrm{d}^3q}{(2\pi)^3} e^{i\boldsymbol{q}\cdot\boldsymbol{r}} \frac{4\pi\alpha_s(q)}{q^2}.$$

For a radial function $f(q)$ the angular integral $\int \mathrm{d}\Omega_q\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}} = 4\pi j_0(qr)$ gives

$$\int \frac{\mathrm{d}^3q}{(2\pi)^3} e^{i\boldsymbol{q}\cdot\boldsymbol{r}} f(q) = \frac{1}{2\pi^2}\int_0^\infty q^2 j_0(qr) f(q)\,\mathrm{d}q.$$

Step 2 (first-order shift). Change $\alpha_s(q) \to \alpha_s(q) + \delta\alpha_s(q)$. The level shift is $\delta M = \int_0^\infty r^2 R^2(r)\,\delta V_V(r)\,\mathrm{d}r$ (Feynman–Hellmann; the potential is central, so only the radial density enters). Insert Step 1 and exchange the integrals:

$$\delta M = -\frac{4}{3}\cdot\frac{4\pi}{2\pi^2}\int_0^\infty \mathrm{d}q\,\delta\alpha_s(q)\underbrace{\int_0^\infty r^2R^2(r) j_0(qr)\,\mathrm{d}r}_{\mathcal{F}(q)} = -\frac{8}{3\pi}\int_0^\infty \mathcal{F}(q)\,\delta\alpha_s(q)\,\mathrm{d}q. \tag{10.22}$$

$\mathcal{F}(q)$ is the *form factor* of the state (the Fourier transform of its density).

Sensitivity of a level to the coupling at scale Q

$$\delta M_i = \int K_i(Q)\,\delta\alpha_s(Q)\,\mathrm{d}\ln Q, \qquad K_i(Q) = -\frac{8}{3\pi}\,Q\,\mathcal{F}_i(Q). \tag{10.23}$$

Checks: $\mathcal{F}_i(0) = 1$ (normalization); for constant $\delta\alpha_s$, $\int_0^\infty \mathcal{F}\,\mathrm{d}q = \int r^2R^2\frac{\pi}{2r}\mathrm{d}r = \frac{\pi}{2}\langle r^{-1}\rangle$, so $\delta M_i = -\frac{4}{3}\langle r^{-1}\rangle_i \delta\alpha_s$, which is the Feynman–Hellmann result for $-\frac{4}{3}\alpha_s/r$.

GEM form factor. Expand $j_0(x) = \sum_k (-1)^k x^{2k}/(2k+1)!$, integrate term by term with (7.13), and use $(2k+1)! = 2^{2k+1}k!\,\Gamma(k+\frac{3}{2})/\sqrt{\pi}$:

$$\int_0^\infty r^{2l+2} e^{-ar^2} j_0(qr)\,\mathrm{d}r = \frac{\Gamma(l+\frac{3}{2})}{2a^{l+3/2}}\,{}_1F_1\Big(l+\tfrac{3}{2};\tfrac{3}{2};-\frac{q^2}{4a}\Big), \tag{10.24}$$

with $a = \nu_{n'} + \nu_n$. By Kummer's transformation ${}_1F_1(l+\frac{3}{2};\frac{3}{2};-x) = e^{-x}\,{}_1F_1(-l;\frac{3}{2};x)$, which is e^{-x} times a polynomial of degree l: e^{-x} for $l=0$ and $e^{-x}(1-\frac{2x}{3})$ for $l=1$. Then $\mathcal{F}(q) = \sum_{n'n} c_{n'}c_n N_{n'l}N_{nl} \times (\dots)$.

Hyperfine kernel. The contact term $\frac{2}{3m^2}\nabla^2 V_V\, \boldsymbol{S}_1\cdot\boldsymbol{S}_2$ has $\nabla^2 V_V = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}(-q^2)\tilde{V}_V = \int \frac{\mathrm{d}^3q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\frac{16\pi}{3}\alpha_s(q)$: no $1/q^2$ suppression. The same steps give, for the S-wave splitting,

$$\delta\Delta_{\mathrm{hf}} = \int K_{\mathrm{hf}}(Q)\,\delta\alpha_s(Q)\,\mathrm{d}\ln Q, \qquad K_{\mathrm{hf}}(Q) = \frac{16}{9\pi m^2}\,Q^3\mathcal{F}(Q)\ \big[\times\,\mathrm{e}^{-Q^2/4\sigma^2}\ \text{if smeared}\big]. \tag{10.25}$$

Numbers (worked-example wave functions). The 16th, 50th and 84th percentiles of $|K_i|$ in $\ln Q$ (GeV):

state	Q_{16}	Q_{50}	Q_{84}		Q_{16}	Q_{50}	Q_{84}
$\eta_c(1S)$	0.22	0.76	1.74	$\eta_c(2S)$	0.11	0.38	1.87
J/ψ	0.19	0.64	1.45	$\psi(2S)$	0.11	0.37	1.72
$1P$ states	0.10	0.34	0.72	hyperfine $1S$ (smeared)	0.76	1.33	2.10
				hyperfine $1S$ (point)	1.13	2.46	7.98

The static masses probe α_s mostly below 2 GeV; the hyperfine splitting probes higher Q, and the smearing cuts it off at $Q\sim 2\sigma$. (Both sum rules above were checked numerically.)

How many features of $\alpha_s(Q)$ are determined? Let $\delta\alpha_s$ be piecewise constant in bins of Q. The Jacobian of the eight masses with respect to the bin values is $\int_{\mathrm{bin}} K_i\,\mathrm{d}\ln Q$ (static plus hyperfine kernels). Free one bin at a time on top of $(m_c, \alpha_s, b, \sigma)$, all fitted, with the worked-example errors. The marginal errors on $\delta\alpha_s$ in that bin are:

bin (GeV)	0–0.25	0.25–0.5	0.5–1	1–2	2–4	>4
$\sigma(\delta\alpha_s)$	1.1	0.54	0.12	0.05	0.13	0.9

Conclusion: the below-threshold charmonium spectrum determines the shape of $\alpha_s(Q)$ only for $0.5\lesssim Q\lesssim 4$ GeV, and only two or three independent features of it. Below 0.5 GeV and above 4 GeV the fitted coupling is whatever the parametrization and priors make it. (These numbers use the NR kernels of Model A as a proxy; the GI factors and the fine-structure terms change the details but not the conclusion. In your GI fit compute the Jacobian directly, §7.11.)

10.9.2 The infrared degeneracy: α_s at small Q versus the string tension

The Fourier transform of a linear potential is $\int \mathrm{d}^3r\,\mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\,br = -8\pi b/q^4$ (as a distribution; obtain it from $\nabla^2(r) = 2/r$ and $\widetilde{1/r} = 4\pi/q^2$). Compare it with the one-gluon form $-\frac{4}{3}\cdot\frac{4\pi\alpha_s(q)}{q^2}$: they are *equal* if

$$\alpha_s(q) = \frac{3b}{2q^2}. \tag{10.26}$$

An infrared-enhanced vector coupling *is* a confining potential (this is how Richardson's potential confines). Consequently, freedom of $\alpha_s(Q)$ at $Q\lesssim 0.5$ GeV trades against b and against constant shifts (c, m_Q): this is the large error of the lowest bins above. In GI the two are separated by convention: α_s freezes at $\alpha_s(0) = \sum_k \alpha_k$ and confinement sits in the scalar $\tilde{S}$. The degeneracy is only partly broken by the fine structure, because vector and scalar exchange give different spin–orbit terms (§3.6). Keep $\alpha_s(0)$ fixed or under a prior when you free the shape.

10.9.3 Parametrization, constraints and label switching

- *Positivity*: $\alpha_k = \mathrm{e}^{a_k}$ guarantees $\alpha_s(Q) > 0$ for all Q.

- *Label switching*: $\alpha_s(Q)$ is invariant under any permutation of the pairs (α_k, γ_k), so χ^2 has $K!$ identical minima. A Markov chain hops between them and the marginal posterior of "α_1" is meaningless. Impose the ordering $\gamma_1 < \gamma_2 < \cdots < \gamma_K$ through the reparametrization

$$\gamma_1 = \mathrm{e}^{g_1}, \qquad \gamma_k = \gamma_{k-1}(1 + \mathrm{e}^{g_k}) \quad (k \geq 2),$$

  with unconstrained g_k.

- *Merging terms*: if $\gamma_k \to \gamma_{k+1}$ the two Gaussians become one with $\alpha_k + \alpha_{k+1}$, and $\alpha_k - \alpha_{k+1}$ becomes a flat direction. A smallest Fisher eigenvalue along such a direction means K is too large.

- *Linearity in α_k*: at fixed (γ_k, σ_{12}) the Hamiltonian is exactly linear, $\mathbf{H} = \mathbf{H}_0 + \sum_k \alpha_k \mathbf{H}_k$, where $\mathbf{H}_k$ contains every Coulomb-generated term built from $-\frac{4}{3}\,\mathrm{erf}(\tau_k r)/r$ alone. Precompute $\mathbf{H}_k$; then $\partial\mathcal{E}/\partial\alpha_k = \boldsymbol{c}^T \mathbf{H}_k \boldsymbol{c}$, and a new $\{\alpha_k\}$ costs only a sum of matrices.

10.9.4 Penalties = priors

The total objective is

$$\chi^2_{\mathrm{tot}} = \chi^2_{\mathrm{data}} + \chi^2_{\mathrm{pert}} + \chi^2_{\mathrm{IR}} + \chi^2_{\mathrm{smooth}}. \tag{10.27}$$

- *Perturbative matching*: $\chi^2_{\mathrm{pert}} = \sum_j \left[\alpha_s^{\mathrm{GI}}(Q_j) - \alpha_s^{\mathrm{pert}}(Q_j)\right]^2/\delta_j^2$ on a logarithmic grid $Q_j \in [2, 10]\,\mathrm{GeV}$. Here α_s^{pert} is the two-loop running (5.1) from $\alpha_s(M_Z) \approx 0.118$ with flavour thresholds, and δ_j is half the spread under the scale variation $\mu \in [Q/2, 2Q]$ (or a fixed 5–10%). Remember that a Gaussian sum falls off faster than any logarithm, so the matching can only hold up to $Q \sim 2\gamma_K$.

- *Infrared value*: $\chi^2_{\mathrm{IR}} = (\sum_k \alpha_k - \alpha_0)^2/\delta_0^2$, e.g. $\alpha_0 = 0.6 \pm 0.15$, or fix $\sum_k \alpha_k$.

- *Smoothness* (Tikhonov): $\chi^2_{\mathrm{smooth}} = \lambda \int \left[\mathrm{d}^2\alpha_s/\mathrm{d}(\ln Q)^2\right]^2 \mathrm{d}\ln Q$, discretized. Choose λ by minimizing the leave-one-out $\sum_i z_i^2$ (10.19).

Each penalty is $-2\ln$ of a Gaussian prior (§10.6). The parameter covariance becomes $\mathsf{C} = (\mathsf{F}_{\mathrm{data}} + \mathsf{F}_{\mathrm{pert}} + \ldots)^{-1}$, with $\mathsf{F}_{\mathrm{pen}} = \mathsf{J}_{\mathrm{pen}}^T \mathsf{W}_{\mathrm{pen}} \mathsf{J}_{\mathrm{pen}}$ built in the same way from the penalty "data".

10.9.5 Derivatives for the Jacobian

Write $g_\tau(r) = \mathrm{erf}(\tau r)/r$, so $\tilde{G} = -\frac{4}{3}\sum_k \alpha_k g_{\tau_k}$, and the GI spin terms use the building blocks $\nabla^2 g_\tau$ (contact), $g_\tau'/r - g_\tau''$ (tensor) and g_τ'/r (spin–orbit) (§4.3). Differentiating with respect to τ turns every block into a Gaussian:

$$\partial_\tau g_\tau = \frac{2}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2} \equiv h_\tau, \qquad \partial_\tau \frac{g_\tau'}{r} = \frac{h_\tau'}{r} = -\frac{4\tau^2}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2},$$

$$\partial_\tau \nabla^2 g_\tau = \nabla^2 h_\tau = \frac{2}{\sqrt{\pi}}(4\tau^4 r^2 - 6\tau^2)\mathrm{e}^{-\tau^2 r^2}, \quad \partial_\tau\Big(\frac{g_\tau'}{r} - g_\tau''\Big) = \frac{h_\tau'}{r} - h_\tau'' = -\frac{8\tau^4 r^2}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}. \tag{10.28}$$

Derivation: $\partial_\tau \,\mathrm{erf}(\tau r) = \frac{2r}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}$ gives h_τ; the derivatives commute with ∂_r; then $h' = -2\tau^2 r h$, $h'' = (4\tau^4 r^2 - 2\tau^2)h$, and combine. Check the third line against $\nabla^2 g_\tau = -\frac{4\tau^3}{\sqrt{\pi}}\mathrm{e}^{-\tau^2 r^2}$ differentiated directly. All of these have analytic Gaussian matrix elements: $\int r^{2l+2+2j}\mathrm{e}^{-(a+\tau^2)r^2}\mathrm{d}r$. The chain rule through $\tau_k = \gamma_k\sigma_{12}/\sqrt{\gamma_k^2 + \sigma_{12}^2}$ needs

$$\frac{\partial\tau_k}{\partial\gamma_k} = \frac{\sigma_{12}^3}{(\gamma_k^2 + \sigma_{12}^2)^{3/2}}, \qquad \frac{\partial\tau_k}{\partial\sigma_{12}} = \frac{\gamma_k^3}{(\gamma_k^2 + \sigma_{12}^2)^{3/2}}. \tag{10.29}$$

Built-in degeneracy: the Coulomb-generated terms depend on γ_k and σ_{12} only through τ_k. Only the smearing of the confining part $\tilde{S}$ (and the mass dependence of σ_{12}) separates them, so expect strong γ_k–σ_0 correlations.

10.9.6 Uncertainty band for the fitted coupling

With $\boldsymbol{\theta}_\alpha = (\alpha_1, \gamma_1, \dots)$ and its covariance block C_α, (10.14) gives

$$\sigma^2(\alpha_s(Q)) = \boldsymbol{g}(Q)^T \mathsf{C}_\alpha \boldsymbol{g}(Q), \qquad \frac{\partial \alpha_s}{\partial \alpha_k} = \mathrm{e}^{-Q^2/4\gamma_k^2}, \qquad \frac{\partial \alpha_s}{\partial \gamma_k} = \frac{\alpha_k Q^2}{2\gamma_k^3} \mathrm{e}^{-Q^2/4\gamma_k^2}. \tag{10.30}$$

Plot the band only over the window where the data constrain it (above), and mark where it is prior-driven.

10.9.7 Alternative: fit a few physical parameters, use Gaussians only numerically

Parametrize $\alpha_s(Q)$ with one or two physical parameters (an overall factor $\lambda\,\alpha_s^{\mathrm{GI}}(Q)$; or a frozen one-loop form $\alpha_s(Q) = \frac{4\pi}{\beta_0 \ln[(Q^2+m_g^2)/\Lambda^2]}$ with an infrared mass m_g). Fit those parameters. To keep the analytic GEM matrix elements, represent the resulting $\alpha_s(Q)$ by a least-squares Gaussian sum $\sum_k \alpha_k \mathrm{e}^{-Q^2/4\gamma_k^2}$ on the window $[0, Q_{\max}]$ (a linear least-squares problem in α_k for fixed γ_k). Identifiability and smoothness are then built in.

10.9.8 Recommended protocol

1. Reproduce published GI masses with the published α_s (code validation).
2. Fit the other GI parameters with α_s fixed: baseline χ^2 and σ_{th}.
3. Free the coupling in steps at fixed σ_{th}: (a) overall factor λ; (b) $K = 1$ (freezing value and one scale); (c) $K = 2$, $K = 3$ with penalties. Compare by $\Delta\chi^2$ (nested), AIC/BIC and leave-one-out pulls; inspect the Fisher eigenvalues at each step.
4. Report the fitted $\alpha_s(Q)$ with its band in 0.5–4 GeV and the dependence on the priors (refit with δ_j doubled and halved).

10.10 Worked example: Model A fitted to charmonium

Every method of this section, applied to the simplest realistic case. You can reproduce all numbers with the formulas of these notes.

Setup. BGS-type Model A for $c\bar{c}$ (the four parameters of the NR column in §12.3):

$$H = 2m_c + \frac{p^2}{m_c} - \frac{4\alpha_s}{3r} + br + \frac{32\pi\alpha_s}{9m_c^2}\Big(\frac{\sigma}{\sqrt{\pi}}\Big)^3 \mathrm{e}^{-\sigma^2 r^2}\, \boldsymbol{S}_1\cdot\boldsymbol{S}_2$$

diagonalized per (L, S). The spin–orbit and tensor terms are added at first order with the $1P$ eigenvector, $\Delta M = \langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle a + \langle S_{12}\rangle t$, $a = \frac{1}{m_c^2}\langle \frac{2\alpha_s}{r^3} - \frac{b}{2r}\rangle$, $t = \frac{\alpha_s}{3m_c^2}\langle r^{-3}\rangle$, with $(\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle, \langle S_{12}\rangle) = (1, -\frac{2}{5}), (-1, 2), (-2, -4)$ for $J = 2, 1, 0$ (§3.6). Note that r^{-3} cannot be put into H non-perturbatively: for $^3P_0$ the combination is attractive $\propto -1/r^3$, and H is unbounded below. Parameters $\boldsymbol{\theta} = (m_c, \alpha_s, b, \sigma)$, $c = 0$. Data: Table 2 ($N = 8$, $\nu = 4$). Basis: $n_{\max} = 30$, $r_1 = 0.02$, $r_{\max} = 30\,\mathrm{GeV}^{-1}$ in every channel, fixed during the fit. Minimizer: Levenberg–Marquardt with a central-difference Jacobian, checked against Feynman–Hellmann.

Theory floor. Iterating the condition $\chi^2_{\min}/\nu = 1$ gives $\sigma_{\mathrm{th}} = 8.3\,\mathrm{MeV}$. With experimental errors only, $\chi^2 = 10\,011$ and the fit is dictated by J/ψ and $\psi(2S)$ (§10.2).

Best fit ($\chi^2 = 4.00$, $\nu = 4$):

	m_c (GeV)	α_s	b (GeV^2)	σ (GeV)
LM fit, $\sqrt{\mathsf{C}_{aa}}$	1.4670 ± 0.0209	0.5339 ± 0.0242	0.1486 ± 0.0071	1.130 ± 0.107
conditional $1/\sqrt{\mathsf{F}_{aa}}$	0.0017	0.0038	0.0010	0.072
bootstrap ($B = 300$), std	0.0225	0.0266	0.0076	0.118
MCMC posterior mean	1.4648	0.5308	0.1492	1.148
MCMC posterior std	0.0203	0.0238	0.0070	0.112

The values recalled in §12.3 for BGS (1.4794, 0.5461, 0.1425, 1.0946) lie within one standard deviation of this fit, although BGS used a different data set.

Residuals ($y-\mu$, MeV): η_c +2.1, J/ψ +4.5, χ_{c0} −13.8, χ_{c1} +0.9, h_c +5.9, χ_{c2} −3.0, $\eta_c(2S)$ −0.4, $\psi(2S)$ +4.1. The largest is χ_{c0}, the state most sensitive to the first-order treatment of the $1/r^3$ terms.

Correlations and Fisher eigenvalues.

$$\rho = \begin{pmatrix} 1 & 0.939 & -0.969 & -0.466 \\ 0.939 & 1 & -0.839 & -0.626 \\ -0.969 & -0.839 & 1 & 0.363 \\ -0.466 & -0.626 & 0.363 & 1 \end{pmatrix} \quad (m_c, \alpha_s, b, \sigma).$$

The relative Fisher matrix has eigenvalues (94.5, 364, 4.19×10^3, 7.88×10^5), i.e. relative precisions $1/\sqrt{\lambda_k}$ = (10%, 5.2%, 1.5%, 0.11%). The best-determined combination is almost pure m_c (the overall mass scale, to 1.7 MeV). The worst is dominated by σ with admixtures of α_s and b: the hyperfine splitting fixes only the product of coupling and smeared density.

The m_c–c degeneracy. Freeing c (same $\sigma_{\rm th}$) gives $m_c = 1.26 \pm 0.13$ GeV, $c = 0.39 \pm 0.23$ GeV, $\rho(m_c, c) = -0.987$, a condition number of 1.3×10^5 for F, and $\chi^2 = 1.67$. Then $\Delta\chi^2 = 2.33$ for one extra parameter (Wilks $p = 0.13$); AIC = 12.00 → 11.67 and BIC = 12.32 → 12.07, differences well below 2. The data do not ask for c, and fixing $c = 0$ is what makes m_c measurable.

Profile of m_c versus the parabola. $\Delta\chi^2_{\rm prof}$ = 1.086, 1.064 at $\mp1\sigma$ and 4.39, 4.22 at $\mp2\sigma$ (parabola: 1 and 4). The quadratic approximation is good to about 10%, with a slight asymmetry.

Leave-one-out. Exact refits versus the one-fit formula (10.19) ($\sum_i h_i = 4.000 = p$):

	η_c	J/ψ	χ_{c0}	χ_{c1}	h_c	χ_{c2}	$\eta_c(2S)$	$\psi(2S)$
leverage h_i	0.895	0.666	0.231	0.182	0.259	0.596	0.530	0.643
$y_i - \mu_i^{(-i)}$ exact (MeV)	+19.4	+14.0	−18.1	+1.1	+7.9	−7.0	−0.8	+10.9
$\Delta_i/(1-h_i)$ (MeV)	+19.9	+13.6	−18.0	+1.1	+8.0	−7.5	−0.8	+11.6
pull z_i	+0.75	+0.96	−1.89	+0.12	+0.81	−0.54	−0.06	+0.77

The linear formula reproduces the refits to $\lesssim 0.7$ MeV. $\eta_c(1S)$ has the highest leverage (it alone fixes σ); χ_{c0} is the state least well predicted by the others.

Sampling methods. Bootstrap standard deviations are 7–10% larger than $\sqrt{\mathsf{C}_{aa}}$, with biases $\leq 0.1\sigma$. MCMC (4 chains × 6000 steps, flat priors, proposal $\frac{2.38^2}{4}\mathsf{C}$, 1000 burn-in): acceptance 0.30, $\hat{R} \leq 1.0005$, $\tau_{\rm int} \approx 11$–$14$, $n_{\rm eff} \approx 1500$–1800 per parameter. All three error estimates agree: this model is close to Gaussian, and C suffices. Expect larger differences in GI fits.

Predictions with errors (parametric errors from (10.14); for masses add $\sigma_{\rm th} = 8.3\,\text{MeV}$ in quadrature):

prediction	value	$\sigma_{\rm par}$	comment
$M(1^3D_1)$	3795.2 MeV	6.5	$\psi(3770)$ at ≈ 3773.7: 2.0σ below, with $\sigma_{\rm th}$
$M(2^3P_2)$	3986.1 MeV	10.0	$\chi_{c2}(3930)$ at ≈ 3922.5: prediction 64 MeV too high
$M(3^3S_1)$	4091.2 MeV	16.7	$\psi(4040)$ at ≈ 4039: prediction 52 MeV too high
$M(J/\psi) - M(\eta_c)$	110.4 MeV	9.9	exp. 112.8; $\rho(M_{J/\psi}, M_{\eta_c}) = 0.10$
$\lvert R_{1S}(0)\rvert^2$ (J/ψ)	$1.171\,\text{GeV}^3$	0.116	$\sigma_{\rm num} = 0.0014$
$\Gamma_{ee}(J/\psi)$, LO, $M = M_{\rm exp}$	11.6 keV	1.1	$M = 2m_c$: 12.9 ± 1.0 keV; exp. ≈ 5.5 keV

The states above threshold deviate by more than the errors, in the direction expected from coupled-channel effects. This is the reason they were not fitted. For Γ_{ee} the QCD correction and the convention for M dominate the error budget (§10.8).

Numerical and truncation errors. Nested doubling $n_{\max} = 30 \to 59$ moves the fitted masses by at most 6×10^{-5} MeV, but $\lvert R_{1S}(0)\rvert^2$ by 0.12% and Γ_{ee} by 0.013 keV. The a-priori relativistic estimate $-\langle p^4\rangle/4m_c^3$ is (MeV): η_c −146, J/ψ −74, 1^3P −36, h_c −41, $\eta_c(2S)$ −120, $\psi(2S)$ −90, with a state-dependent rms spread of 39 MeV, compared with the fitted $\sigma_{\rm th} = 8.3$ MeV (§10.8).

10.11 The complete workflow

1. **Code validation.** Unit tests of §12 (hydrogen, oscillator, reference numbers, $\partial M/\partial c = 1$, Jacobian vs. finite differences).
2. **Data.** States below threshold, PDG values with errors; decide on the theory floor (common or state-dependent).
3. **Basis.** At a nominal $\boldsymbol{\theta}_0$: optimize $\boldsymbol{\xi}$ per channel (§7.12); verify $\sigma_{\rm num} \le 10^{-2} \min s_i$ by nested doubling; freeze.
4. **Fit.** Global search (for many parameters), then Levenberg–Marquardt with the Feynman–Hellmann Jacobian; level tracking by maximum overlap; iterate $\sigma_{\rm th}$ to $\chi^2/\nu = 1$.
5. **Re-check the basis** at $\hat{\boldsymbol{\theta}}$; refit if anything moved by more than $\sim 0.1\sigma$.
6. **Uncertainties.** $\mathsf{C} = \mathsf{F}^{-1}$; correlation matrix; Fisher eigenvalues. If degenerate: fix, add data, or add a prior. Profile likelihood for the key parameters; bootstrap or MCMC if the profile is not parabolic.
7. **Validation.** Leave-one-out pulls; nested-model tests; AIC/BIC across models with a common $\sigma_{\rm th}$.
8. **Predictions.** $\boldsymbol{g}^T\mathsf{C}\boldsymbol{g}$ (or posterior predictive) $\oplus\ \sigma_{\rm th}$ for masses; ambiguity spreads for widths; nested doubling for $\sigma_{\rm num}$.
9. **Report.** Parameters with errors and correlations, $\sigma_{\rm th}$, the error budget table, the model spread, and the basis and quadrature settings used (so others can reproduce).

11 Implementation blueprint: the C++ program

This section is the design document for the program you will write: what it must do, how it is layered, which libraries it uses, the data structures, the function catalogue module by module, the algorithms in pseudocode, the tests, and the order in which to build it. It extends the code

you already have (`solve_gem`, `QuarkoniumSystem`, `solve_generalized_eigenproblem`) rather than replacing its ideas.

What has been verified, and how. Every library call and every short code fragment printed in this section was compiled and run, and the numerical algorithms were implemented once in a throw-away test harness (not given here, so that the implementation stays yours). On your machine (CachyOS, GCC 16.2.1, CMake 4.4.4, Eigen 5.0.1, Ceres 2.2.0, GoogleTest 1.18, SQLite 3.53, spdlog 1.17) the harness reproduces reference Tests 1–4 of §12.2 to all printed digits and the complete worked fit of §10.10 (Test 6: $\sigma_{\text{th}} = 8.34\,\text{MeV}$, $\hat{\boldsymbol{\theta}} = (1.4670, 0.5339, 0.1486, 1.130)$, all errors, correlations and leverages) in 0.10 s. The PDG reader of §11.10 was run against your `data/pdg-2026.0.sqlite`. Statements that were *not* tested are marked as such.

11.1 What the program must do

R1 Compute the spectrum of Models A–D for $c\bar{c}$, $b\bar{b}$ and $b\bar{c}$: all channels (L, S, J) up to a chosen $L_{\max}$, including the coupled channels (tensor $L = J \mp 1$; and ${}^1L_L$–${}^3L_L$ for $m_1 \neq m_2$).

R2 Return eigenvectors and the observables of §8.

R3 Optimize and freeze basis hyperparameters per channel; estimate σ_{num} by nested doubling.

R4 Read masses and errors from the PDG SQLite file; build the data covariance including common systematics.

R5 Fit the physics parameters (Levenberg–Marquardt with Feynman–Hellmann Jacobian), with the theory floor, parameter transformations, bounds and optional penalties (priors).

R6 Uncertainties: covariance, correlations, Fisher eigen-analysis, profiles, parametric bootstrap, Markov chain Monte Carlo with diagnostics.

R7 Validation: leave-one-out, AIC/BIC, likelihood-ratio tests.

R8 Predictions with full error budgets; machine-readable output (JSON) for plotting and for the report.

R9 Reproducibility: every run fully specified by a configuration file plus a random seed; results carry the configuration and the code version.

R10 Tests: unit tests, reference values (§12.2), golden values from your Python code and from Mathematica.

11.2 Architecture

The program is a stack of layers. Each layer uses only the layers below it, so each can be tested on its own.

One χ^2 evaluation, end to end. (i) The fitter hands an unconstrained vector $\boldsymbol{u}$ to the parameter layer, which maps it to physical $\boldsymbol{\theta}$ (exponentials for positive parameters, §10.3). (ii) For every channel needed by the data, the model assembles $\mathbf{H}(\boldsymbol{\theta})$ from cached parameter-independent matrices and the few matrices that depend non-linearly on a parameter (Algorithm 11.3). (iii) The solver diagonalizes (Algorithm 11.5) and labels the levels. (iv) The data layer picks the predicted masses (or splittings, widths) and forms whitened residuals z_i. (v) If a Jacobian is requested, the derivative layer returns $\partial z_i/\partial u_a$ by Feynman–Hellmann (Algorithm 11.7). Steps (ii)–(iii) cost $\approx 0.3\,\text{ms}$ for Model A with four channels of 30 functions (measured).

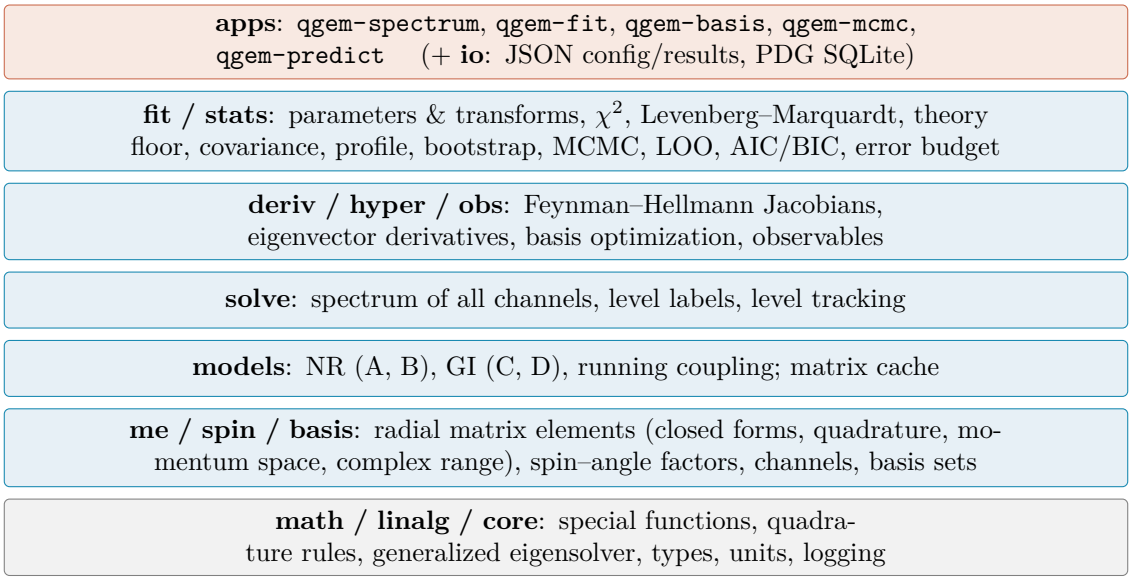


Figure 5: Layers of the program. Arrows are implicit: every layer calls only the layers below it.

11.3 Toolchain and libraries

component	on your machine	used for	CMake target / note
GCC	16.2.1	C++26 compiler (`std::expected`, `std::span`, `std::numbers`, special math)	needs CMake $\geq$ 3.30 for `CXX_STANDARD 26`
CMake	4.4.4	build	`cmake_minimum_required(VERSION 3.30)`
Eigen	**5.0.1**	dense linear algebra, eigensolvers	`Eigen3::Eigen`; *no version* in `find_package` (see below)
spdlog / fmt	1.17 / 12.2	logging, formatting	`spdlog::spdlog`
SQLite	3.53	PDG database	`SQLite3::SQLite3` (CMake $\geq$ 4.1; `SQLite::SQLite3` is deprecated)
GoogleTest	1.18	unit tests	`GTest::gtest_main`
Ceres Solver	2.2.0	optional: LM with bounds, covariance	`Ceres::ceres`
nlohmann-json	*not installed*	configuration, results, golden files	`nlohmann_json::nlohmann_json`; install with `pacman -S nlohmann-json`
Python 3 + numpy, matplotlib	—	plots, comparison with your `gem_solver.py`	outside the C++ build

- *Eigen version.* Your README says Eigen 3.4, but the installed version is 5.0.1. The calls used here (`SelfAdjointEigenSolver`, `GeneralizedSelfAdjointEigenSolver`, `LLT`, `LDLT`) behave identically in both (verified with 3.4.0 and 5.0.1). However `find_package(Eigen3 3.4 NO_MODULE)` *fails* with 5.0.1 (verified): the version check rejects a different major version. Write `find_package(Eigen3 REQUIRED NO_MODULE)` and update the README.
- *No Boost needed.* Everything special is either in `<cmath>` (`std::tgamma`, `std::lgamma`, `std::erf`, `std::erfc`; and in libstdc++ `std::sph_bessel`, verified) or short enough to write and

test yourself: erfcx (§11.10), Gauss–Legendre and generalized Gauss–Laguerre rules by the Golub–Welsch method (Algorithm 11.2). Note that clang's libc++ does not provide most C++17 special functions; on Linux clang uses libstdc++ by default, so this only matters if you switch standard libraries (not tested).

- *Optimizer.* Write your own Levenberg–Marquardt (Algorithm 11.9, ∼100 lines with Eigen); the harness version reproduces the worked fit. Ceres is a good cross-check and gives bounds and covariances for free; a verified minimal use with an analytic Jacobian is in §11.10. Ceres minimizes $\frac{1}{2}\sum r_i^2$, so $\chi^2 = 2\times$`final_cost`. NLopt (derivative-free and global optimizers) is not installed and not needed: basis optimization uses Nelder–Mead or the Feynman–Hellmann gradient, and the global stage of a physics fit uses multistart.
- *SIMD.* `std::simd` is not needed (Eigen vectorizes its kernels). For the record: `<simd>` with `std::simd::vec<float>` compiles with GCC 16.2 and `-std=c++26` on your machine.
- *Random numbers.* Use `std::mt19937_64` with an explicit seed; its output sequence is fixed by the standard. The *distributions* (`std::normal_distribution`, ...) are not: their algorithms are implementation-defined, so the same seed can give different Gaussian samples with another standard library. For bit-reproducible bootstrap/MCMC across compilers, generate Gaussians yourself from the engine (Box–Muller), or record the compiler with the results.

11.4 Repository layout and build

```
Quarkonium-System/
  CMakeLists.txt
  config/        run configurations (JSON): model, parameters, basis, data, seeds
  data/          pdg-2026.0.sqlite (gitignored)
  src/           every name below is a .hpp/.cpp pair unless marked
    core/        types (.hpp)  constants (.hpp)  log  error
    math/        special  quadrature
    linalg/      gep                          <- grows out of your linalg.{hpp,cpp}
    basis/       basis
    me/          radial  smeared  momentum  complex_range  gi
    spin/        channels  factors
    models/      model (.hpp)  nr_model  gi_model  alpha_s  cache
    solve/       spectrum                     <- grows out of your gem_solver.{hpp,cpp}
    obs/         observables  angular (6j, C_fi)
    deriv/       derivatives
    hyper/       basis_opt
    data/        pdg  dataset  assignment
    fit/         params  chi2  lm  floor  uncertainty  multistart
    stats/       bootstrap  mcmc  diagnostics  loo  compare
    report/      budget  json_io
    apps/        qgem_spectrum  qgem_fit  qgem_basis  qgem_mcmc  qgem_predict  (.cpp)
    generated/   qgem_kernels.hpp             <- written by the generator, committed (§13.5.10)
  tests/         test_*.cpp (GoogleTest), golden/*.json
                 test_generated_kernels.cpp   <- written by the generator, committed
  python/        plot_spectrum.py  plot_corner.py  compare_golden.py
  mathematica/   QGEMEngine.wl  generate_kernels.wls  QGEMKernels.cmake
                 derivations/D01..D07  (evaluated notebooks, §13.5.10)
                 your own notebooks, golden-value generators (§13)
  docs/
```

Build the code once as a static library `qgem` and link the apps and the tests against it. A minimal `CMakeLists.txt` (CMake 4.4.4, GCC 16.2.1, Eigen 5.0.1, GoogleTest on your machine):

```
cmake_minimum_required(VERSION 3.30)   # CXX_STANDARD 26 with GCC needs >= 3.30
project(QuarkoniumGEM VERSION 0.1 LANGUAGES CXX)
include(mathematica/QGEMKernels.cmake)   # option to regenerate the Mathematica-derived kernels
set(CMAKE_CXX_STANDARD 26)
set(CMAKE_CXX_STANDARD_REQUIRED ON)
set(CMAKE_CXX_EXTENSIONS OFF)
set(CMAKE_EXPORT_COMPILE_COMMANDS ON)  # for clangd / clang-tidy
if(NOT CMAKE_BUILD_TYPE)
  set(CMAKE_BUILD_TYPE Release)
```

```
endif()

find_package(Eigen3 REQUIRED NO_MODULE)   # no version: accepts 3.4 and 5.x
find_package(spdlog REQUIRED)
find_package(SQLite3 REQUIRED)            # target SQLite3::SQLite3 needs CMake >= 4.1
find_package(nlohmann_json 3.11 REQUIRED)
find_package(Ceres QUIET)                 # optional cross-check
find_package(GTest REQUIRED)

add_compile_options(-Wall -Wextra -Wpedantic)

file(GLOB_RECURSE QGEM_SOURCES CONFIGURE_DEPENDS src/*.cpp)
list(FILTER QGEM_SOURCES EXCLUDE REGEX "src/apps/")
add_library(qgem STATIC ${QGEM_SOURCES})
target_include_directories(qgem PUBLIC src)
target_link_libraries(qgem PUBLIC Eigen3::Eigen spdlog::spdlog SQLite3::SQLite3
                                  nlohmann_json::nlohmann_json)

foreach(app qgem_spectrum qgem_fit qgem_basis qgem_mcmc qgem_predict)
  add_executable(${app} src/apps/${app}.cpp)
  target_link_libraries(${app} PRIVATE qgem)
endforeach()

enable_testing()
file(GLOB TEST_SOURCES CONFIGURE_DEPENDS tests/test_*.cpp)
add_executable(qgem_tests ${TEST_SOURCES})
target_link_libraries(qgem_tests PRIVATE qgem GTest::gtest_main)
qgem_use_generated_kernels(qgem qgem_tests)   # no-op unless QGEM_REGENERATE_KERNELS=ON
include(GoogleTest)
gtest_discover_tests(qgem_tests)
```

(This file was verified on your machine exactly as printed, with a dummy source tree of one library file, the five apps and a test, except for the two `nlohmann_json` lines, which had to be removed because the package is not installed there yet: it configured without warnings, built, and `ctest` passed. It was verified again with the two lines for the generated kernels and the generated header and tests in place: 37 tests passed.) Build with `cmake -S . -B build && cmake --build build -j && ctest --test-dir build`. Your existing CI job runs `clang-format`; add a second job that builds and runs `ctest`.

11.5 Conventions

- *Units*: GeV and GeV^{-1} everywhere inside the program, $\hbar = c = 1$. Drop the `hbar` member of `QuarkoniumSystem`; keep one constant $\hbar c = 0.1973269804\,\mathrm{GeV\,fm}$ for printing radii in fm. Masses read from the PDG file are in MeV: convert once, at the boundary.
- *Types*: `Mat = Eigen::MatrixXd`, `Vec = Eigen::VectorXd`, `CMat = Eigen::MatrixXcd`. Replace `std::vector<std::vector<double>>`: it is an array of separately allocated rows, so matrix products cannot be vectorized and every access is a double indirection.
- *Naming*: one name per mathematical object, matching the notes (`S`, `H`, `nu`, `mr`, `kappa`, `sigma_th`). Functions that return matrices are nouns (`overlap`, `kinetic_nr`); functions that do work are verbs (`solve_gep`, `fit`).
- *Errors*: throw for programming and configuration errors (unknown channel, missing PDG entry). Return `std::expected<T, Error>` for *numerical* failures that a caller may want to survive (a parameter point where the overlap has no eigenvalue above η, a collapsing Hamiltonian): the fitter turns them into $\chi^2 = +\infty$ and shrinks the step. Your current code throws in that case, which would abort a fit.
- *Logging*: `spdlog` at `info` for run-level events, `debug` for per-iteration details, `warn` for anything that changes results silently (levels dropped by η, a level-tracking swap, a parameter at a bound).

- *Determinism*: every stochastic routine takes a seed; parallel code gives each thread its own engine seeded from (master seed, thread index).

11.6 From your current code to the new layout

Your `gem_solver.cpp` already contains the core of Models A/B for one channel. The following changes turn it into the `models/solve` layers; keep the current version and its golden test until the new one reproduces it.

1. *Normalized closed forms.* You compute un-normalized integrals with `std::tgamma` and `std::pow` and divide by the norms at the end. For large l or extreme ranges the intermediate numbers over- or underflow. Use the normalized forms: $S_{ij} = \left(2\sqrt{\nu_i\nu_j}/(\nu_i+\nu_j)\right)^{l+3/2}$, $\langle r^k\rangle_{ij} = S_{ij}\,\frac{\Gamma(l+\frac{k+3}{2})}{\Gamma(l+\frac{3}{2})}(\nu_i+\nu_j)^{-k/2}$, $T_{ij} = \frac{2l+3}{m_r}\frac{\nu_i\nu_j}{\nu_i+\nu_j}S_{ij}$, with `std::lgamma` for the gamma ratio (verified to reproduce Tests 1–2).
2. *Solver.* The LLT path with a 10^{-12} diagonal jitter on failure silently changes the problem. Use canonical orthogonalization (Algorithm 11.5) as the default and log the number of kept directions; keep the Cholesky path as `solve_gep_cholesky` for cross-checks.
3. *Spin.* Replace `int spin` with a `Channel`; replace the hard-coded $-\frac{3}{4}, \frac{1}{4}$ by `s1s2(S)` and add the spin–orbit, tensor and ${}^1L_L$–${}^3L_L$ factors (§11.8).
4. *Hyperfine in P waves.* Your code includes the smeared contact term for $l = 0$ only. With smearing it has small but non-zero P-wave matrix elements (§7.5); the worked example includes it in every channel. Either choice is a model choice: make it a configuration flag and state which one you use.
5. *Smearing default.* `sigma_smear` defaults to $\sqrt{m_r}$ (a legacy of the Python code). Make σ an explicit parameter with no hidden default.
6. *Matrix cache.* Build $\mathbf{S}$, $\mathbf{T}/m_r$ (as $\mathbf{T}_1$ with $m_r = 1$), $\langle r^{-1}\rangle$, $\langle r\rangle$ once per basis; only $\langle\delta_\sigma\rangle$ depends non-linearly on a parameter. This is what makes a model evaluation cost 0.3 ms.

11.7 Core data structures

```
namespace qgem {
using Mat = Eigen::MatrixXd;   using Vec = Eigen::VectorXd;
using CMat = Eigen::MatrixXcd; using cplx = std::complex<double>;
inline constexpr double hbarc = 0.1973269804;   // GeV fm, output only

// ---- spin/channels.hpp
struct Component { int L, S; };                 // one partial wave (L, S)
struct Channel {                                // one block  H c = E S c
  int J, P, C;                                  // J^{PC};  C = 0 if m1 != m2
  std::vector<Component> comps;                 // 1, or 2 if coupled
  std::string label() const;                    // "3S1-3D1", "1P1", "3P0", ...
};

// ---- basis/basis.hpp
struct BasisSpec {                              // per partial wave L
  int n = 30;  double r1 = 0.02, rmax = 30.0;   // real-range set (GeV^-1)
  int nc = 0;  double rho1 = 1.0, rhomax = 20.0, omega = 0.5;   // complex-range set
  double eta = 1e-12;                           // orthogonalization threshold
};
enum class Fn { Real, Cos, Sin };
struct Basis { int L = 0; std::vector<cplx> z; std::vector<Fn> kind; Vec norm; };

// ---- fit/params.hpp
enum class Transform { Identity, Log, OrderedLog };   // OrderedLog: gamma_1 < gamma_2 < ...
struct ParamSpec {
  std::string name;  double value;  bool free = true;
```

```
  Transform tr = Transform::Identity;
  double lo = -INFINITY, hi = INFINITY;
  std::optional<std::pair<double, double>> prior;  // Gaussian (mean, sigma) -> penalty
};
class ParamSet {                                   // free, transformed u <-> physical theta
 public:
  Vec theta(const Vec& u) const;  Vec u(const Vec& theta) const;
  Mat dtheta_du(const Vec& u) const;               // diagonal except for OrderedLog
  int n_free() const;  std::span<const ParamSpec> specs() const;
};

// ---- solve/spectrum.hpp
struct StateId { std::string channel; int n; };  // {"3P0", 0} = lowest 3P0 level
struct Level {
  Channel ch;  int n;  double E, M;              // M = E + mass offset
  Vec c;                                         // c^T S c = 1
  std::vector<double> prob;                      // component probabilities
};
struct Spectrum {
  std::vector<Level> levels;
  const Level& at(const StateId&) const;
};

// ---- data/dataset.hpp
struct Datum {
  std::string label;  long mcid = 0;             // PDG Monte Carlo id
  enum class Kind { Mass, Splitting, Width, Ratio } kind = Kind::Mass;
  std::vector<StateId> states;                   // mass: 1 state; splitting: 2
  double y = 0, err_plus = 0, err_minus = 0;     // GeV
  std::string sys_group;  double sys_common = 0; // common systematic (Ex. 9.1)
  bool in_fit = true;  double extra_floor = 0;   // e.g. states above threshold
};
struct Dataset {
  std::vector<Datum> d;
  Mat V_exp() const;                             // incl. common systematics
};

// ---- fit/fit.hpp
struct FitOptions {
  double lambda0 = 1e-3;  int max_iter = 200;  double ftol = 1e-12, xtol = 1e-10;
};
struct FitResult {
  Vec u, theta;  double chi2 = 0;  int ndof = 0;  double sigma_th = 0;
  Mat Jz;                                        // whitened Jacobian dz/du (N x p)
  Mat C;                                         // covariance of theta
  Vec residuals, pulls, leverage;
  int iterations = 0;  bool converged = false;
};
}  // namespace qgem
```

The model interface. Models differ only in how they assemble **H**; everything else is shared.

```
struct ChannelMatrices { Mat H, S; bool orthonormal = false; };  // GI builds H' directly

// Parameter-independent matrices per (channel, basis). An entry that depends on one
// non-linear parameter (sigma, mu_s, tau_k) stores that value and is rebuilt when it changes.
class MatrixCache;

class Model {
 public:
  virtual ~Model() = default;
  virtual std::string name() const = 0;
  virtual ParamSet parameters() const = 0;
  virtual double mass_offset(const Vec& theta) const = 0;   // m1+m2 (A, B) or 0 (C, D)
  virtual ChannelMatrices build(const Channel&, const Vec& theta, MatrixCache&) const = 0;
  // dH/dtheta_a for all parameters; default implementation: central differences of build()
  virtual std::vector<Mat> dH(const Channel&, const Vec& theta, MatrixCache&) const;
};
class NRModel final : public Model { /* Models A and B: flags screened, spin_mode */ };
class GIModel final : public Model { /* Models C and D: flag screened, alpha_s form */ };
std::unique_ptr<Model> make_model(const nlohmann::json& config);
```

11.8 Function catalogue

Signatures with their contracts as comments. Every matrix function returns a matrix in the normalized basis.

math/special, math/quadrature

```
// Int_0^inf r^p e^{-a r^2} dr = Gamma((p+1)/2) / (2 a^{(p+1)/2}), evaluated with lgamma
double master_integral(double p, double a);
cplx   master_integral(double p, cplx a);        // principal branch, Re a > 0 (complex range)
double erfcx(double x);                          // e^{x^2} erfc x (§11.10)
// I_j(a, tau), j = 0..jmax, and Q_k(a, mu), k = 0..kmax: the recursions of §7.5
void   erf_t_integrals(int jmax, double a, double tau, std::span<double> I);
// Q_k: upward for x = mu/(2 sqrt a) < 1/2, downward ratios otherwise (the remark after the Q_k lemma);
// the generated header has this function as gen::detail::screened_moments
void   screened_moments(int kmax, double a, double mu, std::span<double> Q);
// smeared r and e^{-mu r} with their r-derivatives: eqs. (6.22), (6.24), erfcx form
double smeared_r(double r, double s);              double smeared_r_d(double r, double s);
double smeared_exp(double r, double mu, double s);  double smeared_exp_d(double r, double mu, double s);
double gi_alpha_s(double Q, std::span<const double> alpha, std::span<const double> gamma);
struct QuadRule { Vec x, w; };
const QuadRule& gauss_legendre01(int n);           // cached rule on [0, 1]
const QuadRule& gauss_laguerre(int n, double alpha);  // cached, weight t^alpha e^{-t} on [0, inf)
```

linalg, basis

```
struct GepResult { Vec E; Mat C; Mat X; int n_kept; double s_min, s_max; };
Mat       orthonormalizer(const Mat& S, double eta);   // X with X^T S X = 1, kept directions
GepResult solve_gep(const Mat& H, const Mat& S, double eta = 1e-12);   // Alg. 11.5
GepResult solve_gep_cholesky(const Mat& H, const Mat& S);  // your current method, cross-checks
Basis  make_basis(int L, const BasisSpec&);              // Alg. 11.1
Basis  nested_double(const Basis&);                      // n -> 2n-1: every old function kept
double radial_value(const Basis&, const Vec& c, double r);
Vec    origin_coefficients(const Basis&);   // v_n = N_nL: R(0) = c.v (L=0), R ~ (c.v) r^L
```

me (radial matrix elements)

`bra` and `ket` bases may have different L (tensor coupling, transitions). The closed forms for one pair of basis functions, with their parameter derivatives, are in the generated header `generated/qgem_kernels.hpp` (§13.5.10): `qgem::gen::overlap`, `kinetic`, `r_power`, `gaussian`, `erf_over_r`, `screened_exp`, `smeared_linear`, `smeared_exp`, their spin–orbit and tensor partners, and the complex-range bilinear forms. The functions below fill matrices from them; write them yourself or use the generated ones, and test one against the other.

```
Mat overlap(const Basis&);
Mat kinetic_nr(const Basis&, double mr);                          // T = p^2/(2 mr)
Mat r_power(const Basis& bra, const Basis& ket, double k);        // <r^k>, L+L'+3+k > 0
Mat gaussian(const Basis& bra, const Basis& ket, double beta);    // <e^{-beta r^2}>
Mat erf_over_r(const Basis&, double tau);                         // <erf(tau r)/r>
Mat exp_power(const Basis&, double mu, int k);                    // <r^k e^{-mu r}>
// any V(r): Gauss-Legendre on [0, sqrt(40/a)] per pair, cached rule rescaled
Mat local(const Basis& bra, const Basis& ket, const std::function<double(double)>& V,
          int nq = 128);
// <f(p)>: generalized Gauss-Laguerre with alpha = L + 1/2
Mat momentum_diagonal(const Basis&, const std::function<double(double)>& f, int nq = 64);

// ---- me/gi.hpp: tilde G, tilde S and the GI spin radial functions
struct GILocal { Mat G, Sconf, contact, tensor, so_v, so_s; };
GILocal gi_local(const Basis& bra, const Basis& ket, const GIParams&, bool screened);
// closure: returns (Xb^T Fb Xb)(Xb^T W Xk)(Xk^T Fk Xk), i.e. F'W'F' in orthonormal bases
Mat sandwich_closure(const Mat& Fbra, const Mat& W, const Mat& Fket,
                     const Mat& Xbra, const Mat& Xket);
// two-step: tabulate chi_n = f(p) R_n on radial nodes, then integrate chi W chi
```

```
Mat sandwich_two_step(const Basis&, const std::function<double(double)>& f,
                      const std::function<double(double)>& W, int np, int nr);
```

spin

Values verified symbolically (§13.4.2).

```
double s1s2(int S);              // -3/4 (S=0), +1/4 (S=1)
double ls(int L, int S, int J);  // [J(J+1) - L(L+1) - S(S+1)]/2
// S=1 only; L=0 gives 0. J=L+1: -2L/(2L+3);  J=L: 2;  J=L-1: -2(L+1)/(2L-1)
double s12_diag(int L, int J);
double s12_offdiag(int J);       // <(J-1)1J|S12|(J+1)1J> = 6 sqrt(J(J+1))/(2J+1); 3S1-3D1: 2 sqrt 2
double ls_minus(int L);          // <1L_L| L.(S1-S2) |3L_L> = sqrt(L(L+1)), m1 != m2 only
std::vector<Channel> enumerate_channels(int Lmax, bool equal_masses);
```

models, solve, obs

```
Spectrum solve_spectrum(const Model&, const Vec& theta, std::span<const Channel>,
                        MatrixCache&);
// new level b for every old level a: argmax_b |c_b(new)^T S c_a(old)|
std::vector<int> track_levels(const Mat& S, const Mat& C_old, const Mat& C_new);
double R0_squared(const Level&, const Basis&);              // |R(0)|^2, S waves
double origin_derivative(const Level&, const Basis&);       // R'(0) (P), R''(0) (D) via A_L
double moment(const Level&, const Basis&, double k);        // <r^k>
double radial_overlap(const Level& f, const Level& i, double k);  // <f|r^k|i>, L may differ
// widths of §8; E1 uses C_fi from 6j, M1 <j0(kr/2)> by quadrature
double width_ee(const Level&, double M, double eQ, double qcd_factor);
double width_E1(const Level& i, const Level& f, double eQ);
double width_M1(const Level& i, const Level& f, double eQ, double mQ);
double sixj(int j1, int j2, int j3, int j4, int j5, int j6);   // Racah formula
```

deriv, hyper

```
// N x p Jacobian of the predictions by Feynman-Hellmann (Alg. 11.7)
Mat jacobian_fh(const Model&, const Vec& theta, const Spectrum&, const Dataset&,
                MatrixCache&);
Vec eigvec_derivative(const GepResult&, int a, const Mat& dH, const Mat& dS);  // (7.35)
Mat jacobian_fd(const std::function<Vec(const Vec&)>& f, const Vec& x, const Vec& h);
struct HyperResult { BasisSpec spec; double objective; std::vector<double> nested_shift; };
HyperResult optimize_basis(const Model&, const Vec& theta0, const Channel&,
                           BasisSpec start, int K, std::span<const double> weights);  // Alg. 11.8
double numerical_error(const Model&, const Vec& theta, const Channel&, const BasisSpec&,
                       const StateId&);
```

data, fit, stats, report

```
class PdgDatabase;                                  // §11.10 (verified)
Dataset load_dataset(const nlohmann::json& selection, const PdgDatabase&);
double  open_flavour_threshold(const PdgDatabase&, char flavour);  // 2 M(D0) or 2 M(B0)

class Chi2Problem {                                 // model + data + parameters + floor
 public:
  Chi2Problem(const Model&, Dataset, ParamSet, double sigma_th);
  std::expected<Vec, Error> residuals(const Vec& u);   // whitened z_i
  std::expected<Mat, Error> jacobian(const Vec& u);    // dz/du (Feynman-Hellmann or FD)
  void set_sigma_th(double);  int n_data() const;  int n_free() const;
};
FitResult levenberg_marquardt(Chi2Problem&, Vec u0, const FitOptions&);   // Alg. 11.9
double    solve_theory_floor(const Vec& Delta, const Vec& s_exp, int nu); // Alg. 11.10
FitResult fit_with_floor(Chi2Problem&, Vec u0, const FitOptions&);
FitResult multistart(Chi2Problem&, int n_starts, std::uint64_t seed);
```

```
struct Uncertainty {
  Mat C, rho;  Vec sigma, sigma_conditional, fisher_eigenvalues;  Mat fisher_eigenvectors;
};
Uncertainty analyse(const FitResult&);                                    // Alg. 11.11
std::vector<std::pair<double, double>> profile(Chi2Problem&, const FitResult&, int a,
                                               std::span<const double> grid);
struct Samples { Mat theta; Vec chi2; double acceptance = 0; };
Samples parametric_bootstrap(Chi2Problem&, const FitResult&, int B,
                             std::uint64_t seed);                         // Alg. 11.12
std::vector<Samples> metropolis(Chi2Problem&, const FitResult&, int chains, int steps,
                                int burn, std::uint64_t seed);            // Alg. 11.13
Vec rhat(std::span<const Samples>);
Vec tau_int(const Samples&);
struct LooResult { Vec leverage, loo_residual, pull; };
LooResult leave_one_out(Chi2Problem&, const FitResult&, bool exact_refits);
double aic(const FitResult&);
double bic(const FitResult&);
struct Budget { double value, par, num, th, amb, model_spread; };
// arguments: prediction functor, fit, basis variants, convention variants   (Alg. 11.14)
Budget error_budget(/* ... */);
void   write_json(const std::string& path, const nlohmann::json&);
```

11.9 Algorithms

Notation as in §7–§10. Line comments (▷) give the reason for each step.

Algorithm 11.1 Build a basis for one partial wave L

Input: L; real set $(n, r_1, r_{\max})$; complex set $(n_c, \rho_1, \rho_{\max}, \omega)$

$q \leftarrow (r_{\max}/r_1)^{1/(n-1)}$
for $k = 0, \dots, n-1$ **do**
    append $z = 1/(r_1 q^k)^2$, kind REAL
if $n_c > 0$ **then**
    $q_c \leftarrow (\rho_{\max}/\rho_1)^{1/(n_c-1)}$
    for $k = 0, \dots, n_c - 1$ **do**
        $\eta_k \leftarrow 1/(\rho_1 q_c^k)^2$; append $(\eta_k(1+\mathrm{i}\omega), \mathrm{COS})$ and $(\eta_k(1+\mathrm{i}\omega), \mathrm{SIN})$
for each function i **do** ▷ normalize so that $S_{ii} = 1$
    compute the unnormalized S_{ii} from the master integral (real) or the rules (7.33) (cos/sin); $\mathrm{norm}_i \leftarrow S_{ii}^{-1/2}$

Output: `Basis`; real functions alone have the closed form N_{nL} of §7.2

Algorithm 11.2 Gauss quadrature by Golub–Welsch (Legendre, generalized Laguerre)

Input: n; for Laguerre the exponent α of the weight $t^\alpha \mathrm{e}^{-t}$

build the symmetric tridiagonal Jacobi matrix T:
    Legendre: $\mathsf{T}_{kk} = 0$, $\mathsf{T}_{k,k-1} = k/\sqrt{4k^2-1}$, $\mu_0 = 2$ ▷ on $[-1, 1]$
    Laguerre: $\mathsf{T}_{kk} = 2k + \alpha + 1$, $\mathsf{T}_{k,k-1} = \sqrt{k(k+\alpha)}$, $\mu_0 = \Gamma(\alpha+1)$
diagonalize $\mathsf{T} = \mathbf{V}\,\mathrm{diag}(x_i)\mathbf{V}^T$ ▷ `SelfAdjointEigenSolver`
nodes x_i; weights $w_i \leftarrow \mu_0\, V_{0i}^2$ ▷ first component of each eigenvector
cache the rule by (n, α) ▷ rebuilding it per matrix element costs $10^3\times$

Output: $\sum_i w_i x_i^k = \Gamma(\alpha + k + 1)$ for $k \le 2n-1$ (unit test)

Algorithm 11.3 Assemble one channel of Model A/B

Input: channel with components $\alpha = (L_\alpha, S)$; parameters $\boldsymbol{\theta} = (m_1, m_2, \alpha_s, b, c, \sigma[, \mu_s])$; cache

1: **for** each component α **do**

2: fetch from cache (built once per basis): $\mathbf{S}$, $\mathbf{T}_1$ ($m_r = 1$), $\langle r^{-1}\rangle$, $\langle r\rangle$, $\langle r^{-3}\rangle$ ($L \geq 1$)

3: fetch or rebuild if σ changed: $\mathbf{G}_\sigma = \langle \mathrm{e}^{-\sigma^2 r^2}\rangle$; if μ_s changed: $\langle \mathrm{e}^{-\mu_s r}\rangle$, $\langle \mathrm{e}^{-\mu_s r}/r\rangle$

4: $\mathbf{H}_{\alpha\alpha} \leftarrow \mathbf{T}_1/m_r + c\,\mathbf{S} - \frac{4}{3}\alpha_s\langle r^{-1}\rangle + V_S + \frac{32\pi\alpha_s}{9m_1m_2}\left(\frac{\sigma}{\sqrt{\pi}}\right)^3 \mathtt{s1s2}(S)\,\mathbf{G}_\sigma$ with $V_S = b\langle r\rangle$ (A) or $\frac{b}{\mu_s}(\mathbf{S} - \langle \mathrm{e}^{-\mu_s r}\rangle)$ (B)

5: **if** spin–orbit/tensor non-perturbative (smeared versions only, §4.1) **then** add $o_k(L,S,J)\,\mathbf{W}_k$

6: **if** two components **then** ▷ tensor $L = J \mp 1$, or ${}^1L_L{}^3L_L$ for $m_1 \neq m_2$

7: off-diagonal block: $\mathtt{s12_offdiag}(J) \cdot \langle R_{L_1}|W_T|R_{L_2}\rangle$, or $\mathtt{ls_minus}(L) \cdot \langle B(r)\rangle$

8: $\mathbf{S} \leftarrow$ block-diagonal; $\mathbf{H} \leftarrow \frac{1}{2}(\mathbf{H} + \mathbf{H}^T)$

Output: `ChannelMatrices`; mass offset $m_1 + m_2$ added after diagonalization

Perturbative spin terms (as in the worked example): diagonalize the spin-independent (plus smeared contact) Hamiltonian, then add $\langle \boldsymbol{L}\cdot\boldsymbol{S}\rangle a + \langle S_{12}\rangle t$ with a, t from the eigenvector; their parameter derivatives need (7.35).

Algorithm 11.4 Assemble one channel of Model C/D (GI) in the orthonormalized basis

Input: channel; $\boldsymbol{\theta} = (m_1, m_2, b, c, \sigma_0, s, \epsilon_c, \epsilon_t, \epsilon_{sov}, \epsilon_{sos}[, \{\alpha_k, \gamma_k\}][, \mu_s])$

1: σ_{12} from (6.8); $\tau_k = \gamma_k\sigma_{12}/\sqrt{\gamma_k^2 + \sigma_{12}^2}$

2: **for** each component L **do**

3: $\mathbf{X}_L \leftarrow \mathtt{orthonormalizer}(\mathbf{S}_L, \eta)$ ▷ cached per basis

4: $\mathbf{K} \leftarrow \mathtt{momentum_diagonal}(E_1 + E_2)$; $\mathbf{F}_G \leftarrow \mathtt{momentum_diagonal}\big((1 + p^2/E_1E_2)^{1/2}\big)$

5: $\boldsymbol{\Phi}_x \leftarrow \mathtt{momentum_diagonal}\big((m_am_b/E_aE_b)^{1/2+\epsilon_x}\big)$ for $x \in \{c, t, sov, sos\}$ ▷ one p-quadrature each

6: local matrices $\tilde{\mathbf{G}}$, $\tilde{\mathbf{S}}$ (or $\tilde{\mathbf{S}}_{\mathrm{scr}}$), $\mathbf{W}_c, \mathbf{W}_t, \mathbf{W}_{so(v)}, \mathbf{W}_{so(s)}$ ▷ closed forms or `local`

7: $\mathbf{H}'_{LL} \leftarrow \mathbf{X}^T\mathbf{K}\mathbf{X} + \mathbf{F}'_G\tilde{\mathbf{G}}'\mathbf{F}'_G + \tilde{\mathbf{S}}' + \sum_x o_x(L,S,J)\,\boldsymbol{\Phi}'_x\mathbf{W}'_x\boldsymbol{\Phi}'_x$ ▷ $\mathbf{A}' = \mathbf{X}^T\mathbf{A}\mathbf{X}$

8: coupled block: $\mathbf{H}'_{L_1L_2} \leftarrow o_t\,(\mathbf{X}_1^T\boldsymbol{\Phi}_{t,1}\mathbf{X}_1)(\mathbf{X}_1^T\mathbf{W}_{t,12}\mathbf{X}_2)(\mathbf{X}_2^T\boldsymbol{\Phi}_{t,2}\mathbf{X}_2)$

9: symmetrize; diagonalize $\mathbf{H}'$ (standard problem); $\mathbf{C} \leftarrow \mathbf{X}\mathbf{C}'$ ▷ eigenvalues are masses

Output: closure method (ii) of §7.7; validate against the two-step method (Test 4)

Algorithm 11.5 Generalized eigenproblem by canonical orthogonalization

Input: symmetric $\mathbf{H}$, positive semi-definite $\mathbf{S}$ (numerically), threshold η

1: $\mathbf{S} = \mathbf{U}\,\mathrm{diag}(s_i)\mathbf{U}^T$, ascending s_i ▷ `SelfAdjointEigenSolver`

2: keep $\mathcal{K} = \{i :\ s_i > \eta\, s_{\max}\}$; if $\mathcal{K} = \varnothing$ return error

3: $\mathbf{X} \leftarrow \mathbf{U}_{\mathcal{K}}\,\mathrm{diag}(s_i^{-1/2})_{i\in\mathcal{K}}$ ▷ $\mathbf{X}^T\mathbf{S}\mathbf{X} = \mathbf{1}$

4: $\mathbf{H}' \leftarrow \mathbf{X}^T\mathbf{H}\mathbf{X}$; symmetrize; diagonalize $\mathbf{H}' = \mathbf{V}\,\mathrm{diag}(\mathcal{E}_k)\mathbf{V}^T$

5: $\mathbf{C} \leftarrow \mathbf{X}\mathbf{V}$ ▷ columns satisfy $\boldsymbol{c}_k^T\mathbf{S}\boldsymbol{c}_k = 1$

Output: $\mathcal{E}$ ascending, $\mathbf{C}$, $\mathbf{X}$, $|\mathcal{K}|$, $s_{\min}/s_{\max}$ (log a warning if $|\mathcal{K}| < n$)

Algorithm 11.6 Label levels and track them between parameter points

Input: eigenvectors $\mathbf{C}^{\mathrm{old}}$ (from the previous accepted point), new $\mathbf{C}$, overlap $\mathbf{S}$ (unchanged: fixed basis)

1: $\mathbf{O} \leftarrow |\mathbf{C}^T\mathbf{S}\mathbf{C}^{\mathrm{old}}|$ ▷ $O_{ba} \approx \delta_{ba}$ if nothing crossed

2: **for** each old level a that is assigned to a datum **do**

3: $\quad b(a) \leftarrow \arg\max_b O_{ba}$; if $b(a) \neq a$ log `warn` (level crossing)

4: for coupled channels: label by the dominant component, $P_\alpha = \boldsymbol{c}_\alpha^T \mathbf{S}^\alpha \boldsymbol{c}_\alpha$

Output: map datum → level index at the new point

Algorithm 11.7 Feynman–Hellmann Jacobian of the whitened residuals

Input: spectrum at $\boldsymbol{\theta}$; for each channel the matrices $\partial\mathbf{H}/\partial\theta_a$ (`Model::dH`); data; transform Jacobian $\partial\boldsymbol{\theta}/\partial\boldsymbol{u}$

1: **for** each datum i and each parameter a **do**

2: $\quad$ mass datum with level (ch, k): $\partial\mu_i/\partial\theta_a \leftarrow \boldsymbol{c}_k^T(\partial\mathbf{H}/\partial\theta_a)\boldsymbol{c}_k + \partial(\mathrm{offset})/\partial\theta_a$ ▷ $\mathbf{S}' = 0$

3: $\quad$ splitting: difference of two such terms; perturbative add-ons: chain rule with (7.35)

4: $\partial z_i/\partial\theta_a \leftarrow -(\partial\mu_i/\partial\theta_a)/s_i$; then $\partial z/\partial\boldsymbol{u} = (\partial z/\partial\boldsymbol{\theta})(\partial\boldsymbol{\theta}/\partial\boldsymbol{u})$

Output: $N \times p$ matrix; unit test against `jacobian_fd` (relative agreement $\sim 10^{-7}$)

Algorithm 11.8 Basis hyperparameter optimization and convergence check

Input: model at nominal $\boldsymbol{\theta}_0$; channel; start spec; K target levels with weights $w_k > 0$

1: variables $\boldsymbol{\xi} = (\ln r_1, \ln r_{\max}[, \ln\rho_1, \ln\rho_{\max}, \omega])$; n, n_c fixed ▷ n is not continuous (Property 2)

2: objective $f(\boldsymbol{\xi}) = \sum_{k<K} w_k \mathcal{E}_k(\boldsymbol{\xi})$ ▷ an upper bound for each k (HUM, Ky Fan)

3: minimize f with Nelder–Mead, or with the gradient $\partial\mathcal{E}_k/\partial\xi = \boldsymbol{c}_k^T(\mathbf{H}' - \mathcal{E}_k\mathbf{S}')\boldsymbol{c}_k$ (7.34)

4: nested check: $n \to 2n-1$ (`nested_double`); record the shift of every observable as σ_{num}

5: require $\sigma_{\mathrm{num}} \leq 10^{-2}\min_i s_i$; otherwise increase n and repeat

Output: frozen `BasisSpec` per channel, written to the run configuration

Algorithm 11.9 Levenberg–Marquardt on whitened residuals

Input: `Chi2Problem`; start $\boldsymbol{u}_0$; $\lambda_0 = 10^{-3}$; tolerances

1: $\boldsymbol{z} \leftarrow \mathrm{residuals}(\boldsymbol{u})$; $\chi^2 \leftarrow \boldsymbol{z}^T\boldsymbol{z}$

2: **repeat**

3: $\quad \mathbf{J} \leftarrow \mathrm{jacobian}(\boldsymbol{u})$; $\mathsf{F} \leftarrow \mathbf{J}^T\mathbf{J}$; $\boldsymbol{g} \leftarrow -\mathbf{J}^T\boldsymbol{z}$ ▷ $\mathbf{J} = \partial\boldsymbol{z}/\partial\boldsymbol{u}$, so $\boldsymbol{g} = \mathsf{J}^T\mathsf{W}\boldsymbol{\Delta}$

4: $\quad$ **loop**

5: $\quad\quad$ solve $(\mathsf{F} + \lambda\,\mathrm{diag}\,\mathsf{F})\,\boldsymbol{\delta} = \boldsymbol{g}$ ▷ `ldlt()`

6: $\quad\quad$ $\boldsymbol{z}' \leftarrow \mathrm{residuals}(\boldsymbol{u}+\boldsymbol{\delta})$; if it fails (`std::expected` error) set $\chi'^2 = \infty$

7: $\quad\quad$ **if** $\chi'^2 < \chi^2$ **then** accept; $\lambda \leftarrow \max(\lambda/10, 10^{-12})$; **break**

8: $\quad\quad$ **else** $\lambda \leftarrow 10\lambda$; if $\lambda > 10^{12}$ stop (no progress)

9: **until** $(\chi^2_{\mathrm{old}} - \chi^2)/\chi^2 < 10^{-12}$ and $\max_a |\delta_a|/(|u_a| + 10^{-8}) < 10^{-10}$

10: covariance in $\boldsymbol{u}$: $(\mathbf{J}^T\mathbf{J})^{-1}$; in $\boldsymbol{\theta}$: $\mathsf{C} = \mathbf{K}(\mathbf{J}^T\mathbf{J})^{-1}\mathbf{K}^T$, $\mathbf{K} = \partial\boldsymbol{\theta}/\partial\boldsymbol{u}$

Output: `FitResult`. Bounds: by transforms (log for positive parameters), not by clipping

Note the sign: with $z_i = (y_i - \mu_i)/s_i$ the Jacobian $\partial z/\partial u$ is minus the whitened J, hence $\boldsymbol{g} = -\mathbf{J}^T\boldsymbol{z}$. Getting this sign wrong makes every step go uphill, and the method then shrinks λ forever.

Algorithm 11.10 Theory floor (common to all states)

Input: start $\sigma_{\text{th}}^{(0)}$ (e.g. 10 MeV); $\nu = N - p$

1: **repeat**
2: fit with $s_i^2 = \sigma_{\text{exp},i}^2 + \sigma_{\text{th}}^2$ (Algorithm 11.9, warm start) $\to$ residuals Δ_i
3: solve $g(x) = \sum_i \Delta_i^2/(\sigma_{\text{exp},i}^2 + x) - \nu = 0$ for $x \geq 0$ by bisection on $[0, \sum \Delta_i^2]$ ▷ g is decreasing
4: if $g(0) \leq 0$ set $x = 0$ (no floor needed); $\sigma_{\text{th}} \leftarrow \sqrt{x}$
5: **until** $|\Delta\sigma_{\text{th}}| < 10^{-4}$ MeV ▷ 2–3 iterations in the worked example

Output: σ_{th}, final fit with $\chi^2/\nu = 1$

Algorithm 11.11 Uncertainty analysis at the minimum

1: $\mathsf{F} \leftarrow \mathbf{J}^T\mathbf{J}$ in physical variables; scale $\tilde{\mathsf{F}} = \mathsf{D}_\theta \mathsf{F} \mathsf{D}_\theta$ with $\mathsf{D}_\theta = \text{diag}(\hat{\theta}_a)$
2: eigen-decompose $\tilde{\mathsf{F}}$; condition number $\lambda_{\max}/\lambda_{\min}$; if $> 10^{10}$ warn (degenerate direction, print its eigenvector)
3: $\mathsf{C} \leftarrow \mathsf{F}^{-1}$ (via the eigen-decomposition); $\sigma_a = \sqrt{\mathsf{C}_{aa}}$; $\rho_{ab} = \mathsf{C}_{ab}/(\sigma_a\sigma_b)$
4: conditional errors $1/\sqrt{\mathsf{F}_{aa}}$ (diagnostic only; never quote them)
5: profile of each key parameter: fix θ_a on a grid $\hat{\theta}_a + t\sigma_a$, $t \in [-3, 3]$, refit the rest starting from the neighbouring grid point; record $\Delta\chi^2(t)$; interpolate the $\Delta\chi^2 = 1$ crossings

Output: `Uncertainty`; flag parameters whose profile deviates from t^2 by more than 10% at $|t| = 1$

Algorithm 11.12 Parametric bootstrap

Input: best fit $\hat{\boldsymbol{\theta}}$; data covariance V (with σ_{th}^2 on the diagonal); B; seed

1: Cholesky $\mathsf{V} = \mathsf{L}\mathsf{L}^T$ once
2: **for all** replicas $b = 1, \ldots, B$ **in parallel do** ▷ one engine per thread
3: $\boldsymbol{y}^* \leftarrow \boldsymbol{\mu}(\hat{\boldsymbol{\theta}}) + \mathsf{L}\boldsymbol{\xi}$, $\boldsymbol{\xi} \sim \mathcal{N}(0, 1)$
4: refit from $\hat{\boldsymbol{\theta}}$ with the *same* σ_{th} and basis $\to \hat{\boldsymbol{\theta}}^{*(b)}$; store predictions $P(\hat{\boldsymbol{\theta}}^{*(b)})$

Output: means, standard deviations, 16/84 percentiles, bias $\overline{\boldsymbol{\theta}^*} - \hat{\boldsymbol{\theta}}$

Algorithm 11.13 Metropolis sampling with diagnostics

Input: $\hat{\boldsymbol{\theta}}$, C; $m \geq 4$ chains; n steps; burn-in; seed

1: proposal covariance $\Sigma = \frac{2.38^2}{p}\mathsf{C}$; Cholesky $\Sigma = \mathsf{L}_\Sigma\mathsf{L}_\Sigma^T$
2: **for all** chains j **in parallel do**
3: start at $\hat{\boldsymbol{\theta}} + 3\,\mathsf{L}_C\boldsymbol{\xi}$ (over-dispersed); $\ell \leftarrow -\frac{1}{2}\chi^2_{\text{tot}}$ (data + penalties)
4: **for** $t = 1, \ldots, n$ **do**
5: $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta} + \mathsf{L}_\Sigma\boldsymbol{\xi}$; $\ell' \leftarrow -\frac{1}{2}\chi^2_{\text{tot}}(\boldsymbol{\theta}')$ ($-\infty$ outside bounds or on failure)
6: accept if $\ln U < \ell' - \ell$, $U \sim \mathcal{U}(0, 1)$ ▷ log scale: no overflow
7: record $\boldsymbol{\theta}$ (also when rejected)
8: drop burn-in; $\hat{R}$ per parameter (§10.6); $\tau_{\text{int}} = 1 + 2\sum_{k=1}^{M}\hat{\rho}_k$ with the smallest window $M \geq 5\tau_{\text{int}}(M)$

Output: samples, acceptance (aim 0.2–0.4), $\hat{R} < 1.01$, $n_{\text{eff}} = mn/\tau_{\text{int}} \gtrsim 10^3$

Algorithm 11.14 Validation and error budget of a prediction P

1: leverages $h_i = \text{diag of } \mathbf{J}\mathsf{C}\mathbf{J}^T$ (whitened); check $\sum h_i = p$; LOO pulls $z_i = \Delta_i/(s_i\sqrt{1-h_i})$ (10.19); confirm with exact refits
2: AIC, BIC with a common σ_{th} across the compared models; LRT for nested pairs (Chernoff for boundaries)
3: $\sigma_{\text{par}} \leftarrow \sqrt{\boldsymbol{g}^T\mathsf{C}\boldsymbol{g}}$ with $\boldsymbol{g} = \nabla_\theta P$ (FH or FD), or the bootstrap/posterior spread
4: $\sigma_{\text{num}} \leftarrow |P(2n-1) - P(n)|$ (and quadrature/η variations); Aitken if three nested values
5: $\sigma_{\text{amb}} \leftarrow$ half the envelope over convention variants (mass in widths, α_s scale, ...)
6: model spread: same prediction from all fitted models

Output: `Budget`; written to JSON with the configuration and code version

11.10 Verified implementation details

These fragments are short, library-specific, or numerically delicate, so they are given in full. Each was compiled and run (GCC 14.2/Eigen 3.4 and GCC 16.2/Eigen 5.0.1).

erfcx. Needed for $\mathcal{Q}_0$ and the smeared exponential. Relative error $\le 4 \times 10^{-16}$ against 40-digit arithmetic on $[-26, 200]$ (checked on a grid of 1000 points).

```
double erfcx(double x) {                                  // e^{x^2} erfc(x)
  if (x < 0.0) return 2.0 * std::exp(x * x) - erfcx(-x);  // overflows only for x < -26.6
  if (x < 26.0) return std::exp(x * x) * std::erfc(x);
  const double y = 1.0 / (2.0 * x * x);                   // asymptotic series, six terms
  const double s = 1.0 - y * (1.0 - 3.0 * y * (1.0 - 5.0 * y * (1.0 - 7.0 * y
                              * (1.0 - 9.0 * y * (1.0 - 11.0 * y)))));
  return s / (x * std::sqrt(std::numbers::pi));
}
```

(With only four terms of the series the error at $x = 25$ is 3×10^{-13}: the switch point and the number of terms matter.)

Generalized Gauss–Laguerre (Algorithm 11.2). With $n = 40$, $\alpha = 1.5$ the moments $k \le 10$ are exact to 8×10^{-15}.

```
QuadRule gauss_laguerre_uncached(int n, double alpha) {
  Mat T = Mat::Zero(n, n);
  for (int k = 0; k < n; ++k) T(k, k) = 2.0 * k + alpha + 1.0;
  for (int k = 1; k < n; ++k) T(k, k - 1) = T(k - 1, k) = std::sqrt(k * (k + alpha));
  Eigen::SelfAdjointEigenSolver<Mat> es(T);
  QuadRule q{es.eigenvalues(), Vec(n)};
  for (int i = 0; i < n; ++i)
    q.w(i) = std::tgamma(alpha + 1.0) * es.eigenvectors()(0, i) * es.eigenvectors()(0, i);
  return q;
}
// use: <f(p)>_ij = S_ij / Gamma(L+3/2) * sum_k w_k f( sqrt(x_k / beta_ij) ),
//      beta_ij = (nu_i + nu_j) / (4 nu_i nu_j)
```

The last comment line is the whole relativistic kinetic matrix: with $f = E_1 + E_2$ and 64–80 nodes it reproduces Test 3 to all digits.

Canonical orthogonalization (Algorithm 11.5). The default solver of the program; with $\eta = 10^{-12}$ it reproduces Tests 1–4.

```
GepResult solve_gep(const Mat& H, const Mat& S, double eta) {
  Eigen::SelfAdjointEigenSolver<Mat> es(S);
  const Vec& s = es.eigenvalues();                        // ascending
  int first = 0;
  while (first < s.size() && s(first) <= eta * s(s.size() - 1)) ++first;
  const int m = s.size() - first;
```

```
  if (m == 0)                                          // or return std::unexpected(...)
    throw std::runtime_error("overlap: no eigenvalue above threshold");
  Mat X = es.eigenvectors().rightCols(m)
          * s.tail(m).cwiseSqrt().cwiseInverse().asDiagonal();
  Mat Hp = X.transpose() * H * X;
  Hp = 0.5 * (Hp + Hp.transpose());
  Eigen::SelfAdjointEigenSolver<Mat> eh(Hp);
  return {eh.eigenvalues(), X * eh.eigenvectors(), X, m, s(first), s(s.size() - 1)};
}
```

Reading the PDG file. The schema (verified on your file): `pdgparticle` (name, `mcid`, `pdgid`, quantum numbers), `pdgid` (one row per listed quantity; masses have `data_type='M'` and `parent_pdgid` = the particle), `pdgdata` (values; `value`, `error_positive`, `error_negative`, `unit_text`, `value_type`, `in_summary_table`). Three facts matter: (i) select `in_summary_table = 1`: $\psi(2S)$ has two mass rows, the fit value (3686.097 ± 0.011, type `FC`, in the summary table) and the average (± 0.010, type `AC`, not); (ii) errors can be asymmetric (`error_positive` $\neq$ `error_negative`, e.g. $\Upsilon(10860)$); use the mean of the two, or the larger one, and say which; (iii) for some states the second (common) error is *not* in the error columns: $\chi_{bJ}(1P)$ has `error_positive` $= 0.42$ while `display_value_text` reads `9859.44+-0.42+-0.31`. Parse the display text when it contains two errors and put the second one into the common-systematic group of Example 9.1.

```
class PdgDatabase {
 public:
  explicit PdgDatabase(const std::string& path) {
    if (sqlite3_open_v2(path.c_str(), &db_, SQLITE_OPEN_READONLY, nullptr) != SQLITE_OK)
      throw std::runtime_error("cannot open " + path + ": " + sqlite3_errmsg(db_));
  }
  ~PdgDatabase() { sqlite3_close(db_); }
  PdgDatabase(const PdgDatabase&) = delete;
  PdgDatabase& operator=(const PdgDatabase&) = delete;

  std::optional<PdgValue> mass(long mcid) const {        // summary-table mass, MeV
    static const char* sql =
        "SELECT p.name, d.value, d.error_positive, d.error_negative, d.unit_text, "
        "       d.display_value_text, d.value_type "
        "FROM pdgparticle p "
        "JOIN pdgid i   ON i.parent_pdgid = p.pdgid AND i.data_type = 'M' "
        "JOIN pdgdata d ON d.pdgid_id = i.id "
        "WHERE p.mcid = ?1 AND d.in_summary_table = 1;";
    sqlite3_stmt* st = nullptr;
    if (sqlite3_prepare_v2(db_, sql, -1, &st, nullptr) != SQLITE_OK)
      throw std::runtime_error(sqlite3_errmsg(db_));
    sqlite3_bind_int64(st, 1, mcid);
    std::optional<PdgValue> out;
    while (sqlite3_step(st) == SQLITE_ROW) {
      auto text = [&](int c) {
        auto p = sqlite3_column_text(st, c);
        return p ? std::string(reinterpret_cast<const char*>(p)) : std::string();
      };
      out = PdgValue{text(0), text(4), text(5), text(6), sqlite3_column_double(st, 1),
                     sqlite3_column_double(st, 2), sqlite3_column_double(st, 3)};
    }
    sqlite3_finalize(st);
    return out;
  }
 private:
  sqlite3* db_ = nullptr;
};
```

Monte Carlo ids of the states you will use (from the same file): $\eta_c(1S)$ 441, J/ψ 443, χ_{c0} 10441, χ_{c1} 20443, h_c 10443, χ_{c2} 445, $\eta_c(2S)$ 100441, $\psi(2S)$ 100443, $\psi(3770)$ 30443; $\eta_b(1S)$ 551, $\Upsilon(1S)$ 553, $\chi_{b0,1,2}(1P)$ 10551, 20553, 555, $h_b(1P)$ 10553, $\eta_b(2S)$ 100551, $\Upsilon(2S)$ 100553, $\Upsilon_2(1D)$ 20555, $\chi_{b0,1,2}(2P)$ 110551, 120553, 100555, $h_b(2P)$ 110553, $\Upsilon(3S)$ 200553, $\Upsilon(4S)$ 300553. Thresholds: D^0 421, B^0 511. Some states (e.g. $\chi_{c1}(3872)$) have no `mcid`; select them by `pdgid` instead. The assignment table `mcid` $\rightarrow$ (channel, radial index) is physics input: keep it in the configuration

file, not in code.

Ceres with an analytic Jacobian (optional). Ceres needs a cost function; with eigenvalue-based residuals automatic differentiation is not available, so supply the Feynman–Hellmann Jacobian. Verified on Toy 2 (it returns $\hat{a} = \frac{5}{6}$, $\hat{b} = \frac{3}{2}$, $\sigma_a = 0.912871$, $\sigma_b = 0.707107$, $\rho = -0.774597$):

```
class WhitenedResiduals : public ceres::CostFunction {
 public:
  WhitenedResiduals(Chi2Problem& prob) : prob_(prob) {
    set_num_residuals(prob_.n_data());
    mutable_parameter_block_sizes()->push_back(prob_.n_free());     // one block: u
  }
  bool Evaluate(double const* const* params, double* residuals,
                double** jacobians) const override {
    Eigen::Map<const Vec> u(params[0], prob_.n_free());
    auto z = prob_.residuals(u);                                     // z_i = (y_i - mu_i)/s_i
    if (!z) return false;                                            // Ceres shrinks the step
    Eigen::Map<Vec>(residuals, prob_.n_data()) = *z;
    if (jacobians && jacobians[0]) {                                 // row-major N x p
      auto J = prob_.jacobian(u);  if (!J) return false;
      Eigen::Map<Eigen::Matrix<double, Eigen::Dynamic, Eigen::Dynamic, Eigen::RowMajor>>(
          jacobians[0], prob_.n_data(), prob_.n_free()) = *J;
    }
    return true;
  }
 private:
  Chi2Problem& prob_;
};
// problem.AddResidualBlock(new WhitenedResiduals(prob), nullptr, u.data());
// problem.SetParameterLowerBound(u.data(), a, lo);
// options.trust_region_strategy_type = ceres::LEVENBERG_MARQUARDT;
// chi^2 = 2 * summary.final_cost;  covariance: ceres::Covariance on the block {u, u}
```

(The class above is the verified toy version with the `Chi2Problem` calls substituted; the toy computed the residuals of a straight line in place of `prob_.residuals`.)

A unit test (GoogleTest). Verified with GoogleTest 1.18 and `gtest_discover_tests`:

```
TEST(Spectrum, ConstantShiftMovesEveryLevelByOne) {      // dM/dc = 1 because dH/dc = S
  Basis B = make_basis(0, {.n = 20, .r1 = 0.05, .rmax = 30.0});
  Mat S = overlap(B);
  Mat H = kinetic_nr(B, 0.75) - (2.0 / 3.0) * r_power(B, B, -1) + 0.18 * r_power(B, B, 1);
  Vec E0 = solve_gep(H, S).E, E1 = solve_gep(H + 0.1 * S, S).E;
  EXPECT_NEAR(((E1 - E0).array() - 0.1).abs().maxCoeff(), 0.0, 1e-10);
}
TEST(Spectrum, ReferenceTest1CornellGround) {              // §12.2, Test 1
  Basis B = make_basis(0, {.n = 20, .r1 = 0.05, .rmax = 30.0});
  Mat H = kinetic_nr(B, 0.75) - (2.0 / 3.0) * r_power(B, B, -1) + 0.18 * r_power(B, B, 1);
  EXPECT_NEAR(solve_gep(H, overlap(B)).E(0), 0.2164465, 5e-8);
}
```

(Compiled and passed exactly as printed, against thin adapters with these signatures, with GoogleTest 1.14; the CMake/`gtest_discover_tests` set-up was verified with GoogleTest 1.18 on your machine. The designated initializers `{.n = 20, ...}` require `BasisSpec` to remain an aggregate: no user-declared constructors.)

A run configuration. One JSON file fixes a run completely (the parser calls were verified with nlohmann-json 3.11.3):

```
{
  "system": "ccbar",
  "model": { "type": "NR", "screened": false, "spin": "contact_nonpert_LS_T_perturbative",
             "contact_in_all_L": true },
  "parameters": {
    "m1":      { "value": 1.48,  "free": true,  "transform": "log", "tied_to": null },
```

```
    "m2":      { "value": 1.48,  "free": false, "tied_to": "m1" },
    "alpha_s": { "value": 0.55,  "free": true,  "transform": "log" },
    "b":       { "value": 0.14,  "free": true,  "transform": "log" },
    "c":       { "value": 0.0,   "free": false },
    "sigma":   { "value": 1.1,   "free": true,  "transform": "log" }
  },
  "basis": { "default": { "n": 30, "r1": 0.02, "rmax": 30.0, "eta": 1e-12 } },
  "data": {
    "pdg_file": "data/pdg-2026.0.sqlite",
    "states": [ { "mcid": 441,    "channel": "1S0", "n": 0 },
                { "mcid": 443,    "channel": "3S1", "n": 0 },
                { "mcid": 10441,  "channel": "3P0", "n": 0 },
                { "mcid": 20443,  "channel": "3P1", "n": 0 },
                { "mcid": 10443,  "channel": "1P1", "n": 0 },
                { "mcid": 445,    "channel": "3P2", "n": 0 },
                { "mcid": 100441, "channel": "1S0", "n": 1 },
                { "mcid": 100443, "channel": "3S1", "n": 1 } ],
    "theory_floor": { "mode": "fit_to_chi2_per_dof_1", "start_MeV": 10.0 }
  },
  "fit": { "lambda0": 1e-3, "max_iter": 200, "jacobian": "feynman_hellmann" },
  "uncertainty": { "profile": ["m1", "alpha_s"], "bootstrap": 300,
                   "mcmc": { "chains": 4, "steps": 6000, "burn": 1000 } },
  "seed": 20261011,
  "output": "results/ccbar_modelA.json"
}
```

11.11 Testing strategy

level	what	reference
math	erfcx, $\mathcal{Q}_k$, I_j, quadrature moments, master integral	mpmath or Mathematica values (§13.4.12)
generated kernels	golden values, finite-difference derivatives, finiteness on the scan grid, spin table (`test_generated_kernels.cpp`, written by the generator)	30-digit Mathematica, rel. 10^{-12} (§13.5.10)
matrix elements	closed forms vs. `local` quadrature; $S_{ii} = 1$; symmetry; complex-range rules vs. quadrature	agreement $\leq 10^{-12}$ (harness: $\leq 10^{-13}$)
spin	factors vs. the table of §3.6; $(2J+1)$-weighted sums	exact
solver	hydrogen, oscillator, Test 1, nested monotonicity, $\partial M/\partial c = 1$	§12.2
models	Tests 1–4; virial theorem; limits $\mu_s \to 0$, GI $\to$ semi-relativistic	§12.2
derivatives	FH vs. central differences for every parameter	relative 10^{-6}
observables	Test 5	§12.2
fit	Toy 2 (exact); worked example (Test 6)	§9.6, §10.10
statistics	$\sum h_i = p$; LOO formula vs. refits; bootstrap vs. C on Toy 2; $\hat{R}$ on two identical chains	exact or statistical
regression	your current golden test (bb, cc spectra, rel. 10^{-8})	keep it

Golden files: store reference numbers in `tests/golden/*.json` together with how they were produced (Python script, Mathematica notebook, or these notes), so a failing test tells you which side to suspect.

11.12 Performance and parallelism

Measured in the harness: one Model A evaluation (four channels, $n = 30$) 0.3 ms on your machine; the full worked fit including the floor iteration and a central-difference Jacobian 0.10 s; `solve_gep` 0.2 ms for $n = 40$ and 1.3 ms for $n = 100$ (test container). Consequences:

- Model A/B fits are cheap; 10^5 MCMC steps take about half a minute single-threaded.
- The dominant cost is wasted work, not arithmetic: building a 40×40 matrix by quadrature took 7 s when the Gauss–Legendre rule was rebuilt for every element and 6 ms with the rule cached and rescaled. Cache quadrature rules and parameter-independent matrices.
- GI is more expensive (one p-quadrature per momentum factor and parameter change). Precompute what does not depend on the parameter being varied; at fixed (γ_k, σ_{12}) the coupling enters linearly (§10.9).
- Parallelize at the outermost level: channels within one evaluation, replicas in the bootstrap, chains in MCMC. Use OpenMP (`find_package(OpenMP)`, `OpenMP::OpenMP_CXX`) or `std::jthread`; Eigen's const operations are thread-safe, but give every thread its own `MatrixCache` and random engine (not tested here).

11.13 Milestones

Build in this order; each milestone has a test that tells you it is done.

M0 *Refactor.* Eigen types, normalized closed forms, canonical orthogonalization, a CMake build and GoogleTest tests. *Done when* your existing golden test passes in the new structure.

M1 *Model A/B spin-independent. Done when* Test 1 (all 18 numbers) and the hydrogen/oscillator tests pass.

M2 *Spin terms and channels* (contact, $\boldsymbol{L} \cdot \boldsymbol{S}$, tensor, coupled blocks, ${}^1L_L{}^{-3}L_L$). *Done when* Test 2 and the centre-of-gravity theorem pass.

M3 *Observables. Done when* Test 5 passes.

M4 *Derivatives. Done when* FH agrees with finite differences for every parameter of Models A/B.

M5 *PDG data and fitting* (LM, floor, covariance). *Done when* Toy 2 and the worked example (Test 6) are reproduced.

M6 *Uncertainties and validation* (profile, bootstrap, MCMC, LOO, AIC/BIC). *Done when* the numbers of §10.10 are reproduced within sampling error.

M7 *Basis optimization* and nested convergence. *Done when* $\sigma_{\text{num}} \leq 10^{-2} \min s_i$ is checked automatically for every fitted state.

M8 *Relativistic kinetic energy and smeared local GI terms. Done when* Test 3 passes.

M9 *Non-local GI terms* (closure and two-step). *Done when* Test 4 passes with both methods.

M10 *Full Models C and D. Done when* published GI masses are reproduced with the published parameters (after checking them against the paper, §12.3).

M11 *Complex-range Gaussians. Done when* the table of §7.10 is reproduced.

M12 *Running-coupling fit* (§10.9) and the error budget of every prediction.

12 Tests and reference numbers

The program design, the algorithms and the build are in §11. This section collects what you test the program against: model-independent checks and reference numbers. The milestones of §11.13 say which test closes which stage.

12.1 Unit tests that do not depend on the physics model

1. **Hydrogen-like**: $H = p^2/2m_r - \kappa/r$ has $E_n = -\frac{m_r\kappa^2}{2n^2}$, $n = n_r + l + 1$. With $m_r = 1$, $\kappa = \frac{2}{3}$: $E = -0.2\overline{2}, -0.0\overline{5}, -0.02469$ for $1S, 2S, 3S$ and the same values starting from $n = 2$ (P) and $n = 3$ (D).
2. **Harmonic oscillator**: $V = \frac{1}{2}m_r\omega^2 r^2$ gives $E = \omega(2n_r + l + \frac{3}{2})$ exactly (the matrix element of r^2 is analytic).
3. **Virial theorem** and **Feynman–Hellmann** (§4.1).
4. **Symmetry** $\max|\mathbf{H} - \mathbf{H}^T|$; **normalization** $\boldsymbol{c}^T\mathbf{S}\boldsymbol{c} = 1$.
5. **Variational monotonicity** in $n_{\max}$ (nested bases).
6. **Limits**: $\mu_s \to 0$ (B→A, D→C). In C, setting $\epsilon_x = -\frac{1}{2}$ (all $\Phi \to 1$), dropping the factor $(1 + p^2/E_1E_2)^{1/2}$ and letting $\sigma_{12} \to \infty$ ($\tau_k \to \gamma_k$, $\tilde{S} \to br + c$) leaves $E_1 + E_2 + G(r) + br + c$ plus the Breit–Fermi spin terms built from G: a semi-relativistic model you can test separately.
7. **Centre-of-gravity theorem** for the ${}^3P_J$ multiplet in first order (§3.6).
8. **Derivatives**: $\partial M/\partial c = 1$ for every level (because $\partial\mathbf{H}/\partial c = \mathbf{S}$); Feynman–Hellmann derivatives and (7.35) against central finite differences (§7.11).
9. **Statistics code**: $\sum_i h_i = p$ for the leverages; leave-one-out formula (10.19) against explicit refits; bootstrap and C agree for a nearly linear toy problem (§10.10).

12.2 Reference numbers for testing your code

The following numbers were computed with an independent implementation of the formulas in these notes and cross-checked with a second method (finite differences with Richardson extrapolation for local non-relativistic problems; a sine discrete-variable representation, in which any $f(p)$ is diagonal, for S-wave relativistic problems). The parameters are *illustrative*, not a fit to data.

Test 1: spin-independent Models A and B. $m_1 = m_2 = 1.5\,\mathrm{GeV}$ ($m_r = 0.75\,\mathrm{GeV}$), $\alpha_s = 0.5$ ($\kappa = \frac{2}{3}$), $b = 0.18\,\mathrm{GeV}^2$, $c = 0$; Model B with $\mu_s = 0.10\,\mathrm{GeV}$. Entries are $\mathcal{E} = M - 2m$ in GeV. Bases: $n_{\max} = 20$, $r_1 = 0.05$, $r_{\max} = 30\,\mathrm{GeV}^{-1}$; $n_{\max} = 40$, $r_1 = 0.02$, $r_{\max} = 50\,\mathrm{GeV}^{-1}$; threshold $\eta = 10^{-13}$.

	Model A (Cornell)			Model B (screened, $\mu_s = 0.1$)		
State	GEM $n = 20$	GEM $n = 40$	FD	GEM $n = 20$	GEM $n = 40$	FD
1S	0.2164465	0.2164462	0.2164461	0.1823122	0.1823119	0.1823119
2S	0.8673229	0.8672945	0.8672945	0.7227575	0.7227531	0.7227531
3S	1.3260325	1.3240175	1.3240174	1.0349623	1.0349507	1.0349507
1P	0.6993398	0.6993328	0.6993328	0.6062354	0.6062335	0.6062335
2P	1.1753880	1.1753865	1.1753865	0.9492210	0.9491315	0.9491315
3P	1.5780260	1.5719843	1.5719842	1.1895326	1.1887053	1.1887053
1D	1.0093770	1.0093090	1.0093090	0.8498083	0.8498066	0.8498066
2D	1.4216081	1.4212850	1.4212847	1.1135569	1.1134266	1.1134266
3D	1.7927322	1.7838549	1.7838539	1.3109787	1.3085364	1.3085363

Observe: low states converge already with $n_{\max} = 20$; the third radial excitation needs the larger basis (it is more extended). The screened spectrum is compressed at high n.

Test 2: expectation values and first-order spin splittings (Model A, same parameters, $n_{\max} = 40$ basis).

Quantity	Value	Comment
$1S$: $\sqrt{\langle r^2 \rangle}$	$1.981259\,\mathrm{GeV}^{-1} =$ $0.390956\,\mathrm{fm}$	rms radius
$1S$: $\lvert R(0)\rvert^2$	$1.3796907\,\mathrm{GeV}^3$	$\lvert\psi(0)\rvert^2 = \lvert R(0)\rvert^2/4\pi$
$1S$: $\langle\delta_\sigma\rangle$, $\sigma = 1.1\,\mathrm{GeV}$	$0.0371741\,\mathrm{GeV}^3$	Gaussian formula
$M(^3S_1) - M(^1S_0)$, smeared	$92.2753\,\mathrm{MeV}$	$\frac{32\pi\alpha_s}{9m^2}\langle\delta_\sigma\rangle$
$M(^3S_1) - M(^1S_0)$, point δ	$272.5315\,\mathrm{MeV}$	shows the strong σ dependence
$1S$: $2\langle T\rangle$	$0.84184803\,\mathrm{GeV}$	virial: $\kappa\langle r^{-1}\rangle + b\langle r\rangle = 0.84184807\,\mathrm{GeV}$
$1P$: $\langle r^{-3}\rangle$, $\langle r^{-1}\rangle$	$0.1120235\,\mathrm{GeV}^3$, $0.3812009\,\mathrm{GeV}$	
$1P$: $a = \frac{1}{2m^2}\langle 3\kappa r^{-3} - br^{-1}\rangle$	$34.5402\,\mathrm{MeV}$	spin–orbit strength
$1P$: $t = \frac{\kappa}{4m^2}\langle r^{-3}\rangle$	$8.2980\,\mathrm{MeV}$	tensor strength
$^3P_2$, $^3P_1$, $^3P_0$ shifts	$+31.2210$, -17.9441, $-102.2726\,\mathrm{MeV}$	$(2J+1)$-weighted sum $= 0$

Test 3: relativistic kinetic energy (S waves). $H = 2\sqrt{p^2 + m^2} - \kappa\frac{\mathrm{erf}(\tau r)}{r} + br$ with $m = 1.5$, $\kappa = \frac{2}{3}$, $\tau = 2.0\,\mathrm{GeV}$, $b = 0.18$. Eigenvalues M (GeV): GEM ($n_{\max} = 40$) and DVR agree to all digits shown:

	1S	2S	3S
M	3.2060876	3.8167930	4.2405517
NR comparison ($p^2/2m_r$, same V), $\mathcal{E}$	0.2498198	0.8870021	1.3401780

Test 4: a GI-type sandwiched Coulomb term (S waves).

$$H = 2E(p) + f(p)\Big[-\kappa\frac{\mathrm{erf}(\tau r)}{r}\Big]f(p) + br, \qquad f = \Big(1 + \frac{p^2}{E^2}\Big)^{1/2}, \quad E = \sqrt{p^2 + m^2},$$

same parameters as Test 3.

Method	1S	2S	3S
DVR (reference)	3.0335728	3.6863728	4.1247389
GEM $n_{\max} = 30$, closure	3.0335728	3.6863728	4.1247409
GEM $n_{\max} = 30$, two-step	3.0335727	3.6863728	4.1247408

Both GEM treatments of the non-local operator agree with the DVR to $\lesssim 2\times10^{-6}$ GeV.

Test 5: observables (Model A of Test 1, spin-independent, $n_{\max}=40$ basis). $e_Q=\frac{2}{3}$, $\alpha=1/137.036$, $M=$ model masses ($M_{1S}=3.2164462$, $M_{2S}=3.8672945$, $M_{1P}=3.6993328$ GeV), leading-order formulas of §8:

Quantity	Value	Comment
$\lvert R_{1S}(0)\rvert^2$, $\lvert R_{2S}(0)\rvert^2$	1.3796907, 0.8293325 GeV^3	$R(0)=\sum_n c_n N_{n0}$
$\Gamma(n^3S_1\to e^+e^-)$, $n=1,2$	12.6252, 5.2496 keV	$4\alpha^2e_Q^2\lvert R(0)\rvert^2/M^2$
f_V, $n=1,2$	640.01, 452.53 MeV	$f_V^2=3\lvert R(0)\rvert^2/\pi M$
$\Gamma(n^1S_0\to\gamma\gamma)$, $n=1,2$	16.8336, 6.9994 keV	same $\lvert R(0)\rvert^2$ and M
$\langle1P\lvert r\rvert2S\rangle$; $k(2S\to1P)$	2.614908 GeV^{-1}; 164.314 MeV	$k=(M_i^2-M_f^2)/2M_i$
$\Gamma(2^3S_1\to1^3P_J\gamma)$, $J=0,1,2$	14.5753, 43.7260, 72.8767 keV	ratio 1 : 3 : 5
$\lvert\langle1S\rvert r\rvert1P\rangle\rvert$; $k(1P\to1S)$	1.940281 GeV^{-1}; 451.370 MeV	
$\Gamma(1^3P_J\to1^3S_1\gamma)$	499.032 keV (all J)	$C_{fi}=\frac{1}{3}$
M1: k, $\langle j_0(kr/2)\rangle_{1S}$	90.9611 MeV, 0.99864790	$\Delta_{\rm hf}=92.2753$ MeV from Test 2
$\Gamma(1^3S_1\to1^1S_0\gamma)$	1.4426 keV	$\frac{4}{3}\frac{e_Q^2\alpha}{m^2}k^3\langle j_0\rangle^2$
$\langle v^2\rangle$ per quark, $1S$	0.2806	$\langle p^2\rangle/m^2$
$\langle p^4\rangle_{1S}$, $\langle p^4\rangle_{2S}$	1.16717, 1.43262 GeV^4	converged; the direct p^4 matrix gives 1.16371, 1.43054 at $n_{\max}=40$ (cusp, Property 4)

The $6j$ coefficients C_{fi} were checked against exact Wigner $6j$ symbols. For $\langle p^4\rangle$ of an exact eigenstate, $\langle p^4\rangle=\|p^2\psi\|^2=(2m_r)^2\langle(\mathcal{E}-V)^2\rangle$, which converges faster than the direct matrix (it needs only $\langle r^{-2}\rangle$, $\langle r^{-1}\rangle$, …).

Test 6: the full statistics pipeline. Reproduce the worked example of §10.10: $\sigma_{\rm th}=8.3$ MeV, $\hat{\boldsymbol\theta}=(1.4670, 0.5339, 0.1486, 1.130)$, the correlation matrix, the leverages and the predictions. Agreement to the digits shown tests the whole chain from matrix elements to error propagation.

12.3 Literature parameter sets (recalled from memory — verify before use)

	Godfrey–Isgur (Model C)	Barnes–Godfrey–Swanson NR (Model A, $c\bar c$)
quark masses (GeV)	$m_{u,d}=0.220$, $m_s=0.419$, $m_c=1.628$, $m_b=4.977$	$m_c=1.4794$
b (GeV^2)	0.18	0.1425
c (GeV)	-0.253	-0.2506 (if a constant is used)
α_s	Gaussian sum (5.7)	0.5461
smearing	$\sigma_0=1.80$ GeV, $s=1.55$	$\sigma=1.0946$ GeV
$\epsilon_c,\epsilon_{sov},\epsilon_t,\epsilon_{sos}$	-0.168, -0.035, $+0.025$, $+0.055$	—

These values are quoted from memory of the original tables and must be checked against the papers before you use them for physics results. For Models B and D take the parameters from the specific paper you reproduce (e.g. Li–Chao; Wang et al. for the modified GI model).

13 Deriving and checking everything yourself in Mathematica

This section is a guide for writing your own derivation notebooks: how to organize them, a small verification toolkit, the functions you need for each chapter, a recipe per chapter with code you can type in, and the traps. It does not replace the derivations by hand; it is the second half of a loop:

1. derive the result on paper;
2. state the claim in Wolfram Language, as an expression that must be `True` or `0`;
3. let Mathematica decide it symbolically, or, when that is hopeless, at random high-precision points;
4. spot-check numerically against an independent route (quadrature, a different formula, a limit);
5. optionally generate C++ code or golden numbers for the program (§13.4.12).

For the matrix elements the loop can also run the other way round: Mathematica derives them from nothing but the basis function and the operator, and you check its results (§13.5). Done for the whole catalogue of Models A–D, this is a generator that writes the C++ kernels with their tests (§13.5.10). Every code fragment below was run in Mathematica 15.0.1 (your installation) and the outputs quoted are the ones it produced.

13.1 Organizing the notebooks

One notebook per chapter.

notebook	main claims to verify	key functions
`02_TwoBody`	CM separation, radial equation, Frobenius exponents, units	`Series`, `Laplacian`, `UnitConvert`
`03_Interaction`	colour factors, Fourier transforms, Breit–Fermi currents, $\partial_i\partial_j V$, spin–angle factors	`PauliMatrix`, `KroneckerProduct`, `ClebschGordan`, `ThreeJSymbol`
`04_Models`	GI non-relativistic limit, Φ expansions, spin–orbit split	`Series`, `Simplify`
`05_AlphaS`	RGE solutions, APT and Richardson limits, Gaussian sum $\to$ erf	`DSolveValue`, `Series`, `Integrate`
`06_Smearing`	radial convolution, smeared $1/r$, r, $\mathrm{e}^{-\mu_s r}$, derivatives	`Integrate`, `NIntegrate`
`07_GEM`	master integral, S, T, all potential matrix elements, transfer lemma, momentum space, complex range, numerical GEM	`Integrate`, `Eigensystem`
`08_Observables`	colour factors, C_{fi}, widths	`SixJSymbol`
`09_10_Statistics`	toys of §9, LM by hand, bootstrap, MCMC	`LinearModelFit`, `RandomVariate`
`99_Golden`	reference numbers and generated code for C++	`CForm`, `Export`

Inside a notebook. Use cell styles as structure: *Title* (Alt+1), *Section* (Alt+4), *Subsection* (Alt+5), *Text* (Alt+7) for the statement of each claim with an equation number from these notes, then *Input* cells. One claim per subsection: statement (Text), construction (Input), `check[...]` (Input), numerical spot check (Input). Start every notebook with `ClearAll["Global‘*"]` and load your helpers.

Shared helpers in a package. Put `check`, `checkN` and recurring definitions (Pauli matrices, the GEM basis) into `mathematica/QGEMUtils.wl` and load them with

```
Get[FileNameJoin[{NotebookDirectory[], "QGEMUtils.wl"}]]
```

Keep the `.wl` files under git; treat `.nb` files as lab notebooks. Notebooks store outputs and layout, so their diffs are unreadable; a package diff is plain code.

The worked notebooks. `mathematica/derivations/` holds seven evaluated notebooks, D01–D07, that do the matrix-element part of the work of `07_GEM` for you, step by step: every section states what is wanted, how you would do it by hand, lets the engine `mathematica/QGEMEngine.wl` do it, says what the result means, and checks it against a numerical integral that uses no closed form. The table at the end of §13.5.10 lists their contents. Read them next to §7.5–§7.10; the exercises ask you to change one line and predict the result first. Your own `07_GEM` then only needs what they leave out: the derivations by hand that you want to check.

13.2 The verification toolkit

```
$checks = <||>;
check[label_String, expr_, assum_ : True] := Module[{v = FullSimplify[expr, assum]},
  $checks[label] = TrueQ[v] || v === 0;
  Print[label, ": ", If[$checks[label], "OK", Row[{"FAILED, left with ", v}]]]; v];

checkN[label_String, diff_, vars_List, n_Integer : 5] := Module[{vals},
  vals = Table[Abs[N[diff /. Thread[vars -> RandomReal[{1/3, 3}, Length[vars],
            WorkingPrecision -> 40]], 30]], {n}];
  $checks[label] = Max[vals] < 10^-20;
  Print[label, ": ", If[$checks[label], "OK", Row[{"FAILED, max |diff| = ", Max[vals]}]]];
  Max[vals]];
```

How to use them. Write every claim as *left side minus right side* (or as an equation) and hand it to `check` with the assumptions it needs, e.g. `check["eq. (7.10)", lhs - rhs, nu > 0 && l >= 0]`. At the end of a notebook, `$checks` lists every claim and whether it passed.

Why random-point checks are trustworthy. `FullSimplify` can fail to prove a true identity (it returns something that is not visibly zero). Then evaluate the difference at a few random points with 40-digit numbers. A non-zero analytic expression vanishes only on a set of measure zero, so it is essentially impossible for a wrong identity to give $|\text{diff}| < 10^{-20}$ at five random points; and with 40-digit arithmetic, round-off cannot fake a non-zero result. Use `checkN` whenever `check` does not return `OK`, before you start doubting the derivation.

Assumptions: local, not global. Do not set the global variable `$Assumptions`. Pass assumptions to each call instead: the third argument of `check`, the option `Assumptions ->` of `Integrate`, or a block `Assuming[cond, ...]`. A global `$Assumptions = m > 0 && ...` silently applies to every `Simplify` that a built-in function calls internally, including ones where your symbol `m` is a loop variable taking negative values: the angular-momentum code below then prints `Refine::fas` warnings (verified). Keep symbol names in assumptions distinct from iterator names.

13.3 Functions by task

task	functions	remarks
expansions	`Series`, `Normal`, `SeriesCoefficient`	introduce a bookkeeping parameter (`eps`) to expand in p/m
integrals	`Integrate[..., Assumptions -> ...]`, `Inactive[Integrate]`, `NIntegrate[..., WorkingPrecision -> 30]`	give exact inputs when you raise the precision
differential operators	`D`, `Grad`, `Laplacian[f, {r, th, ph}, "Spherical"]`	the coordinate chart handles the measure for you
equations	`Solve`, `Reduce`, `DSolveValue`	substitute the solution back to check it
linear algebra	`Eigensystem[{H, S}]`, `CholeskyDecomposition`, `LinearSolve`, `KroneckerProduct`, `PauliMatrix`, `LeviCivitaTensor`, `Tr`	generalized eigenvalues are *not* sorted ascending
angular momentum	`ClebschGordan`, `ThreeJSymbol`, `SixJSymbol`, `SphericalHarmonicY`	guard against unphysical arguments (messages)
special functions	`Gamma`, `Erf`, `Erfc`, `Hypergeometric1F1`, `SphericalBesselJ`	
statistics	`LinearModelFit`, `NonlinearModelFit`, `ChiSquareDistribution`, `Quantile`, `CDF`, `RandomVariate`, `MultinormalDistribution`	default fit errors are rescaled (§13.4.11)
units	`Quantity`, `UnitConvert`	
code generation	`CForm`, `Export[..., "RawJSON"]`	§13.4.12

13.4 Recipes, chapter by chapter

Each recipe gives the claims, a tested fragment and its output. Build the rest of each notebook in the same pattern.

13.4.1 Units and the two-body problem (§1, §2)

```
N@UnitConvert[Quantity[1, "ReducedPlanckConstant"] Quantity[1, "SpeedOfLight"]/
            Quantity[1, "Gigaelectronvolts"], "Femtometers"]          (* 0.197327 fm *)
check["CM separation", (P m1/(m1 + m2) + p)^2/(2 m1) + (P m2/(m1 + m2) - p)^2/(2 m2)
        - (P^2/(2 (m1 + m2)) + p^2 (m1 + m2)/(2 m1 m2))];
check["NR expansion of E1+E2", Normal@Series[Sqrt[k^2 + m1^2] + Sqrt[k^2 + m2^2], {k, 0, 4}]
        - (m1 + m2 + k^2 (m1 + m2)/(2 m1 m2) - k^4/8 (1/m1^3 + 1/m2^3)), m1 > 0 && m2 > 0];
lapCheck[l_] := With[{Y = SphericalHarmonicY[l, 1, th, ph]},
  Simplify[Laplacian[R[r] Y, {r, th, ph}, "Spherical"]/Y
           - (R''[r] + 2/r R'[r] - l (l + 1)/r^2 R[r])] === 0];
check["radial Laplacian, l = 1, 2, 3", And @@ (lapCheck /@ {1, 2, 3})];
```

All three print `OK`. Extension: the Frobenius analysis at the origin. Insert $u = r^s(1 + a_1 r)$ into the radial equation for $u = rR$, collect powers of r, and solve order by order:

```
eqU[u_] := -D[u, {r, 2}]/(2 mr) + (-kap/r + b r + l (l + 1)/(2 mr r^2) - En) u;
ser = Expand[Simplify[eqU[r^s (1 + a1 r)]/r^(s - 2), r > 0]];
indicial = Coefficient[ser, r, 0]                  (* (l + l^2 + s - s^2)/(2 mr): s = l+1 or s = -l *)
Solve[Coefficient[ser /. s -> l + 1, r, 1] == 0, a1]   (* a1 -> -(kap mr)/(1 + l): the cusp for l = 0 *)
```

13.4.2 Spin and angular momentum (§3.5)

Build everything as explicit matrices; then the spin–angle factors are matrix elements between explicitly coupled states. This verifies the operator-equivalent results of §3.5.1 independently.

```
sig = PauliMatrix /@ {1, 2, 3};  id2 = IdentityMatrix[2];
s1 = KroneckerProduct[#, id2] & /@ sig;  s2 = KroneckerProduct[id2, #] & /@ sig;
S1dotS2 = Sum[s1[[i]] . s2[[i]], {i, 3}]/4;
Eigenvalues[S1dotS2]                                   (* {-3/4, 1/4, 1/4, 1/4} *)

(* angular momentum j in the basis m = j, j-1, ..., -j *)
jz[j_] := DiagonalMatrix[Table[m, {m, j, -j, -1}]];
jp[j_] := Table[If[a == b + 1, Sqrt[j (j + 1) - b (b + 1)], 0], {a, j, -j, -1}, {b, j, -j, -1}];
jvec[j_] := {(jp[j] + ConjugateTranspose[jp[j]])/2, (jp[j] - ConjugateTranspose[jp[j]])/(2 I), jz[j]};

(* coupled states |(L S) J M> in the space |L mL> (x) |s1> (x) |s2> *)
chi[S_, mS_] := Flatten@Table[If[a + b == mS, ClebschGordan[{1/2, a}, {1/2, b}, {S, mS}], 0],
                              {a, 1/2, -1/2, -1}, {b, 1/2, -1/2, -1}];
ket[L_, S_, J_, M_] := Sum[If[Abs[M - mL] <= S, ClebschGordan[{L, mL}, {S, M - mL}, {J, M}]
      Flatten[KroneckerProduct[UnitVector[2 L + 1, L - mL + 1], chi[S, M - mL]]], 0], {mL, -L, L}];
LdotS[L_] := Sum[KroneckerProduct[jvec[L][[i]], (s1[[i]] + s2[[i]])/2], {i, 3}];
Table[Conjugate[ket[1, 1, J, J]] . LdotS[1] . ket[1, 1, J, J], {J, 0, 2}]   (* {-2, -1, 1} *)
```

The tensor operator without angular integrals. Matrix elements of $\hat{r}_q = C^{(1)}_q(\hat{r})$ between $|L'm'\rangle$ and $|Lm\rangle$ follow from the Gaunt formula,

$$\langle L'm'|C^{(1)}_q|Lm\rangle = (-1)^{m'}\sqrt{(2L'+1)(2L+1)}\begin{pmatrix} L' & 1 & L \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} L' & 1 & L \\ -m' & q & m \end{pmatrix}.$$

Products $\hat{r}_i\hat{r}_j$ only need intermediate $L'' = L \pm 1$, because $\hat{r}_j|Lm\rangle$ has only those components. This is exact and fast (0.3 s for everything below), whereas symbolic angular integrals take seconds per matrix element.

```
C1[Lp_, L_, q_] := C1[Lp, L, q] = Table[If[Abs[Lp - L] == 1 && m - mp + q == 0,
   (-1)^mp Sqrt[(2 Lp + 1) (2 L + 1)] ThreeJSymbol[{Lp, 0}, {1, 0}, {L, 0}]
   ThreeJSymbol[{Lp, -mp}, {1, q}, {L, m}], 0], {mp, Lp, -Lp, -1}, {m, L, -L, -1}];
rhat[Lp_, L_] := {(C1[Lp, L, -1] - C1[Lp, L, 1])/Sqrt[2], I (C1[Lp, L, -1] + C1[Lp, L, 1])/Sqrt[2],
                  C1[Lp, L, 0]};
rr[Lp_, L_] := Table[Sum[rhat[Lp, Lm][[i]] . rhat[Lm, L][[j]], {Lm, Select[{L - 1, L + 1}, # >= 0 &]}],
                     {i, 3}, {j, 3}];
S12[Lp_, L_] := S12[Lp, L] = Simplify[3 Sum[KroneckerProduct[rr[Lp, L][[i, j]],
     KroneckerProduct[sig[[i]], sig[[j]]]], {i, 3}, {j, 3}]
     - If[Lp == L, KroneckerProduct[IdentityMatrix[2 L + 1], 4 S1dotS2], 0]];
me[Lp_, L_, J_] := Simplify[Conjugate[ket[Lp, 1, J, J]] . S12[Lp, L] . ket[L, 1, J, J]];
{Table[me[1, 1, J], {J, 0, 2}], Table[me[2, 2, J], {J, 1, 3}], me[0, 2, 1],
 Table[me[3, 3, J], {J, 2, 4}], me[1, 3, 2]}
(* {{-4, 2, -2/5}, {-2, 2, -4/7}, 2 Sqrt[2], {-8/5, 2, -2/3}, 6 Sqrt[6]/5} *)
```

These are the diagonal values $-\frac{2(L+1)}{2L-1}$, 2, $-\frac{2L}{2L+3}$ for $J = L-1, L, L+1$ and the off-diagonal $\frac{6\sqrt{J(J+1)}}{2J+1}$ of §3.5. The antisymmetric spin–orbit operator for unequal masses works the same way:

```
LSminus[L_] := Sum[KroneckerProduct[jvec[L][[i]], (s1[[i]] - s2[[i]])/2], {i, 3}];
Table[Conjugate[ket[L, 0, L, L]] . LSminus[L] . ket[L, 1, L, L], {L, 1, 3}]
                                              (* {Sqrt[2], Sqrt[6], 2 Sqrt[3]} *)
```

13.4.3 Colour (§3.1, §8)

Mathematica has no built-in Gell-Mann matrices; type them in (and check the normalization).

```
gm = {{{0,1,0},{1,0,0},{0,0,0}}, {{0,-I,0},{I,0,0},{0,0,0}}, {{1,0,0},{0,-1,0},{0,0,0}},
      {{0,0,1},{0,0,0},{1,0,0}}, {{0,0,-I},{0,0,0},{I,0,0}}, {{0,0,0},{0,0,1},{0,1,0}},
```

```
      {{0,0,0},{0,0,-I},{0,I,0}}, {{1,0,0},{0,1,0},{0,0,-2}}/Sqrt[3]};
T = gm/2;
Table[Tr[T[[a]] . T[[b]]], {a, 8}, {b, 8}] == IdentityMatrix[8]/2      (* True *)
F1F2 = Sum[KroneckerProduct[T[[a]], -Conjugate[T[[a]]]], {a, 8}];     (* quark (x) antiquark *)
singlet = Flatten[IdentityMatrix[3]]/Sqrt[3];
Conjugate[singlet] . F1F2 . singlet                                    (* -4/3 *)
dabc = Table[2 Tr[(T[[a]] . T[[b]] + T[[b]] . T[[a]]) . T[[c]]], {a, 8}, {b, 8}, {c, 8}];
{Total[Flatten[dabc^2]], Total[Flatten[dabc^2]]/(16*3)}                (* {40/3, 5/18} *)
```

The other eight eigenvalues of `F1F2` are $+\frac{1}{6}$ (the octet). The last line is the three-gluon colour factor $\frac{5}{18}$.

13.4.4 Fourier transforms (§3.2, §5)

For radial functions do the angular integral by hand once and let Mathematica do the radial one. A Yukawa factor regulates the Coulomb and linear cases.

```
ft3[f_] := 4 Pi/q Integrate[r Sin[q r] f, {r, 0, Infinity}, Assumptions -> q > 0 && mu > 0];
invft3[g_] := 1/(2 Pi^2 r) Integrate[q Sin[q r] g, {q, 0, Infinity}, Assumptions -> r > 0 && gam > 0];
Simplify[ft3[Exp[-mu r]/r]]                                  (* 4 Pi/(mu^2 + q^2) *)
Limit[Simplify[ft3[r Exp[-mu r]]], mu -> 0]                  (* -8 Pi/q^4 *)
Simplify[invft3[4 Pi/q^2 Exp[-q^2/(4 gam^2)]]]               (* Erf[gam r]/r *)
```

13.4.5 Breit–Fermi (§3.4.3)

Expand the normalized current in the momenta by scaling them with `eps` and keeping second order:

```
sdot[v_] := Sum[v[[i]] sig[[i]], {i, 3}];
pv = {p1, p2, p3}; ppv = {k1, k2, k3}; qv = ppv - pv;              (* p, p', q = p' - p *)
Ep = Sqrt[m^2 + eps^2 pv . pv]; Epp = Sqrt[m^2 + eps^2 ppv . ppv];
j0 = (Sqrt[(Ep + m) (Epp + m)] IdentityMatrix[2]
      + eps^2 sdot[ppv] . sdot[pv]/Sqrt[(Ep + m) (Epp + m)])/(2 Sqrt[Ep Epp]);
target = (1 - qv . qv/(8 m^2)) IdentityMatrix[2] + I sdot[Cross[qv, pv]]/(4 m^2);
Simplify[(Normal@Series[j0, {eps, 0, 2}] /. eps -> 1) - target, m > 0]   (* zero matrix: eq. (j0) *)
```

And the decomposition of $\partial_i\partial_j V$ (3.12) for an arbitrary radial V:

```
xs = {x, y, z}; rr0 = Sqrt[x^2 + y^2 + z^2];
hess = Table[D[V[rr0], xs[[i]], xs[[j]]], {i, 3}, {j, 3}];
claim = Table[KroneckerDelta[i, j]/3 (V''[rr0] + 2/rr0 V'[rr0])
   + (xs[[i]] xs[[j]]/rr0^2 - KroneckerDelta[i, j]/3) (V''[rr0] - V'[rr0]/rr0), {i, 3}, {j, 3}];
Simplify[hess - claim] === ConstantArray[0, {3, 3}]              (* True *)
```

`Cross` does not accept vectors whose components are matrices (like $\boldsymbol{\sigma}$); for $\boldsymbol{\sigma}\times\boldsymbol{q}$ write the sum with `LeviCivitaTensor[3]` explicitly.

13.4.6 Running coupling (§5)

```
sol = DSolveValue[{a'[t] == -b0/(4 Pi) a[t]^2, a[t0] == a0}, a[t], t]
          (* 4 a0 Pi/(4 Pi + a0 b0 (t - t0)): one loop with t = ln Q^2 *)
Simplify[D[sol, t] + b0/(4 Pi) sol^2]                        (* 0: always substitute back *)
apt = 4 Pi/b0 (1/Log[x] + 1/(1 - x));                        (* x = Q^2/Lambda^2 *)
Normal@Series[apt, {x, 1, 0}]                                (* 2 Pi/b0: finite at the Landau pole *)
```

(`DSolveValue` printed two internal `Part::pkspec1` messages and still returned the right solution; substituting back is what makes such a result trustworthy. `Limit[apt, x -> 1]` also works but prints `Power::infy` while probing the point; `Series` is clean.)

13.4.7 Smearing (§6)

The radial convolution formula (6.9) as a function:

```
smear[f_] := sg/(Sqrt[Pi] r) Integrate[rp (f /. r -> rp) (Exp[-sg^2 (r - rp)^2] - Exp[-sg^2 (r + rp)^2]),
                                       {rp, 0, Infinity}, Assumptions -> r > 0 && sg > 0];
Simplify[smear[1/r], r > 0 && sg > 0]                          (* Erf[r sg]/r *)
rtil = Exp[-sg^2 r^2]/(Sqrt[Pi] sg) + (r + 1/(2 sg^2 r)) Erf[sg r];    (* eq. (smlin) *)
check["smeared r", smear[r] - rtil, r > 0 && sg > 0];          (* OK *)
```

For the screened exponential the symbolic integral is slow; compare the closed form (6.24) with `NIntegrate` at a point (they agree to all 30 digits):

```
smexp[r_, mu_, s_] := With[{a = mu/(2 s^2), b = mu/(2 s)},
   Exp[b^2]/(2 r) ((r - a) Exp[-mu r] Erfc[b - s r] + (r + a) Exp[mu r] Erfc[b + s r])];
With[{r0 = 7/10, mu0 = 3/10, s0 = 13/10},
  s0/(Sqrt[Pi] r0) NIntegrate[x Exp[-mu0 x] (Exp[-s0^2 (r0 - x)^2] - Exp[-s0^2 (r0 + x)^2]),
    {x, 0, Infinity}, WorkingPrecision -> 30] - smexp[r0, mu0, s0]]
```

The trap of §6 in one line: smearing $f' = 1$ gives 1, but `D[rtil, r] /. {r -> 1/2, sg -> 1}` is 0.358.

13.4.8 GEM: analytic matrix elements (§7.5, §7.6, §7.10)

```
Integrate[r^(2 l + 2 + k) Exp[-a r^2], {r, 0, Infinity}, Assumptions -> a > 0 && 2 l + 3 + k > 0]
                                                  (* a^((-3 - k - 2 l)/2) Gamma[3/2 + k/2 + l]/2 *)
nrm2[l_, nu_] := 2 (2 nu)^(l + 3/2)/Gamma[l + 3/2];
Rb[n_, l_] := Sqrt[nrm2[l, nu[n]]] r^l Exp[-nu[n] r^2];
ovl = Integrate[r^2 Rb[1, l] Rb[2, l], {r, 0, Infinity}, Assumptions -> nu[1] > 0 && nu[2] > 0 && l >= 0];
check["overlap", ovl/(2 Sqrt[nu[1] nu[2]]/(nu[1] + nu[2]))^(l + 3/2) - 1, nu[1] > 0 && nu[2] > 0 && l >=
    ↪ 0];
radialOp[f_, l_] := -D[f, {r, 2}] - 2/r D[f, r] + l (l + 1)/r^2 f;
check["kinetic lemma", radialOp[r^l Exp[-v r^2], l]/(r^l Exp[-v r^2]) - (2 v (2 l + 3) - 4 v^2 r^2)];
```

(The overlap check printed `FullSimplify::time` for a sub-step and still returned `OK`.) The recursions of §7.5 against quadrature:

```
Ij[0, a_, tau_] := ArcSinh[tau/Sqrt[a]];
Ij[j_, a_, tau_] := (2 (j - 1) Ij[j - 1, a, tau] + tau (a + tau^2)^(-j + 1/2))/((2 j - 1) a);
Table[N[Ij[j, 13/10, 7/3] - NIntegrate[(13/10 + t^2)^(-j - 1/2), {t, 0, 7/3}, WorkingPrecision -> 30],
    ↪ 20],
     {j, 1, 4}]                                                (* all 0 to 30 digits *)
Q[0, a_, mu_] := Sqrt[Pi/a]/2 Exp[mu^2/(4 a)] Erfc[mu/(2 Sqrt[a])];
Q[1, a_, mu_] := (1 - mu Q[0, a, mu])/(2 a);
Q[k_, a_, mu_] := ((k - 1) Q[k - 2, a, mu] - mu Q[k - 1, a, mu])/(2 a);
Table[N[Q[k, 13/10, 3/10] - NIntegrate[r^k Exp[-13/10 r^2 - 3/10 r], {r, 0, Infinity},
     WorkingPrecision -> 30], 20], {k, 0, 6}]                  (* all below 10^-28 *)
```

Momentum space, the transfer lemma and complex ranges:

```
hank[l_] := Simplify[Sqrt[2/Pi] Integrate[r^2 SphericalBesselJ[l, p r] r^l Exp[-v r^2], {r, 0, Infinity},
                    Assumptions -> v > 0 && p > 0], v > 0 && p > 0];
Table[Simplify[hank[l]/((2 v)^(-l - 3/2) p^l Exp[-p^2/(4 v)]), v > 0 && p > 0], {l, 0, 2}]   (* {1, 1, 1}
    ↪ *)

rt[r_, s_] := Exp[-s^2 r^2]/(Sqrt[Pi] s) + (r + 1/(2 s^2 r)) Erf[s r];   (* transfer lemma, l = 1, f = r
    ↪ *)
lhs = NIntegrate[r^4 Exp[-13/10 r^2] rt[r, 17/10], {r, 0, Infinity}, WorkingPrecision -> 30];
rhs = With[{s = 17/10}, -D[(s^2/(a + s^2))^(3/2) 1/(2 (a s^2/(a + s^2))^2), a] /. a -> 13/10];
lhs - rhs                                                      (* ~ 10^-29 *)

z = 13/10 (1 + I 7/10);                                        (* complex range *)
NIntegrate[r^3 Exp[-z r^2], {r, 0, Infinity}, WorkingPrecision -> 30] - Gamma[2]/(2 z^2)   (* ~ 10^-19 *)
ClearAll[z]                                    (* see the pitfall about stale globals *)
```

13.4.9 GEM: numerics, HUM and Feynman–Hellmann (§7.4–§7.11)

Build the matrices from the closed forms, in exact arithmetic, and solve with high precision:

```
basisNu[n_, r1_, rmax_] := Table[1/(r1 (rmax/r1)^((k - 1)/(n - 1)))^2, {k, n}];
mats[nus_, l_, mr_, kap_, b_] := Module[{N2, S, T, Vc, Vl, a},
  N2 = 2 (2 nus)^(l + 3/2)/Gamma[l + 3/2];  a = Outer[Plus, nus, nus];
  S  = Outer[Times, Sqrt[N2], Sqrt[N2]] Gamma[l + 3/2]/(2 a^(l + 3/2));
  T  = (2 l + 3)/mr Outer[Times, nus, nus]/a S;
  Vc = Outer[Times, Sqrt[N2], Sqrt[N2]] Gamma[l + 1]/(2 a^(l + 1));
  Vl = Outer[Times, Sqrt[N2], Sqrt[N2]] Gamma[l + 2]/(2 a^(l + 2));
  {S, T - kap Vc + b Vl, Vl}];
{S, H, Vl} = mats[basisNu[20, 1/20, 30], 0, 1, 2/3, 0];        (* hydrogen-like: mr = 1, kap = 2/3 *)
Sort@Eigenvalues[N[{H, S}, 40]] // Take[#, 3] &                (* -0.2222219, -0.0555555, -0.0246913 *)
Eigenvalues[N[{H, S}]] // Take[#, -3] &                        (* NOT the lowest three: unsorted! *)
```

Exact values are $-\frac{2}{9}$, $-\frac{1}{18}$, $-\frac{2}{81}$; the 3×10^{-7} error of the ground state is the Coulomb cusp (§7.12, Property 4). Canonical orthogonalization in a few lines:

```
canon[H_, S_, eta_] := Module[{s, U, keep, X}, {s, U} = Eigensystem[S];
   keep = Flatten@Position[s, x_ /; x > eta Max[s]];
   X = Transpose[U[[keep]]] . DiagonalMatrix[1/Sqrt[s[[keep]]]];
   {Sort@Eigenvalues[Transpose[X] . H . X], Length[keep]}];
canon[N[H], N[S], 10^-12]              (* {{-0.2222219, -0.0555555, -0.0246913, ...}, 20}: 20 kept *)
```

Feynman–Hellmann against a finite difference (Cornell, $m_r=\frac{3}{4}$, $\kappa=\frac{2}{3}$, $b=0.18$, 30 functions):

```
nus = basisNu[30, 1/50, 30];
lowest[b_] := Module[{S, H, Vl, vals, vecs}, {S, H, Vl} = mats[nus, 0, 3/4, 2/3, b];
   {vals, vecs} = Eigensystem[N[{H, S}, 40]];
   With[{k = Ordering[vals, 1][[1]]}, {vals[[k]], vecs[[k]], S, Vl}]];
{e0, c0, S0, V0} = lowest[18/100];  c0n = c0/Sqrt[c0 . S0 . c0];
{c0n . V0 . c0n, (lowest[18/100 + 10^-8][[1]] - lowest[18/100 - 10^-8][[1]])/(2 10^-8)}
                                   (* {1.770472817, 1.770472817}; E_1S = 0.2164461569 *)
```

HUM by experiment: repeat `lowest` with the nested sequence $n=6,11,21,41$ (fixed r_1, $r_{\max}$) and with $n=10,\dots,16$; reproduce the table of Property 2 in §7.12.

13.4.10 Observables (§8)

```
Cfi[Lp_, Jp_, L_, J_, S_] := Max[L, Lp] (2 Jp + 1) SixJSymbol[{Lp, Jp, S}, {J, L, 1}]^2;
{Table[Cfi[1, Jp, 0, 1, 1], {Jp, 0, 2}], Table[Cfi[0, 1, 1, J, 1], {J, 0, 2}],
 Table[Cfi[1, Jp, 2, 1, 1], {Jp, 0, 2}]}
(* {{1/9, 1/3, 5/9}, {1/3, 1/3, 1/3}, {2/9, 1/6, 1/90}} *)
```

These are $\frac{2J+1}{9}$ for ${}^3S_1\to{}^3P_J$, $\frac{1}{3}$ for ${}^3P_J\to{}^3S_1$, and the ${}^3D_1\to{}^3P_J$ values of the Key result.

13.4.11 Statistics (§9, §10)

Toy 2 with the built-in fitters. **Read this before using them**: by default `LinearModelFit` and `NonlinearModelFit` treat the weights as *relative* and rescale the parameter errors by $\sqrt{\chi^2_{\min}/\nu}$. For known errors you must say so with `VarianceEstimatorFunction -> (1 &)`.

```
data = {{0, 1}, {1, 2}, {2, 4}}; sy = {1, 1, 1};
fitDefault = LinearModelFit[data, x, x, Weights -> 1/sy^2];
fitKnown = LinearModelFit[data, x, x, Weights -> 1/sy^2, VarianceEstimatorFunction -> (1 &)];
{fitDefault["ParameterErrors"], fitKnown["ParameterErrors"], fitKnown["CorrelationMatrix"]}
(* {{0.3727, 0.2887}, {0.9129, 0.7071}, {{1, -0.7746}, {-0.7746, 1}}} *)
N@Table[Quantile[ChiSquareDistribution[k], {0.6827, 0.9545}], {k, 1, 4}]   (* Delta chi^2 table *)
```

The default errors are smaller by $\sqrt{\chi^2/\nu}=\sqrt{1/6}=0.408$: correct only if you believe the scatter of three points more than the stated errors. A second trap: `fit["MeanPredictionBands", ConfidenceLevel -> 0.6827]` uses Student-t quantiles with ν degrees of freedom even with

`VarianceEstimatorFunction -> (1 &)`; at $x = 3$ it returned the half-width 2.81 instead of $\sqrt{\boldsymbol{g}^T C \boldsymbol{g}} = 1.53$ ($t_{0.841,\nu=1} = 1.84$). Compute bands yourself from `fit["CovarianceMatrix"]`.

A Metropolis sampler and a parametric bootstrap in a dozen lines; the outputs quoted are for `SeedRandom[7]`:

```
chi2[{a_, b_}] := Total[((data[[All, 2]] - (a + b data[[All, 1]]))/sy)^2];
Cfit = fitKnown["CovarianceMatrix"];
prop = MultinormalDistribution[{0, 0}, 2.38^2/2 Cfit];
mh[start_, n_] := Module[{th = start, c2 = chi2[start], out = ConstantArray[0., {n, 2}],
                          acc = 0, new, c2new},
   Do[new = th + RandomVariate[prop]; c2new = chi2[new];
      If[RandomReal[] < Exp[-(c2new - c2)/2], th = new; c2 = c2new; acc++]; out[[i]] = th, {i, n}];
   {out, acc/n}];
SeedRandom[7]; {chain, acc} = mh[{0.8, 1.5}, 40000]; kept = chain[[5001 ;;]];
{N@acc, StandardDeviation[kept], Correlation[kept[[All, 1]], kept[[All, 2]]]}
                                                  (* {0.355, {0.926, 0.701}, -0.772} *)
boot = Table[With[{yb = (5/6 + 3/2 data[[All, 1]]) + RandomVariate[NormalDistribution[0, 1], 3]},
    LinearModelFit[Transpose[{data[[All, 1]], yb}], x, x, Weights -> 1/sy^2,
      VarianceEstimatorFunction -> (1 &)]["BestFitParameters"]], {2000}];
StandardDeviation[boot]              (* {0.900, 0.706}; sqrt(diag C) = {0.913, 0.707} *)
```

Extensions: the leverages are `Diagonal[X . Cfit . Transpose[X]]` with the design matrix `X = {{1,0},{1,1},{1,2}}`; compare with the leave-one-out refits of §9.12. Then reproduce the theory-floor iteration of the worked example with `NonlinearModelFit` around a function that calls your GEM solver.

13.4.12 Golden numbers and generated code

Your `mathematica/gem_matrix_elements.wl` already shows the right pattern: reduce every integrand to master integrals, compare with the formulas in the C++ code, and print `CForm`. Two additions make it a test generator for the C++ program:

```
expr = (2 l + 3)/mr nu1 nu2/(nu1 + nu2) (2 Sqrt[nu1 nu2]/(nu1 + nu2))^(l + 3/2);
ToString[CForm[expr]]
(* "(Power(2,1.5 + l)*(3 + 2*l)*nu1*nu2*Power(Sqrt(nu1*nu2)/(nu1 + nu2),1.5 + l))/(mr*(nu1 + nu2))" *)
golden = <|"test" -> "Coulomb 1S, mr = 1, kappa = 2/3", "basis" -> {20, 0.05, 30},
           "E" -> N[First@Sort@Eigenvalues[N[{H, S}, 40]], 16]|>;      (* E -> -0.2222219245613235 *)
Export[FileNameJoin[{NotebookDirectory[], "..", "tests", "golden", "coulomb_1s.json"}], golden, "RawJSON"]
```

`CForm` writes `Power`, `Sqrt` with capitals; replace them by `std::pow`, `std::sqrt` (your existing file does this with `StringReplace`). Export golden numbers with 16 significant digits; the C++ test compares at a tolerance that reflects the method (e.g. 10^{-12} for closed forms, 10^{-8} for whole spectra).

13.5 Letting Mathematica derive the matrix elements itself

The recipes above run in one direction: you derive, Mathematica checks. Here the direction is reversed. Mathematica is given only the basis function and the operator, and it derives every matrix element from scratch: normalization, overlap, kinetic energy, every potential, the spin-dependent terms, the momentum-space factors and the complex-range functions. You then check *its* results against the notes and against your own hand derivation. The same derived expressions then become numbers (the GEM spectrum) and C++ code, which closes the loop with the program of §11. Everything below was run on your Mathematica 15.0.1; timings are from that machine.

13.5.1 The engine

```
ClearAll["Global`*"];
```

```
phi[l_, nu_] := r^l Exp[-nu r^2];                    (* the only input: unnormalized basis function *)
Nrm[l_, nu_] := 1/Sqrt[Integrate[r^2 phi[l, nu]^2, {r, 0, Infinity},
                                 Assumptions -> nu > 0 && l > -3/2]];
R[l_, nu_] := Nrm[l, nu] phi[l, nu];                 (* normalized: Mathematica found N_nl itself *)

(* an operator is a function of (ket function, its l) *)
local[V_] := Function[{f, l}, V f];                  (* any potential acts by multiplication *)
kinetic = Function[{f, l}, -(D[f, {r, 2}] + 2/r D[f, r] - l (l + 1)/r^2 f)/(2 mr)];

deriveME[op_, l1_, l2_, cond_ : True] := Integrate[r^2 R[l1, nu1] op[R[l2, nu2], l2], {r, 0, Infinity},
   Assumptions -> nu1 > 0 && nu2 > 0 && l1 >= 0 && l2 >= 0 && cond];
pos = nu1 > 0 && nu2 > 0 && l >= 0;
```

Why it is built this way.

- Nothing about Gaussians is typed in except $r^l \mathrm{e}^{-\nu r^2}$. The normalization comes from `Integrate`, the kinetic operator from the radial Laplacian, every potential from its formula. No master integral, no recursion, no formula from the notes is used.
- The operator acts on the ket *before* the integral, so the kinetic matrix element is computed as $\int r^2 R_{n'}\,(T R_n)\,\mathrm{d}r$: the Galerkin definition of §7.4 itself, not a formula for it.
- `l1`, `l2` can be symbols (results for every l at once) or numbers. Bra and ket can have different l (transitions, tensor coupling).
- `cond` passes the extra assumptions a potential needs ($\tau > 0$, $\mu > 0$, convergence conditions).

13.5.2 Closed forms for every l at once

```
S  = Simplify[deriveME[local[1], l, l], pos]
Tn = FunctionExpand@Simplify[deriveME[kinetic, l, l, mr > 0]/S, pos && mr > 0]
Rk = FunctionExpand@Simplify[deriveME[local[r^k], l, l, 2 l + 3 + k > 0]/S, pos && 2 l + 3 + k > 0]
Gs = Simplify[deriveME[local[Exp[-be r^2]], l, l, be > 0]/S, pos && be > 0]
```

Output (about 15 s for all four):

$$S = 2^{\frac{3}{2}+l}(\nu_1\nu_2)^{\frac{3+2l}{4}}(\nu_1+\nu_2)^{-\frac{3}{2}-l}, \qquad \frac{T}{S} = \frac{(3+2l)\,\nu_1\nu_2}{m_r(\nu_1+\nu_2)},$$

$$\frac{\langle r^k\rangle}{S} = \frac{\Gamma\left(\frac{3}{2}+\frac{k}{2}+l\right)}{\Gamma\left(\frac{3}{2}+l\right)}(\nu_1+\nu_2)^{-k/2}, \qquad \frac{\langle \mathrm{e}^{-\beta r^2}\rangle}{S} = \left(\frac{\beta+\nu_1+\nu_2}{\nu_1+\nu_2}\right)^{-\frac{3}{2}-l}.$$

These are the formulas of §7.5, found by the program. `FunctionExpand` is what turns $\Gamma(l+\frac{5}{2})/\Gamma(l+\frac{3}{2})$ into $l+\frac{3}{2}$. The `FullSimplify::time` messages printed along the way come from inside `Integrate`; they do not affect the result.

13.5.3 Comparing with the notes

Now the check runs the other way: type the notes' formulas once and let Mathematica compare.

```
notes = <|"S" -> (2 Sqrt[nu1 nu2]/(nu1 + nu2))^(l + 3/2),
          "T/S" -> (2 l + 3)/mr nu1 nu2/(nu1 + nu2),
          "rk/S" -> Gamma[l + (k + 3)/2]/Gamma[l + 3/2] (nu1 + nu2)^(-k/2),
          "gauss/S" -> ((nu1 + nu2)/(nu1 + nu2 + be))^(l + 3/2)|>;
derived = <|"S" -> S, "T/S" -> Tn, "rk/S" -> Rk, "gauss/S" -> Gs|>;
Table[key -> FullSimplify[FunctionExpand[derived[key] - notes[key]],
         pos && mr > 0 && be > 0 && 2 l + 3 + k > 0], {key, Keys[notes]}]
(* {"S" -> 0, "T/S" -> 0, "rk/S" -> 0, "gauss/S" -> 0} *)
```

If a difference is not 0, decide who is wrong: evaluate both at a random point and compare with `NIntegrate` of the original integrand (the `checkN` pattern of §13.2).

13.5.4 Potentials whose closed form depends on l

For $\operatorname{erf}(\tau r)/r$ Mathematica finds a result for symbolic l (in 2.4 s), but in terms of a hypergeometric function, ${}_2F_1\!\left(\frac{1}{2}, \frac{3}{2}+l; \frac{3}{2}; -\frac{\tau^2}{\nu_1+\nu_2}\right)$. For fixed l it gives algebraic expressions, which is what you implement. Derive a table for $l = 0, 1, 2$ and compare with the recursion of §7.5:

```
erfL = Table[Simplify[deriveME[local[Erf[tau r]/r], ll, ll, tau > 0]/(S /. l -> ll), pos && tau > 0],
             {ll, 0, 2}]
Ij[0, a_, tau_] := ArcSinh[tau/Sqrt[a]];
Ij[j_, a_, tau_] := (2 (j - 1) Ij[j - 1, a, tau] + tau (a + tau^2)^(-j + 1/2))/((2 j - 1) a);
Table[FullSimplify[erfL[[ll + 1]] - 2 (nu1 + nu2)^(ll + 3/2)/Sqrt[Pi] Ij[ll + 1, nu1 + nu2, tau],
                   pos && tau > 0], {ll, 0, 2}]
    ↪ (* {0, 0, 0} *)
```

Mathematica's own result for $l = 1$, for example, is

$$\left.\frac{\langle \operatorname{erf}(\tau r)/r \rangle}{S}\right|_{l=1} = \frac{2\tau\sqrt{\nu_1+\nu_2}\,(3\nu_1+3\nu_2+2\tau^2)}{3\sqrt{\pi}\,(\nu_1+\nu_2+\tau^2)^{3/2}}.$$

The screened potentials (B, D) work the same way; Mathematica produces erfc and exponentials (7 s for $l = 0, 1, 2$ of both integrals), which agree with the $\mathcal{Q}_k$ recursion:

```
Qn[0, a_, mu_] := Sqrt[Pi/a]/2 Exp[mu^2/(4 a)] Erfc[mu/(2 Sqrt[a])];
Qn[1, a_, mu_] := (1 - mu Qn[0, a, mu])/(2 a);
Qn[k_, a_, mu_] := ((k - 1) Qn[k - 2, a, mu] - mu Qn[k - 1, a, mu])/(2 a);
NN[ll_] := Nrm[ll, nu1] Nrm[ll, nu2];
scr  = Table[Simplify[deriveME[local[Exp[-mu r]], ll, ll, mu > 0], pos && mu > 0], {ll, 0, 2}];
scrR = Table[Simplify[deriveME[local[Exp[-mu r]/r], ll, ll, mu > 0], pos && mu > 0], {ll, 0, 2}];
{Table[FullSimplify[scr[[ll + 1]] - NN[ll] Qn[2 ll + 2, nu1 + nu2, mu], pos && mu > 0], {ll, 0, 2}],
 Table[FullSimplify[scrR[[ll + 1]] - NN[ll] Qn[2 ll + 1, nu1 + nu2, mu], pos && mu > 0], {ll, 0, 2}]}
                                                                    (* {{0, 0, 0}, {0, 0, 0}} *)
```

For the smeared potentials of GI, let Mathematica first derive the smeared function (the radial convolution of §6) and then its matrix element; compare with the transfer lemma (7.22):

```
smear[f_] := sg/(Sqrt[Pi] r) Integrate[rp (f /. r -> rp) (Exp[-sg^2 (r - rp)^2] - Exp[-sg^2 (r + rp)^2]),
                                       {rp, 0, Infinity}, Assumptions -> r > 0 && sg > 0];
rtil  = Simplify[smear[r], r > 0 && sg > 0];                    (* derives eq. (smlin) *)
smLin = Table[Simplify[deriveME[local[rtil], ll, ll, sg > 0], pos && sg > 0], {ll, 0, 1}];
transfer[ll_, If_] := Module[{aa}, (-1)^ll D[(sg^2/(aa + sg^2))^(3/2) If[aa sg^2/(aa + sg^2)], {aa, ll}]
                                   /. aa -> nu1 + nu2];
Ir[ap_] := Integrate[r^3 Exp[-ap r^2], {r, 0, Infinity}, Assumptions -> ap > 0];
Table[FullSimplify[smLin[[ll + 1]] - NN[ll] transfer[ll, Ir], pos && sg > 0], {ll, 0, 1}]   (* {0, 0} *)
```

13.5.5 Different l in bra and ket

Transitions ($\langle f|r|i\rangle$) and the tensor coupling (S–D) need $l' \neq l$; the same `deriveME` handles them. The check uses the master integral with $\ell = \frac{l+l'}{2}$:

```
masterN[l1_, l2_, k_] := Nrm[l1, nu1] Nrm[l2, nu2] Gamma[(l1 + l2)/2 + (k + 3)/2]/
                         (2 (nu1 + nu2)^((l1 + l2)/2 + (k + 3)/2));
Simplify[{deriveME[local[r], 1, 0], deriveME[local[r^-3], 0, 2], deriveME[local[r^-3], 1, 1]}
         - {masterN[1, 0, 1], masterN[0, 2, -3], masterN[1, 1, -3]}, nu1 > 0 && nu2 > 0]   (* {0, 0, 0} *)
```

13.5.6 Spin-dependent matrix elements, derived end to end

Three things are derived here: the radial functions, from the potentials by differentiation (the Breit–Fermi formulas (3.17)–(3.19) for equal masses); their matrix elements, by the engine; and the spin–angle factors, from explicit coupled states. For the last, use the code of §13.4.2 (`ket`, `S1dotS2`, `jvec`, `rr`) with one function per operator:

```
opMat["SS", Lp_, L_] := If[Lp == L, KroneckerProduct[IdentityMatrix[2 L + 1], S1dotS2], 0];
opMat["LS", Lp_, L_] := If[Lp == L, Sum[KroneckerProduct[jvec[L][[i]], (s1[[i]] + s2[[i]])/2], {i, 3}],
    ↪ 0];
opMat["T", Lp_, L_]  := 3 Sum[KroneckerProduct[rr[Lp, L][[i, j]], KroneckerProduct[sig[[i]], sig[[j]]]],
                           {i, 3}, {j, 3}] -
                        If[Lp == L, KroneckerProduct[IdentityMatrix[2 L + 1], 4 S1dotS2], 0];
factor[op_, {Lp_, Sp_}, {L_, Sp_}, J_] :=
  Simplify[Conjugate[ket[Lp, Sp, J, J]] . opMat[op, Lp, L] . ket[L, Sp, J, J]];
Table[{op, factor[op, {1, 1}, {1, 1}, #] & /@ {0, 1, 2}}, {op, {"SS", "LS", "T"}}]
(* {{"SS", {1/4, 1/4, 1/4}}, {"LS", {-2, -1, 1}}, {"T", {-4, 2, -2/5}}} *)

VV = -4/3 als Erf[tau r]/r;  VS = bb r;                          (* smeared Coulomb + linear *)
lap[f_] := D[r^2 D[f, r], r]/r^2;
Wss = Simplify[2/(3 m^2) lap[VV]];                                  (* contact, eq. (HSS) *)
Wls = Simplify[3/(2 m^2) D[VV, r]/r - D[VS, r]/(2 m^2 r)];        (* spin-orbit, eq. (HLS), m1 = m2 *)
Wt  = Simplify[1/(12 m^2) (D[VV, r]/r - D[VV, {r, 2}])];          (* tensor, eq. (HT) *)
cond = tau > 0 && m > 0 && als > 0;
rad = Association@Table[w -> Simplify[deriveME[local[ToExpression["W" <> w]], 1, 1, cond],
                                      nu1 > 0 && nu2 > 0 && cond], {w, {"ss", "ls", "t"}}];
HSD[J_] := factor["SS", {1, 1}, {1, 1}, J] rad["ss"] + factor["LS", {1, 1}, {1, 1}, J] rad["ls"] +
           factor["T", {1, 1}, {1, 1}, J] rad["t"];        (* operator at the END of the line! *)
```

Everything takes 5 s, and the result is a closed-form $\langle R_{n'1}|H_{SD}|R_{n1}\rangle$ for each ${}^3P_J$. For $J=0$ Mathematica returns

$$\frac{32\alpha_s\tau^3(\nu_1\nu_2)^{5/4}\sqrt{2/\pi}}{9m^2(\nu_1+\nu_2+\tau^2)^{5/2}}-\frac{64\alpha_s\tau^5(\nu_1\nu_2)^{5/4}\sqrt{2/\pi}}{9m^2(\nu_1+\nu_2)(\nu_1+\nu_2+\tau^2)^{5/2}}+\frac{16(\nu_1\nu_2)^{5/4}\sqrt{2/\pi}}{3m^2(\nu_1+\nu_2)^2}\Big(b-\frac{4\alpha_s(\nu_1+\nu_2)\tau^3}{(\nu_1+\nu_2+\tau^2)^{3/2}}\Big)$$

(`als` is α_s in the code). Check: at $\nu_1=1.3$, $\nu_2=0.7$, $\tau=1.7$, $m=1.5$, $\alpha_s=0.5$, $b=0.18$ the closed forms give -0.786485, -0.216756, $+0.381060$ for $J=0,1,2$, equal to direct `NIntegrate` of the full operator to 30 digits. Along the way Mathematica also derives $\nabla^2\frac{\operatorname{erf}(\tau r)}{r}=-\frac{4\tau^3}{\sqrt{\pi}}\mathrm{e}^{-\tau^2r^2}$ and the tensor function T_τ of (6.20): print `Wss` and `Wt` and compare with §6.

13.5.7 Momentum space: relativistic kinetic energy and GI factors

Let Mathematica derive the momentum-space basis, check that it reproduces S and T, and then find the reduced one-dimensional integral for any $f(p)$ by the substitution $p=\sqrt{t/\beta}$:

```
Rp[l_, nu_] := Simplify[Sqrt[2/Pi] Integrate[r^2 SphericalBesselJ[l, p r] R[l, nu], {r, 0, Infinity},
                       Assumptions -> nu > 0 && p > 0], nu > 0 && p > 0];
Table[{Simplify[Integrate[p^2 Rp[ll, nu1] Rp[ll, nu2], {p, 0, Infinity}, Assumptions -> nu1 > 0 && nu2 >
    ↪ 0]
              - (S /. l -> ll), nu1 > 0 && nu2 > 0],
      Simplify[Integrate[p^2 Rp[ll, nu1] p^2/(2 mr) Rp[ll, nu2], {p, 0, Infinity},
                         Assumptions -> nu1 > 0 && nu2 > 0]
              - (Tn S /. l -> ll), nu1 > 0 && nu2 > 0 && mr > 0]}, {ll, 0, 2}]      (* all {0, 0} *)
integrand[ll_] := p^2 Rp[ll, nu1] Rp[ll, nu2] f[p];
ClearAll[t];
Table[Simplify[(((integrand[ll] /. p -> Sqrt[t/be]) D[Sqrt[t/be], t]) /. be -> (nu1 + nu2)/(4 nu1 nu2))
        / ((S /. l -> ll) t^(ll + 1/2) Exp[-t] f[Sqrt[t 4 nu1 nu2/(nu1 + nu2)]]),
      nu1 > 0 && nu2 > 0 && t > 0], {ll, 0, 2}]                                    (* {2/Sqrt[
    ↪ Pi], 4/(3 Sqrt[Pi]), 8/(15 Sqrt[Pi])} *)
```

The last output is $1/\Gamma(l+\frac{3}{2})$: Mathematica has derived $\langle f(p)\rangle=\frac{S}{\Gamma(l+3/2)}\int_0^\infty t^{l+1/2}\mathrm{e}^{-t}f(\sqrt{t/\beta})\,\mathrm{d}t$, the generalized Gauss–Laguerre form of §7.6 with weight exponent $\alpha=l+\frac{1}{2}$. For $f=E_1+E_2$ or the GI factors there is no closed form; this reduction *is* the result to implement. (The `ClearAll[t]` matters: an earlier `t = AbsoluteTiming[...]` left a number in `t`, and the substitution then failed with `D::ivar`.)

13.5.8 Complex-range functions

Give Mathematica the real functions c_n, s_n of §7.10 and let it integrate them directly; compare with the rule (7.33) built from the complex master integral:

```
cfun[l_, e_, w_] := r^l Exp[-e r^2] Cos[w e r^2];
sfun[l_, e_, w_] := r^l Exp[-e r^2] Sin[w e r^2];
ca = e1 > 0 && e2 > 0 && w > 0;
direct = {Integrate[r^2 cfun[0, e1, w] cfun[0, e2, w], {r, 0, Infinity}, Assumptions -> ca],
          Integrate[r^2 sfun[0, e1, w] sfun[0, e2, w], {r, 0, Infinity}, Assumptions -> ca],
          Integrate[r^2 cfun[1, e1, w] sfun[1, e2, w]/r, {r, 0, Infinity}, Assumptions -> ca]};
B[l_, k_, z1_, z2_] := Gamma[l + (k + 3)/2]/(2 (z1 + z2)^(l + (k + 3)/2));
z1 = e1 (1 + I w); z2 = e2 (1 + I w);
rule = {1/2 Re[B[0, 0, z1, z2] + B[0, 0, z1, Conjugate[z2]]],
        1/2 Re[B[0, 0, z1, Conjugate[z2]] - B[0, 0, z1, z2]],
        -1/2 Im[B[1, -1, z1, z2] - B[1, -1, z1, Conjugate[z2]]]};
N[(direct - rule) /. {e1 -> 13/10, e2 -> 7/10, w -> 1/2}, 20]          (* zero to 70 digits *)
```

(14 s.) A symbolic comparison leaves expressions with `Arg` that `FullSimplify` does not reduce to zero; for complex arguments use the numerical comparison at a few points.

13.5.9 Closing the loop: numbers, C++, and back

From derived formulas to a spectrum. Turn the derived expressions into functions and run the GEM. With the Cornell parameters of Test 1 this reproduces the reference numbers, so the whole chain from $r^l e^{-\nu r^2}$ to the spectrum was derived by Mathematica:

```
toFun[expr_] := Function[{ll, a, b}, Evaluate[expr /. {l -> ll, nu1 -> a, nu2 -> b}]];
{fS, fT, fRk} = toFun /@ {S, Tn S, Rk S};
basisNu[n_, r1_, rmax_] := Table[1/(r1 (rmax/r1)^((k - 1)/(n - 1)))^2, {k, n}];
spectrum[ll_, nus_, pars_] := Module[{Sm, Hm},
  Sm = Outer[fS[ll, #1, #2] &, nus, nus];
  Hm = Outer[fT[ll, #1, #2] - kap (fRk[ll, #1, #2] /. k -> -1)
             + bb (fRk[ll, #1, #2] /. k -> 1) &, nus, nus];
  Sort@Eigenvalues[N[{Hm /. pars, Sm}, 30]]];
pars = {mr -> 3/4, kap -> 2/3, bb -> 18/100};
{Take[spectrum[0, basisNu[20, 1/20, 30], pars], 3], Take[spectrum[1, basisNu[20, 1/20, 30], pars], 2]}
(* {{0.2164465, 0.8673229, 1.3260325}, {0.6993398, 1.1753880}}: Test 1, Model A *)
```

From derived formulas to C++.

```
cpp[expr_] := StringReplace[ToString[CForm[expr]], {"Power(" -> "std::pow(", "Sqrt(" -> "std::sqrt(",
    "Pi" -> "std::numbers::pi", "Erfc(" -> "std::erfc(", "Erf(" -> "std::erf(",
    "Gamma(" -> "std::tgamma("}];
cpp[erfL[[2]]]
(* "(2*std::sqrt(nu1 + nu2)*tau*(3*nu1 + 3*nu2 + 2*std::pow(tau,2)))/
    (3.*std::sqrt(std::numbers::pi)*std::pow(nu1 + nu2 + std::pow(tau,2),1.5))" *)
```

`TeXForm[erfL[[2]]]` gives the same expression in LaTeX, to paste next to your hand derivation.

And back: checking a formula from your C++ code. Your `src/gem_solver.cpp` uses the basis $u = rR$, un-normalized integrals $J(p, a) = \Gamma(\frac{p+1}{2})/(2a^{(p+1)/2})$ and divides by the norms at the end. Type its formulas in and compare them with the derivation:

```
Jm[p_, a_] := Gamma[(p + 1)/2]/(2 a^((p + 1)/2));
cppT = 1/(2 mr) ((l + 1)^2 Jm[2 l, nu1 + nu2] - 2 (l + 1) (nu1 + nu2) Jm[2 l + 2, nu1 + nu2]
                 + 4 nu1 nu2 Jm[2 l + 4, nu1 + nu2]) + l (l + 1)/(2 mr) Jm[2 l, nu1 + nu2];
cppNorm[nu_] := Sqrt[Jm[2 l + 2, 2 nu]];
{FullSimplify[FunctionExpand[cppT/(cppNorm[nu1] cppNorm[nu2]) - Tn S], pos && mr > 0],
 FullSimplify[FunctionExpand[-(4/3) as Jm[2 l + 1, nu1 + nu2]/(cppNorm[nu1] cppNorm[nu2])
                             - (-(4/3) as Rk S /. k -> -1)], pos]}                    (* {0, 0} *)
```

Your kinetic and Coulomb formulas agree with the derivation for every l. Do the same for each new matrix element you add to the C++ program, then export the derived expressions at a few random points as golden numbers (§13.4.12).

13.5.10 The generator: all kernels of Models A–D, derived, checked and written as C++

The pages above derive one kernel at a time. The script `mathematica/generate_kernels.wls` does it for the whole catalogue of the four models and writes the C++ header that the program of §11 includes; seven notebooks explain every derivation step by step. All of it is in your repository:

file	content
`mathematica/QGEMEngine.wl`	the engine of §13.5.1, plus the transfer lemmas (7.22), (7.23), the screened moments as an abstract function `Qm` with $\partial_a \mathcal{Q}_k = -\mathcal{Q}_{k+2}$, $\partial_\mu \mathcal{Q}_k = -\mathcal{Q}_{k+1}$, spin–angle factors, momentum space, numerical checks, C++ emission
`mathematica/generate_kernels.wls`	the generator (run with `wolframscript`; about 2.5 min)
`mathematica/QGEMKernels.cmake`	CMake option `QGEM_REGENERATE_KERNELS`
`src/generated/qgem_kernels.hpp`	the kernels, `namespace qgem::gen` (generated, committed)
`tests/test_generated_kernels.cpp`	GoogleTest for them (generated, committed)
`mathematica/generated_kernels_report.txt`	what was derived, how long it took, every check and scan number
`mathematica/generated_kernels.wl`	every derived expression, readable with `Get`
`mathematica/derivations/D01..D07*.nb`	the derivation notebooks, evaluated

Four steps per C++ function.

1. *Derive* with the engine, from the definitions of the basis function and the operator: `Integrate` for local potentials, the transfer lemmas for smeared ones, `D` for parameter derivatives. Nothing from these notes is typed in. Local kernels are brought to the form $N_{l_1}(\nu_1) N_{l_2}(\nu_2)\, J(a)$, $a = \nu_1 + \nu_2$; the script verifies that J depends on a only.
2. *Check* against an independent numerical integral of the *original* integrand at a random point, in 30-digit arithmetic (tolerance 10^{-14}). Derivatives are checked by differentiating the integrand, not the closed form. Smeared kernels are checked in two stages: the smeared function, derived by the radial convolution (or given in the catalogue entry when `Integrate` cannot do that integral, as for $1/r^2$ in D07), against the numerical convolution integral at five radii, then the matrix element. The two spin–orbit derivations (transfer lemma and integration by parts) must agree. Complex-range kernels are checked at complex exponents.
3. *Scan* the emitted expression in machine precision on a grid ($\nu \in [10^{-3}, 10^{4}]\,\mathrm{GeV}^2$, $\tau \in [0.2, 20]$, $\sigma \in [0.5, 10]$, $\mu_s \in [0.01, 0.3]$, $\beta \in [0.05, 100]\,\mathrm{GeV}$ or GeV^2 as appropriate), against 30-digit values. The run fails if the error divided by the natural size of the matrix element ($\int r^2 |R_1 R_2|\,\mathrm{d}r$ times a typical value of the operator) exceeds 10^{-12} anywhere.
4. *Emit* C++ (`CForm` plus replacements) and golden values for the tests. If any check or scan failed, nothing is written and the exit code is 1. The random check and golden points are seeded from each function's name, so two runs write identical files, and adding a kernel changes only the lines of the new kernel.

The catalogue. $l \le 4$ by default (`--lmax` changes it):

C++ name (`qgem::gen::`)	operator	l	derivatives
`norm`, `overlap`, `kinetic`	1, $p^2/2m_r$	any l	m_r
`r_power(l1,l2,k,...)`	r^k	any l_1, l_2	—
`gaussian(l1,l2,...)`	$\mathrm{e}^{-\beta r^2}$	any l_1, l_2	β
`contact_smeared`, `erf_contact`	δ_σ, $\nabla^2 \frac{\mathrm{erf}(\tau r)}{r}$	any l	σ, τ
`erf_over_r`	$\mathrm{erf}(\tau r)/r$	0..4	τ
`erf_spinorbit`, `erf_tensor`	g'/r, $g'/r - g''$	1..4; tensor also $(0,2),(1,3),(2,4)$	τ
`screened_exp`, `screened_exp_over_r`	$\mathrm{e}^{-\mu_s r}$, $\mathrm{e}^{-\mu_s r}/r$	0..4	μ_s
`smeared_linear`, `smeared_linear_so`	$\tilde{r}$, $\tilde{r}\,'/r$	0..4, 1..4	σ
`smeared_exp`, `smeared_exp_so`	$\widetilde{\mathrm{e}^{-\mu_s r}}$, $\widetilde{\mathrm{e}^{-\mu_s r}}\,'/r$	0..4, 1..4	σ, μ_s
`momentum_prefactor`, `momentum_beta`	reduction of $\langle f(p) \rangle$ (§7.6)	any l	—
`kin_rel`, `coulomb_factor`, `phi_factor`	E_1+E_2, $(1+\frac{p^2}{E_1 E_2})^{1/2}$, $\Phi_{ab}^{(\epsilon)}$	—	masses, ϵ
`gi_tau`, `gi_sigma12`	τ_k, σ_{12}	—	σ, γ; m_i, σ_0, s
`spin_factor(op,L',S',L,S,J)`	$\boldsymbol{S}_1 \cdot \boldsymbol{S}_2$, $\boldsymbol{L} \cdot \boldsymbol{S}$, S_{12}, $\boldsymbol{L} \cdot (\boldsymbol{S}_1 - \boldsymbol{S}_2)$	$L \leq 4$	—
`complex_range::B_*`	overlap, kinetic, r^k, Gaussian, $\mathrm{erf}(\tau r)/r$, $\tilde{r}$	any l or 0..4	—

In total 142 functions, 70 of them parameter derivatives (`d_<kernel>_d<param>`), and 68 spin–angle factors. The worst check deviation is 7×10^{-20} (tolerance 10^{-14}); the worst machine-precision error over all scans is 7×10^{-15} of the natural size (tolerance 10^{-12}). Relative errors reach 10^{-11} only where a derivative passes through zero ($\partial_\epsilon \Phi$ at $p \to 0$, $\partial_\sigma \widetilde{\mathrm{e}^{-\mu_s r}}$).

What the scan found. The first version wrote the screened kernels as Mathematica returns them, with erfcx times polynomials. The scan rejected them: relative errors up to 3×10^{-7} for $l = 4$ at $\nu = 10^{-3}$, $\mu_s = 0.3$, the cancellation of the upward recursion described after the $\mathcal{Q}_k$ lemma (§7.5). The generator now keeps the moments as `Qm`; with the derivative rules above the transfer lemma turns every screened kernel into a combination of $\mathcal{Q}_k(a', \mu_s)$ with coefficients of one sign, and the C++ computes the $\mathcal{Q}_k$ by `detail::screened_moments` (upward for $x < \frac{1}{2}$, downward ratios otherwise). The smeared spin–orbit kernels by integration by parts lost up to five digits for $a \gg \sigma^2$; the spin–orbit transfer lemma (7.23) replaced them. Both changes are derived and illustrated in D03.

The header. One local and one screened kernel, exactly as generated:

```
// <l nu1| erf(tau r)/r |l nu2>: one Gaussian term of the smeared running Coulomb potential G~ of GI. [l =
    ↪ 1]
// closed form vs numerical integral at {nu1 -> 2.60, nu2 -> 0.600, tau -> 0.900}: rel. dev. 0
// machine-precision scan over 196 points: max rel. error 6.7e-16, max error / size 8.2e-16
inline double erf_over_r_l1(double nu1, double nu2, double tau) {
  const double a = nu1 + nu2;
  const double nn = gen::norm(1, nu1) * gen::norm(1, nu2);
  return nn * ((3*a*tau + 2*std::pow(tau,3))/(4.*std::pow(a,2)*std::pow(a + std::pow(tau,2),1.5)));
}

// <l nu1| [exp(-mu r)]~'(r)/r |l nu2>: scalar spin-orbit of Model D, by the spin-orbit transfer lemma with
    ↪ J_f(a') = -mu Q_3(a', mu). [l = 2]
// closed form vs numerical integral at {nu1 -> 1.70, nu2 -> 2.30, sigma -> 2.50, mu -> 1.90}: rel. dev. 0
    ↪ (smeared function itself checked against the convolution integral: 1.4e-24); spin-orbit transfer
    ↪ lemma vs integration by parts: 0
// machine-precision scan over 441 points: max rel. error 1.3e-15, max error / size 7.5e-16
inline double smeared_exp_so_l2(double nu1, double nu2, double sigma, double mu) {
  const double a = nu1 + nu2;
  const double nn = gen::norm(2, nu1) * gen::norm(2, nu2);
  const double ap = (a*std::pow(sigma,2))/(a + std::pow(sigma,2));
```

```
  double Q[6];
  detail::screened_moments(5, ap, mu, Q);
  return nn * (-((mu*Q[5]*std::pow(sigma,9))/std::pow(a + std::pow(sigma,2),4.5)) - (5*mu*Q[3]*std::pow(
    ↪ sigma,5))/(2.*std::pow(a + std::pow(sigma,2),3.5)));
}
```

Compare the first with §7.5: $\Gamma(\frac{5}{2})/\sqrt{\pi} = \frac{3}{4}$ times $\int_0^\tau (a+t^2)^{-5/2}\mathrm{d}t = \frac{\tau(3a+2\tau^2)}{3a^2(a+\tau^2)^{3/2}}$. The second is (7.23) with $l = 2$ and $J_f = -\mu_s \mathcal{Q}_3$: both terms have the same sign. The kernels for one l sit in `qgem::gen::detail`; you call the dispatchers, which take l as an `int` and throw `std::out_of_range` beyond the generated $l_{\max}$. `erf_tensor(l1, l2, ...)` handles $l_2 = l_1$ and $l_2 = l_1 \pm 2$ (the swapped order by symmetry). A matrix function of §11.8 is then a loop:

```
#include "generated/qgem_kernels.hpp"
Mat erf_over_r(const Basis& b, double tau) {          // me/radial.cpp
  Mat m(b.size(), b.size());
  for (int i = 0; i < b.size(); ++i)
    for (int j = 0; j < b.size(); ++j) m(i, j) = qgem::gen::erf_over_r(b.L, b.nu(i), b.nu(j), tau);
  return m;
}
```

The comment above each function in the header gives its own check and scan numbers.

The tests. `tests/test_generated_kernels.cpp` holds, per family, golden values (30-digit evaluations; the first point of each kernel is the one where the closed form was checked against `NIntegrate`; tolerance 10^{-12} or the measured machine error, whichever is larger) and a central-difference check of every derivative; then the screened moments at x from 0.24 to 16, finiteness of every kernel on the whole scan grid, the spin table against the closed forms of §3.5, $S_{nn} = 1$, the swap symmetry of the S–D tensor kernel (`erf_tensor(2, 0, ...)` against `erf_tensor(0, 2, ...)`), and the dispatch errors. With GCC 16.2.1 (`-O3 -Wall -Wextra -Wpedantic`) the header compiles without warnings and all 35 tests (466 `EXPECT` statements, several of them in loops over the scan grid) pass.

CMake. Add two lines to the `CMakeLists.txt` of §11.4:

```
include(mathematica/QGEMKernels.cmake)        # after project(...)
...
qgem_use_generated_kernels(qgem qgem_tests)    # after add_library / add_executable
```

The generated files are committed, so an ordinary build needs no Mathematica. To regenerate:

```
cmake -S . -B build -DQGEM_REGENERATE_KERNELS=ON [-DQGEM_KERNEL_LMAX=4]
cmake --build build                  # runs wolframscript if the generator or the engine changed
```

Verified on your machine with the full file of §11.4: the default build configures without warnings and `ctest` passes (37 tests); with the option on, touching the generator reran it (1 min for $l_{\max} = 1$), rebuilt and passed, and a second build did not rerun it. A failed check stops the build (exit code 1; the report lists the failures and no C++ is written). Should Mathematica abort a run itself, `wolframscript` may still return 0, so the CMake command deletes the report before the run and requires it afterwards: a run that wrote no report stops the build too.

The derivation notebooks. They are in `mathematica/derivations/`, each evaluated, with outputs saved; Evaluation → Evaluate Notebook reruns one in 20–60 s. Every section has the same pattern: what we want, the idea (by hand), Mathematica does it, what it means, check.

notebook	content
`D01_Engine_and_Basic_Kernels`	engine, normalization, master integral, S, T (two derivations), r^k for any l_1, l_2, Gaussians and δ_σ, Feynman–Hellmann, Test 1 spectrum and wave functions
`D02_Coulomb_and_Spin_Radial_Functions`	$\mathrm{erf}(\gamma r)/r$ from the Fourier transform, $1/\tau^2 = 1/\gamma^2 + 1/\sigma^2$, $\langle \mathrm{erf}(\tau r)/r \rangle$ and the I_j recursion, ∂_τ, GI chain rule, contact/spin–orbit/tensor functions and their limits, ${}^3P_J$ example
`D03_Confinement_Linear_Screened_Smeared`	screened moments and their numerics, radial convolution, $\tilde{r}$, $\widetilde{e^{-\mu_s r}}$, both transfer lemmas, cost of cancellation
`D04_Spin_Angle_Factors_and_Channels`	coupled states, $\boldsymbol{L}\cdot\boldsymbol{S}$, S_{12} (diagonal, $L \pm 2$), sum rules, $\boldsymbol{L}\cdot(\boldsymbol{S}_1 - \boldsymbol{S}_2)$, J^{PC} channels, a coupled ${}^3S_1$–${}^3D_1$ calculation
`D05_Momentum_Space_and_GI_Factors`	$\tilde{R}_l(p)$, the Gauss–Laguerre reduction, Golub–Welsch, $\langle E_1 + E_2 \rangle$, GI factors and derivatives, closure for fWf
`D06_Complex_Range_Gaussians`	analytic continuation, cos/sin rules, a spectrum: complex ranges win from $4S$ upward
`D07_Kernel_Generator`	the generator step by step on one kernel, reading the header, tests and report, adding a kernel (smeared $1/r^2$)

13.6 Pitfalls (each one met while preparing this section)

1. *Global `$Assumptions` leak into built-ins.* With `m > 0` in `$Assumptions`, the Gaunt code above (where `m` is a `Table` iterator running over negative values) printed `Refine::fas`. Use local assumptions.

2. *Stale global values.* After `z = 13/10 (1 + I 7/10)` three cells earlier, the $\partial_i\partial_j V$ check with `xs = {x, y, z}` failed with `D::ivar` (z was a number). `ClearAll` what you no longer need, or use `Module`/`Block`.

3. *Generalized eigenvalues are not sorted.* `Eigenvalues[{H, S}]` returns them ordered by decreasing absolute value. Always `Sort` (and keep eigenvectors aligned with `Ordering`).

4. *Precision.* `NIntegrate[..., WorkingPrecision -> 30]` with machine-number inputs (`0.7`) prints `NIntegrate::precw` and is not more accurate; use exact inputs (`7/10`). Build ill-conditioned matrices like **S** from exact numbers and convert with `N[..., 40]`.

5. *Default fit errors and bands* (§13.4.11).

6. *`Cross` on matrix-valued vectors* does not work; write ϵ_{ijk} sums with `LeviCivitaTensor`.

7. *Spherical measure.* In a hand-written angular integral, forget $\sin\theta$ and Gauss's law gives $-2\pi^2$ instead of -4π. Prefer `Laplacian[..., "Spherical"]` or the Gaunt route.

8. *Unphysical angular-momentum arguments.* `ThreeJSymbol`, `ClebschGordan` return 0 with a message (`ClebschGordan::phy`, `::tri`) for $m_1 + m_2 + m_3 \neq 0$ or non-triangular j's. Guard with `If`, as in `C1` above, so that real problems are not drowned in noise.

9. *Internal messages with correct results.* `Integrate`, `DSolveValue`, `FullSimplify` sometimes print messages (`MinValue::lpsub`, `Part::pkspec1`, `FullSimplify::time`) and still return the right answer. Never trust or distrust a result because of a message: check it.

10. *Reserved names.* `E`, `I`, `N`, `C`, `D`, `K`, `O`, `Gamma`, `Beta` are built-in. Use `En`, `kap`, `nu`. Avoid `Subscript` for variables (`m1`, not m_1 typed as a subscript).

11. *A line break can end a statement.* In a notebook cell or a `.wl` file, a newline after a *complete* expression starts a new one. Breaking `f[x_] := a + b` before the `+` defines `f[x_] := a` and then evaluates `+ b` on its own, silently. This happened here: a tensor operator split before its `- If[...]` gave $\langle S_{12} \rangle = (-3, 3, \frac{3}{5})$ instead of $(-4, 2, -\frac{2}{5})$. Break lines after an operator or inside an open bracket, never before an operator.

12. `=` *versus* `:=`. `f[x_] = expr` evaluates `expr` once, with whatever global values exist at that moment; `:=` evaluates at every call. Use `f[x_] := f[x] = ...` for memoization (as in `C1`).

13. *Symbolic angular integrals are slow.* Integrating $Y^*_{L'm'}\, S_{12}\, Y_{Lm}$ symbolically took $\sim 7\,\mathrm{s}$ per $L = 2$ block; the Gaunt construction does all $L \le 3$ blocks in 0.3 s.

14. *Do not evaluate notebooks through the front end from a script.* Calling `NotebookEvaluate` inside `UsingFrontEnd` from a kernel without a running front end can hang for minutes. Evaluate notebooks interactively, or run `.wl` files with `wolframscript`. Rendering a finished notebook is fine: `UsingFrontEnd[Export["x.pdf", nb]]` from `wolframscript` works and is a quick way to proofread.

15. `|` *binds tighter than* `->`. `a -> b | c -> d` means `a -> ((b | c) -> d)`. A pattern for bold-or-italic text written this way matched nothing, silently. Parenthesize: `(a -> b) | (c -> d)`.

16. `Cases` *never looks at level 0.* In the generator, `Cases[J, Qm[k_, _, _] :> k, Infinity]` found nothing for the screened kernels, whose `J` is a single `Qm[n, a, mu]`: levels start at 1. Wrap the expression, `Cases[\{J\}, ...]`, or use the level specification `\{0, Infinity\}`.

17. *Machine overflow is invisible in Mathematica.* When a machine number overflows, Mathematica silently continues with arbitrary-precision numbers: `Exp[800.]` gives 2.7×10^{347}. In C++ the same expression gives `inf`, and `inf/inf` gives `NaN`. A machine-precision scan in Mathematica therefore cannot detect overflow; the generated tests evaluate every kernel on the scan grid in C++ and require finite results.

18. `Simplify` *with* `z > 0` *may break the complex continuation.* A form valid for positive exponents need not be the analytic continuation (principal branches of powers and roots). Check every complex-range formula at complex points (D06).

19. *One kernel at a time.* On a single-machine licence a second kernel, for example `wolframscript` while a notebook is open, can fail to start with "No valid password found". Run the generator and batch evaluations one after another, with the notebook kernel quit.

A Special functions and conventions

Gamma function. $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(\tfrac12) = \sqrt{\pi}$, $\Gamma(n+1) = n!$, $\Gamma(l+\tfrac32) = \frac{(2l+1)!!}{2^{l+1}}\sqrt{\pi}$; $(2l+1)!! = 1\cdot 3\cdots(2l+1)$.

Error functions. $\operatorname{erf} x = \frac{2}{\sqrt{\pi}}\int_0^x \mathrm{e}^{-t^2}\mathrm{d}t$, $\operatorname{erfc} x = 1 - \operatorname{erf} x$, $\operatorname{erfcx} x = \mathrm{e}^{x^2}\operatorname{erfc} x$. Series $\operatorname{erf} x = \frac{2}{\sqrt{\pi}}(x - \frac{x^3}{3} + \frac{x^5}{10} - \dots)$; asymptotics $\operatorname{erfc} x \simeq \frac{\mathrm{e}^{-x^2}}{x\sqrt{\pi}}(1 - \frac{1}{2x^2} + \dots)$, $\operatorname{erfcx} x \simeq \frac{1}{x\sqrt{\pi}}$; $\operatorname{erfc}(-x) = 2 - \operatorname{erfc} x$; $\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{erf} x = \frac{2}{\sqrt{\pi}}\mathrm{e}^{-x^2}$.

Spherical harmonics. $\int Y^*_{l'm'} Y_{lm}\mathrm{d}\Omega = \delta_{ll'}\delta_{mm'}$, $Y_{00} = 1/\sqrt{4\pi}$, $\boldsymbol{L}^2 Y_{lm} = l(l+1)Y_{lm}$; $r^l Y_{lm}(\hat{r})$ is a harmonic homogeneous polynomial of degree l.

Spherical Bessel functions. $j_0(x) = \frac{\sin x}{x}$, $j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}$, $j_2(x) = (\frac{3}{x^3} - \frac{1}{x})\sin x - \frac{3}{x^2}\cos x$; $j_l(x) \simeq \frac{x^l}{(2l+1)!!}$ for small x. Rayleigh expansion: $\mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{r}} = 4\pi\sum_{lm}\mathrm{i}^l j_l(pr) Y^*_{lm}(\hat{p}) Y_{lm}(\hat{r})$; it implies the partial-wave transform $\tilde{R}_l(p) = \sqrt{2/\pi}\int r^2 j_l(pr) R_l(r)\mathrm{d}r$ (up to the phase $(-\mathrm{i})^l$).

Gaussian Fourier transforms.

$$\int \mathrm{d}^3 r\, \mathrm{e}^{-\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}} \mathrm{e}^{-ar^2} = \Big(\frac{\pi}{a}\Big)^{3/2} \mathrm{e}^{-q^2/4a}, \qquad \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r} - sq^2} = (4\pi s)^{-3/2} \mathrm{e}^{-r^2/4s}.$$

B Table of integrals used

Integral	Result
$\int_0^\infty r^n \mathrm{e}^{-ar^2}\mathrm{d}r$	$\dfrac{\Gamma(\frac{n+1}{2})}{2a^{(n+1)/2}}$
$\int_0^\infty r^{2\ell+2}\mathrm{e}^{-ar^2}\frac{\mathrm{erf}(\tau r)}{r}\mathrm{d}r$	$\dfrac{\Gamma(\ell+\frac{3}{2})}{\sqrt{\pi}}\displaystyle\int_0^\tau \frac{\mathrm{d}t}{(a+t^2)^{\ell+3/2}}$
$\int_0^\tau (a+t^2)^{-3/2}\mathrm{d}t$	$\dfrac{\tau}{a\sqrt{a+\tau^2}}$
$\int_0^\infty r^k \mathrm{e}^{-ar^2-\mu r}\mathrm{d}r \equiv \mathcal{Q}_k$	$\mathcal{Q}_0 = \frac{1}{2}\sqrt{\pi/a}\,\mathrm{erfcx}\,\frac{\mu}{2\sqrt{a}}$, recursion in §7.5
$\int_{-r}^\infty x^2 \mathrm{e}^{-\sigma^2 x^2}\mathrm{d}x$	$-\dfrac{r\mathrm{e}^{-\sigma^2 r^2}}{2\sigma^2} + \dfrac{\sqrt{\pi}}{4\sigma^3}\big[1+\mathrm{erf}(\sigma r)\big]$
$\int_0^\infty u\,\mathrm{e}^{-\sigma^2(u-c)^2}\mathrm{d}u$	$\dfrac{\mathrm{e}^{-\sigma^2 c^2}}{2\sigma^2} + \dfrac{\sqrt{\pi}c}{2\sigma}\,\mathrm{erfc}(-\sigma c)$
$\int \mathrm{d}\Omega'\, \mathrm{e}^{2\sigma^2 \boldsymbol{r}\cdot\boldsymbol{r}'}$	$\dfrac{\pi}{\sigma^2 r r'}\big(\mathrm{e}^{2\sigma^2 r r'} - \mathrm{e}^{-2\sigma^2 r r'}\big)$
$\int \frac{\mathrm{d}^3 q}{(2\pi)^3}\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{r}}\frac{4\pi}{q^2}\mathrm{e}^{-q^2/4\gamma^2}$	$\dfrac{\mathrm{erf}(\gamma r)}{r}$
$\int_0^\infty \frac{\sin qr}{q}\mathrm{d}q$	$\dfrac{\pi}{2}$

C Addendum (v1.0.0): grading, decays and Bayesian inference

The program as released grades its fits by standard methods; the iterated floor of Alg. 11.10 remains a diagnostic.

Theory floor. With residuals Δ and $V(\sigma) = V_{\mathrm{exp}} + \sigma^2 W$, the floor solves $\Delta^\top V(\sigma)^{-1}\Delta = N - p$ with the full PDG covariance: the Paule–Mandel random-effects estimator. Its interval is the χ^2 test inverted (the Q-profile interval): every σ with $\chi^2_{N-p}(\frac{1-\mathrm{CL}}{2}) \le \Delta^\top V(\sigma)^{-1}\Delta \le \chi^2_{N-p}(\frac{1+\mathrm{CL}}{2})$.

Grades. One mass error $\sigma_\star$ per system is declared for all models: the Paule–Mandel estimator pooled over the physical set, $\sum_m \Delta_m^\top V_m(\sigma_\star)^{-1}\Delta_m = \sum_m \nu_m$. Each model is refitted at $\sigma_\star$ and graded by Pearson's χ^2/ν and $P(\chi^2_\nu \ge \chi^2)$. The measured annihilation widths of the fitted states are a held-out benchmark: with the NLO prediction Γ of the mass-only fit, $\chi^2_\Gamma = (\Gamma_{\mathrm{PDG}} - \Gamma)^\top (V_{\mathrm{exp}} + V_{\mathrm{th}} + J_\Gamma C J_\Gamma^\top)^{-1}(\Gamma_{\mathrm{PDG}} - \Gamma)$.

Widths. $\Gamma(V \to e^+e^-) = 4\alpha^2 e_Q^2 A^2/M^2$ with $A = R_S(0) + \frac{5}{2\sqrt{2}\,m_Q^2}R_D''(0)$ (S–D mixed states included); NLO multiplies by $1 - \frac{16\alpha_s}{3\pi}$ (e^+e^-, ${}^3P_2$), $1 - \frac{(20-\pi^2)\alpha_s}{3\pi}$ (${}^1S_0$), $1 + (\frac{\pi^2}{3} - \frac{28}{9})\frac{\alpha_s}{\pi}$ (${}^3P_0$),

with the $\overline{\text{MS}}$ coupling (two-loop, $\alpha_s(M_Z) = 0.1180$) at $\mu = M/2$. Theory error: half the band $\mu = M/4 \ldots M$ (correlated within a process) $\oplus \langle v^2 \rangle \Gamma$.

Bayesian inference. The posterior of $(u, \ln\sigma)$ is $\propto \exp[-\frac{1}{2}(\chi^2 + \ln\det V)]$ with flat priors in u and in $\ln\sigma$ on $[10^{-5}, 0.05]$ GeV; posterior predictive p-values use $T = \chi^2$ and $T = \max_i |z_i|$, both exact per draw.

Joint fit. $c\bar{c}$ and $b\bar{b}$ share the string tension (and, optionally, the coupling normalisation or the GI slope s); each floor solves $\chi^2_s = N_s - \sum_{i \in s} h_{ii}$ with the leverages h of the joint fit, and the universality test is $\chi^2_{\text{joint}} - \chi^2_{\text{separate}} \sim \chi^2_k$ at the separate floors.

Selection. A model wins only if no test is rejected at $\alpha = 0.05$ after Holm's correction (masses and decays, both systems) and it passes the physics checks: radial nodes, $m_b - m_c$ against heavy-quark symmetry, $\sqrt{b}$ against the lattice, the $1P$ centre of gravity, convergence, $\langle v^2 \rangle < 0.5$; among those, the lowest $\chi^2_{\text{masses}} + \chi^2_{\text{decays}} + 2p$.

References

[1] E. Eichten, K. Gottfried, T. Kinoshita, K. D. Lane, T.-M. Yan, Phys. Rev. D **17**, 3090 (1978); Phys. Rev. D **21**, 203 (1980). [Cornell model]

[2] S. Godfrey, N. Isgur, Phys. Rev. D **32**, 189 (1985). [GI model, Model C]

[3] S. Capstick, N. Isgur, Phys. Rev. D **34**, 2809 (1986). [relativized baryons; same conventions]

[4] T. Barnes, S. Godfrey, E. S. Swanson, Phys. Rev. D **72**, 054026 (2005). [NR and GI charmonium side by side]

[5] E. Eichten, F. Feinberg, Phys. Rev. D **23**, 2724 (1981); D. Gromes, Z. Phys. C **26**, 401 (1984). [spin-dependent potentials, Gromes relation]

[6] W. Lucha, F. F. Schöberl, D. Gromes, Phys. Rep. **200**, 127 (1991). [review: Breit–Fermi reduction, Lorentz structure]

[7] V. B. Berestetskii, E. M. Lifshitz, L. P. Pitaevskii, *Quantum Electrodynamics* (Pergamon), sections on the Breit equation.

[8] B.-Q. Li, K.-T. Chao, Phys. Rev. D **79**, 094004 (2009); Commun. Theor. Phys. **52**, 653 (2009). [screened potential, Model B]

[9] W.-J. Deng et al., Phys. Rev. D (2017) [charmonium and bottomonium in linear and screened NR models; arXiv 2016].

[10] J.-Z. Wang et al., "Higher bottomonium zoo", Eur. Phys. J. C (2018). [modified GI, Model D]

[11] W.-X. Zhang et al., "Truncation of the radial ladder in heavy quarkonia" (2026). [screened GI solved with GEM]

[12] J. L. Richardson, Phys. Lett. B **82**, 272 (1979).

[13] D. V. Shirkov, I. L. Solovtsov, Phys. Rev. Lett. **79**, 1209 (1997).

[14] G. S. Bali et al., Phys. Rev. D **71**, 114513 (2005); J. Bulava et al., Phys. Lett. B **793**, 493 (2019). [string breaking on the lattice]

[15] I. W. Herbst, Commun. Math. Phys. **53**, 285 (1977). [boundedness of $\sqrt{p^2 + m^2} - \kappa/r$]

[16] M. Kamimura, Phys. Rev. A **38**, 621 (1988). [origin of GEM]

[17] E. Hiyama, Y. Kino, M. Kamimura, Prog. Part. Nucl. Phys. **51**, 223 (2003). [GEM review]

[18] E. Hiyama, M. Kamimura, Front. Phys. **13**, 132106 (2018). [GEM review incl. complex-range Gaussians]

[19] D. Ebert, R. N. Faustov, V. O. Galkin, Phys. Rev. D **67**, 014027 (2003); Eur. Phys. J. C **71**, 1825 (2011).

[20] S. F. Radford, W. W. Repko, Phys. Rev. D **75**, 074031 (2007).

[21] N. Brambilla et al., Eur. Phys. J. C **71**, 1534 (2011). [quarkonium review]

[22] E. W. Hobson, *The Theory of Spherical and Ellipsoidal Harmonics* (Cambridge, 1931). [Hobson's theorem]

[23] K. W. H. Stevens, Proc. Phys. Soc. A **65**, 209 (1952). [operator equivalents]

[24] A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton, 1957). [reduced matrix elements, $6j$ symbols]

[25] E. A. Hylleraas, B. Undheim, Z. Phys. **65**, 759 (1930); J. K. L. MacDonald, Phys. Rev. **43**, 830 (1933). [HUM theorem]

[26] K. Fan, Proc. Natl. Acad. Sci. USA **35**, 652 (1949). [trace minimum principle]

[27] T. Kato, Commun. Pure Appl. Math. **10**, 151 (1957). [cusp condition]

[28] R. Van Royen, V. F. Weisskopf, Nuovo Cimento A **50**, 617 (1967). [leptonic widths and decay constants]

[29] W. Kwong, P. B. Mackenzie, R. Rosenfeld, J. L. Rosner, Phys. Rev. D **37**, 3210 (1988). [QCD corrections to annihilation decays]

[30] E. J. Eichten, C. Quigg, Phys. Rev. D **52**, 1726 (1995). [wave functions at the origin]

[31] S. J. Brodsky, R. F. Lebed, Phys. Rev. Lett. **102**, 213401 (2009). [true muonium]

[32] Particle Data Group, *Review of Particle Physics*, chapters "Statistics" and "Quantum chromodynamics"; pdgLive for masses.

[33] K. Levenberg, Q. Appl. Math. **2**, 164 (1944); D. W. Marquardt, J. SIAM **11**, 431 (1963).

[34] W. H. Press et al., *Numerical Recipes*, 3rd ed. (Cambridge, 2007), chapter on modelling of data.

[35] N. Metropolis et al., J. Chem. Phys. **21**, 1087 (1953); W. K. Hastings, Biometrika **57**, 97 (1970).

[36] A. Gelman, D. B. Rubin, Stat. Sci. **7**, 457 (1992); G. O. Roberts, A. Gelman, W. R. Gilks, Ann. Appl. Probab. **7**, 110 (1997).

[37] B. Efron, Ann. Stat. **7**, 1 (1979). [bootstrap]

[38] S. S. Wilks, Ann. Math. Stat. **9**, 60 (1938); H. Chernoff, Ann. Math. Stat. **25**, 573 (1954). [likelihood-ratio tests, boundary case]

[39] H. Akaike, IEEE Trans. Autom. Control **19**, 716 (1974); G. Schwarz, Ann. Stat. **6**, 461 (1978). [AIC, BIC]